

TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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GUIDE TO THE TECHNICAL SUPPLEMENT

This supplement is a current part of *Neural Bellman Operators*. It contains the general verification results, the complete capital-model proofs, the paired-transfer and occupation accounts, observation and implementation arguments, and the full computational evidence. The separate *Economic Applications* companion contains the full original application models and their derivations and proof arguments. All three scientific documents have live cross-references.

The first part develops the mathematical results by subject. The computational part preserves the previous complete experiment and its post-freeze analytical refinement together, followed by the current scientific records. Evidence from different simulation banks and confidence families retains its original scope. In particular, the previous dimension-ten paired refinement is not described as a prospectively fixed experiment, and fresh observations on the same policies are not new training streams.

Supplementary equations and mathematical results use the prefix S. The applications companion uses A. Every one of the 285 labels in the reviewed R15 article and supplement is present in one of the three current scientific documents. The source and proof preservation map in Section [S.29](#) includes complete proof sections as well as explicit proof environments.

S.1. RETAINED THEORY AND COMPARISON ACCOUNTS

The following complete sections retain the earlier theory summaries, computational design, conditional risk analysis, and catalogue closure. Their statements, proofs, labels, adverse findings, and confidence accounts are unchanged. They are placed together here to keep the article's main sequence on the economic deci-

sion, policy certificate, and executed services. The original continuous-time and application derivations follow in full.

S.1.1. *Evaluation, feasible improvement, and approximation*

The algorithm produces a continuation approximation and a feasible policy. Its economic interpretation requires an error account for each. The general results in technical Section S.5 bound the consequence of a fixed-policy evaluation error and an incomplete feasible maximization under the stated comparison and boundary conditions. The positive-weight Bellman formulation also provides a monotone approximation route when classical derivatives are unavailable. The complete statements and proofs remain current in the supplement.

The following capital economy permits a more operational account. Its payoff identity follows the implemented action through the original diffusion. Its finite Bellman identity measures continuation error along the candidate's actual action change. This specialization makes it possible to compare a learned continuation with a cached simulation continuation using the same economic decision and a common assessment design.

S.1.2. *A common continuation for different economic decisions*

The capital economy permits a direct study of continuation reuse. Consider a finite-period decision in which a temporary consumption weight, consumption externality, or consumption cap changes at the current date. Subsequent preferences, technology, Gaussian innovations, terminal dispersion cost, and reference consumption remain fixed. For task $\theta = (w, \eta, u)$, the action-dependent payoff is

$$Q_\theta(y, a) = A_0\{w\overline{\log a} - \eta\bar{a}^2/2\} - W_0\bar{a} + S\{x_a(y)\},$$

$$x_a(y) = y + h\{c1 + f(y) - a\}, \quad S(x) = \mathbb{E}F(x; \xi), \quad a \in [\underline{a}, u]^d. \quad (\text{S.1})$$

Here a known current term independent of a is suppressed. The random variable $F(x; \xi)$ includes the first Gaussian innovation at postdecision mean x , all subsequent reference rewards, and the terminal payoff. The stored period weights and innovation amplitudes define this finite capital problem. Its decision values have their

own interpretation, separate from the continuous-diffusion value in the preceding payoff theorem.

Proposition S.14 establishes that S is identical across these tasks. NBO therefore fits one scalar continuation and queries its gradient when the current economic problem changes. This reuse has a precise information boundary. A stored payoff or costate remains an exact rollout observation at the argument where it was computed. Reusing the same innovation array at another argument requires reevaluating its state recursion and payoff. The Raw comparators may cache valid observations and innovations under the same rule. The comparison concerns the accuracy and complete cost of these representations of the common continuation.

The scalar representation also imposes structure on costate fitting. Fix postdecision arguments and positive weights before drawing rollout labels. Stack the true gradients, multiplied by the square roots of the weights, in q . Let the corresponding labels be $z = q + \varepsilon$, with $\mathbb{E}\varepsilon = 0$ and finite covariance Σ . For differentiable scalar features fixed independently of these labels, let P be the Euclidean orthogonal projection onto their stacked weighted gradient columns. A least-squares scalar critic has fitted gradients Pz , and

$$\mathbb{E}\|Pz - q\|^2 - \mathbb{E}\|z - q\|^2 = \|(I - P)q\|^2 - \text{tr}\{(I - P)\Sigma\}. \quad (\text{S.2})$$

Proposition S.15 proves the identity without requiring independent coordinates or Gaussian label noise. The variance removed outside the gradient dictionary is the statistical benefit; approximation error is its cost. When the true gradient belongs to the dictionary and the removed variance is positive, the scalar critic strictly reduces this risk. The same fitted object can then serve every task in (S.1).

This calculation gives a conditional statistical account of the scalar critic. A fully trained nonlinear network generally chooses its features using its labels and need not coincide with this fixed projection. A Raw procedure using the same projection has the same identity. The economic comparisons consequently evaluate all five complete procedures on independent payoff observations, including their approximation errors, action choices, and computational work. Technical Section S.12 gives

the full proofs, direct conditional payoff range, and separate accuracy account over an entire scalar intervention interval.

S.1.3. *Signed continuation assessment*

The occupation account can retain the sign of the continuation prediction error. At a sampled candidate state and date, evaluate the reference continuation from the candidate and reference postdecision states using the same future innovations. Add the current reward difference before forming the interval. The conditional expectation is the reference Bellman advantage; averaging over the candidate's occupation distribution gives the finite policy gain.

Proposition S.12 makes this statement precise. The predicted improvement and its signed continuation error cancel sample by sample. A lower endpoint for their combined statistic therefore need not pay separately for the full range of the fitted predictor. Proposition S.13 supplies an economic range using the action tube, bounded production derivatives, and antithetic future paths. It retains the original clipping and arithmetic account and the separate transfer to the continuous-time payoff. The complete derivation is in technical Section S.11.7.

This calculation addresses the precision of an economic identity for the selected policy. The comparison of a learned critic with a Raw continuation remains a distinct empirical question. The repeated-decision design makes that comparison directly, including the cost and benefit of reusing each continuation representation.

S.2. COMPUTATIONAL DESIGN

The experiments distinguish a decision problem, a complete algorithm, a selected policy, and an independent payoff bank. Each confidence statement conditions on the objects fixed before its bank. The earlier complete method distribution contains sixteen equally weighted algorithm streams at each dimension. Generating new payoff paths for those policies does not create new training configurations. A finite repeated-decision economy likewise has its own value target and is not silently identified with the continuous diffusion.

S.2.1. *Reusable continuation and new decisions*

The central comparison studies the common continuation in (S.1) across twelve temporary preference, externality, and consumption-cap tasks. Each task is evaluated at thirty-two fixed states, giving 384 equally weighted queries in each dimension. Four capital calibrations retain the original low-volatility primitives, a two-year quarterly horizon, a four-year half-year horizon, and an intermediate three-year quarterly horizon. The two- and four-year calibrations each have eight decision periods. The new model and its stored period coefficients define a finite repeated-decision economy; no continuous-time transfer is asserted for these menu values.

Five complete procedures share the permitted continuation information: the NBO scalar critic, an unrestricted vector costate predictor, a materialized Raw actor, a direct-policy actor, and query-level Raw sample-average optimization. The vector predictor uses the same postdecision labels as NBO. The Raw actor retains labels at reference actions, whereas direct policy and query-level Raw may retain innovations and reevaluate the continuation when their actions change. Thus the comparisons assess complete representation and optimization procedures, with all approximation and action-selection errors included.

Each calibration and dimension has sixteen streams fixed before confirmation. All methods retain three work stages, using 128, 512, and 2,048 replay states for the fitted procedures and 4, 16, and 64 antithetic paths for query-level Raw. No stage is selected on the final payoff bank. The 216 primary statements cover every method's reference gain and every direct NBO contrast at all stages and cells. Independent confirmation uses 512 antithetic pairs per query and family error 0.018. The fixed materiality and equivalence margins are both 10^{-4} ; a positive interval can also lie inside the equivalence band. A separate family with error 0.002 assesses 128 scalar-intervention accuracy bounds. Technical Section S.23 gives the complete calibration, fitting, inference, and work contracts.

S.2.2. *Prospective confirmation of previously selected policies*

The confirmation retains all 128 frozen policies from the previous experiment: four methods, sixteen declared streams, and two dimensions. Before generating its new bank, it fixes the original paired-transfer theorem, outward constants, clipping account, reporting code, and an independent family error allocation of 0.01. The new bank has 8,192 paths on 2,048 decision cells. It is independent of the original assessment bank and of the completed training. Its target is the mean payoff of the original finite policy distribution and its specified direct comparisons.

The deterministic transfer is audited separately in technical Section S.11.6. The confirmation retains the original protected constants. It does not substitute a smaller high-precision diagnostic for an interval enclosure. Earlier online stopping decisions remain those made on their original checkpoint banks, irrespective of later confirmation.

S.2.3. *A strengthened neural-HJB implementation*

This separate continuous-economy family trains new policies at four capital calibrations—high volatility, the original low volatility, a longer horizon, and joint economic stress—in dimensions ten and fifty. Each cell has eight new streams, giving 64 trials, 256 method-group processes, and 320 policy outputs. A new 8,192-path, 2,048-cell bank assesses each trial. All 648 conditional stream and finite-stream-mean events share error allocation 0.01. The stress design changes coupling, terminal dispersion preferences, initial heterogeneity, and the feasible actor radius together; its comparison does not isolate the effect of any one primitive. This family is distinct from both the fixed-policy replication and the sixteen-stream finite menu.

The stronger comparator evaluates the diffusion trace exactly at dimensions ten and fifty. Four learning-rate and collocation configurations are fixed in advance. Each uses Adam followed by L-BFGS; independent residual diagnostics select the configuration and its checkpoint. The final payoff bank does not select the fitted network. The implemented policy records both an optimized action root and a distilled actor, with separately reported payoff and work for each output. Complete work includes every fitted configuration, selection, distillation, action

deployment, and verification. This is a fixed-envelope comparison: a stop based on independent residual checks is not an online economic-target stopping rule. The previous stochastic-trace implementation and its complete outcomes remain in technical Section S.18.

S.2.4. Precision of the signed occupation account

The signed Bellman experiment uses independent candidate-occupation and continuation banks. Its protected statistic combines the current reward and paired future-return difference before clipping and inference. Its allocation is fixed using the proved economic range and a power calculation. It reports the numerical transfer and every remaining allowance alongside the stochastic radius. The result is interpreted as an assessment of the selected policy’s Bellman account; direct critic–Raw comparisons retain their own payoff, continuation-risk, and work estimands.

S.2.5. Economic and inferential scope

The confidence family is stated separately for each new experiment. A union of results from different families is asserted only with its explicit allocation. Previous publication evidence retains its original probability account. All results retain their economic parameters, state population, implemented controller, and complete cost boundary. A direct comparison interval that spans the declared margin remains unresolved. A protected lower gain over the schedule, a bound on full-class regret, a finite-grid residual, and an empirical refinement discrepancy answer different economic questions.

S.3. CONTINUATION RISK AT THE ECONOMIC ACTION

The payoff comparison establishes the economic performance of the complete NBO procedure. To assess the fitted continuation itself, we compare its prediction with a cached simulation predictor at the same action pair. For query j , let x_j^+ be the postdecision state of the final NBO action and x_j^0 that of the reference action. The object to be predicted is $c_j = S(x_j^+) - S(x_j^0)$ in the finite economy of Section S.1.2. The scalar network supplies $p_{Nj} = \widehat{S}_N(x_j^+) - \widehat{S}_N(x_j^0)$; cached Raw supplies p_{Rj} . Both

are evaluated at these same states. Conditional on the complete training record, these states, and the realized prediction caches, define

$$R_M = J^{-1} \sum_{j=1}^J (p_{Mj} - c_j)^2, \quad M \in \{N, R\}. \quad (\text{S.3})$$

Conditioning permits nonlinear fitting and learned features. It does not replace the fitted network by a fixed linear dictionary.

A fresh common-future-path contrast Z_{jk} has conditional mean c_j . The difference of its two squared prediction errors is

$$T_{jk} = (p_{Nj} - p_{Rj})(p_{Nj} + p_{Rj} - 2Z_{jk}). \quad (\text{S.4})$$

Its conditional mean averages to $R_N - R_R$, and its conditional variance is $4(p_{Nj} - p_{Rj})^2 \text{Var}(Z_{jk})$. The common squared-noise term cancels exactly. Centering the remaining continuation residual by its known terminal quadratic component gives a bounded statistic with explicit clipping and arithmetic allowances. Propositions S.26 and S.28 give the complete identities and finite-sample interval. The same scalar differences define a population contrast-training loss (Corollary S.27); the policies reported here are the unchanged fitted policies, not policies retrained with that loss.

The independent assessment fixes all predictions and geometric bounds before drawing its audit innovations. It retains four calibrations, both dimensions, all sixteen streams, and all 384 queries per stream. Each query receives 256 independent antithetic pairs. A sign pair is averaged into one observation; the same audit is reused across the three Raw budgets and is not counted three times as independent information. There are 12,582,912 distinct pair observations over the eight cells. The comparison conditions on each realized Raw cache, rather than averaging over unexecuted caches. It also conditions on NBO's chosen action pairs; it is not an occupation-risk claim for every feasible action or for Raw's separately chosen actor.

Table S.1 reports all 24 comparisons, not only those favoring learning. Eight intervals have upper endpoint below -10^{-8} and establish lower NBO contrast risk. Four have positive lower endpoint and favor Raw; twelve include zero. At the four-path budget, every Quarterly, Long, and Intermediate cell favors NBO. At sixteen paths, the ten-dimensional Quarterly and Intermediate cells also favor NBO.

TABLE S.1

CONDITIONAL CONTINUATION-RISK DIFFERENCES

Design	d	Raw: 4 paths	Raw: 16 paths	Raw: 64 paths
Low	10	$[-0.0242, 0.00950]$	$[-0.00938, 0.00650]$	$[-0.00491, 0.00486]$
Low	50	$[0.00817, 0.0552]$	$[0.0106, 0.0564]$	$[0.0114, 0.0566]$
Quarterly	10	$[-104, -62.7]$	$[-25.8, -4.52]$	$[-5.94, 8.78]$
Quarterly	50	$[-38.0, -14.9]$	$[-10.2, 3.59]$	$[-2.68, 7.75]$
Long	10	$[-8360, -4760]$	$[-1650, 619]$	$[125, 1900]$
Long	50	$[-5220, -2690]$	$[-1300, 27.5]$	$[-326, 654]$
Intermediate	10	$[-151, -73.0]$	$[-43.7, -4.20]$	$[-12.9, 10.4]$
Intermediate	50	$[-72.2, -15.1]$	$[-20.5, 13.5]$	$[-4.37, 17.2]$

Notes: All intervals are for $R_N - R_R$, in units of 10^{-8} . Negative intervals favor NBO. Endpoints are rounded outward to three significant digits; the complete precision is retained in the report. The 24-event family has simultaneous coverage at least .99. Predictions, action pairs, and realized Raw caches are fixed before the independent assessment.

In contrast, the fifty-dimensional Low cell favors Raw at every budget, and Long in ten dimensions favors the 64-path Raw predictor. This pattern identifies where the observed fitted continuation improves prediction, and where additional simulation removes that advantage.

The preassigned risk margin 10^{-8} is the square of the payoff unit 10^{-4} , not a payoff difference. Proposition S.25 bounds average binary-decision loss by $\sqrt{R_M}$, but a smaller risk need not give a better action: prediction errors with different signs can reverse that ordering. A difference of risks is not an absolute-risk certificate. Accordingly, the risk result and the actual five-method payoff result provide distinct evidence for NBO; neither is substituted for the other.

The risk family has failure probability at most .01. Combining it with the menu payoff family, whose allocation is .018, gives simultaneous coverage at least .972 by a union bound, without requiring independence between families. Combining it instead with all four earlier scientific families, including their scalar events, gives coverage at least .94. The individual family guarantees remain unchanged. The source freeze, pre-audit receipts, and exact report replay appear in Section S.28.

S.4. DECISION ACCURACY ACROSS THE COMPLETE BUDGET CATALOGUE

A final-budget comparison alone does not determine which computed procedure meets an economic tolerance at the lowest cost. We therefore use every prespecified menu stage. In each economic cell, the catalogue contains the reference and the five methods at each of three stages. Write V_i for candidate i 's mean payoff over the same complete sixteen-stream population and 384 queries. The target in this section is $g_i = \max_j V_j - V_i$ over these sixteen candidates. This is a catalogue of computed vector decisions, not the full continuous vector-action set or the adapted diffusion class.

The original simultaneous intervals bound five reference-relative gains and four direct NBO contrasts at each stage. If $L_{ab} \leq V_a - V_b \leq U_{ab}$, put directed edges $b \rightarrow a$ of length U_{ab} and $a \rightarrow b$ of length $-L_{ab}$. Let D_{ij} be the shortest-path length from i to j . The same confidence event then gives

$$\max_j \{-D_{ji}\} \leq g_i \leq \bar{g}_i := \max_j D_{ij}. \quad (\text{S.5})$$

Proposition S.29 proves that these bounds are sharp in the uncertainty set defined by the supplied intervals. No new simulation, normal approximation, or additional probability allocation is used. For a tolerance τ , let $\mathcal{C}_\tau = \{i : \bar{g}_i \leq \tau\}$. Selecting the least recorded cost within $\mathcal{C}_\tau$ preserves the simultaneous accuracy guarantee. It identifies the cheapest *certified* candidate, not necessarily the cheapest candidate whose unknown true regret is below τ .

The fifty-dimensional Intermediate design gives a positive NBO accuracy-work result against the complete budget catalogue. NBO's final-stage regret upper bound is 0.000091391 (rounded upward), below 10^{-4} , at mean construction and query cost 12.6993 seconds (rounded upward). It is the least-cost candidate with this certificate. The cheaper final Raw and vector actors have regret lower bounds above 0.0001874 and 0.0001064, respectively. The cheaper four-path SAA and second-stage NBO candidates remain unresolved: their regret intervals straddle 10^{-4} . Thus the certified choice does not establish actual minimum work over these unresolved alternatives.

TABLE S.2

LEAST ACCOUNTED COST WITH A CATALOGUE ACCURACY CERTIFICATE

Design	d	Certified choice	$\bar{g}$	Seconds	NBO (3): $\bar{g}$
Low	10	SAA (1)	0.03935	3.47	0.03883
Low	50	SAA (1)	0.03563	3.97	0.05945
Quarterly	10	Raw actor (2)	0.7543	4.85	0.1848
Quarterly	50	Vector (3)	0.7678	11.15	0.5235
Long	10	SAA (3)	0	30.42	1.984
Long	50	SAA (3)	0.4893	67.04	2.688
Intermediate	10	Vector (2)	0.8382	6.52	0.4282
Intermediate	50	NBO (3)	0.9140	12.70	0.9140

Notes: Regret bounds are in units of 10^{-4} ; the target is at most one. Parentheses identify the original stage. The catalogue is the reference plus fifteen method-stage procedures, not the full action space. Early-stage seconds are prefix allocations; stage-three seconds are complete construction/query clocks. Shared assessment and the complete comparative bill are reported separately. The selected cost is least among certified candidates only.

Table S.2 retains the choices in every other design. The final NBO candidate is within 10^{-4} of the best catalogue payoff in six cells. Both Long cells instead select final-stage SAA; elsewhere a cheaper Raw actor, vector predictor, or low-budget SAA has the certificate. The lower-cost alternatives are part of the result, not exclusions from the comparison.

Early-stage costs are the original construction-prefix allocations, not independent clocks for a prospectively stopped procedure. Final-stage costs are measured complete three-stage construction and query processes. The shared confirmation process is charged separately and in full; adding this same bill to each candidate does not alter its ordering. The total cost of having conducted the comparison includes *all* five construction processes and that shared assessment, not just the selected candidate. Section S.28 reports these complete bills and every candidate. The closure and resulting selection are a post-freeze deterministic reanalysis of an already simultaneous family, not a newly executed stopping rule or a fresh confirmatory experiment.

S.5. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (\text{S.6})$$

The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM S.1—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the boundary payoff error is at most B . Any reflected boundary condition is imposed exactly. Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,*

$$|v(t, x) - V^a(t, x)| \leq e^{Ls} B + A_L(s) e, \quad (\text{S.7})$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls} B + A_L(s) (e + \eta), \quad (\text{S.8})$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls} B + A_L(s) (2e + \eta). \quad (\text{S.9})$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled residual is not yet the uniform hypothesis of the theorem. Section S.7.2.4 addresses that distinction.

S.5.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For

a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (\text{S.10})$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$ is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$ gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM S.2—Discrete NBO certificate: *Suppose (S.10) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (\text{S.11})$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM S.3—Viscosity-consistent neural Bellman approximation: Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem S.2. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy

$$\delta_h + \epsilon_h = o(h), \quad (\text{S.12})$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural. The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

Appendix S.7 gives the differential-residual counterexample, independent domain and action coverage, the complete-action envelope, and the guarded finite-model evaluation map. Appendix S.8 proves the results, including finite-horizon error accumulation.

S.6. BELLMAN EVALUATION AND DIRECT METHOD COMPARISONS

The NBO map has two separate empirical obligations. Its evaluation block must produce a useful continuation-value derivative, and the resulting policy must justify the resources used to construct it. Verification of a policy does not by itself answer the second question. We therefore connect evaluation error to Hamiltonian error and compare candidate-generation methods by a direct paired payoff statistic. The economic problem and the feasible competitor class remain unchanged.

S.6.1. *What the critic contributes*

Use the normalized inner product $\langle x, z \rangle_d = d^{-1} x^\top z$ and norm $\|x\|_d^2 = \langle x, x \rangle_d$. At a state of the capital economy, set $q = dD_y V^\pi$. The part of the Hamiltonian depending on consumption is

$$H_q(m) = d^{-1} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \langle q, m \rangle_d.$$

On the original action box, and on any convex tube contained in it, H_q is μ -strongly concave for $\mu = \bar{m}_{\text{box}}^{-2}$. Here $\bar{m}_{\text{box}}$ denotes the primitive upper consumption-share bound, not the cross-sectional mean. Let m_q maximize H_q , let $\hat{q}$ be a critic costate, and let $\tilde{m}$ be within δ of the maximum of $H_{\hat{q}}$ over the same action set.

PROPOSITION S.4—Evaluation error and feasible improvement: *Under the preceding conditions,*

$$H_q(m_q) - H_q(\tilde{m}) \leq \left(\sqrt{\delta} + \frac{\|q - \hat{q}\|_d}{\sqrt{2\mu}} \right)^2. \quad (\text{S.13})$$

For a random rollout costate Z and a square-integrable predictor $g(X)$,

$$\mathbb{E} \|Z - g(X)\|_d^2 = \mathbb{E} \|Z - \mathbb{E}[Z | X]\|_d^2 + \mathbb{E} \|\mathbb{E}[Z | X] - g(X)\|_d^2. \quad (\text{S.14})$$

These statements do not identify the costate of an Euler rollout with the continuous-time costate; a discretization or evaluation bias must be added when that identification is required.

Equation (S.13) identifies a reason to regress costates rather than only value levels: Hamiltonian loss is quadratic in a suitably measured costate error when the

actor solves its improvement problem. Equation (S.14) identifies both the variance reduction available from conditional prediction and its regression bias. Neither identity promises that a trained critic reduces mean-square error relative to raw rollout derivatives. The existing comparison therefore includes raw-costate improvement without a learned critic, value-only and costate-only critics, changes in block-update counts, and direct policy optimization. These are substantive alternatives, not alternative labels for NBO. The strong-concavity and conditional-projection arguments are established mathematical principles; their role here is to specify a testable evaluation–improvement mechanism, not to claim a new principle of policy iteration.

The continuous-time performance-difference account in Proposition S.8 integrates evaluation and action errors under the candidate’s occupation law. Its full statement, proof, and alternative residual-to-costate conditions are in the supplement. Training-design prediction errors alone do not meet that account. The current finite-step mechanism below instead states its continuation target exactly and carries its welfare comparison to the continuous economy through the separate payoff-transfer bound.

S.6.2. *Direct paired economic comparisons*

Let a and b be two fitted sampled controllers, fixed before the final simulation bank. For each independent final path, compute their schedule-relative statistics $X_{a,h}$ and $X_{b,h}$ with exactly the same initial state and Brownian increments, and form

$$D_{ab,h} = X_{a,h} - X_{b,h}.$$

The common schedule’s numerical trajectory cancels. The implementation checks equality of its terminal arrays and the initial-state and innovation hashes. The deterministic transfer allowance is

$$E_{ab,h} = E_{a,h}^{\text{policy}} + E_{b,h}^{\text{policy}} + E_{a,h}^{\text{stat}} + E_{b,h}^{\text{stat}},$$

with outward arithmetic and a separate subtraction-roundoff cushion. In particular, it does not add the schedule transfer error twice. If b_a, b_b are the two clipping

thresholds and τ_a, τ_b their expectation-tail allowances, use $b_{ab} = b_a + b_b$ and $\tau_{ab} = \tau_a + \tau_b$.

THEOREM S.5—Simultaneous method and population comparisons: *Condition on all fitting and validation data. Suppose the final paths follow the declared independent sampling law and the conditional arithmetic and diffusion-transfer contract of Theorem S.31. Apply the empirical Bernstein construction in (S.107) directly to $D_{ab,h}$, with threshold b_{ab} and allowances $E_{ab,h}, \tau_{ab}$. The resulting endpoints L_{ab}, U_{ab} satisfy*

$$L_{ab} \leq J(a) - J(b) \leq U_{ab}$$

simultaneously with the other declared one-sided statements at their joint confidence level. A difference of two schedule-relative lower endpoints is not used.

The same construction applies to $J_{\mathcal{P}}(a) = \sum_j p_j J_{y_j}(a)$ for a prespecified finite initial-state population $\mathcal{P}$. Draw its profile independently on every final path, average the conditional deterministic transfer bounds with the exact population weights, and use a common clipping threshold and tail bound valid for all profiles. The resulting guarantee is integrated over $\mathcal{P}$; it is not uniform over a continuum of initial states or over unobserved training seeds.

The comparison does not require NBO to dominate a direct method. It makes that empirical question answerable by the quantity actually of interest. A lower endpoint below zero is inconclusive about superiority unless the upper endpoint is also below zero. Certified improvement over the schedule and certified improvement over another trained policy remain different statements. Likewise, a smaller policy-regret upper endpoint need not equal the unknown actual regret.

S.7. POLICY EVALUATION, APPROXIMATION, AND COMPLETE ACTION COVERAGE

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective under joint differentiation satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (\text{S.15})$$

Thus the block-specific critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a_\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv} D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (\text{S.16})$$

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (24) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

S.7.0.1. *Conditioning and stopping*

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio

tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual, and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

S.7.1. Policy Evaluation and the Recursive Policy Gradient

Let a_ϕ be differentiable in a scalar parameter direction, and suppose V^{a_ϕ} is sufficiently differentiable for the following calculation. The terminal payoff is independent of the policy parameter. Assume differentiation under the expectation is justified, that the differentiated state and utility equations are integrable, and that the boundary conditions of the differentiated problem are zero. Denote the directional derivative of the policy value by w .

Differentiating the fixed-policy equation gives

$$0 = w_t + \mathcal{L}^{a_\phi} w + f_y(t, x, a_\phi, V^{a_\phi}) w + D_a \mathcal{H}^{a_\phi}[V^{a_\phi}] D_\phi a_\phi.$$

The operator derivative includes the dependence of both drift and covariance on the action. The derivative of the generator acting on the changing value is contained in $\mathcal{L}^{a_\phi} w$, while the action derivative of the generator acting on the fixed value is contained in $D_a \mathcal{H}$. No Hessian-action channel is discarded.

Applying the linear Feynman–Kac formula to the differentiated equation gives

$$w(t, x) = \mathbb{E}_{t,x}^{a_\phi} \int_t^\tau \exp \left(\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv \right) D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds.$$

This formula concerns the performance derivative from the stated initial state. A domain-uniform Hamiltonian objective uses a different weighting. Such an objective is useful for approximate pointwise improvement, but it becomes a lifetime policy gradient only with the appropriate occupation and adjoint weights. Trajectories sampled by the current actor are detached in the implemented domain objective. Differentiating through their sampling law would define a different gradient estimator and must be analyzed separately.

The critic’s residual derivative is the derivative of the entire fixed-policy equation. In particular, its level is determined by the recursive evaluation equation together with boundary data. A separate negative-Hamiltonian gradient in the critic is not an innocuous level anchor: in the scalar stationary example it moves the minimizer away from the Bellman root.

S.7.2. Representation, Boundary Conditions, and Error Measurement

S.7.2.1. Smooth approximation and what it establishes

A derivative-approximation result concerns a smooth target on a compact domain and a family of networks whose capacity can increase. It does not imply that a fixed network attains zero residual, that optimization finds its best approximation, or that a nonsmooth target has classical derivatives everywhere. In a smooth problem, increasing the representation class can reduce approximation error in the value and derivatives. In a nonsmooth problem, NBO instead evaluates a consistent positive-weight Bellman equation, or solves an explicitly regularized equation before taking a justified limit.

The hard-terminal architecture is $v(t, x) = G(x) + (T - t)N(t, x)$. It exactly enforces $v(T, x) = G(x)$ independently of training. A lateral Dirichlet face needs an additional compatible lifting and vanishing factor. For a reflected state, the Neumann condition is a different constraint. A loss on only interior points leaves all of these boundary questions unanswered. The regression test evaluates the terminal architecture at randomly chosen states and checks exact equality, not merely a small mean terminal error.

The recursive Epstein–Zin domain is $bV > 0$. An exponential output ensures positivity, while an interval transformation can keep bV between declared positive barriers. If $bG > 0$, one terminal-compatible positive representation is

$$bv(t, x) = \exp [\log(bG(x)) + (T - t)N(t, x)].$$

Its terminal value is exact. It does not by itself impose a lateral exit payoff or a two-sided barrier. These additional conditions must be built into the architecture or

measured explicitly. Clipping an invalid utility argument inside a fractional power changes the equation and is not an acceptable hidden numerical repair.

S.7.2.2. From residual samples to a uniform bound

Let r be K -Lipschitz on a domain in $\mathbb{R}^d$. If every point is within distance h_V of a validation point, the triangle inequality gives $\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kh_V$. If K is estimated from samples rather than bounded, the resulting expression is itself a diagnostic, not a certified uniform bound.

An integral bound needs more assumptions. Suppose the sampling density is at least $p_0 > 0$, and each ball of radius r_0 or less centered in the domain intersects the domain in volume at least $c_D r^d$. Let $M = \|r\|_\infty$ and assume $M/(2K) \leq r_0$. Near a point approaching the maximum, Lipschitz continuity gives $|r| \geq M/2$ on a ball of radius $M/(2K)$. Hence

$$\mathbb{E}r^2 \geq \frac{p_0 c_D}{2^{d+2} K^d} M^{d+2}, \quad M \leq \left(\frac{2^{d+2} K^d \mathbb{E}r^2}{p_0 c_D} \right)^{1/(d+2)}.$$

For larger M , use the capped radius r_0 and the corresponding two-case inequality. A concentration argument is additionally needed to replace the population mean by an empirical one. This calculation explains why a dimension-free uniform certificate cannot be read directly from an average training loss.

The action gap is treated similarly. If an action net has covering radius h_A and the Hamiltonian is K_A -Lipschitz in the action, the unobserved supremum exceeds the net maximum by at most $K_A h_A$. Thus the net gap plus this term is an upper bound. A collection of successful deviations without a covering or duality argument is only a lower bound on exploitability.

S.7.2.3. Why an integrated differential residual does not select the value

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (\text{S.17})$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$ exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

S.7.2.4. Coverage and independent maximization

A verification distribution is declared independently of the actor's occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} , then

$$\|r\|_{\infty} \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (\text{S.18})$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

S.7.2.5. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + h r_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION S.6—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (\text{S.19})$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem S.3.

S.7.2.6. Continuous-action proposals and a budget-safe Bellman enclosure

The learned map and its verifier have different computational roles. In the differential implementation, a feasible actor supplies an action and the critic evaluates that action's generator. In the positive-weight implementation, write

$$(T_n^a w)_i = r_{n,i}(a) + q_n \{P_{n,i}(a)w + b_{n,i}(a)\}, \quad (T_n w)_i = \sup_{a \in \mathcal{A}_i} (T_n^a w)_i. \quad (\text{S.20})$$

Here $P_{n,i}(a)$ has nonnegative entries with sum at most one. The term b contains prescribed stopping payoffs, and $q_n > 0$. Continuous controls need not enter this expression concavely. Reflection, interpolation cells, and stopping decisions can change as the action changes. A supremum, rather than an attained maximizer, suffices below.

Our continuous-action implementation fits the actor by differentiating its own evaluated continuation payoff. It does not construct an all-action target or receive an exact maximizing label. A bounded tanh output enforces the control box. A

local improvement step can subsequently compare that proposal with finitely many continuous perturbations. Such a step is recorded as part of a *hybrid* policy; it is not identified with the raw network and is not assumed globally optimal. Critic fitting uses the selected feasible action and the preceding continuation critic. Training ends before any exhaustive reference or action-cover verification begins.

For independent verification, freeze the complete policy $\pi = (\pi_n)$ and cover each action box by closed subboxes. At a state and time, let c_i be an outward lower bound obtained by evaluating any feasible action using a fixed continuation upper array. A box can be discarded only when its verified upper bound does not exceed $c_i + \tau$. Every box not so discarded is either subdivided or retained with its upper bound. Exhausting a time, depth, or box budget therefore enlarges the reported bound; it never converts an unresolved search into a success.

PROPOSITION S.7—Budget-safe action and policy envelopes: *Suppose the terminal arrays satisfy $L_N \leq g \leq U_N$. At each time n , evaluate the fixed-policy map outward to obtain $L_n \leq T_n^{\pi_n} L_{n+1}$. For every state, maintain a complete action cover whose box bounds enclose $T_n^a U_{n+1}$ for all actions in that box. Set $U_{n,i}$ to the maximum of the outward-rounded pruning threshold and all unresolved box upper bounds. Then, regardless of which boxes remain unresolved,*

$$L_n \leq V_n^\pi \leq V_n^* \leq U_n, \quad 0 \leq V_n^* - V_n^\pi \leq U_n - L_n. \quad (\text{S.21})$$

All inequalities are on the stated nodal economy. No concavity, exact training labels, or convergence of neural optimization is required. If $U_n - T_n U_{n+1} \leq e_n \mathbf{1}$, then its optimal-value excess is at most

$$\left(\prod_{j=n}^{N-1} q_j \right) \|U_N - g\|_\infty + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) e_k.$$

The proof, including stopping faces, is in the supplement. A stored unresolved bracket supplies an upper bound for e_n ; it is not replaced by the requested tolerance. The policy lower envelope and the Bellman upper envelope are computed separately. The latter may be shared across several frozen policies on the same economy, while

each policy requires its own evaluation. This separation prevents an expensive verification oracle from becoming undisclosed actor supervision.

In the preference application, the continuous box has three coordinates and the state grid has two. Bilinear interpolation is enclosed cell by cell. On each rectangle, its extrema occur at cell corners; a second implementation uses cell-slope enclosures. If a box crosses a wealth stopping face, the verifier takes the union of the interior continuation and the prescribed exit payoff. Their numerical values need not coincide at off-grid exit locations. Treating the entire action objective as globally differentiable at this switch would invalidate a derivative-only bound. Reflection is handled by including every crossed fold. Positive consumption and a preference interval strictly above one keep the powers and denominators in their stated domains.

The complete-action bound is not a continuous-state or continuous-time theorem. The verifier interprets stored primitive coefficients and nodal abscissae as exact real inputs and encloses subsequent operations outward. Training and verification agree to numerical precision on the inherited action grid, but the certificate's precise object is this declared nodal economy, not an unspecified sequence of floating-point evaluations. The supplemental transfer result states separately the state interpolation, diffusion approximation, boundary monitoring, and action errors needed to recover the continuous economy. None is inferred from a sampled residual or silently assigned zero.

S.8. PROOFS

S.8.0.1. *Proof of Theorem S.1*

Extend the aggregator Lipschitz-continuously outside its invariant interval for this comparison argument. The extension leaves all policy utilities and the candidate residual unchanged. For a nonnegative number c , define $d_c(t) = Be^{L(T-t)} + cA_L(T-t)$. Then $d_c(T) = B$, $d_c(t) \geq B$, and $-d'_c = Ld_c + c$. Since a spatially constant shift leaves the generator unchanged and f is L -Lipschitz in utility,

$$F[v + d_c] \geq F[v] - d'_c - Ld_c = F[v] + c, \quad F[v - d_c] \leq F[v] + d'_c + Ld_c = F[v] - c.$$

Because $F[v] = r^a[v] - g^a[v] \in [-e - \eta, e]$, $v + d_{e+\eta}$ is a supersolution and $v - d_{e+\eta}$ is a subsolution. They dominate the required stopping data on every boundary face. A spatially constant shift also preserves an exactly satisfied homogeneous reflected boundary condition. Comparison proves (S.8).

Apply the same argument to the fixed-policy operator $F^a[v] = -v_t - \mathcal{H}^a[v]$, with $c = e$, to obtain (S.7). The triangle inequality gives (S.9), and policy admissibility gives its nonnegative lower bound. The construction uses $L \geq 0$ and $d_c(t) \geq B$ specifically to cover lateral stopping boundaries; it does not incorrectly discount their errors as if every path stopped only at T . For nonsmooth candidates the same argument applies when their residual inequalities are verified in the viscosity sense. The theorem's differential implementation, however, requires the stated smooth candidate. Q.E.D.

S.8.0.2. Exact policy improvement

Suppose $v = V^a$ solves the fixed-policy problem and a feasible $\tilde{a}$ satisfies $\mathcal{H}^{\tilde{a}}[v] \geq \mathcal{H}^a[v]$ pointwise. Then $F^{\tilde{a}}[v] \leq 0$ with the same boundary payoff. Policy comparison gives $V^{\tilde{a}} \geq V^a$. This is a policy-space statement. A single gradient step on a domain-averaged actor loss does not automatically meet its pointwise premise. The measured action gap and policy-evaluation errors are retained precisely because approximate neural improvement can violate that premise.

S.8.0.3. Proof of Theorem S.2

For a fixed state and action, let z and $\tilde{z}$ solve (S.10) for w and $\tilde{w}$. Strict monotonicity of $z - hf(z)$ gives

$$|z - \tilde{z}| \leq \frac{|P_h^a(w - \tilde{w})|}{1 + \lambda h} \leq q_h \|w - \tilde{w}\|_\infty.$$

The same monotonicity implies order preservation. Taking the maximum over actions preserves both order and the Lipschitz constant. Banach's fixed-point argument supplies the two fixed points on the invariant complete value domain.

Now $\|v - v_h^a\| \leq \|v - T_h^a v\| + q_h \|v - v_h^a\|$, which gives the first bound. Also $\|v - T_h v\| \leq \delta + \epsilon$, giving the second. The triangle inequality gives the third. All norms

are over every state of the specified finite model. An omitted action changes T_h and must be accounted for as a separate error, not hidden in δ . Q.E.D.

S.8.0.4. Proof of Theorem S.3

First, (S.11) gives

$$\|v_h - v_h^*\|_\infty \leq \frac{\delta_h + \epsilon_h}{1 - q_h} = \frac{1 + \lambda h}{\lambda} \frac{\delta_h + \epsilon_h}{h} \longrightarrow 0.$$

It remains to identify the limit of the exact monotone scheme. Stability makes the upper and lower half-relaxed limits finite. At a strict local maximum of the upper limit minus a smooth test function, select nearby maximizing grid points. Monotonicity compares the neighboring scheme values to that test function plus the vanishing contact offset. Local consistency then gives the viscosity subsolution inequality. The corresponding minima give the supersolution inequality for the lower limit. Consistent boundary data and the assumed barriers give the required boundary inequalities. Comparison orders the upper limit below the lower limit. The reverse order follows by definition, so the two limits coincide with the unique continuous viscosity solution. The vanishing distance between v_h and v_h^* transfers that convergence to the neural arrays. The discrete policy-value conclusion follows from the third bound. Interpreting a discrete policy as a continuous-time implementation additionally requires convergence of its controlled interpolation and payoff; the theorem does not infer this merely from convergence of its state arrays. This is the usual monotone-scheme selection argument, with the neural approximation error made explicit; see Barles and Souganidis (1991), Crandall et al. (1992). Q.E.D.

S.8.0.5. Finite-horizon Bellman error accumulation

Let T_n and T_n^a be the time- n Bellman and fixed-policy maps, with Lipschitz constants q_n (which may exceed one). For arrays v_n , define $\delta_n = \|v_n - T_n^a v_{n+1}\|$ and $\epsilon_n = \|T_n v_{n+1} - T_n^a v_{n+1}\|$. If the terminal error is B_N , backward induction gives

$$\|v_n - v_n^*\| \leq \left(\prod_{j=n}^{N-1} q_j \right) B_N + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) (\delta_k + \epsilon_k).$$

The policy-evaluation bound replaces $\delta_k + \epsilon_k$ by δ_k . Their sum bounds regret. Empty products equal one. For the implicit Lipschitz recursive step, $q_n \leq (1 - hL)^{-1}$; for additive exponential discount, $q_n = e^{-\rho h}$. Thus a per-step error $o(h)$ and consistent terminal data suffice on a fixed finite horizon, without asserting a stationary contraction in an inapplicable utility regime.

S.8.0.6. Derivation of the temporal-self condition

A self at (t, x) replaces a^* by a fixed feasible local action a for duration ϵ and then returns to a^* . For each exponential component, the deviation value is

$$\mathbb{E} \left[\int_t^{t+\epsilon} e^{-\rho_j(s-t)} u(a) ds + e^{-\rho_j \epsilon} W_j(t + \epsilon, S_{t+\epsilon}^a) \right].$$

Itô expansion gives its first-order increment relative to $W_j(t, x)$ as $\epsilon[u(a) + (W_j)_t + \mathcal{L}^a W_j - \rho_j W_j] + o(\epsilon)$. Weighting the two components and subtracting their evaluation equations under a^* leaves

$$\epsilon\{u(a) - u(a^*) + (\mathcal{L}^a - \mathcal{L}^{a^*})V\} + o(\epsilon).$$

Nonpositive first-order gain for every feasible action is exactly (A.19). This derivation assumes the smoothness and local integrability needed for the expansion; it is an equilibrium deviation condition, not an assertion of optimal commitment over a finite interval. The homothetic derivation follows by integrating $e^{-[\rho_j - b(r-m)]s}$ and applying the consumption first-order condition. Q.E.D.

S.8.1. Proofs for the Guarded and Signed Map

S.8.1.1. Proof of Proposition A.1

At a replaced node the output equals the computed target. At any other node its distance from that target is at most τ_h . The triangle inequality gives the one-step conclusion. For a frozen policy sequence, let e_n be the sup-norm difference between the guarded and exact policy values at date n . Positivity of the transition weights and their mass bound imply $e_n \leq \tau_h + \eta_h + q_h e_{n+1}$, with $e_N = 0$ for exact terminal data. Backward induction gives the geometric sum. As $N = T/h$ and $q_h \leq 1$, the bound is at most $T(0.1h + \eta_h/h)$ under the stated schedule. No property of the

optimizer was used. The assertion concerns the guarded nodal values. An off-node
neural interpolant, an improved policy, and a diffusion approximation each require
their separate error account. In fixed-precision arithmetic, $\eta_h = o(h)$ is a maintained
refinement condition, not a fact that can be inferred as h tends to zero indefinitely. $\square$

S.8.1.2. *Proof of Theorem A.2*

On a regular box, the payoff restricted to a line segment is continuous and piece-
wise continuously differentiable, with finitely many interpolation and reflection
knots, or is the limit of such segments. Integration of its almost-everywhere deriva-
tive yields

$$Q_w(a) - Q_w(c) = \int_0^1 \nabla Q_w(c + s(a - c)) \cdot (a - c) ds.$$

Every partial derivative is included in its interval hull. Maximizing the resulting
interval products proves (A.30). Taking the minimum with another valid upper
enclosure preserves validity. If a partial derivative is strictly positive throughout the
box, integration along that coordinate shows that replacing it by its upper endpoint
cannot lower the payoff; the negative case is analogous. Repeating the argument
proves face domination and the dimension statement.

A subdivision replaces a parent by children whose union contains it. A face
reduction replaces a set by a subset with the same supremum. Removal occurs
only after that supremum has a bound no larger than the incumbent plus tolerance.
Feasible point evaluations can only increase the incumbent, so an earlier removed
box stays bounded by the final incumbent plus tolerance. The remaining boxes
keep their upper bounds without any success inference from an interrupted search.
Induction over all operations proves (A.31). At a stopping switch the line-segment
continuity premise can fail; the implementation therefore uses the natural union
enclosure and does not invoke the derivative argument there.

For the local quadratic observation, a K -Lipschitz gradient with a zero-gradient
point in the box has norm at most Ks throughout a box of diameter s . Integration
over a segment of length at most s bounds the variation by Ks^2 . This conditional
observation does not extend to an arbitrary switching box, nor does it assert that all
nodes possess such a box. $\square$

1 S.8.1.3. Derivative hulls for the preference economy 1

2 Write $y = \log X$, and let F be the continuation interpolant evaluated at a cubature 2
3 transition. On an interior leg, its action derivatives are 3

$$4 \quad \partial_c F = -hF_y, \quad \partial_p F = F_y \{h(0.06 - 0.04p) + s_z\}, \quad \partial_\theta F = hF_u r_u, \quad 5$$

6 where s_z is the stored portfolio shock coefficient and $r_u \in \{-1, 1\}$ is the reflection 6
7 slope; both slopes enter the hull at a fold. The flow derivatives are 7

$$8 \quad h \exp\{(1-u)y - u \log c\}, \quad 0, \quad -hk\theta. \quad 9$$

10 Outside a wealth stopping face, clipped wealth is constant and F is the exact 10
11 stopping payoff. Its y derivative with respect to the transition is zero, while 11

$$12 \quad F_u = \frac{\exp\{(1-u)y\} \{1 - (1-u)y\}}{(1-u)^2}. \quad 13$$

14 Every crossed bilinear cell contributes its one-sided derivative range, restricted 14
15 by the other coordinate's interval. Interval evaluation includes these ranges and 15
16 the flow terms. A leg whose wealth interval meets both sides of a stopping switch 16
17 disables the signed bound for that box. Point and interval evaluations share the 17
18 stated nodal primitives but are computed independently of the actor's derivative 18
19 graph. 19
20
21

22 S.8.1.4. Proof of Proposition A.3 22

23 Let $\bar{m} = d^{-1} \sum_i m_i$. Jensen's inequality gives $d^{-1} \sum_i \log m_i \leq \log \bar{m}$. Moreover, 23
24 $d^{-1} \sum_i \tanh((BY)_i) \leq 1$. The mean-state drift is consequently no larger than $b_+ - \bar{m}$. 24
25 The dispersion penalty at the terminal date is nonnegative and can be dropped 25
26 for an upper bound. Fubini's theorem applied to the integrated expected mean 26
27 state and terminal mean state yields the kernel $w(t)$. The Brownian terms have 27
28 zero mean; bounded controls and bounded nonlinear drift ensure the required 28
29 integrability. Thus every admissible policy has expected payoff at most 29
30

$$31 \quad \int_0^T e^{-\rho t} \max_{m \in [\underline{m}, \bar{m}]} \{\log m - \zeta m^2 / 2 + w(t)(b_+ - m)\} dt. \quad 31$$

The integrand is strictly concave in m . Its derivative is $1/m - \zeta m - w(t)$, whose positive root gives $m_0(t)$ before clipping. This proves the upper bound.

For the feasible common schedule, the expected mean-state drift is at least $b_- - m_0(t)$. Define the normalized dispersion norm $\|Z\|_{2,d}^2 = \mathbb{E}[d^{-1} \sum_i Z_i^2]$. Centred common controls and common noise vanish from the dispersion equation. For numbers in $[-1, 1]$, their cross-sectional variance is at most one. Hence Minkowski's inequality bounds the norm of the integrated centred nonlinear drift by κT . The centred idiosyncratic Brownian part has norm $\sigma_I \sqrt{(1 - 1/d)T}$. A second application of Minkowski and squaring give the upper bound D_d on expected terminal dispersion. Subtracting $\chi e^{-\rho T} D_d$ from the lower mean-payoff calculation proves $J(m_0) \geq L_d$. Feasibility gives $J(m_0) \leq V^*$. The derivation never restricts the dense matrix B beyond the stated bounded-tanh form. $\square$

S.8.1.5. *Compensation and arithmetic scope*

For each realized state and positive consumption, CRRA felicity is increasing in consumption, including when $u > 1$ and its level is negative. The financed supplement leaves the state law and every other payoff term unchanged. Its expected payoff is therefore nondecreasing in e . Positivity of the nodal transition operator propagates a lower interval payoff at each date. If its lower endpoint exceeds the optimal upper envelope at every active starting node and date, the corresponding financed supplement compensates the loss throughout that nodal economy. Bisection searches for a sufficient supplement; only the final verified inequality provides the stated upper bound.

All implemented interval conclusions use the inherited binary64 execution contract, with outward elementary operations and separately bounded transcendental evaluations. They concern exact real interpretations of stored primitive coefficients and grid nodes. They are not a formal verification of the interpreter, hardware, or all library internals. In the capital bound, square-root seeds are accepted only after interval squaring verifies both endpoint inequalities, and the time integral is bounded by interval rectangles rather than by an adaptive quadrature tolerance. Random stress tests are regression evidence, not substitutes for these inclusion arguments.

S.8.2. Additional Proofs and Numerical Contracts

S.8.2.1. Costate error and action quality

Let $\hat{m}$ maximize $H_{\hat{q}}$ on the stated convex feasible set, and let $e = q - \hat{q}$. Strong concavity and the variational first-order condition at $\hat{m}$ imply

$$H_{\hat{q}}(\hat{m}) - H_{\hat{q}}(m) \geq \frac{\mu}{2} \|m - \hat{m}\|_d^2$$

for every feasible m , including boundary maximizers. Therefore $\|\tilde{m} - \hat{m}\|_d \leq \sqrt{2\delta/\mu}$. Decomposing the true Hamiltonian difference yields

$$\begin{aligned} H_q(m_q) - H_q(\tilde{m}) &\leq \delta - \frac{\mu}{2} \|m_q - \hat{m}\|_d^2 + \langle e, \tilde{m} - m_q \rangle_d \\ &\leq \delta + \|e\|_d \sqrt{2\delta/\mu} + \frac{\|e\|_d^2}{2\mu}. \end{aligned}$$

The last step completes the square in $\|m_q - \hat{m}\|_d$. This is (S.13). The Hessian of H_q is the negative diagonal matrix with entries $1/(dm_i^2)$ minus the positive semidefinite mean-penalty matrix. It is bounded above by $-\mu I/d$, proving the strong-concavity premise.

For (S.14), write $Z - g(X) = [Z - \mathbb{E}(Z | X)] + [\mathbb{E}(Z | X) - g(X)]$. The cross term has zero expectation by conditioning on X . Integrability justifies this orthogonal decomposition. If the conditional Euler costate differs from the continuous policy costate by $b(X)$, that term remains an additional bias; it is not removed by the projection identity. Neither the sampled costate loss nor a successful optimizer return is a uniform upper bound on this bias.

S.8.2.2. Cancellation, clipping and finite-population inference

For each candidate, the exact payoff identity used in Proposition S.10 consists of a policy production term, its consumption term, its terminal dispersion, and the corresponding reference terms. When two candidates use the identical reference path, subtraction cancels every exact reference term. Comparing the two policy terms with their numerical approximations therefore requires the two policy transfer errors and the statistic-roundoff accounts, but no reference-state transfer error. The

implementation retains conservative original statistic budgets even where their reference quadrature components also cancel.

For any real x, y and positive thresholds b_a, b_b ,

$$(|x - y| - b_a - b_b)_+ \leq (|x| - b_a)_+ + (|y| - b_b)_+.$$

Taking expectations proves the summed clipping-tail allowance without requiring independence between policies. Applying the empirical Bernstein inequality to the clipped difference and then to its negative gives the two one-sided bounds. A union bound covers all declared tests. Our allocation counts both directions and every method, profile design and auxiliary evaluation; it does not spend new confidence mass for deterministic transformations such as (S.118).

For the finite population, draw I with the fixed probabilities p_j , independently of each future innovation path. The pair (I, W) is then independent across paths. Conditional transfer gives

$$\left| \mathbb{E}[X_{a,h}] - \sum_j p_j \{J_{y_j}(a) - J_{y_j}(m_0)\} \right| \leq \sum_j p_j E_{a,h}(y_j).$$

The clipping threshold is the maximum of the profile-specific thresholds; its expectation-tail allowance is bounded by the maximum of their allowances. The prescribed implementation uses equal probabilities over nine fixed profiles. Both averaging and the final subtraction are interval-enclosed. Random empirical profile frequencies are not substituted for the population weights when calculating deterministic error budgets. They are part of the sampling variation of the final statistic.

For the paired method difference, the same profile is assigned to both policies on each path. Common random numbers across different methods or training seeds do not invalidate a finite-family union bound. They also do not create independent replications of the optimizer. The ten fixed training streams are reported individually and descriptively; no confidence assertion about an unspecified distribution of future hyperparameter choices is inferred.

1 S.8.2.3. *Observed-history implementation* 1

2 Under the observation assumptions, M is a continuous martingale with bracket 2
 3 $\langle M \rangle_t = Qt$. Let $A = \Sigma^\top Q^{-1}$. The matrix $A\Sigma$ is the orthogonal projector onto the row 3
 4 space of Σ , and $nn^\top$ is the projector onto its orthogonal complement. Independence 4
 5 of the private Brownian motion gives 5

$$6 \quad \langle AM + nB^0 \rangle_t = (AQA^\top + nn^\top)t = I_{d+1}t. \quad 7$$

8
 9 The multivariate Lévy characterization proves the Brownian assertion. Also $\Sigma A = I_d$ 9
 10 and $\Sigma n = 0$, so $\Sigma \tilde{W} = M$. On each decision cell, the action is determined at its left 10
 11 endpoint by the stored internal state and past recovered increments. With that held 11
 12 action, the physical state equation has a unique strong solution because the produc- 12
 13 tion drift is globally Lipschitz and bounded. Induction over the cells establishes 13
 14 uniqueness in law for the joint physical and internal process. Driven by the Brown- 14
 15 ian motion just constructed, it therefore has the same law as the controller originally 15
 16 defined using directly supplied innovations. Applying the specified componentwise 16
 17 clipping to the recovered vector preserves this distributional equivalence. 17

18 The proposition uses only private randomization and observed state history; it 18
 19 does not reveal the original null-space economic driver. Randomized controls are 19
 20 admissible in the enlarged filtration. Since the original deterministic performance 20
 21 envelope holds against all such adapted controls and independent randomization 21
 22 cannot improve an unconstrained expected-payoff supremum beyond the best 22
 23 deterministic randomization realization, the comparison with the original optimum 23
 24 is preserved. The result is not an implementability theorem for discrete noisy state 24
 25 measurements. The exact drift integral is stated explicitly, and errors in approximat- 25
 26 ing it must be treated separately. 26

28 S.8.2.4. *Independent scalar HJB approximation* 28

29
 30 For $d = 1$, the terminal dispersion penalty vanishes but production remains 30
 31 nonlinear in log capital. The reference solves the same primitive action-box problem, 31
 32 not a Riccati or homothetic reduction. Put $f(y) = \kappa \tanh(By)$ and $\nu = (\sigma_I^2 + \sigma_C^2)/2$. 32

On a uniform grid of width Δy , define the positive rates

$$\ell_i(m) = \frac{v}{\Delta y^2} + \frac{(m - c - f(y_i))_+}{\Delta y}, \quad r_i(m) = \frac{v}{\Delta y^2} + \frac{(c + f(y_i) - m)_+}{\Delta y}.$$

The implicit Bellman step solves

$$\frac{v_i - v_i^+}{h} + \rho v_i = \max_m \{ \log m + y_i - \zeta m^2 / 2 + \ell_i(m)(v_{i-1} - v_i) + r_i(m)(v_{i+1} - v_i) \}.$$

For a fixed action vector the coefficient matrix is strictly diagonally dominant with positive diagonal and nonpositive off-diagonal entries. Its inverse is nonnegative. Howard updates maximize over each drift-sign interval; on either interval the positive root of $1/m - \zeta m = p$ is clipped to its feasible endpoints, and the two candidates are compared. This is an independent monotone matrix computation, not a trained critic used as its own reference.

At the two outer boundaries, the prescribed affine asymptotes replace production by its limiting saturation value. If $w(t)$ is the coefficient on initial log capital and $m_0(t)$ the common schedule, their intercept is the numerical integral of the corresponding deterministic discounted flow. The reference reports the grid, time step, boundary distance, linear-system residual, policy-iteration convergence and domain sensitivity. Fitted actors are evaluated as Markov feedbacks on this finite grid. These comparisons do not conflate that controller with the innovation-driven implementation. Grid differences and asymptotic boundary values are not asserted to be rigorous continuous-domain error enclosures.

S.8.2.5. *From evaluation error to economic improvement*

Proposition S.4 controls a decision at a given state. The distribution of states at which that decision is used matters for the economic conclusion. We make this link explicit. In this subsection, π is a fixed Markov policy with a sufficiently regular policy value V^π to justify Itô's formula and the expectations below. Specifically, $V^\pi \in C^{1,2}([0, T) \times \mathbb{R}^d)$ is continuous at the terminal date with $V^\pi(T, \cdot) = g$. Along each candidate considered, the drift and terminal terms in the discounted Itô identity

are integrable and its stochastic integral is a zero-mean martingale. The condition

$$\mathbb{E} \int_0^T e^{-2\rho t} \|\Sigma^\top D_y V^\pi(t, Y_t^a)\|_2^2 dt < \infty$$

is sufficient for the latter. The parameters of π are frozen. Let a be any admissible candidate, including a history-dependent candidate, and let Y^a be its physical state. Set $q^\pi = dD_y V^\pi$ and define

$$g^\pi(t, y) = \max_{m \in \mathcal{A}} H_{q^\pi(t, y)}(m) - H_{q^\pi(t, y)}(\pi(t, y)) \geq 0,$$

where $\mathcal{A}$ is the same convex action set for both maximizations and both policies.

One may always take the primitive action box. For a costate predictor $\hat{q}$, let δ_t bound the $H_{\hat{q}(t, Y_t^a)}$ maximization gap of a_t over that set. Write $\bar{q}_h(t, y)$ for the conditional mean of the frozen-policy rollout costate at mesh h . Define, under the candidate's state law,

$$\mathcal{G}_a = \mathbb{E} \int_0^T e^{-\rho t} g^\pi(t, Y_t^a) dt,$$

$$\mathcal{D}_a = \mathbb{E} \int_0^T e^{-\rho t} \delta_t dt,$$

$$\mathcal{E}_{a,h}^2 = \mathbb{E} \int_0^T e^{-\rho t} \|\hat{q}(t, Y_t^a) - \bar{q}_h(t, Y_t^a)\|_d^2 dt,$$

$$\mathcal{B}_{a,h}^2 = \mathbb{E} \int_0^T e^{-\rho t} \|\bar{q}_h(t, Y_t^a) - q^\pi(t, Y_t^a)\|_d^2 dt.$$

The quantity $\mathcal{E}_{a,h}$ concerns regression accuracy; $\mathcal{B}_{a,h}$ retains the rollout discretization bias.

PROPOSITION S.8—Evaluation, action quality, and payoff: *Suppose the displayed quantities are finite and the consumption Hamiltonian is μ -strongly concave on $\mathcal{A}$. Then*

$$J_y(a) - J_y(\pi) \geq \mathcal{G}_a - \left\{ \sqrt{\mathcal{D}_a} + \frac{\mathcal{E}_{a,h} + \mathcal{B}_{a,h}}{\sqrt{2\mu}} \right\}^2. \quad (\text{S.22})$$

In particular, an upper bound on the action-loss term and a lower bound on $\mathcal{G}_a$ establish a sufficient condition for policy improvement. The proposition applies to any costate estimator;

a method comparison still requires the independently evaluated payoffs and their associated work.

PROOF: The fixed-policy equation for V^π , evaluated along Y^a , gives the performance-difference identity

$$J_y(a) - J_y(\pi) = \mathbb{E} \int_0^T e^{-\rho t} \{H_{q^\pi(t, Y_t^a)}(a_t) - H_{q^\pi(t, Y_t^a)}(\pi(t, Y_t^a))\} dt. \quad (\text{S.23})$$

The diffusion and the state-dependent part of felicity are common to the two actions, so their differences cancel. The terminal terms agree. Localization followed by the assumed integrability justifies the martingale expectation. Subtract the true greedy Hamiltonian inside the integrand and use Proposition S.4. Minkowski's inequality for the measure $e^{-\rho t} dt d\Pr$, first for $\sqrt{\delta_t}$ and the costate norm, and then for the regression and bias components of the costate, gives

$$\mathbb{E} \int_0^T e^{-\rho t} \left(\sqrt{\delta_t} + \frac{\|q^\pi - \hat{q}\|_d}{\sqrt{2\mu}} \right)^2 dt \leq \left\{ \sqrt{\mathcal{D}_a} + \frac{\mathcal{E}_{a,h} + \mathcal{B}_{a,h}}{\sqrt{2\mu}} \right\}^2.$$

Combining these statements proves the result. Q.E.D.

Independent costate diagnostics. Condition on a fitted policy, its critic, and a held-out state X . Let $\bar{Z}_1, \bar{Z}_2$ be means from independent rollout banks with the same conditional mean $\bar{q}_h(X)$. Conditional independence and zero conditional mean of the two errors imply

$$\mathbb{E}[\langle \hat{q}(X) - \bar{Z}_1, \hat{q}(X) - \bar{Z}_2 \rangle_d \mid X] = \|\hat{q}(X) - \bar{q}_h(X)\|_d^2.$$

With r independent replicates per bank and conditional single-rollout variance $v_h(X) = \mathbb{E}[\|Z - \bar{q}_h(X)\|_d^2 \mid X]$,

$$\frac{1}{2} \mathbb{E}[\|\bar{Z}_1 - \bar{Z}_2\|_d^2 \mid X] = \frac{v_h(X)}{r}.$$

The first sample average is not constrained to be positive. Both identities refer to the actual held-out state distribution. They estimate the quantities in (S.22) only when that distribution is the candidate's discounted state occupation law, or when

a justified change-of-measure account is supplied. Nested-grid differences diagnose time-step sensitivity; without a remainder bound they do not bound $\mathcal{B}_{a,h}$.

A continuous-time costate error account. There is also a direct sufficient condition for controlling the true costate, independent of a rollout bias estimate. For a smooth candidate value v , define its fixed-policy residual and terminal error by

$$r^\pi[v] = -v_t - \mathcal{L}^\pi v - u(\cdot, \pi) + \rho v, \quad b_v = v(T, \cdot) - g.$$

Suppose $b^\pi = c1 + f - \pi$ is continuously differentiable with $\|D_y b^\pi\|_2 \leq L_b$, and suppose differentiating the following Feynman–Kac expectations is justified. If $\|dD_y b_v\|_d \leq C_T$ and $\|dD_y r^\pi[v](s, \cdot)\|_d \leq C_r(s)$ everywhere, then

$$\|dD_y v(t, y) - q^\pi(t, y)\|_d \leq e^{(L_b - \rho)(T-t)} C_T + \int_t^T e^{(L_b - \rho)(s-t)} C_r(s) ds. \quad (\text{S.24})$$

To see this, write $v - V^\pi$ as the expected discounted terminal error plus the expected discounted residual integral. The state Jacobian $J_{t,s}$ satisfies $\dot{J}_{t,s} = D_y b^\pi(s, Y_s) J_{t,s}$, $J_{t,t} = I$, and hence $\|J_{t,s}\|_2 \leq e^{L_b(s-t)}$. Differentiate the expectation and apply that bound. A hard terminal condition gives $C_T = 0$. Equation (S.24) is a sufficient mathematical account for a smooth continuous policy. A sampled residual, a rollout grid difference, or the derivatives of a different deployed implementation do not supply its global hypotheses.

S.9. CAPITAL ERROR BOUNDS AND NUMERICAL ACCOUNTS

This appendix supplies the complete supporting constants and proofs for the capital model and policy-specific theorem in the main article. The economic state, objective, action box, benchmark schedule, and implemented controller are those in (S.99)–(S.101) and (S.104). The analytical anchor, diffusion transfer, and clipping account concern different error terms; none is a training convergence assertion.

S.9.1. The analytical anchor and tube bound

Let $\beta \geq \kappa\|B\|_2$, let s_0 bound the initial cross-sectional standard deviation, and set

$$Q_d = s_0 + \kappa T + \sigma_I \sqrt{(1 - 1/d)T}, \quad (\text{S.25})$$

$$K(s) = \beta \int_s^T e^{-\rho t} w(t) e^{\beta(t-s)} dt + 2\chi e^{-\rho T} Q_d e^{\beta(T-s)}, \quad (\text{S.26})$$

$$\mu(t) = \frac{2\{\log[m_0(t)/M] + [M - m_0(t)]/m_0(t)\}}{[M - m_0(t)]^2}. \quad (\text{S.27})$$

The quantity $\mu(t)$ is the smallest tangent-remainder curvature of log utility on the entire primitive action interval, relative to $m_0(t)$. It is not a Hessian bound estimated from training points. In particular, it can be substantially sharper than the uniform lower curvature M^{-2} .

THEOREM S.9—Curvature-controlled policy guarantee: *Suppose $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, the schedule in (S.101) is interior, and $\|\Pi y\|_d \leq s_0$. Let $V^*(y) = \sup_m J_y(m)$, where the supremum is over the full original admissible class. Then*

$$0 \leq V^*(y) - J_y(m_0) \leq G_d := \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds. \quad (\text{S.28})$$

For $\epsilon \geq 0$ satisfying $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$, every admissible policy m^ϕ with $|m_i^\phi(t, Y) - m_0(t)| \leq \epsilon$ for all coordinates, times, and states satisfies

$$0 \leq V^*(y) - J_y(m^\phi) \leq G_d + D_{d,\epsilon}, \quad (\text{S.29})$$

where, with $A_\beta(T) = \int_0^T e^{\beta t} dt$,

$$\begin{aligned} D_{d,\epsilon} = & \int_0^T \left[\epsilon K(s) + \frac{\epsilon^2 e^{-\rho s}}{2} \{[m_0(s) - \epsilon]^{-2} + \zeta\} \right] ds \\ & + \chi e^{-\rho T} \epsilon^2 A_\beta(T)^2. \end{aligned} \quad (\text{S.30})$$

These bounds hold for every initial mean, every initial state with the stated dispersion bound, and arbitrary adapted competitors in the original action box. They require no neural training convergence or discretization error estimate.

The proof uses synchronous state comparison, an exact discounted mean-state identity, and completion of a quadratic bound on an arbitrary policy deviation. The complete proof appears in Section S.9.2. The result is a model-specific numerical principle: curvature limits the benefit from changing consumption, whereas

a spectral bound limits the propagation of that change through dense nonlinear production. Their balance yields a scalar integral certificate for a high-dimensional policy. Bounded nonlinearity, the additive diffusion structure, and the chosen felicity are essential assumptions, rather than generic properties of all HJB equations.

S.9.1.1. Computable scope and cost

An SVD is used only to propose a scalar λ . Positive pivots in an outward interval $LDL^\top$ calculation for $\lambda I - B^\top B$ certify the spectral majorant. A failed pivot triggers enlargement, not acceptance of the SVD output. The remaining integrals use outward rectangle sums, polynomial exponential and logarithm bounds, and square-root endpoints checked by interval squaring. For a fixed majorant with verified pivots, this certificate requires $O(d^3)$ arithmetic operations and $O(d^2)$ storage for the matrix check, followed by $O(N)$ scalar quadrature work. This is not a dimension-independent training theorem or a worst-case bound on the number of attempted majorants.

The guarantee holds already at initialization. Training is intended to improve the feasible policy inside the tube, not to manufacture its certificate. The anchor itself has the tighter bound G_d and requires no training. A comparison omitting it would overstate the role of the actor. Neither a small $G_d + D_{d,\epsilon}$ nor faster actor evaluation proves a net economic advantage over that benchmark.

S.9.2. Proof of the Continuous-Capital Guarantee

All norms in this proof use the normalization $d^{-1} \sum_i z_i^2$. This normalization is important: the spectral norm of B remains the ordinary Euclidean operator norm, while the stochastic moment of cross-sectional dispersion is of order one rather than order d .

S.9.2.1. Logarithmic tangent remainder

For $a > 0$ and $x > -a$, twice integrating the second derivative gives

$$\frac{x}{a} - \log(1 + x/a) = x^2 \int_0^1 \frac{1-r}{(a+rx)^2} dr.$$

For $x \geq 0$ the integral decreases as x increases. For $x < 0$ it is at least $1/(2a^2)$, its value at zero. Consequently, for every $m \in [\underline{m}, M]$ and interior $a < M$,

$$\log(m/a) - \frac{m-a}{a} \leq -\frac{\mu(a)}{2}(m-a)^2, \quad \mu(a) = \frac{2\{\log(a/M) + (M-a)/a\}}{(M-a)^2}. \quad (\text{S.31})$$

On the smaller interval $|m-a| \leq \epsilon < a$, the same integral also yields

$$0 \leq \frac{m-a}{a} - \log(m/a) \leq \frac{(m-a)^2}{2(a-\epsilon)^2}. \quad (\text{S.32})$$

These are global inequalities on the specified intervals, including their endpoints; no empirical concavity check is used.

S.9.2.2. Synchronous comparison and baseline moments

Construct Y^m and Y^0 on the same probability space with identical initial state and Brownian drivers, where the second process uses m_0 . With $\Delta Y = Y^m - Y^0$ and $\Delta m = m - m_0 \mathbf{1}$, additive noise cancels. Since componentwise hyperbolic tangent is one-Lipschitz,

$$\|\Delta Y_t\|_d \leq \int_0^t \{\beta \|\Delta Y_s\|_d + \|\Delta m_s\|_d\} ds.$$

Gronwall's inequality followed by Minkowski's integral inequality implies

$$(\mathbb{E} \|\Delta Y_t\|_d^2)^{1/2} \leq \int_0^t e^{\beta(t-s)} r(s) ds, \quad r(s) = (\mathbb{E} \|\Delta m_s\|_d^2)^{1/2}. \quad (\text{S.33})$$

All terms are integrable because controls and the nonlinear production term are bounded. For the baseline, common consumption, common drift constants, and common noise vanish under Π . Furthermore, $\|\Pi \tanh(BY)\|_d \leq 1$ and $\mathbb{E} \|\Pi W_T\|_d^2 = (1 - 1/d)T$. Minkowski's inequality therefore gives

$$(\mathbb{E} \|\Pi Y_T^0\|_d^2)^{1/2} \leq \|\Pi y\|_d + \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \leq Q_d. \quad (\text{S.34})$$

This estimate is uniform in the initial mean. It does not suppose that the optimal policy is symmetric or that the dense economy separates.

S.9.2.3. An exact payoff-difference identity

Put $h(Y) = \overline{\tanh(BY)}$. Integrating the mean-state equation inside the discounted flow and terminal mean, and interchanging integrable expectations and time integrals, yields

$$\begin{aligned} J_y(m) - J_y(m_0) &= \int_0^T e^{-\rho t} \mathbb{E} \left[\overline{\log(m_t/m_0(t))} - w(t) \overline{\Delta m_t} - \frac{\zeta}{2} \{\bar{m}_t^2 - m_0(t)^2\} \right] dt \\ &\quad + \kappa \int_0^T e^{-\rho t} w(t) \mathbb{E}[h(Y_t^m) - h(Y_t^0)] dt \\ &\quad - \chi e^{-\rho T} \mathbb{E} \left[2 \langle \Pi Y_T^0, \Pi \Delta Y_T \rangle_d + \|\Pi \Delta Y_T\|_d^2 \right]. \end{aligned} \quad (\text{S.35})$$

The martingale mean terms have zero expectation. The schedule satisfies $m_0(t)^{-1} - w(t) - \zeta m_0(t) = 0$. Thus all terms linear in $\overline{\Delta m_t}$ cancel. Equation (S.31) bounds the first integral above by $-\frac{1}{2} \int e^{-\rho t} \mu(t) r(t)^2 dt$; the remaining mean-adjustment square is nonpositive and may be dropped in an upper bound.

Next, $|h(Y^m) - h(Y^0)| \leq \|B\|_2 \|\Delta Y\|_d$. Equations (S.33) and (S.34), Cauchy-Schwarz, and a change in integration order bound the potentially positive production and terminal-dispersion terms by $\int_0^T K(s) r(s) ds$. Dropping the nonpositive terminal square, we obtain

$$J_y(m) - J_y(m_0) \leq \int_0^T \left[K(s) r(s) - \frac{1}{2} e^{-\rho s} \mu(s) r(s)^2 \right] ds \leq \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds.$$

Taking the supremum over the original adapted controls proves (S.28). Attainment of the supremum is not needed.

For a tube policy, $r(s) \leq \epsilon$. Reverse the payoff difference in (S.35). Equation (S.32) and $\mathbb{E}(\overline{\Delta m})^2 \leq \mathbb{E}\|\Delta m\|_d^2$ bound the consumption and adjustment loss by

$$\frac{\epsilon^2}{2} \int_0^T e^{-\rho s} \{[m_0(s) - \epsilon]^{-2} + \zeta\} ds.$$

The production and terminal cross terms contribute at most $\epsilon \int K(s) ds$. The terminal square now has positive sign and is bounded by $\chi e^{-\rho T} \epsilon^2 A_\beta(T)^2$ using (S.33). This proves $J_y(m_0) - J_y(m^\phi) \leq D_{d,\epsilon}$. Adding (S.28) proves (S.29). Feasibility supplies the nonnegative lower bound. Q.E.D.

S.9.2.4. Interval spectral and deployment checks

For the stored exact-real matrix B , outward products and sums enclose each entry of $B^\top B$. The Schur complement recurrence for an $LDL^\top$ factorization is enclosed entry by entry. If every pivot interval has strictly positive lower endpoint, every exact pivot is positive, so $\lambda I - B^\top B$ is positive definite. Hence $\|B\|_2 < \sqrt{\lambda}$. The proposed numerical singular value is not part of this proof; an unsuccessful interval factorization supplies no spectral bound.

If $\ell \leq m_0(t) \leq u$, then $\text{up}(u - \epsilon) \geq m_0(t) - \epsilon$ and $\text{down}(\ell + \epsilon) \leq m_0(t) + \epsilon$. Every finite output after projection onto (S.103) therefore meets the theorem's tube condition under the declared arithmetic contract, regardless of errors in the neural proposal. The proof does not verify the interpreter or processor. The arithmetic assumptions, overflow checks, saved weights, and input-clock convention remain explicit components of the computational claim.

S.9.3. A network-independent diffusion-transfer bound

Here the word independent concerns the transfer constants, not the economic statistic. Let

$$q_i = \sigma_I^2 \|B_{i\cdot}\|_2^2 + \sigma_C^2 \left(\sum_j B_{ij} \right)^2,$$

$$H = \kappa \left(d^{-1} \sum_i q_i \right)^{1/2}, \quad Q = \frac{\kappa}{2} \left(d^{-1} \sum_i q_i^2 \right)^{1/2}.$$

The letter Q here denotes a generator bound, not the dispersion bound Q_d in (S.25). For a policy in a radius- ϵ interval at sampling dates, put

$$M_\epsilon = \max_{m \in [m_0(0) - \epsilon, m_0(T) + \epsilon]} |c - m| + \kappa, \quad K_\epsilon = \beta M_\epsilon + Q.$$

The terminal dispersion majorant is

$$S_\epsilon = s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}.$$

The formula uses the monotonicity of this economy's m_0 . For more general schedules their extrema replace the endpoints. Define

$$u_\epsilon(t) = he^{\beta t} \{K_\epsilon t/2 + H\sqrt{t/3}\}, \quad (\text{S.36})$$

$$\begin{aligned} E_\epsilon(h) = & \beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt \\ & + \chi e^{-\rho T} u_\epsilon(T) \{2S_\epsilon + u_\epsilon(T)\}. \end{aligned} \quad (\text{S.37})$$

PROPOSITION S.10—Transfer of paired policy values: *For exact arithmetic and untruncated internal innovations, the preceding construction satisfies*

$$|\Delta_\phi - \mathbb{E}X_{\phi,h}| \leq E_\epsilon(h) + E_0(h).$$

With clipped internal innovations, forward rounding, and enclosed scalar integration, replace each u_ϵ by its stated enlarged bound and add the statistic's quadrature and arithmetic error. Denote the resulting saved upper bound by $E_{\phi,h}^{\text{all}}$. It is $O(h)$ apart from these explicit numerical terms and does not involve a network derivative bound. The economic process remains the original continuous diffusion.

The complete proof appears in Section S.9.5. Its key step is to retain the martingale cancellation in the accumulated local drift error. Bounding each local stochastic error separately by its absolute magnitude would yield only an $O(\sqrt{h})$ account and would obscure the small policy differences studied here. This proposition concerns the sampled implementation (S.104); it does not prove this rate for arbitrary state-observed neural feedback.

S.9.4. The deterministic clipping and tail account

Paired statistics are not uniformly bounded, because the terminal dispersion difference contains Gaussian shocks. This is dealt with explicitly, rather than ignored in a bounded concentration inequality. For each declared initial state and radius, the supplement constructs constants C_ϕ, A_ϕ such that

$$\Pr\{|X_{\phi,h}| > C_\phi + A_\phi(\sqrt{d-1} + v)\} \leq e^{-v^2/2}, \quad v \geq 0.$$

They are obtained from synchronous state separation and the norm of the projected reference innovations. Neither an empirical output range nor an empirical maximum enters their construction. With $v_* = 8$, put $b_\phi = C_\phi + A_\phi(\sqrt{d-1} + v_*)$ and $\tau_\phi = A_\phi e^{-v_*^2/2}/v_*$. Then clipping $X_{\phi,h}$ to $[-b_\phi, b_\phi]$ changes its expectation by at most τ_ϕ .

S.9.5. Proofs for Policy-Specific Verification

This section proves the continuous-time transfer, clipping, and statistical statements used in the main text. Norms are normalized by dimension throughout. The internal controller and the economic state are distinct. This distinction is maintained in every comparison below.

S.9.5.1. Generator and stochastic local-error bounds

Let $f_i(y) = \kappa \tanh(B_i y)$, $b(y, m) = c\mathbf{1} + f(y) - m$, and let Σ have idiosyncratic columns $\sigma_I e_i$ and common column $\sigma_C \mathbf{1}$. For the ordinary spectral majorant $\beta \geq \kappa \|B\|_2$,

$$\|f(y) - f(z)\|_d \leq \beta \|y - z\|_d, \quad \|b(y, m)\|_d \leq M_\epsilon.$$

Each row of $Df(y)\Sigma$ has squared norm at most $\kappa^2 q_i$, so its normalized squared Hilbert–Schmidt norm is at most H^2 . Also

$$\left| \frac{1}{2} \text{tr}\{\Sigma \Sigma^\top D^2 f_i(y)\} \right| \leq \frac{\kappa}{2} q_i,$$

because $|\tanh''| \leq 1$. The generator of f under any held admissible action therefore satisfies

$$\|\mathcal{A}^m f(y)\|_d \leq \beta M_\epsilon + Q = K_\epsilon. \tag{S.38}$$

The controls used during the comparison are adapted, but fixed over each cell. The same bound applies to the reference’s deterministic, time-varying common control, with $\epsilon = 0$ and the time-dependent generator.

First take exact arithmetic and untruncated internal innovations. On cell k , Itô's formula and stochastic Fubini give

$$\begin{aligned} \int_{t_k}^{t_{k+1}} \{f(Y_s) - f(Y_{t_k})\} ds &= \int_{t_k}^{t_{k+1}} (t_{k+1} - v) \mathcal{A}^{a_k} f(Y_v) dv \\ &\quad + \int_{t_k}^{t_{k+1}} (t_{k+1} - v) Df(Y_v) \Sigma dW_v =: D_k + M_k. \end{aligned}$$

The deterministic term has L^2 norm at most $K_\epsilon h^2/2$. The stochastic terms are martingale increments over disjoint cells. Itô's isometry, not a sum of their absolute norms, gives

$$\left(\mathbb{E} \left\| \sum_{j < k} M_j \right\|_d^2 \right)^{1/2} \leq Hh \sqrt{t_k/3}.$$

Set $e_k = Y_{t_k} - Z_k$. The chosen actions cancel from the recursions. The triangle inequality yields

$$(\mathbb{E} \|e_k\|_d^2)^{1/2} \leq \beta h \sum_{j < k} (\mathbb{E} \|e_j\|_d^2)^{1/2} + h \{K_\epsilon t_k/2 + H \sqrt{t_k/3}\}.$$

The inhomogeneous term is nondecreasing in t_k . Discrete Gronwall and $(1 + \beta h)^k \leq e^{\beta t_k}$ prove (S.36). Importantly, no derivative of a_k with respect to its internal state appears in this recursion. Replacing a_k by a policy evaluated separately at Y_{t_k} and Z_k would destroy this cancellation and would be a different proposition.

For $t_k \leq t < t_{k+1}$, conditioning Itô's formula gives

$$\|\mathbb{E}\{f(Y_t) - f(Y_{t_k})\}\|_d \leq K_\epsilon(t - t_k).$$

Combining this inequality with the grid-point strong error bounds the absolute bias in the weighted mean-production sum by

$$\beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt.$$

The common components of drift, action, and Brownian noise vanish after applying II. The policy's coordinate deviations from its common schedule at grid dates are

at most ϵ . Therefore

$$(\mathbb{E}\|\Pi Y_T\|_d^2)^{1/2} \leq s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T} = S_\epsilon.$$

For any coupled Y, Z , Cauchy–Schwarz gives

$$\left| \mathbb{E}\{\|\Pi Y\|_d^2 - \|\Pi Z\|_d^2\} \right| \leq (\mathbb{E}\|Y - Z\|_d^2)^{1/2} \{2(\mathbb{E}\|\Pi Y\|_d^2)^{1/2} + (\mathbb{E}\|Y - Z\|_d^2)^{1/2}\}.$$

This proves the terminal term in (S.37). Applying the calculation once to the policy and once to the reference proves Proposition S.10 in exact arithmetic.

S.9.5.2. *Clipped innovations and numerical integration*

Let g be a standard normal variable and $\text{clip}_z(g)$ its projection onto $[-z, z]$. Symmetry gives zero mean for $g - \text{clip}_z(g)$, and

$$\begin{aligned} \mathbb{E}\{g - \text{clip}_z(g)\}^2 &\leq \mathbb{E}[g^2 \mathbf{1}\{|g| > z\}] = 2\{z\varphi(z) + \Pr(g > z)\} \\ &\leq 2\varphi(z)(z + z^{-1}) =: v_z. \end{aligned}$$

Thus the accumulated internal-noise discrepancy has normalized L^2 norm at most $\sqrt{\sigma_I^2 + \sigma_C^2} \sqrt{T v_z}$. It is added to the same Gronwall recurrence, giving the conservative enlargement $e^{\beta T} \sqrt{\sigma_I^2 + \sigma_C^2} \sqrt{T v_z}$ at every grid point. This bounds a change in the controller’s internal update, not a replacement of the economic innovation law.

The scalar weights A_k, B_k are integrated from their exponential expressions. C_k , the reference action integral, and the within-cell production weight are enclosed on a dyadic subdivision with 64 panels per cell. Their midpoint errors are explicitly accumulated. If δ_M bounds the sum of the absolute errors in the integrated reference actions, $e^{\beta T} \delta_M$ enlarges the reference state error. The errors in A_k, B_k, C_k are multiplied by global bounds on consumption, its logarithm, and bounded production, and are added to the statistic error. These terms are not inferred from a difference between two grids.

S.9.5.3. *Arithmetic contract and forward-error budget*

All stored binary64 model coefficients denote exact real inputs to the economic problem. The computation assumes round-to-nearest elementary arithmetic, the

usual dot-product forward-error bound, and correctly rounded input conversion. It does not assume correct rounding of a platform exponential, logarithm, or hyperbolic tangent when proving a payoff endpoint. Network library functions produce proposals only; the interval schedule and inward projection enforce feasibility independently of proposal accuracy. The simulation drift and log utility use the polynomial kernels described next. The Python interpreter, numerical libraries, compiler, hardware, and random-number generator are not machine-formally verified.

For $|x| \leq 20$, write $r = x/256$ and approximate e^r by its degree-14 Taylor polynomial in Horner form. Nine squarings approximate e^{2x} . The Taylor remainder is bounded by $e^{|r|}|r|^{15}/15!$. The coefficient representation and Horner operations satisfy the standard γ_{30} forward bound, with $\gamma_j = ju/(1 - ju)$ and $u = 2^{-52}$, a conservative choice relative to binary64 unit roundoff. Propagating relative error through nine squarings bounds the exponential relative error by less than 2^{-35} on this range. The derivative of $1 - 2/(1 + z)$ with respect to $\log z$ is $2z/(1 + z)^2 \leq 1/2$. Including the final arithmetic and saturation outside $[-20, 20]$ therefore bounds absolute tanh error by 2^{-34} . The saturation error itself is below $2e^{-40}$.

For positive consumption, binary exponent extraction writes $x = 2^{e-1}(2m)$, with $m \in [1/2, 1)$. Then

$$\log x = 2 \sum_{j=0}^{\infty} \frac{q^{2j+1}}{2j+1} + (e-1) \log 2, \quad q = \frac{2m-1}{2m+1} \in [0, 1/3].$$

The implementation retains 18 terms and evaluates $\log 2$ with the same series at $q = 1/3$. The geometric remainder and Horner forward error, including the bounded exponents on the primitive consumption interval, are below 2^{-34} . No empirical comparison with the platform logarithm serves as this error bound. The optional compiled kernels use identical binary64 coefficients, operation order, and disabled fast-math transformations; the array implementation is retained as a fallback.

Clipped innovations yield a deterministic internal state cap. A conservative cap for the implemented ranges is

$$S_{\text{cap}} = 2\{\|y\|_{\infty} + M_{\epsilon}T + (\sigma_I + \sigma_C)10T/\sqrt{h} + 1\}.$$

Using $\gamma_{d+2} S_{\text{cap}} \max_i \sum_j |B_{ij}|$ for the matrix-product error, the code propagates drift, innovation-combination, and state-addition errors through the same discrete Gronwall bound. It verifies the bootstrap condition that this enlargement is less than one. This condition, together with the doubled cap, bounds the entire numerical path. Separately, it accumulates the errors of all statistic sums, log evaluations, and terminal squares. The exact formulas and intermediate constants are returned by `constants`; each final record saves the resulting state cap, forward-roundoff bound, scalar-integration bound, and total bias endpoint. Scalar transcendental evaluations used only to assemble conservative budgets receive additional 10^{-12} bias and 10^{-10} clipping cushions on the declared bounded parameter range; they do not replace the enclosed spectral or integral calculations.

The sampling contract is ideal independent standard-normal innovations rounded to binary64 before the specified clipping, with the forward-error allowance for their conversion. Pseudorandom draws are a reproducible implementation of this contract, not a proof of ideal independence. The resulting certificate is conditional numerical mathematics and conditional statistical inference. It must not be described as a machine-formal guarantee of a particular hardware random stream.

S.9.5.4. *A global clipping envelope*

Let Z^0 denote the internal reference trajectory. In exact arithmetic, synchronous coupling gives

$$\|Z_k^\phi - Z_k^0\|_d \leq D_k, \quad D_k = \epsilon_{\text{eff}} \{e^{\beta t_k} - 1\} / \beta,$$

where ϵ_{eff} also covers schedule movement within one cell, integrated-action error, and both forward-rounding increments. For the retained parameters, $|m'_0| \leq 1$, so $\epsilon + h$ suffices before these numerical enlargements. Define D_T analogously. Production differences have absolute sum at most $\beta \sum_k B_k D_k$. The consumption deficit is at most

$$\frac{\epsilon_{\text{eff}}^2}{2} \{[m_0(0) - \epsilon_{\text{eff}}]^{-2} + \zeta\} \int_0^T e^{-\rho t} dt.$$

The code verifies the strictly positive denominator. Terminal differences are bounded by $\chi e^{-\rho T} \{D_T^2 + 2D_T \|\Pi Z_T^0\|_d\}$.

The projected reference state satisfies

$$\|\Pi Z_T^0\|_d \leq s_0 + \kappa T + e_{\text{round}} + \frac{\sigma_I \sqrt{T}}{\sqrt{d}} H_0,$$

where H_0 is the Euclidean norm of the projected sum of the standardized clipped idiosyncratic innovations, normalized to unit total variance. As a function of all underlying independent standard Gaussians, H_0 is one-Lipschitz. Independence, symmetry, and the variance reduction under clipping give $\mathbb{E}H_0 \leq \sqrt{d-1}$. Gaussian concentration therefore gives $\Pr\{H_0 > \sqrt{d-1} + v\} \leq e^{-v^2/2}$. Collecting the deterministic terms defines C_ϕ and setting $A_\phi = 2\chi e^{-\rho T} D_T \sigma_I \sqrt{T/d}$ gives the claimed tail envelope. Integrating that envelope beyond $v_* > 0$ gives

$$\mathbb{E}[(|X_{\phi,h}| - b_\phi)_+] \leq A_\phi \int_{v_*}^{\infty} e^{-v^2/2} dv \leq A_\phi e^{-v_*^2/2} / v_*.$$

The construction does not require the learned network to be Lipschitz as a function of shocks; only its feasible action range is used in the deterministic separation bound.

For completeness, the Gaussian concentration fact can be obtained from the Gaussian logarithmic Sobolev inequality. If P_t is the Ornstein–Uhlenbeck semigroup, entropy differentiation and the identity $\nabla P_t = e^{-t} P_t \nabla$ imply $\text{Ent}(e^g) \leq \frac{1}{2} \mathbb{E}[e^g \|\nabla g\|^2]$ by Cauchy–Schwarz and integration of e^{-2t} . For $g = \lambda H_0$, the one-Lipschitz bound gives $\log \mathbb{E} e^{\lambda(H_0 - \mathbb{E}H_0)} \leq \lambda^2/2$ by integrating the entropy inequality in λ . Chernoff optimization yields the stated one-sided tail. Smooth approximation and truncation justify the calculation for the norm and clipping functions.

S.9.5.5. *Finite families, selection, and interpretation*

Conditioning on the fitted weights and their validation selection makes each final-test statistic a fixed function of one fresh path. Each row is therefore an independent sample across paths, even when its Brownian bank is shared with other policies. Applying [Maurer and Pontil \(2009, Theorem 4\)](#) to both signs and taking a union bound proves the simultaneous statement. The implementation allocates 2,000 one-sided statements, so at most 1,000 two-sided rows may be reported under this allocation. The audit rejects an overrun. It also checks source weights, path counts,

raw-array hashes, every recomputed mean and variance, and all displayed table inputs.

A certificate may fail to establish improvement. That is distinct from an exception in the numerical code. Both outcomes are retained. A zero correction, a deliberately saturated poor correction, and trained corrections are compared under the same accounting. Small training loss or successful completion is never substituted for a positive lower improvement endpoint. If a candidate is chosen after inspecting several final rows, the simultaneous guarantee continues to hold only within the predeclared finite family; an undeclared search requires a new allocation and independent data.

S.10. PROOFS AND COMPUTABLE ACCOUNTS FOR THE BELLMAN BRIDGE

Throughout this section, $\langle u, v \rangle_d = d^{-1} u^\top v$ and $\|u\|_d^2 = \langle u, u \rangle_d$. Conditioning on fitted parameters and selection is understood. The initial distribution has finite support and maximal dispersion s_0 . The same arguments apply under any initial distribution with the stated moment and tail bounds.

S.10.1. *Finite performance difference and the action bridge*

Bounded drift, bounded controls, clipped innovations, and a finite horizon give finite state moments of every order. For the deterministic reference π_k , the finite composition of smooth transition maps has derivatives bounded by a polynomial in the initial state and innovations. The running state reward is bounded and smooth, and the terminal reward is quadratic. Dominated differentiation therefore establishes (S.111) and its first two derivatives. This argument concerns the finite continuation, so no continuous-time PDE regularity or derivative-discretization estimate is needed.

Let $Q_k^\pi(z, b) = \ell_k(z, b) + S_k^\pi\{x_b(z)\}$. The reference recursion is $V_k^\pi(z) = Q_k^\pi(z, \pi_k)$. For a possibly history-dependent candidate, condition on its history at k , then telescope $V_{k+1}^\pi(Z_{k+1}) - V_k^\pi(Z_k)$. The two terminal functions agree and the initial law is common, giving

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) = \sum_k \mathbb{E}\{Q_k^\pi(Z_k, a_k) - Q_k^\pi(Z_k, \pi_k)\}. \quad (\text{S.39})$$

Put $e_k = \hat{q}_k - q_k^\pi$ and $v_k = a_k - \pi_k \mathbf{1}$. Since $x_{a_k} - x_{\pi_k} = -hv_k$, the fundamental theorem of calculus implies, pathwise,

$$\begin{aligned} & [\hat{Q}_k(Z_k, a_k) - \hat{Q}_k(Z_k, \pi_k)] - [Q_k^\pi(Z_k, a_k) - Q_k^\pi(Z_k, \pi_k)] \\ &= -h \int_0^1 \langle e_k(X_{k,u}), v_k \rangle_d du. \end{aligned}$$

Cauchy–Schwarz with respect to h times counting measure on the grid, the candidate probability law, and uniform measure on $[0, 1]$ proves Proposition S.32. The sign of the line integral is immaterial to its absolute bound. Independence of future reference innovations from the candidate history is used in the recursion.

S.10.2. A computable action gap

The normalized action Hessian of the approximate continuation problem is

$$dD_b^2 \hat{Q}_k(z, b) = -A_k \text{diag}(b_i^{-2}) - \frac{\zeta A_k}{d} \mathbf{1} \mathbf{1}^\top + h^2 dD^2 \hat{S}_k\{x_b(z)\} \preceq \Gamma_k I_d.$$

The second-order integral remainder on the convex action box proves (S.114). Maximizing its scalar quadratic in each coordinate gives the stated endpoint or projection rule, including $\Gamma_k = 0$. The value is nonnegative because $b = a_k$ is allowed.

For the quadratic conclusion, let b_k maximize $\hat{Q}_k$. When $-\Gamma_k \geq \mu_0 h$, strong concavity and the constrained first-order inequality at b_k give

$$\|a_k - b_k\|_d^2 \leq \frac{2\delta_k}{\mu_0 h}, \quad \hat{Q}_k(Z_k, b_k) - \hat{Q}_k(Z_k, \pi_k) \geq \frac{\mu_0 h}{2} \|b_k - \pi_k \mathbf{1}\|_d^2.$$

Use $\|b_k - \pi_k \mathbf{1}\|_d^2 \geq \|a_k - \pi_k \mathbf{1}\|_d^2 / 2 - \|a_k - b_k\|_d^2$ and subtract the actor's gap. Summation yields $\mathcal{M}_h \geq \mu_0 \mathcal{A}_h / 4 - 2\mathcal{D}_h$. Consequently

$$\sqrt{\mathcal{A}_h \mathcal{E}_h} \leq 2\sqrt{(\mathcal{M}_h + 2\mathcal{D}_h) \mathcal{E}_h / \mu_0} \leq \mathcal{M}_h / 2 + \mathcal{D}_h + 2\mathcal{E}_h / \mu_0.$$

Together with (S.113), this proves (S.115). If $a_k = b_k$, the stronger relation $\mathcal{M}_h \geq \mu_0 \mathcal{A}_h / 2$ proves the stated improvement. No claim of global concavity of the original stochastic control functional enters this argument.

S.10.3. Continuation targets on one global grid

The following recursion computes an unbiased normalized derivative target without a matrix-valued state Jacobian. Start at the supplied postdecision mean x , add innovation k , and then simulate the reference states at $j = k + 1, \dots, N$ with the same global grid and the disclosed π_j . With $\gamma_T = 2\chi e^{-\rho T}$, set

$$\begin{aligned} p_N &= -\gamma_T \Pi Z_N, \\ p_j &= B_j Df(Z_j)^\top \mathbf{1} + \{I + hDf(Z_j)\}^\top p_{j+1}, \quad j = N - 1, \dots, k + 1. \end{aligned} \tag{S.40}$$

The output is p_{k+1} ; for $k = N - 1$ it is p_N . There is no derivative of the reference action because the reference depends only on the date. Every running consumption term is likewise constant in the starting state. The chain rule proves (S.40), and the dominated-differentiation argument above shows its expectation is $q_k^\pi(x)$.

The implemented scalar critic has the explicit decomposition, at $s = t_{k+1}$,

$$\hat{S}_k(x) = -\chi e^{-\rho T} \{ \|\Pi x\|_d^2 + \sigma_I^2 (1 - 1/d)(T - s + h) \} + (T - s) \{ F_0(s) + F_1(s, x)/d \}. \tag{S.41}$$

The two networks have distinct parameters. Thus

$$\hat{q}_k(x) = -\gamma_T \Pi x + (T - s) D_x F_1(s, x) :$$

fitting the time level cannot change the spatial costate. Using one instead of the exact clipped-normal second moment in the known state-independent noise term changes neither an action difference nor the costate. The terminal costate is exact. Fitting on a 64-cell grid and applying this scalar predictor on a 2,048-cell grid does not make their continuation problems equal; their difference remains in the assessed $\mathcal{E}_h$.

For two hidden tanh layers, write $F_1(s, x) = w_3^\top \tanh\{W_2 \tanh(W_1(s, x) + b_1) + b_2\} + b_3$. Let W_{1x} contain the state columns of W_1 . Any verified spectral upper bounds $l_1 \geq \|W_{1x}\|_2$, $l_2 \geq \|W_2\|_2$, $l_3 \geq \|w_3\|_2$ give

$$\begin{aligned} \|\hat{q}_k(x) + \gamma_T \Pi x\|_d &\leq C_{\theta,k} := (T - s) l_1 l_2 l_3 / \sqrt{d}, \\ dD^2 \hat{S}_k(x) &\preceq \Lambda_k I_d, \quad \Lambda_k = (T - s) l_3 l_1^2 (l_2^2 + l_2). \end{aligned} \tag{S.42}$$

The Hessian bound uses $|\tanh'| \leq 1$ and $|\tanh''| \leq 1$. The negative quadratic lift may be omitted in this upper bound; an absolute normalized Hessian bound is $\gamma_T + \Lambda_k$. These are computable bounds from the saved parameters, not fitted derivative maxima.

S.10.4. Antithetic cancellation and nonasymptotic ranges

We give explicit ranges for the statistics used in (S.116). All displayed constants are upper endpoints when implemented. The raw costate comparison uses precisely one bank with r independent antithetic pairs. The executed choice is $r = 2$, or four paths per bank. Set

$$\beta \geq \kappa \|B\|_2, \quad n_k = N - k - 1, \quad F_k = (1 + h\beta)^{n_k}, \quad \tau_k = (N - k)h,$$

$$P_k = \beta \sum_{j=k+1}^{N-1} B_j (1 + h\beta)^{j-k-1}.$$

A larger enclosed bound $e^{\beta n_k h}$ may replace F_k . A smaller proved bound on $\|Df(z)^\top \mathbf{1}\|_d$ may replace the leading β in P_k . Suppose $\|\Pi(a_k - m_0(t_k)\mathbf{1})\|_d \leq \epsilon$. Define $s_d = \sqrt{d-1}$ and

$$C_k = C_{\theta,k} + P_k + \gamma_T F_k \kappa n_k h$$

$$+ \gamma_T (F_k - 1) \left\{ s_0 + (\kappa + \epsilon)T + \frac{\sigma_I}{\sqrt{d}} (\sqrt{t_k} + \sqrt{\tau_k}) s_d \right\},$$

$$a_k^{\text{tail}} = \gamma_T (F_k - 1) \frac{\sigma_I}{\sqrt{d}} (\sqrt{t_k} + \sqrt{\tau_k/r}), \quad C_E = \max_k C_k, \quad A_E = \max_k a_k^{\text{tail}}. \quad (\text{S.43})$$

Then the ideal statistic E satisfies

$$\Pr\{|E| > T(C_E + A_E v)^2\} \leq 3e^{-v^2/2}, \quad v \geq 0. \quad (\text{S.44})$$

For the executed choice $v_* = 8$, a clipping threshold and an absolute expectation allowance are

$$b_E = T(C_E + A_E v_*)^2, \quad \tau_E = 6TA_E e^{-v_*^2/2} (C_E/v_* + A_E). \quad (\text{S.45})$$

The same proof bounds E^{raw} by $2T(C_Z + A_Z v)^2$, where C_Z, A_Z omit $C_{\theta,k}$ from (S.43). Its expectation allowance is twice the corresponding value of (S.45). For

$E - E^{\text{raw}}$, one can add these two ranges and allowances; equivalently the direct bound $T(C_E + A_E v)^2 + 2T(C_Z + A_Z v)^2$ uses the same three-event union. Numerical errors are enclosed separately.

To prove these claims, let J be the derivative of a reference terminal state with respect to x . Then $\|J\|_2 \leq F_k$ and $\|J - I\|_2 \leq F_k - 1$. Writing the terminal state as its initial value, the accumulated production, common deterministic drift, and total innovation gives, after subtracting $q_0(x) = -\gamma_T \Pi x$, the production-adjoint term plus

$$-\gamma_T \{(J^\top - I)\Pi x + J^\top \sum_{j>k} h \Pi f(Z_j) + (J^\top - I)\Pi \xi + \Pi \xi\}.$$

The average of the last term is exactly zero within an antithetic bank: the clipping map is odd. The production norm is at most $\kappa n_k h$. The first term is bounded by P_k . Common noise disappears from $\Pi \xi$, although it still affects the states and their Jacobians; the bounds on those terms are global.

For the candidate prefix, let $H_0 = \|\Pi \sum_{j<k} \text{clip}_z(G_{j,I})\|_2 / \sqrt{k}$ when $k > 0$, with zero contribution at $k = 0$. The map defining H_0 is 1-Lipschitz on its standard Gaussian input and $\mathbb{E}H_0 \leq s_d$. The average of the analogous future norms from r independent antithetic base paths has mean at most s_d and Lipschitz constant $r^{-1/2}$. Gaussian concentration therefore bounds the failure of $H_0 \leq s_d + v$ and of the two future bounds $H_l \leq s_d + v / \sqrt{r}$ by $3e^{-v^2/2}$ in total. On their intersection,

$$\|\Pi X_{k,U}\|_d \leq s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{t_k/d} (s_d + v),$$

and each bank error has norm at most $C_E + A_E v$. Cauchy–Schwarz proves (S.44). Integrating its tail, using $\int_{v_*}^\infty e^{-v^2/2} dv \leq e^{-v_*^2/2} / v_*$, proves (S.45). Subtracting two bank residuals proves the raw and paired-error bounds.

For completeness, the predicted improvement has a substantially smaller range than the sum of two global value bounds. Let $\Delta \geq \sup_k \|a_k - \pi_k \mathbf{1}\|_d$ and let l_k, u_k enclose the candidate, reference, and line segments between them. Put

$$L_\ell = \max_k \max_{m \in \{l_k, u_k\}} |(A_k/h)(m^{-1} - \zeta m) - B_k/h|, \quad C_\theta = \max_k C_{\theta,k},$$

and

$$C_M = T\Delta \left[L_\ell + C_\theta + \gamma_T \{s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{T/d} s_d\} \right],$$

$$A_M = T\Delta \gamma_T \sigma_I \sqrt{T/d}.$$

The mean value theorem and the prefix event above imply $\Pr\{|M| > C_M + A_M v\} \leq e^{-v^2/2}$. Thus $b_M = C_M + A_M v_*$ and $\tau_M = A_M e^{-v_*^2/2}/v_*$ are valid. Also $0 \leq A \leq T\Delta^2$. If w_A bounds the normalized diameter of the action boxes and $\Gamma_+ = \max_k \max(0, \Gamma_k/h)$, then

$$D \leq Tw_A \{L_\ell + C_\theta + \gamma_T \|\Pi x_{a_K}(Z_K)\|_d\} + T\Gamma_+ w_A^2/2.$$

This gives b_D, τ_D from the same affine Gaussian-tail calculation. All ranges remain valid under mixtures over the disclosed initial profiles and, using the maximum constants, over fitted streams.

S.10.5. *Statistics, arithmetic, and payoff transfer*

Conditional independence of the two continuation banks gives

$$\mathbb{E}[E \mid K, Z_K, U] = T \|\hat{q}_K(X_{K,U}) - q_K^\pi(X_{K,U})\|_d^2,$$

and the usual two-replicate identity gives

$$\mathbb{E}[E^{\text{raw}} \mid K, Z_K, U] = T \mathbb{E}[\|Z_1 - q_K^\pi(X_{K,U})\|_d^2 \mid K, Z_K, U].$$

Uniform sampling of K proves all the mean identities in (S.116). Equal independent samples from each declared stream identify the finite uniform mean. Independence across streams and the nonidentical-variable version of the empirical Bernstein inequality, [Maurer and Pontil \(2009, Theorem 11\)](#), justify pooling. Dependence among M, A, E, D on one path does not invalidate a union bound across their confidence statements. Clipping changes each expectation by at most its proved tail allowance.

The finite-grid identities above concern exact real transition and reward maps with the disclosed coefficients. A protected numerical audit must additionally enclose its scalar weights, forward-state error, neural values and derivatives, adjoint

recurrence, and statistic arithmetic. The controller's rounded actions are held fixed when its rounded state is compared with the exact internal state, so this state account does not require a derivative of the actor. For example, if the action bridge has normalized representation error at most e_x and the normalized critic and true continuation costates have Lipschitz bounds $L_{\hat{q}}, L_q$ on the enclosing domain, Minkowski's inequality gives

$$\sqrt{\mathcal{E}_h^{\text{exact}}} \leq \sqrt{\mathcal{E}_h^{\text{represented}}} + \sqrt{T}(L_{\hat{q}} + L_q)e_x.$$

A deterministic clipped-noise state enclosure may be used to multiply these small arithmetic errors. It is not used as the statistical range in place of (S.43). The audit saves the resulting cushions and evaluates the endpoint with outward rounding. An ordinary platform derivative without such an account remains a finite-grid diagnostic, rather than a protected confidence endpoint.

The executed account makes these numerical terms explicit. Let $C_{\max}$ enclose every coordinate of a candidate prefix followed by a reference continuation, including its numerical representatives. A sufficient choice is

$$C_{\max} = 4 \left\{ \max_y \|y\|_{\infty} + M_{\epsilon}T + (\sigma_I + \sigma_C)zT/\sqrt{h} + 1 \right\}.$$

The maximum is over the disclosed initial profiles. This deterministic bound is used for numerical error only. Put $u = 2^{-52}$, $\gamma_n = nu/(1 - nu)$, and $\epsilon_{\tanh} = 2^{-34}$. The inherited polynomial kernels have uniform tanh and consumption-log error at most $\epsilon_{\tanh}$. The row dot-product error is bounded by $\gamma_{d+2}C_{\max} \max_i \sum_j |B_{ij}|$. Multiplication by κ , addition of the drift constants and action, and the protected state update give the saved one-step bound ξ_h . The source evaluates all its positive terms outward, with γ_{32} for the scalar update and an additional $16u(\sigma_I + \sigma_C)z\sqrt{h}$ for represented innovations. Thus

$$e_Z = Ne^{\beta T}\xi_h, \quad e_x = (1 + h\beta)e_Z + 2\xi_h + \gamma_{16}(C_{\max} + hM_{\epsilon})$$

are sufficient prefix and bridge representation bounds. The last term also covers the line interpolation and a coupling of the binary uniform representative to a continuous uniform variable within 2^{-53} .

Let $b_r = \max_i \|B_i\|_2$ and $b_1 = \max_i \sum_j |B_{ij}|$. A sufficient operator error for the drift derivative used in the adjoint is

$$\varepsilon_{Df} = \beta \left\{ b_r \sqrt{d} e_Z + \gamma_{d+2} C_{\max} b_1 + 2\varepsilon_{\tanh} + \gamma_5 (1 + \varepsilon_{\tanh})^2 \right\}. \quad (\text{S.46})$$

The first term uses $|\tanh''| \leq 1$ to propagate the state error; the last two enclose the computed $1 - \tanh^2$. If $\tilde{p}_{j+1}$ is a computed adjoint and e_{j+1} encloses its normalized error, a valid next error is

$$e_j = (1 + h\beta)e_{j+1} + (h\|\tilde{p}_{j+1}\|_d + B_j^+) \varepsilon_{Df} + \beta r_{B_j} + \omega_j, \quad (\text{S.47})$$

$$\omega_j = \gamma_{d+24} [\|\tilde{p}_{j+1}\|_d + \kappa \|B\|_2 \{h\|\tilde{p}_{j+1}\|_d + |\bar{B}_j|\} (1 + 4\varepsilon_{\tanh})] + 10^{-24}.$$

Here B_j^+ and r_{B_j} are its scalar upper endpoint and midpoint radius. The terminal error combines an interval evaluation of $-\gamma_T \Pi \tilde{Z}_N$ with $\gamma_T e_Z$. The coefficient γ_{d+24} covers the products, row dot product, and addition in $\tilde{p} + \{(h\tilde{p} + \bar{B}_j)\kappa(1 - \widetilde{\tanh^2})\}B$. The relative dot-product bound uses $\|B\|_2$, bounded outward by its Frobenius norm. The small absolute term dominates gradual-underflow errors in this fixed $d \leq 50$ update: the primitive coefficients and step are bounded, $|B_{ij}| \leq 1$, and even a sum of all its elementary underflow errors is less than 10^{-24} . The positive recurrence (S.47) itself is evaluated by interval operations, so no additional empirical error or terminal correction is inferred from a comparison of grids. Averaging the four path errors and enclosing the bank's summation gives its recorded error.

An explicit costate Lipschitz bound, used only to multiply e_x , is obtained by putting $H_f = \kappa \|B\|_2^2$ and $\mathcal{B} = \sum_j B_j$. The Euclidean bilinear second derivative of the reference state flow is at most $H_f T e^{2\beta T}$, while the Hessian of $\bar{f}$ has norm at most H_f/d . The chain rule therefore gives

$$L_q = H_f e^{2\beta T} \mathcal{B} (1 + \beta \sqrt{d} T) + \gamma_T e^{2\beta T} (1 + \sqrt{d} C_{\max} H_f T). \quad (\text{S.48})$$

For the neural costate one may use $L_{\hat{q}} = \gamma_T + T l_3 l_1^2 (l_2^2 + l_2)$. The error in the predicted advantage caused by the prefix replacement is at most $T \Delta L_{\hat{q}} (1 + h\beta) e_Z$. This uses the difference of the two continuation gradients before taking an absolute bound. The corresponding actor-gap error is at most $Tw_{\mathcal{A}} L_{\hat{q}} (1 + h\beta) e_Z$. The squared-

error account uses the previously displayed Minkowski bound with $\sqrt{T}(L_q + L_{\hat{q}})e_x$.

The source saves all these constants and the verified positive-curvature indicator.

Finally, numerical enclosures of sample values can be used more accurately than

a uniform arithmetic allowance. Suppose the exact clipped sample lies in $[L_i, U_i]$,

let m_i be the disclosed binary midpoint, and let r_i enclose $\max(m_i - L_i, U_i - m_i)$.

Projection onto the subspace orthogonal to the constant vector is nonexpansive.

Consequently the exact sample mean differs from $\bar{m}$ by at most $\bar{r}$, and its sample

standard deviation is at most

$$s_m + \sqrt{\frac{n}{n-1}} \left(\frac{1}{n} \sum_i r_i^2 \right)^{1/2}.$$

With $\ell = \log(4/\alpha_{\text{event}})$, it suffices to add

$$\bar{r} + \sqrt{\frac{2\ell}{n-1}} \left(\frac{1}{n} \sum_i r_i^2 \right)^{1/2}$$

to each side of the midpoint empirical Bernstein interval. This construction en-

closes the unknown exact sample moments. Its data-dependent numerical radii do

not replace the independently proved population clipping ranges. For the costate

product, for example, bank and critic norm errors η_1, η_2 give the sample radius

$T\{\eta_1 \|\tilde{e}_2\|_d + \eta_2 \|\tilde{e}_1\|_d + \eta_1 \eta_2\}$, in addition to the interval dot product. All mean

and variance corrections are evaluated outward. The raw costate comparison uses

the identical represented queries; the welfare endpoint additionally includes their

exact-state representation account.

Finally, subtraction of the reference transformed reward gives exactly

$$\mathbb{E}X_{a,h} = \mathcal{R}_h(a) - \mathcal{R}_h(\pi) - D_{\text{hold}}$$

at the level of expectations, where $X_{a,h}$ is the paired statistic in Proposition S.10. The

reference internal drift uses precisely $h\pi_k = \bar{M}_k$; the difference from the continuous

schedule is already included in that proposition's scalar quadrature and reference-

state account. Pointwise optimality of $m_0(t)$ in the consumption-only part implies

$D_{\text{hold}} \geq 0$. Applying Proposition S.32, the simultaneous endpoints, and the paired

payoff-transfer allowance proves Theorem S.34. Applying Proposition S.33 on the

same event proves its optional quadratic bound. The nonnegative parts of upper risk and action endpoints are valid because their true expectations are nonnegative. The maximum of two valid lower bounds is again valid on that event.

S.11. A PAIRED TRANSFER TO CONTINUOUS-TIME WELFARE

The direct comparison of two executed policies permits a sharper transfer than the sum of their separate discretization bounds. This section establishes that transfer for the original capital diffusion and the original held-action implementation. It changes neither the simulated observations nor the confidence event. The construction uses only the economic primitives, the action tube, and the assessment grid.

S.11.1. *The compared policies and the exact finite statistic*

Condition on all training and selection information. Let a_k, b_k be the two implemented actions on $[t_k, t_{k+1})$, where $t_k = kh$ and $Nh = T$. They may depend on the entire innovation history through t_k , including the rounded internal states used by their implementations. Both satisfy

$$\|a_k - m_0(t_k)\mathbf{1}\|_\infty \leq \varepsilon, \quad \|b_k - m_0(t_k)\mathbf{1}\|_\infty \leq \varepsilon. \quad (\text{S.49})$$

The inward guard in the executed verifier enforces these inequalities. As in the capital implementation, this tube is contained in the primitive positive consumption box. The two policies receive the same initial profile and Brownian innovations. Their physical states satisfy

$$dY_t^v = \{c\mathbf{1} + f(Y_t^v) - v_k\} dt + \Sigma dW_t, \quad f(y) = \kappa \tanh(By), \quad v \in \{a, b\}.$$

Here $\Sigma = (\sigma_I I_d, \sigma_C \mathbf{1})$ and $c = \text{productivity} - (\sigma_I^2 + \sigma_C^2)/2$. The actions in these equations are the actions actually returned by the original implementation. In particular, they are not recomputed at a different state or on a different grid.

For the proof, introduce an auxiliary Euler state with exact arithmetic and unclipped Brownian increments,

$$Z_{k+1}^v = Z_k^v + h\{c\mathbf{1} + f(Z_k^v) - v_k\} + \Sigma(W_{t_{k+1}} - W_{t_k}), \quad Z_0^v = Y_0^v. \quad (\text{S.50})$$

This is an analytical coupling. It is not a second executed policy. Its actions remain those generated by the original clipped, rounded controller. Consequently, no Lipschitz or differentiability assumption on the actor is needed below.

Use $\langle x, y \rangle_d = d^{-1} x^\top y$, $\|x\|_d^2 = \langle x, x \rangle_d$, and $\Pi = I_d - d^{-1} \mathbf{1} \mathbf{1}^\top$. Put

$$\psi(y) = \overline{f(y)}, \quad w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad \omega(t) = e^{-\rho t} w(t),$$

and define the exact scalar weights

$$A_k = \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, \quad B_k = \int_{t_k}^{t_{k+1}} \omega(t) dt, \quad J_k = \int_{t_k}^{t_{k+1}} \omega(t)(t - t_k) dt. \quad (\text{S.51})$$

Let $X_h^{a,b}$ be the following exact-real random variable:

$$\begin{aligned} X_h^{a,b} = & \sum_{k=0}^{N-1} \left[B_k \{ \psi(Z_k^a) - \psi(Z_k^b) \} + A_k \{ \overline{\log a_k} - \overline{\log b_k} \} - B_k (\bar{a}_k - \bar{b}_k) \right] \\ & - \frac{\zeta}{2} \sum_{k=0}^{N-1} A_k (\bar{a}_k^2 - \bar{b}_k^2) - \chi e^{-\rho T} \{ \|\Pi Z_N^a\|_d^2 - \|\Pi Z_N^b\|_d^2 \}. \end{aligned} \quad (\text{S.52})$$

Integration by parts applied to $e^{-\rho t} w(t) \overline{Y_t^v}$ shows that $J(a) - J(b)$ equals the expectation of (S.52), with the production sum replaced by $\int_0^T \omega(t) \{ \psi(Y_t^a) - \psi(Y_t^b) \} dt$ and the terminal states replaced by their physical counterparts. The consumption terms in (S.52) are already integrated exactly because the implemented actions are held on each cell. The common initial and exogenous-drift terms cancel.

S.11.2. Computable global coefficients

In this section $\|\cdot\|_2$ for a matrix is its Euclidean operator norm. Choose a verified bound $b_* \geq \|B\|_2$ and set

$$\beta = \kappa b_*, \quad L_2 = \frac{4}{3\sqrt{3}}, \quad q_i = \sigma_I^2 \|B_i\|_2^2 + \sigma_C^2 (B_i \mathbf{1})^2, \quad r_B = \max_i \|B_i\|_2.$$

B_i denotes row i of B ; the q_i in this section are covariance coefficients, not costates.

Let

$$C_0 = \max_{0 \leq k < N} |c - m_0(t_k)|, \quad M_\varepsilon = C_0 + \kappa + \varepsilon, \quad R_i = C_0 |B_i \mathbf{1}| + (\kappa + \varepsilon) \sum_j |B_{ij}|.$$

Any upper bounds for these quantities may be substituted. In the declared economy, m_0 is increasing and $c < m_0(0)$, so the calculation bounds C_0 by $m_0(T) - c$ using interval endpoints.

The scalar production map admits the Lipschitz bound

$$|\psi(x) - \psi(y)| \leq \beta_\psi \|x - y\|_d, \quad \beta_\psi = \min \left\{ \beta, \kappa \left[\frac{1}{d} \sum_{i,j} \max\{(BB^\top)_{ij}, 0\} \right]^{1/2} \right\}. \quad (\text{S.53})$$

Indeed, $dD\psi = \kappa B^\top u$ with $0 \leq u_i \leq 1$. Bounding $u^\top BB^\top u$ either by $b_*^2 d$ or by the sum of the positive entries proves (S.53). The implementation replaces every displayed quantity by an outward upper bound before taking the minimum.

Define

$$\begin{aligned} H_m &= \kappa L_2 b_*^2, & H_x &= \kappa L_2 b_* r_B \sqrt{d}, \\ G &= \kappa \left(\frac{1}{d} \sum_i q_i \right)^{1/2}, & H_\Sigma &= \kappa L_2 \|\text{diag}(\sqrt{q})B\|_2, \\ K &= \kappa \{C_0 \|B\mathbf{1}\|_d + b_*(\kappa + \varepsilon)\} + \frac{\kappa L_2}{2} \|q\|_d, \\ L_v &= \kappa L_2 \|\text{diag}(R)B\|_2 + \beta^2 + \kappa \|\text{diag}(q)B\|_2, \\ L_m &= H_m M_\varepsilon + \beta_\psi \beta + \kappa \min\{b_* \|q\|_d, \|\text{diag}(q)B\|_2\}. \end{aligned} \quad (\text{S.54})$$

All of these are functions of disclosed primitives and the action tube. They do not depend on fitted weights, simulated outcomes, or an observed maximum of a derivative.

To record the deterministic error recursion, let $r = 2\varepsilon$ and put

$$\begin{aligned} D(t) &= r \frac{e^{\beta t} - 1}{\beta}, & I_1(t) &= \int_0^t D(s) ds, & I_2(t) &= \int_0^t D(s)^2 ds, \\ \eta(t) &= h e^{\beta t} \left\{ \frac{Kt}{2} + G \sqrt{\frac{t}{3}} \right\}. \end{aligned} \quad (\text{S.55})$$

The continuous extension $D(t) = rt$ applies if $\beta = 0$. Set $\xi_0 = 0$ and, successively for $k = 1, \dots, N$, calculate

$$F_k = \frac{h}{2} \{L_v I_1(t_k) + \beta r t_k\} + h H_\Sigma \sqrt{I_2(t_k)/3} + h H_x \sum_{j < k} D(t_j) \eta(t_j),$$

$$\xi_k = F_k + h \beta \sum_{j < k} \xi_j. \quad (\text{S.56})$$

This positive recursion retains the forcing accumulated through the current date. It does not propagate a martingale through a future random Jacobian.

The initial distribution assigns probabilities p_ℓ to deterministic profiles y_ℓ . Define

$$S_\ell = \|\Pi y_\ell\|_d + (\kappa + \varepsilon)T + \sigma_I \sqrt{(1 - 1/d)T}, \quad \bar{S} = \sum_\ell p_\ell S_\ell. \quad (\text{S.57})$$

The mean in this expression is an average of conditional bounds. It is not asserted to be the L^2 norm of the state under the mixture of profiles.

PROPOSITION S.11—Paired payoff transfer: *For the two held policies in (S.49), define*

$$\begin{aligned} \mathfrak{B}_h^{\text{ideal}} = & \sum_{k=0}^{N-1} B_k \{ \beta_\psi \xi_k + H_m D(t_k) \eta(t_k) \} \\ & + \sum_{k=0}^{N-1} J_k \left\{ \frac{L_m}{h} [I_1(t_{k+1}) - I_1(t_k)] + \beta_\psi r \right\} \\ & + \chi e^{-\rho T} \{ (2\bar{S} + 2D(T)) \xi_N + 2D(T) \eta(T) \}. \end{aligned} \quad (\text{S.58})$$

Then

$$\left| J(a) - J(b) - \mathbb{E} X_h^{a,b} \right| \leq \mathfrak{B}_h^{\text{ideal}}. \quad (\text{S.59})$$

Suppose δ_{num} bounds, in conditional L^2 at every node, the distance between each auxiliary Euler state and its original rounded, clipped internal state. Suppose ϵ_{stat} bounds the complete arithmetic error of each stored policy-minus-reference statistic, and ϵ_{sub} bounds their final subtraction. Then the original stored direct difference $\widehat{X}_h^{a,b}$ satisfies

$$\left| J(a) - J(b) - \mathbb{E} \widehat{X}_h^{a,b} \right| \leq \mathfrak{B}_h^{\text{pair}}, \quad (\text{S.60})$$

where

$$\begin{aligned} \mathfrak{B}_h^{\text{pair}} &= \mathfrak{B}_h^{\text{ideal}} + 2\beta_\psi \delta_{\text{num}} \sum_k B_k \\ &\quad + 2\chi e^{-\rho T} \delta_{\text{num}} (2\bar{S} + \delta_{\text{num}}) + 2\epsilon_{\text{stat}} + \epsilon_{\text{sub}}. \end{aligned} \tag{S.61}$$

The conclusion is uniform over both policies satisfying the displayed tube.

S.11.3. Proof of Proposition S.11

Regularity and integrability. The drift is globally Lipschitz in the state, with constant β , and bounded uniformly over admissible actions. Thus the physical equations have unique strong solutions with finite moments on the finite horizon. The actions are bounded and measurable at the left endpoint of their cell. The first three derivatives of f are bounded. All stochastic integrals used below are consequently square integrable, so their expectations vanish and Itô's isometry applies without an unaccounted localization limit. The quadratic terminal payoff and every product below are integrable.

Derivative and generator bounds. The elementary identities for $\tanh$ give $|\tanh'| \leq 1$, $|\tanh''| \leq L_2$, and $|\tanh'''| \leq 2$. For a held action v , write

$$\mathcal{L}^v u(y) = Du(y) \{c\mathbf{1} + f(y) - v\} + \frac{1}{2} \text{tr} \{ \Sigma \Sigma^\top D^2 u(y) \}.$$

For vector-valued u , this definition applies to each component. Since $v = m_0(t_k)\mathbf{1} + e$ with $\|e\|_\infty \leq \varepsilon$,

$$\|\mathcal{L}^v f(y)\|_d \leq K, \quad d^{-1/2} \|Df(y)\Sigma\|_{\text{HS}} \leq G.$$

To see the first inequality, bound $\|Df(y)(c\mathbf{1} + f(y) - v)\|_d$ by $\kappa\{C_0\|B\mathbf{1}\|_d + b_*(\kappa + \varepsilon)\}$. The i th second-order term is $\kappa q_i \tanh''(B_i y)/2$, whose normalized norm is at most $\kappa L_2 \|q\|_d/2$.

For two states x, y and two actions v, w in the same tube, splitting the drift part into the change in Df and the change in drift gives

$$\|\mathcal{L}^v f(x) - \mathcal{L}^w f(y)\|_d \leq L_v \|x - y\|_d + \beta \|v - w\|_d,$$

$$|\mathcal{L}^v \psi(x) - \mathcal{L}^w \psi(y)| \leq L_m \|x - y\|_d + \beta_\psi \|v - w\|_d. \quad (\text{S.62})$$

In the first line, the change in Df is bounded by $\kappa L_2 \|\text{diag}(R)B\|_2 \|x - y\|_d$, because $|B_i(c\mathbf{1} + f(x) - v)| \leq R_i$. The remaining first-order terms are bounded by $\beta^2 \|x - y\|_d + \beta \|v - w\|_d$. The change in the second-order term is bounded by $\kappa \|\text{diag}(q)B\|_2 \|x - y\|_d$.

For the second line, the scalar Hessian obeys $\|D^2\psi\|_2 \leq H_m/d$. Its contribution to the drift difference is at most $H_m M_\epsilon \|x - y\|_d$; the remaining drift terms contribute $\beta_\psi \beta \|x - y\|_d + \beta_\psi \|v - w\|_d$. The change in its second-order term is bounded by either $\kappa b_* \|q\|_d \|x - y\|_d$ or $\kappa \|\text{diag}(q)B\|_2 \|x - y\|_d$. This proves the coefficients in (S.54).

The normalized Hilbert–Schmidt bound for the paired diffusion derivative is

$$\begin{aligned} \frac{1}{d} \|\{Df(x) - Df(y)\} \Sigma\|_{\text{HS}}^2 &= \frac{\kappa^2}{d} \sum_i q_i \{\tanh'(B_i x) - \tanh'(B_i y)\}^2 \\ &\leq \frac{\kappa^2 L_2^2}{d} \|\text{diag}(\sqrt{q})B(x - y)\|_2^2 \leq H_\Sigma^2 \|x - y\|_d^2. \end{aligned} \quad (\text{S.63})$$

This calculation accounts for the dimension normalization explicitly. The scalar production Hessian likewise has no extra $\sqrt{d}$ factor. For the vector four-point bound, an operator-norm estimate does require

$$\|Df(x) - Df(y)\|_2 \leq \kappa L_2 b_* \max_i |B_i(x - y)| \leq H_x \|x - y\|_d.$$

The individual and paired state errors. Fix an initial profile. Let $\Delta Y_t = Y_t^a - Y_t^b$, $\Delta Z_k = Z_k^a - Z_k^b$, and $e_k^v = Y_{t_k}^v - Z_k^v$. The common Brownian increments cancel from each state difference. Since $\|a_k - b_k\|_d \leq r$, pathwise Gronwall gives

$$\|\Delta Y_t\|_d \leq D(t), \quad \|\Delta Z_k\|_d \leq r \frac{(1 + h\beta)^k - 1}{\beta} \leq D(t_k). \quad (\text{S.64})$$

For a single policy the accumulated local drift error through t_k is

$$R_k^v = \sum_{j < k} \int_{t_j}^{t_{j+1}} \{f(Y_s^v) - f(Y_{t_j}^v)\} ds.$$

Apply Itô's formula to $f(Y_s^v)$ within each cell and interchange the integrals. If $\theta_h(s) = t_{j+1} - s$ for $s \in [t_j, t_{j+1})$, Minkowski's inequality and Itô's isometry yield

$$\|R_k^v\|_{L^2(d)} \leq K \int_0^{t_k} \theta_h(s) ds + G \left\{ \int_0^{t_k} \theta_h(s)^2 ds \right\}^{1/2} = h\{Kt_k/2 + G\sqrt{t_k/3}\}.$$

Here $\|U\|_{L^2(d)} = (\mathbb{E}\|U\|_d^2)^{1/2}$, with expectation conditional on the fixed initial profile. The nodal error recursion and discrete Gronwall therefore give

$$\|e_k^v\|_{L^2(d)} \leq \eta(t_k). \quad (\text{S.65})$$

Apply the same representation to $R_k^a - R_k^b$. Equations (S.62) and (S.63) imply

$$\begin{aligned} \|R_k^a - R_k^b\|_{L^2(d)} &\leq \int_0^{t_k} \theta_h(s) \{L_v D(s) + \beta r\} ds \\ &\quad + H_\Sigma \left\{ \int_0^{t_k} \theta_h(s)^2 D(s)^2 ds \right\}^{1/2} \\ &\leq \frac{h}{2} \{L_v I_1(t_k) + \beta r t_k\} + h H_\Sigma \sqrt{I_2(t_k)/3}. \end{aligned}$$

For the last inequality, on each cell θ_h and θ_h^2 decrease while D and D^2 increase. The integral of a decreasing function times an increasing function is no greater than the product of their cell averages times the cell length. Their kernel averages are $h/2$ and $h^2/3$, respectively. The martingale isometry is applied to the single accumulated stochastic integral with predictable integrand. There is no isometry assertion for a future random flow multiplying a past noise increment.

For the difference of the nodal drift errors, the four-point identity and the derivative bound give

$$\begin{aligned} &\|f(Y_{t_j}^a) - f(Y_{t_j}^b) - f(Z_j^a) + f(Z_j^b)\|_d \\ &\leq \beta \|e_j^a - e_j^b\|_d + H_x \|\Delta Z_j\|_d \|e_j^b\|_d. \end{aligned}$$

One obtains this inequality by first changing ΔY_{t_j} to ΔZ_j at the base point $Y_{t_j}^b$, and then changing that base point to Z_j^b with ΔZ_j fixed. Combining it with (S.64) and (S.65) proves inductively that

$$\|e_k^a - e_k^b\|_{L^2(d)} \leq \xi_k. \quad (\text{S.66})$$

All bounds are uniform in the initial profile.

Production at and between decision dates. The corresponding scalar four-point inequality uses H_m in place of H_x and β_ψ in place of β . Thus

$$\mathbb{E}|\psi(Y_{t_k}^a) - \psi(Y_{t_k}^b) - \psi(Z_k^a) + \psi(Z_k^b)| \leq \beta_\psi \xi_k + H_m D(t_k) \eta(t_k).$$

Multiplication by B_k accounts for the first line of (S.58).

For variation inside a cell, take expectations in Itô's formula for $\psi(Y_t^a) - \psi(Y_t^b)$. The martingale has mean zero by the integrability already established. Put $\Psi_k(s) = \int_s^{t_{k+1}} \omega(t) dt$. The expected weighted within-cell error is bounded in absolute value by

$$\int_{t_k}^{t_{k+1}} \Psi_k(s) \{L_m D(s) + \beta_\psi r\} ds \leq J_k \left\{ \frac{L_m}{h} [I_1(t_{k+1}) - I_1(t_k)] + \beta_\psi r \right\}.$$

Here Ψ_k decreases, D increases, and $\int_{t_k}^{t_{k+1}} \Psi_k(s) ds = J_k$. This proves the second line of (S.58).

Terminal dispersion and the initial population. For a fixed profile y_ℓ , projecting the physical state equation removes the scalar schedule, the exogenous scalar drift, and the common shock. Minkowski's inequality yields $\|\Pi Y_T^b\|_{L^2(d)} \leq S_\ell$. The same bound holds for the auxiliary Euler state ΠZ_N^b and for either policy separately. Use the exact terminal identity

$$\begin{aligned} & \|\Pi Y_T^a\|_d^2 - \|\Pi Y_T^b\|_d^2 - \|\Pi Z_N^a\|_d^2 + \|\Pi Z_N^b\|_d^2 \\ &= 2\langle \Pi Y_T^b, \Pi(e_N^a - e_N^b) \rangle_d + 2\langle \Pi e_N^b, \Pi \Delta Z_N \rangle_d \\ & \quad + \langle \Pi(\Delta Y_T + \Delta Z_N), \Pi(e_N^a - e_N^b) \rangle_d. \end{aligned}$$

Conditional Cauchy–Schwarz and the pathwise bounds for both differences give the upper bound $(2S_\ell + 2D(T))\xi_N + 2D(T)\eta(T)$ on the absolute expectation. Average this bound over the profiles. Because D, η, ξ are uniform in the initial state, the result contains $\bar{S} = \sum p_\ell S_\ell$. In particular, the calculation uses the mean initial spread after applying Cauchy–Schwarz separately for each profile; it does not replace a mixture second moment by a mean of square roots. Multiplication by $\chi e^{-\rho T}$ proves the remaining line of (S.58), and hence (S.59).

The stored statistic. Let $\tilde{Z}_k^v$ denote the actual rounded, clipped internal state. The inherited forward-arithmetic and normal-clipping bounds, with the actions held fixed as above, provide a uniform value $\delta_{\text{num}} \geq \|Z_k^v - \tilde{Z}_k^v\|_{L^2(d)}$ conditional on each profile. Scalar production then contributes at most $2\beta_\psi \delta_{\text{num}} \sum B_k$ for the two policies. For each policy,

$$\mathbb{E} \left| \|\Pi Z_N^v\|_d^2 - \|\Pi \tilde{Z}_N^v\|_d^2 \right| \leq \delta_{\text{num}} (2S_\ell + \delta_{\text{num}}).$$

Averaging over the profiles and adding the two bounds gives the terminal arithmetic term in (S.61).

Each original stored gain is a policy-minus-reference statistic. Under the verified common-noise and initial-state identities, their exact-real reference terms coincide, and cancel in the direct difference. Charging the complete stored-statistic error for each policy is sufficient even though some numerical terms also cancel. The final floating-point subtraction receives its original additional allowance. This proves (S.60). Q.E.D.

S.11.4. Outward evaluation and reuse of the original confidence event

The numerical account uses the original $N = 2048$ grid, $T = 1$, and $\varepsilon = 0.1$. Every coefficient is evaluated with the inherited interval arithmetic. The coupling norm is accepted only after positive interval LDL pivots certify a matrix upper bound. For an interval-valued weighted matrix C , the computation uses

$$\|C\|_2 \leq \|C_{\text{mid}}\|_2 + \|C - C_{\text{mid}}\|_F,$$

with a verified norm for the binary64 midpoint and an outward Frobenius bound for its interval radius. Componentwise upper bounds for q and R are legitimate in the weighted norms, since increasing nonnegative diagonal weights can only increase the Euclidean norm of each image.

Positive power series avoid cancellation in D , I_1 , and I_2 near the first node. They are

$$D(t) = r \sum_{m \geq 0} \frac{\beta^m t^{m+1}}{(m+1)!},$$

$$I_1(t) = r \sum_{m \geq 0} \frac{\beta^m t^{m+2}}{(m+2)!},$$

$$I_2(t) = r^2 \sum_{m \geq 0} \frac{(2^{m+2} - 2)\beta^m t^{m+3}}{(m+3)!}.$$

The implementation retains $m = 0, \dots, 28$. Writing $z_* = \beta T < 1$, the successive ratios in each remaining tail are bounded by $z_*/30$, $z_*/31$, and $3z_*/32$, respectively. If u is the last retained term and $r_* < 1$ is its tail-ratio bound, the omitted positive tail is at most $ur_*/(1 - r_*)$. All sums, products, roots, differences of integral enclosures, and the positive recursion (S.56) are rounded outward. The scalar B_k, J_k are their original interval enclosures.

For the implementation terms, the calculation takes the maximum, over the nine initial profiles, of the inherited forward-roundoff plus normal-clipping strong-error bound. It also retains twice the complete original stored-statistic arithmetic allowance and the original 10^{-12} allowance for the direct subtraction. The latter includes no proposed cancellation of scalar quadrature or evaluation errors. The calculation is conditional on the same binary64 and ideal-iid-normal sampling contract as the original verifier.

This account applies when both compared outputs are held controllers. If both outputs select the analytical schedule, their difference remains exactly zero. If exactly one selects that schedule, the original complete schedule-relative transfer is retained. A continuously varying analytical schedule is not silently treated as a held controller.

Finally, let the original empirical-Bernstein event bound the mean of the same clipped direct-difference observations, with its original clipping threshold and tail allowance. Both the original deterministic transfer and (S.60) hold uniformly for the two held controllers. Their minimum may therefore be used on that same event. This operation spends no additional probability budget, changes no sample or stopping rule, and makes no claim of a smaller empirical variance. The original endpoints remain available together with the refined endpoints and the deterministic accounts that distinguish them.

S.11.5. *Proof of Proposition S.22*

Condition on the complete executed policy ensemble, including all online checking and stopping decisions. The final confirmation arrays are independent of that information. For $C > 0$, set $Z_{sj} = (Y_{sj} + C)/(2C)$. These variables lie in $[0, 1]$ and are independent; their distributions may differ across streams. Apply Maurer and Pontil (2009, Theorem 11) to the pooled vector and to its complement, allocating $\delta/2$ to each sign. After rescaling, the sampling radius is the first two terms in (S.86). Equal path counts give

$$\mathbb{E}\tilde{Y} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbb{E}Y_{s1}.$$

The triangle inequality and the stream-specific transfer accounts bound the distance between this expression and the desired finite-distribution mean by $\bar{\beta}$. This proves the interval claim. When $C = 0$, the clipped observations vanish identically and the transfer account alone suffices. The conditional bound also holds unconditionally by integration over the executed ensemble.

At an online checkpoint, condition instead on the preceding execution history. The policy at that checkpoint is fixed under this conditioning, and its new checking bank is independent. The same argument applies to that registered checkpoint event. A union bound over the predeclared events covers the selected stopping time. Final method inference uses independent confirmation arrays, so it does not treat the array that triggered a favorable stop as an ordinary fixed-policy sample. Summing the four family allocations establishes the stated 0.05 error bound. On the simultaneous stream event, (S.87) follows by counting the streams whose lower or upper endpoints cross the target. Q.E.D.

S.11.6. *An independent audit of the paired transfer*

Before the new confirmation bank was generated, we reviewed the proof and implementation of Proposition S.11 independently of the original reporting calculation. The audit follows the actual held actions through the physical and auxiliary states, checks the scalar and vector dimension normalizations separately, verifies the predictable accumulated martingale account, and applies terminal Cauchy–Schwarz

conditional on each initial profile before averaging. It also checks the mixed-fallback case and retention of the original numerical allowances. No algebraic or scope discrepancy was found under the stated contract.

A second implementation evaluates 21 reported quantities in each dimension at 90 decimal digits. It uses closed exponential expressions for the three positive-series functions and exact integrated payoff weights, together with a separately implemented cumulative error recursion. All 42 quantities are enclosed by the original outward endpoints. The resulting diagnostic totals are approximately 5.7524×10^{-5} and 7.8466×10^{-5} ; their modest difference from the protected endpoints reflects the original quadrature enclosures. High precision by itself is not an interval proof. The confirmation therefore retains the original rigorous upper bounds, $5.8033751287117856 \times 10^{-5}$ and $7.918471815054792 \times 10^{-5}$, without reducing them on the basis of this diagnostic. The six original deterministic checks, including fresh interval LDL verification of the stored spectral proposals, also pass.

This is an internal independent derivation and implementation audit, recorded with the source identities. It does not retrospectively change the status of the R15 refinement. The new confirmation is separately labelled and uses the transfer, constants, and reporting decisions fixed before its independent noise bank. Since it retains the previously selected policies, its target remains their declared finite stream distribution; it does not create a new population of training configurations.

S.11.7. *A signed occupation account with shared continuation paths*

Fix the finite economy in (S.109)–(S.110) and condition on the fitted candidate, the continuation predictor, and their selection. Let $F_k(x; \xi)$ be the reference continuation payoff along one future innovation array, starting at the postdecision mean x , including the innovation at k , excluding the current reward, and using the original global grid thereafter. Thus $\mathbb{E}F_k(x; \xi) = S_k^\pi(x)$. The two endpoint rollouts below use the same array ξ . For r independent arrays, average each array and its exact negative, and denote this average by $\bar{F}_k(x)$. The innovation clipping map is odd. Antithetic pairing preserves the marginal law and hence the expectation of the continuation.

Draw an independent trajectory under the selected candidate and an independent uniform date K . At that date write $\delta = a_K - \pi_K \mathbf{1}$ and $x_a = x_{a_K}(Z_K)$, $x_\pi = x_{\pi_K}(Z_K)$.

Define

$$G = N\{\ell_K(Z_K, a_K) - \ell_K(Z_K, \pi_K) + \bar{F}_K(x_a) - \bar{F}_K(x_\pi)\}, \quad (\text{S.67})$$

$$C = N\{\hat{S}_K(x_a) - \hat{S}_K(x_\pi) - \bar{F}_K(x_a) + \bar{F}_K(x_\pi)\}. \quad (\text{S.68})$$

The difference of the two continuation payoffs is evaluated by differencing their running production rewards and their terminal dispersion rewards, using common innovations throughout. In exact arithmetic, $G = M - C$ for every sample. This cancellation is applied before clipping or constructing a confidence interval.

PROPOSITION S.12—Signed Bellman assessment: *Under the original integrability assumptions,*

$$\mathbb{E}G = \mathcal{R}_h(a) - \mathcal{R}_h(\pi), \quad \mathbb{E}C = \mathcal{M}_h - \{\mathcal{R}_h(a) - \mathcal{R}_h(\pi)\}.$$

Consequently, a fresh simultaneous lower endpoint $L_G \leq \mathbb{E}G$, including its numerical and clipping allowances, implies

$$J(a) - J(m_0) \geq L_G - D_{\text{hold}} - E_{a,h}^{\text{all}}.$$

No positive-part replacement of C is made. No separate endpoints for M and C are subtracted to obtain L_G .

PROOF: Conditional on the candidate history through K , antithetic averaging is unbiased at each endpoint, and their shared innovations affect covariance only. Hence the conditional expectation of G/N is the reference Bellman advantage $Q_K^\pi(Z_K, a_K) - Q_K^\pi(Z_K, \pi_K)$. Uniform sampling of K and the finite performance-difference identity (S.39) give the first assertion. The second follows from the sample identity $C = M - G$. The same finite-reference holding deficit and continuous-payoff transfer used in Theorem S.34 give the final assertion. Q.E.D.

The proposition identifies the contribution of the fitted continuation to the advantage it predicts, while retaining the signed error of that prediction. A positive

L_G verifies the selected candidate's occupation Bellman account. It does not, by itself, rank the learned critic against a Raw estimator or attribute a package-level gain uniquely to continuation learning. Those comparisons require the separately specified common-data experiment.

S.11.8. A range that contains no fitted network norm

The antithetic difference in (S.67) has a range based only on the economy and the action tube. Let Δ be a proved uniform bound on $\|a_k - \pi_k \mathbf{1}\|_d$, and retain the original bound ε on the candidate's deviation from the scalar schedule. Let s_0 be the maximal initial dispersion. Put $\gamma = 2\chi e^{-\rho T}$, $s_d = \sqrt{d-1}$, $t_k = kh$, $n_k = N - k - 1$, $\tau_k = (N - k)h$, and

$$F_k = (1 + h\beta)^{n_k}, \quad P_k = \beta_\psi \sum_{j=k+1}^{N-1} B_j (1 + h\beta)^{j-k-1}, \quad H_k = s_0 + (\kappa + \varepsilon)(t_k + h).$$

Here $\beta \geq \kappa \|B\|_2$ and $\beta_\psi \geq \sup_x \|Df(x)^\top \mathbf{1}\|_d$ are independently proved bounds, as in the paired-transfer account. The bound H_k covers the prefix and the current postdecision bridge. Projection removes the scalar reference action and the exogenous scalar drift; the centered candidate action has norm at most ε . A separately proved state-representation allowance is added to H_k when the implemented bridge endpoints require it.

Define, for the single future bank of r antithetic pairs,

$$\begin{aligned} C_{Z,k} &= P_k + \gamma F_k \kappa n_k h + \gamma (F_k - 1) \left\{ H_k + \frac{\sigma_I s_d}{\sqrt{d}} (\sqrt{t_k} + \sqrt{\tau_k}) \right\}, \\ A_{Z,k} &= \gamma (F_k - 1) \frac{\sigma_I}{\sqrt{d}} (\sqrt{t_k} + \sqrt{\tau_k/r}). \end{aligned} \tag{S.69}$$

Let $L_{\ell,k}$ be a proved bound on the normalized action-gradient norm of ℓ_k/h throughout the original action box. For example,

$$L_{\ell,k} = \max_{m \in \{l_k, u_k\}} |(A_k/h)(m^{-1} - \zeta m) - B_k/h|$$

is valid when $[l_k, u_k]$ bounds every coordinate of the action and its line segment to the reference. The derivative is decreasing in its own consumption coordinate and in

the cross-sectional mean, so the two diagonal endpoints enclose every component.

Set

$$\begin{aligned} C_{G,k} &= T\Delta \left[L_{\ell,k} + \gamma \left(H_k + \frac{\sigma_{I^S d} \sqrt{t_k}}{\sqrt{d}} \right) + C_{Z,k} \right], \\ A_{G,k} &= T\Delta \left[\frac{\gamma \sigma_I \sqrt{t_k}}{\sqrt{d}} + A_{Z,k} \right], \quad C_G = \max_k C_{G,k}, \quad A_G = \max_k A_{G,k}. \end{aligned} \quad (\text{S.70})$$

PROPOSITION S.13—Antithetic paired-continuation range: *For the exact-real statistic in (S.67),*

$$\Pr\{|G| > C_G + A_G v\} \leq 2e^{-v^2/2}, \quad v \geq 0.$$

For a fixed $v_* > 0$, valid clipping and expectation-tail allowances are

$$b_G = C_G + A_G v_*, \quad \tau_G = 2A_G e^{-v_*^2/2} / v_*.$$

The same formulas may be evaluated conditionally on a prespecified date or initial-profile stratum before averaging its protected endpoint. They require no maximum of an observed fitted derivative or simulated outcome. All executed arithmetic errors are added separately.

PROOF: For each fixed innovation array, the mean-value identity gives

$$\bar{F}_k(x_a) - \bar{F}_k(x_\pi) = -h \int_0^1 \langle \delta, \bar{q}_k(x_\pi - uh\delta) \rangle_d du,$$

where $\bar{q}_k = dD\bar{F}_k$. This is an identity used for the proof; computing the statistic requires no derivatives.

The adjoint recursion in (S.40) gives $\|J\|_2 \leq F_k$, $\|J - I\|_2 \leq F_k - 1$, and production-adjoint norm at most P_k . For the terminal derivative, subtract $q_0(x) = -\gamma \Pi x$. The exact residual consists of the production adjoint and

$$-\gamma \left\{ (J^\top - I) \Pi x + J^\top \sum_{j>k} h \Pi f(Z_j) + (J^\top - I) \Pi \xi + \Pi \xi \right\}.$$

The average of the final term is exactly zero in every antithetic pair. The other terms are bounded by P_k , $\gamma F_k \kappa n_k h$, and $\gamma(F_k - 1)(\|\Pi x\|_d + \|\Pi \xi\|_d)$.

The normalized norm of the projected prefix innovation sum has mean at most s_d and Lipschitz constant one as a function of its standard Gaussian inputs. The average of the corresponding future norms from r independent base paths has mean at most s_d and Lipschitz constant $r^{-1/2}$. Odd coordinatewise clipping is one-Lipschitz. Gaussian concentration and a union bound therefore give a common event of failure probability at most $2e^{-v^2/2}$ on which, uniformly in u ,

$$\|\Pi x_u\|_d \leq H_k + \frac{\sigma_I \sqrt{t_k}}{\sqrt{d}}(s_d + v), \quad \|\bar{q}_k(x_u) + \gamma \Pi x_u\|_d \leq C_{Z,k} + A_{Z,k}v.$$

The future event is uniform in the endpoint and the bridge parameter: only the common innovation norms and global drift/adjoint bounds enter. No union over bridge points or integration nodes is required. For $k = 0$ the prefix contribution is zero.

The known terminal derivative integrates exactly:

$$-h \int_0^1 \langle \delta, -\gamma \Pi x_u \rangle_d du = h\gamma \langle \delta, \Pi(x_a + x_\pi)/2 \rangle_d.$$

Combining its bound, the residual bound, and $N|\ell_k(z, a) - \ell_k(z, \pi)| \leq T\Delta L_{\ell,k}$ proves the displayed tail inequality. Integrating that inequality above b_G and using $\int_{v_*}^\infty e^{-v^2/2} dv \leq e^{-v_*^2/2}/v_*$ proves the expectation allowance. The bound concerns the exact-real statistic; it does not remove roundoff, protected-function, or state-representation allowances from an implementation. Q.E.D.

The asymmetric stage range used in the implementation. The implemented account further avoids symmetrizing the known current reward. Define

$$f_k(m) = A_k \log m - B_k m - \zeta A_k m^2/2,$$

$$L_k^{\text{stage}} = N \min\{f_k(l_k) - f_k(\pi_k), f_k(u_k) - f_k(\pi_k)\},$$

and

$$U_k^{\text{stage}} = N \left| A_k(\pi_k^{-1} - \zeta \pi_k) - B_k \right| \Delta_k, \quad \Delta_k \geq \sup \|a_k - \pi_k \mathbf{1}\|_d.$$

These are respectively lower and upper bounds on the current stage advantage. For the lower bound, the chord of log on $[l_k, u_k]$ is a lower bound, and its coordinate

average depends only on the cross-sectional mean. Subtracting the linear term and the mean quadratic gives a concave function of that mean, whose minimum is at an endpoint. For the upper bound, apply the concave reward's tangent support at the uniform reference action and Cauchy–Schwarz.

Put

$$C_{q,k} = C_{Z,k} + \gamma \left(H_k + \sigma_{Is_d} \sqrt{t_k/d} \right), \quad A_{q,k} = A_{Z,k} + \gamma \sigma_I \sqrt{t_k/d}.$$

On the same two-event intersection,

$$L_k^{\text{stage}} - T\Delta_k(C_{q,k} + A_{q,k}v) \leq G \leq U_k^{\text{stage}} + T\Delta_k(C_{q,k} + A_{q,k}v).$$

The minimum lower endpoint and maximum upper endpoint at the fixed v_* therefore give a valid asymmetric clipping interval. Centering that interval at an exact binary64 representative and enclosing both distances from its center changes neither the estimand nor the event. The absolute expectation allowance is $2 \max_k (T\Delta_k A_{q,k}) e^{-v_*^2/2} / v_*$. Thus the tighter date-specific prefix and asymmetric stage ranges retain the same proof and spend no additional probability budget.

S.12. REUSABLE CONTINUATION IN A FAMILY OF CAPITAL DECISIONS

The learned Bellman object can be reused when current economic decisions change but their continuation does not. We examine that property within the capital economy. The finite-period experiment changes the current weight on consumption, the current adjustment cost, or a temporary consumption constraint. Every subsequent primitive, innovation law, terminal preference, and reference policy is common across these decisions. The stored numerical coefficients define the finite economy exactly; the experiment does not identify a quarterly Bellman problem with a continuous-time diffusion without a further approximation argument.

Write the first postdecision mean as $x_a(y) = y + h\{c1 + f(y) - a\}$. For a current task θ , let

$$Q_\theta(y, a) = r_\theta(y, a) + S\{x_a(y)\}, \quad S(x) = \mathbb{E}F(x; \xi),$$

where F contains the current innovation and all subsequent reference rewards, excluding the current reward. The finite query catalog, task catalog, action boxes, and complete distribution over training streams are fixed before confirmation.

PROPOSITION S.14—The common continuation object: *If two current tasks differ only in their current reward or current feasible set, their postdecision continuation S is identical. A cached continuation observation at the same x and the same innovation array is therefore valid for either task. Cached innovations may also be reused when x changes, but the corresponding state recursion and payoff must then be reevaluated. A cached payoff or derivative at a different x is not an exact observation at the new argument.*

PROOF: Conditional on x , every future transition kernel, reward, terminal payoff, and reference action is the same. Iterating these common kernels gives the same law and payoff functional, and hence the same expectation. The last assertion follows because the transition recursion depends on its starting state even when its innovations are fixed. Q.E.D.

This distinction determines the comparison. NBO fits one scalar continuation to common rollout labels and queries its derivative for different current choices. The vector-costate fit, cached Raw actor, task-conditioned direct actor, and cached sample-average optimizer are all allowed their own legitimate reuse. In particular, the Raw competitor does not regenerate an innovation array merely because an optimizer revisits a state. The experiment compares the economic decisions and the work required by these complete procedures. It does not infer that every form of continuation reuse requires a neural scalar critic.

PROPOSITION S.15—A fixed-dictionary critic’s bias and variance: *Fix postdecision arguments and positive weights before drawing a new costate-label array. Let q be the vector of true continuation gradients, after multiplication by the square roots of these weights, and let $z = q + \varepsilon$, with $\mathbb{E}\varepsilon = 0$ and finite covariance Σ . Fix differentiable scalar features independently of this label array, and let P be the Euclidean orthogonal projection onto their stacked, weighted gradient columns. The least-squares scalar critic has fitted gradients Pz ,*

and

$$\mathbb{E}\|Pz - q\|^2 - \mathbb{E}\|z - q\|^2 = \|(I - P)q\|^2 - \text{tr}\{(I - P)\Sigma\}.$$

Thus its risk reduction equals the label variance removed outside the gradient dictionary less its approximation cost. If q belongs to that dictionary, the critic weakly reduces this risk, strictly whenever $\text{tr}\{(I - P)\Sigma\} > 0$. The same fitted scalar function and its labels may be reused for every current task having the common continuation in Proposition S.14.

PROOF: Write $Pz - q = -(I - P)q + P\varepsilon$. Orthogonality removes the cross term, and $P = P^\top = P^2$ gives $\mathbb{E}\|P\varepsilon\|^2 = \text{tr}(P\Sigma)$. The raw-label risk is $\text{tr}(\Sigma)$, proving the identity and its consequences. Neither step uses a Gaussian law or independence of the coordinates of a label. Common continuation makes the target gradient and its scalar feature representation independent of the current task; evaluating an already fitted function for another task therefore does not require fitting it again. *Q.E.D.*

The identity isolates a possible statistical contribution of learning the continuation. It also specifies its price: a gradient dictionary can impose enough approximation error to outweigh its variance reduction. The twelve retained random-feature diagnostics and the nonlinear comparisons address this tradeoff; the identity does not determine their empirical signs. Its exact scope is a dictionary fixed independently of the labels used for this least-squares fit and risk at the declared arguments. The fully trained nonlinear NBO critic is assessed by the independent economic comparisons below, rather than being identified with that linear projection. A Raw procedure that uses the same projection receives the same risk calculation. Task reuse eliminates repeated fitting of a common object, while evaluation, optimization, and any new state recursions remain part of each method's recorded work.

S.12.1. A direct conditional payoff statistic

For one fixed query and two already selected actions a, b , let $\bar{F}(x) = \{F(x; \xi) + F(x; -\xi)\}/2$, where a new base array ξ is independent of fitting and all other confirmation pairs. Put $\gamma = 2$ times the stored terminal-dispersion penalty and

define

$$q_0^{ab} = -\frac{\gamma}{2} \{ \|\Pi x_a\|_d^2 - \|\Pi x_b\|_d^2 \}, \quad R^{ab} = \bar{F}(x_a) - \bar{F}(x_b) - q_0^{ab}.$$

The current reward and q_0^{ab} are known conditional on this query and its actions. Thus

$$Q_\theta(y, a) - Q_\theta(y, b) = r_\theta(y, a) - r_\theta(y, b) + q_0^{ab} + \mathbb{E}R^{ab}. \quad (\text{S.71})$$

The implementation may center at an enclosed binary64 representative of q_0^{ab} . Its exact discrepancy is included in the range; the same representative is added back to the known part of the payoff.

Let N denote the number of dates, $n = N - 1$, and let ν_I, ν_C denote the stored idiosyncratic and common innovation amplitudes per date. Set

$$H = (1 + h\beta)^n, \quad P = \beta_\psi \sum_{j=1}^{N-1} B_j (1 + h\beta)^{j-1}, \quad D = \|x_a - x_b\|_d,$$

and $s = \max\{\|\Pi x_a\|_d, \|\Pi x_b\|_d\}$. The following coefficients contain no learned parameter:

$$\begin{aligned} C &= P + \gamma H \kappa n h + \gamma(H - 1) \{s + \nu_I \sqrt{N(1 - 1/d)}\}, \\ A &= \gamma(H - 1) \nu_I \sqrt{N/d}. \end{aligned} \quad (\text{S.72})$$

PROPOSITION S.16—Query-conditional antithetic range: *For one independent antithetic pair,*

$$\Pr\{|R^{ab}| > D(C + Av)\} \leq e^{-v^2/2}, \quad v \geq 0.$$

Consequently, clipping at the prospectively fixed $v_* > 0$ admits the expectation allowance $DAe^{-v_*^2/2}/v_*$. These statements hold conditional on the selected actions, so their geometry may enter the range before the independent pair is generated.

PROOF: Apply the pathwise mean-value identity along the straight segment from x_b to x_a . The reference adjoint has norm growth at most H ; its accumulated production component has norm at most P . Subtracting the known terminal derivative $-\gamma \Pi x$ leaves the production adjoint, a bounded accumulated-production term,

and $(J^\top - I)$ times the centered starting state and centered future innovation. The unmultiplied terminal innovation cancels exactly in an antithetic pair. The starting state is fixed. The norm of the projected future innovation has mean at most $\nu_I \sqrt{N(1 - 1/d)}$ and, as a function of its standard Gaussian array, normalized Lipschitz constant at most $\nu_I \sqrt{N/d}$. Gaussian concentration gives one event, uniform over the entire straight segment. Integration proves the range, and integration of its tail proves the expectation allowance. There is no prefix-state event because the query is conditioned upon. *Q.E.D.*

One observation is one antithetic pair, not its two dependent paths. Different query-pair cells use independent arrays. Equal pair counts at every query and at every declared training stream identify their uniform finite average. For each contrast, the report pools only its independent pair observations, clips each query at its proved range, uses the maximum range in the independent nonidentical empirical-Bernstein bound, and averages numerical and tail allowances. Dependence across methods is retained by forming the direct contrast before inference. The primary family comprises 216 two-sided events and uses probability budget 0.018. The separate scalar-accuracy family below uses 0.002. Their union remains within the fixed total 0.02 budget.

Arithmetic and Gaussian clipping. The numerical verifier uses the preserved protected polynomial for tanh, outward coefficient operations, and forward state-error recursions with multiplier $1 + h\beta$. Fixed dyadic spectral proposals for the two declared coupling matrices are each certified by a fresh outward LDL calculation. Postdecision representatives are interval midpoints, so reconstructing their error accounts does not depend on the last bits of a numerical singular-value or matrix-product routine. A deterministic state cap is computed from the fixed query, bounded drift and reference actions, and the executed Gaussian cap. Production error is bounded by β_ψ times the state error plus its protected-function and dot-product allowances. A terminal state error e at a cap M contributes at most $(\gamma/2)e(2M + e)$, before the separately charged scalar reductions. The source records every component of this account.

The finite economy retains untruncated Gaussian innovations. For cap z ,

$$\mathbb{E}\{G - \text{clip}_z(G)\}^2 \leq 4\phi(z)/z^3.$$

This follows by writing the two-sided tail integral as $2\phi(z) \int_0^\infty u^2 e^{-zu - u^2/2} du$ and dropping $e^{-u^2/2}$. Multiplication by $v_I^2 + v_C^2$ and the same state-error recursion gives an L^2 clipping allowance. The production Lipschitz bound and conditional terminal Cauchy-Schwarz then transfer the implemented expectation to the original finite Gaussian economy. The clipping and arithmetic accounts are distinct.

S.12.2. Accuracy over an entire scalar intervention class

A separate comparison supplies an absolute reference on a restricted decision class while retaining the high-dimensional state. At one fixed state and current task, set $a(s) = a_L + s(a_R - a_L)$, $0 \leq s \leq 1$. The endpoints are the common first-period reference consumption plus or minus the fixed radius, intersected with the original task box. All subsequent actions follow the same reference. Write $q(s) = Q_\theta\{y, a(s)\}$ and $r = a_R - a_L$.

For the first and second variations of a future state with respect to s , start with $V_1 = h\|r\|_d$ and $W_1 = 0$. With the global scalar and vector Hessian coefficients H_m, H_x from the paired transfer, recursively set

$$V_{j+1} = (1 + h\beta)V_j, \quad W_{j+1} = (1 + h\beta)W_j + hH_x V_j^2.$$

Let

$$S_N = \max\{\|\Pi x_{a_L}\|_d, \|\Pi x_{a_R}\|_d\} + \kappa(N-1)h + v_I \sqrt{N(1-1/d)}.$$

Then an upper continuation curvature is

$$\Lambda = \sum_{j=1}^{N-1} B_j(H_m V_j^2 + \beta_\psi W_j) + \gamma S_N W_N.$$

For $u_i = \max(a_{L,i}, a_{R,i})$, current utility weight w and adjustment coefficient η , put

$$\mu = A_0 \left\{ \overline{w r_i^2 / u_i^2} + \eta \bar{r}^2 \right\} - \Lambda.$$

PROPOSITION S.17—Scalar Bellman curvature and a protected optimality gap:

The preceding coefficients imply $q''(s) \leq -\mu$ throughout $[0, 1]$. If $[g_L, g_U]$ is a valid interval for $q'(s_0)$, then

$$\max_{0 \leq s \leq 1} q(s) - q(s_0) \leq \max_{\substack{g \in \{g_L, g_U\} \\ 0 \leq s \leq 1}} \{g(s - s_0) - \mu(s - s_0)^2/2\}.$$

The maximum is a scalar quadratic calculation. It remains valid when $\mu \leq 0$; in that case it does not assert strong concavity.

PROOF: Differentiating the finite transition recursion gives the displayed variation bounds. The production chain rule contributes at most $H_m V_j^2 + \beta_\psi W_j$. The terminal second derivative is $-\gamma\{\|\Pi v_N\|_d^2 + \langle \Pi Y_N, \Pi w_N \rangle_d\}$. Dropping its negative squared term and using the deterministic bound on w_N and the moment bound on Y_N gives $\gamma S_N W_N$. Subtracting the minimum curvature of current utility proves the first claim. Taylor's integral remainder gives the quadratic support at s_0 . Maximizing that support over the gradient interval and feasible scalar displacement proves the second. Q.E.D.

The gradient interval is obtained without identifying an automatic derivative with an exact expectation. On a fixed neighboring interval $[s_-, s_+]$ containing s_0 , a fresh paired-payoff calculation bounds $\{q(s_+) - q(s_-)\}/(s_+ - s_-)$. A global absolute second-derivative bound L adds the deterministic allowance

$$\frac{L\{(s_0 - s_-)^2 + (s_+ - s_0)^2\}}{2(s_+ - s_-)}.$$

This is the integral of the derivative's Lipschitz bound; it also covers one-sided differences at a boundary. The declared scalar candidate is selected from the fitted continuation before this bank. Binary64 construction of the affine actions receives its own payoff allowance. All 128 scalar events use 65,536 independent antithetic pairs each and the fixed probability allocation. The resulting reference covers every scalar action in the interval, including unevaluated ones. Its scope is the specified finite decision class; it is not a high-dimensional full-adapted-control or continuous-time near-optimality assertion.

S.13. STOCHASTIC HESSIAN OBJECTIVES AND COMPLEXITY

Let $B = \Sigma^\top D^2 v \Sigma$, a symmetric matrix in Brownian-factor space. For Gaussian probes, Isserlis' identity gives $\mathbb{E}(z^\top B z) = \text{tr } B$ and $\text{Var}(z^\top B z) = 2 \text{tr}(B^2) = 2 \|B\|_F^2$. Averaging K independent probes divides the variance by K . Rademacher probes instead have variance $2 \sum_{i \neq j} B_{ij}^2 / K$. The variance depends on the chosen covariance factor and probe law, both of which must be specified.

Write $\hat{q}_j = q + \epsilon_j$, with conditionally mean-zero independent errors. Then

$$\mathbb{E}[(d - \hat{q}_1/2)(d - \hat{q}_2/2)] = (d - q/2)^2 + \frac{1}{4} \mathbb{E}[\epsilon_1 \epsilon_2] = (d - q/2)^2.$$

Using one bank twice would retain its variance term. In the scalar Gaussian example, a single-bank squared residual has expectation $1 - h + 3h^2/4$, minimized at $h = 2/3$, whereas the exact residual $(1 - h/2)^2$ is minimized at $h = 2$. The independent product restores the latter population objective. The saved Monte Carlo diagnostic uses 400,000 pairs and records both empirical estimates; it does not claim that the sampling error is identically zero.

The core implementation uses exact Hessians to make the covariance contraction independently testable. A Hessian-vector implementation can replace it without changing the conditional-independence requirement. The cost of a Hessian-vector product depends on the network graph, not just the state dimension. Total cost also includes sampling, optimization, domain coverage, action maximization, and stopping at an agreed error level. A quadratic-neural representation with a Lyapunov solve has different approximation and optimization costs from a generic multilayer network. The quadratic laboratory is explicitly the former; the new dense nonlinear experiments actually train generic networks and Hessian-vector estimators.

S.13.1. *Implemented Error Accounts and Arithmetic*

The implemented map returns a candidate and a verification record, rather than assigning authority to an optimizer's exit code. In the differential consumption economy, the record identifies the binary64 weights, both stopping faces, all dyadic leaves, global consumption maximization, and the error bounds for the continuous diffusion. In NDU it instead identifies the finite state-action-time model, its positive

transition weights, and time-indexed evaluation and improvement errors. The dynamic-game record fixes the entire rival policy before an independent backward best-response solve. An unchecked continuous-action or time-discretization term is represented as unavailable in the machine-readable output, not as zero.

For finite time levels, let $\delta_t = \|v_t - T_t^a v_{t+1}\|_\infty$ and $\epsilon_t = \|T_t v_{t+1} - T_t^a v_{t+1}\|_\infty$. If both maps are β_t -Lipschitz, the triangle inequality gives

$$\|v_t - V_t^a\| \leq \delta_t + \beta_t \|v_{t+1} - V_{t+1}^a\|, \quad \|v_t - V_t^*\| \leq \delta_t + \epsilon_t + \beta_t \|v_{t+1} - V_{t+1}^*\|.$$

Summing the two inequalities yields the regret account. The same recurrence with a player's unilateral gap bounds the gain from a dynamic deviation when that player's policy evaluation is exact. Accounts must be checked at each time level: a time-zero bound need not dominate a later-time bound when discounting and errors differ. The coarse-game regression exercises this distinction and verifies that a missing pure equilibrium is reported rather than silently treated as exact.

S.13.1.1. Full-domain interval implementation

An interval $[l, u]$ encloses an exact real quantity. Each elementary operation expands its rounded endpoints with the adjacent binary64 number. Affine layers split positive and negative weights and include a conservative dot-product forward-error allowance. The arithmetic contract requires binary64 round-to-nearest arithmetic, the stated dot-product error bound, and absence of exceptional overflow or flush-to-zero behavior. Stored parameters denote their exact binary64 real values; the verifier does not assert an unrecorded decimal model identity.

For an exponential argument x , range reduction gives $r = x/2^s$ with $|r| \leq 1/8$. A degree-fourteen Taylor polynomial has remainder bounded by

$$e^{|r|} |r|^{15}/15! < 2(1/8)^{15}/15! < 2^{-80}.$$

Interval Horner evaluation and repeated squaring enclose e^x . The identity $\tanh x = 1 - 2/(e^{2x} + 1)$ then supplies $\tanh$ values and derivative channels $1 - \tanh^2 x$ and $-2 \tanh x(1 - \tanh^2 x)$. Powers of two and the range-reduction domain are checked in the code.

For logarithms, exact binary decomposition gives $x = 2^k m$, with $1 \leq m < 2$. Writing $z = (m - 1)/(m + 1)$,

$$\log m = 2 \sum_{j=0}^{24} \frac{z^{2j+1}}{2j+1} + R, \quad |R| \leq \frac{2|z|^{51}}{51(1-z^2)} < 2^{-72}$$

for the outward-enclosed range $|z| < 0.334$. The same expansion encloses $\log 2$. Neither construction presumes correctly rounded library $\tanh$ or logarithm functions. The saved test suite checks transcendental endpoints against eighty-digit arithmetic, differentiates the trained network independently, and checks all certificate hashes and leaf adjacencies. These tests supplement, but do not replace, the arithmetic contract.

The scalar action objective $h(c) = \log c - pc$ has second derivative $-1/c^2$. If C encloses both the actor and its maximizing alternative, strong concavity on their joining segment gives

$$h(c^*) - h(c) \leq h'(c)(c^* - c) - (c^* - c)^2/(2C^2) \leq C^2 h'(c)^2/2.$$

The argument includes constrained endpoints. Taking the minimum with the natural interval action-gap enclosure markedly reduces verification cost, without using a sampled gradient as a bound. Residual targets are decided on every leaf; reaching a maximum subdivision depth does not change a failed target into a pass.

S.13.1.2. *Application-specific hypotheses*

The stopped consumption diffusion is uniformly nondegenerate on its compact domain; controls and coefficients are bounded and its endpoint data are continuous. The smooth networks and strictly positive consumption map meet the regularity and log-domain requirements of the verification argument. Both stopping-face errors and the full action interval are checked. The upwind reference error is not a premise of its continuous certificate.

The NDU computation is a positive-weight, finite-action, finite-horizon economy with prescribed discrete stopping and folded preference transitions. Positivity, all terminal and wealth-face values, and every finite action are evaluated. The continuous reflected PDE's corner compatibility, continuous-action coverage error,

and diffusion approximation error are not inferred from these tests. Expanding a feasible box changes the constrained economy; densifying a fixed box does not. The sensitivity record keeps these operations separate.

The dense capital computation uses bounded nonlinear drift in log coordinates, bounded positive consumption fractions, constant diffusion, and finite horizon. Its flow has at most linear log-state growth and its terminal payoff at most quadratic growth; these are integrable under the declared bounded-drift diffusion. The fitted box is a sampling domain, not a state constraint. Its sampled differential residuals do not check comparison barriers or uniform errors on $\mathbb{R}^d$. Euler payoff estimates carry both sampling error and unbounded-state/time-discretization issues not covered by the reported Monte Carlo standard error.

The dynamic capital game is a fully enumerated finite approximation with stopped joint boundaries. Stage equilibria and dynamic fixed-rival best responses are numerical statements about that model. The continuous-control best-response comparison assumptions in the main theorem remain logically distinct. The retained homothetic recursive-utility and temporal-self examples use their stated utility domains, bequests, and integrability conditions; initial evaluation study does not relabel them as unrestricted neural experiments.

S.14. OBSERVATION AND IMPLEMENTATION

S.14.1. *Observation and the Economic Implementation*

The capital state and the innovation-driven numerical controller are distinct. We now specify an observation model under which the controller uses the history of observed capital rather than latent economic shocks. The model assumes continuous noiseless observation of every capital coordinate, known diffusion and drift coefficients, knowledge of the controller's own actions, and independent private randomization. It does not assume that a finite set of noisy observations reveals Brownian increments.

Write $\Sigma = [\sigma_I I_d, \sigma_C \mathbf{1}]$, $Q = \Sigma \Sigma^\top$, and let n be a unit vector spanning the one-dimensional null space of Σ . Since $\sigma_I > 0$, Q is invertible. The observed state path

and known actions determine

$$M_t = Y_t - Y_0 - \int_0^t \{c\mathbf{1} + f(Y_s) - a_s\} ds.$$

Let B^0 be a scalar Brownian motion independent of the economic shocks and initial state, used only for private randomization.

PROPOSITION S.18—Implementation from observed state history: *Define*

$$\tilde{W}_t = \Sigma^\top Q^{-1} M_t + n B_t^0. \quad (\text{S.73})$$

Then $\tilde{W}$ is a $(d + 1)$ -dimensional Brownian motion in the enlarged observation filtration and $\Sigma \tilde{W}_t = M_t$. Feeding its cell increments into the prescribed numerical controller, including the componentwise clipping rule, yields the same closed-loop law and payoff as the innovation-driven controller. Thus its existing transfer and policy-specific performance statements apply under this observation model.

Private randomization in (S.73) is essential for this exact equivalence: the right-inverse term alone has rank d , whereas the implemented clipping acts separately on $d + 1$ drivers. The construction does not recover the original unobservable null component path by path. It replaces that component by an independent one with the correct law. Nor is the resulting controller a deterministic function of the state at a single decision date; its memory and independent randomization are part of its definition. The integrated drift in the proposition is an observation-model primitive. A sensor grid, drift quadrature or measurement error would need a further transfer allowance and is not silently treated as exact.

S.14.1.1. *Information and the comparison class*

The comparison class is defined on a canonical product space. Let $W = (W^1, \dots, W^{d+1})$ be the economic Brownian motion and let R be an independent random element on a standard Borel space. The element R may contain a private Brownian path and a sequence of independent sensor errors. Write $\mathbb{F}^W$ for the usual augmentation of the economic Brownian filtration and $\mathbb{F}^{W,R}$ for the corresponding product filtration; the latter reveals only the portion of R specified by its observation clock. An admissible control is a progressively measurable, nonanticipative

functional of the available paths, taking values in the primitive action box. Bounded controls and the globally Lipschitz drift give a unique strong state process and an integrable payoff. These conventions apply to the comparison class in Theorems S.9 and S.31.

PROPOSITION S.19—Independent randomization and the optimum: *Let $V_W^*(y)$ and $V_{W,R}^*(y)$ be the expected-payoff suprema over the preceding economic and product filtrations. Then*

$$V_{W,R}^*(y) = V_W^*(y).$$

The anchor envelope G_d and the policy-specific regret conclusions therefore compare the observed-history and finite-observation policies with the same economic optimum. The equality concerns the supremum; it neither requires an optimal policy nor asserts that a single fixed randomizer realization attains that supremum.

PROOF: The inequality $V_{W,R}^* \geq V_W^*$ follows by ignoring R . For the reverse inequality, represent an admissible randomized control by its nonanticipative measurable functional $a(t, W_{[0,t]}, R_{[0,t]})$ before completion of the filtrations. Fixing a full realization r of R leaves a progressively measurable functional of W . For almost every r , Fubini's theorem preserves feasibility and the state equation; strong uniqueness identifies the resulting state with the controlled state for that section. Independence leaves the law of W equal to Wiener measure after conditioning on $R = r$. Thus the conditional payoff is at most $V_W^*(y)$ for almost every r . Integrating over r proves the reverse inequality. Null-set modifications in the augmented filtration do not change the state law or payoff. Alternatively, the anchor proof may be repeated directly on the product filtration: its synchronous comparison and martingale identities use only adaptedness, bounded actions, and the Brownian property, all of which are preserved. Q.E.D.

Exact continuous observation. For precision in Proposition S.18, work first in the product filtration generated by W and an independent scalar Brownian motion B^0 . Let $\mathbb{G}$ be the usual augmentation of $\sigma(Y_s, B_s^0 : 0 \leq s \leq t)$, with the initial profile included. The stored numerical state and held actions are functions of this history.

Consequently

$$M_t = Y_t - Y_0 - \int_0^t \{c\mathbf{1} + f(Y_s) - a_s\} ds$$

is $\mathbb{G}$ -adapted. Since it equals ΣW_t , it is a $\mathbb{G}$ -martingale: conditional expectation first in the product filtration and then in its subfiltration $\mathbb{G}$ proves this claim. With $A = \Sigma^\top (\Sigma \Sigma^\top)^{-1}$ and a unit null vector n , the process $AM + nB^0$ is a continuous $\mathbb{G}$ -local martingale starting at zero. Its bracket is $I_{d+1}t$, because $A\Sigma$ and $nn^\top$ are complementary orthogonal projections and $[M, B^0] = 0$. The multivariate Lévy characterization gives a Brownian motion in the stated observation filtration. The identity $\Sigma(AM + nB^0) = M$ holds pathwise.

This argument also establishes causality. At a decision date, the already observed drift integral determines the preceding innovation increment and hence the next internal state. The next action is then fixed before the next Brownian increment. The held-action state equation is uniquely solvable on that cell. Induction identifies the joint law of physical state, internal state, actions, and recovered innovations with the prescribed innovation controller. The independent null component supplies the correct law when clipping is applied separately to $d + 1$ coordinates; it does not recover the original latent null component. The exact observation result is separate from the finite measurement implementation below.

S.14.1.2. *Finite observations with a continuous-economy transfer*

At dates $t_k = kh$, a finite-observation controller receives $O_k = \bar{Y}_{t_k} + \eta_k$ and holds $\bar{a}_k = \hat{\pi}_k(O_k)$ until the next date. The sensor error is measurable at t_k in the product filtration just specified and satisfies $\|\eta_k\|_{2,d} \leq \nu_{\text{obs}}$, where $\|X\|_{2,d} = (\mathbb{E}\|X\|_d^2)^{1/2}$. Both the sensor state $\bar{Y}$ and the comparison parent's physical state Y obey the original continuous diffusion. No Brownian coordinate or continuous drift integral is supplied to the sensor controller.

The protected parent has internal state $Z_0 = y$ and selects $a_k = \hat{\pi}_k(Z_k)$. Its update is the specified clipped-innovation Euler update, with an implementation error of normalized norm at most $\tilde{\zeta}_h$ per cell. This includes innovation conversion and recurrence arithmetic. Both action maps use the same inward guard and lie within ϵ of the common schedule $m_0(t_k)$ coordinatewise. For a real reference map π_k ,

suppose

$$\|\pi_k(x) - \pi_k(z)\|_d \leq L\|x - z\|_d, \quad \sup_x \|\hat{\pi}_k(x) - \pi_k(x)\|_d \leq \delta_\pi.$$

Global matrix-norm products and forward error propagation through the protected activation supply L and δ_π for the saved weights. Measurement error is measured before the protected map; input representation within that map is included in δ_π .

Retain the production coefficient κ of the economic model and let $\beta \geq \kappa\|B\|_2$. Put $\bar{\sigma} = (\sigma_I^2 + \sigma_C^2)^{1/2}$ and let M bound $\|c1 + f(y) - m\|_d$ for all feasible held actions. Define the generator constants H and $K = \beta M + Q_f$ as in the diffusion-transfer argument of Theorem S.31, where $H^2 \geq d^{-1} \sup_y \|Df(y)\Sigma\|_F^2$, $q_i = (B\Sigma\Sigma^\top B^\top)_{ii}$, and $Q_f = (\kappa/2)(d^{-1} \sum_i q_i^2)^{1/2}$ is the generator bound, not a diffusion covariance. Let $z > 0$ be the clipping threshold and let $v_{\text{clip}} \geq \mathbb{E}(G - \text{clip}(G, -z, z))^2$ for a standard normal G . The bound $2\varphi(z)(z + z^{-1})$ is valid. Set

$$b_k = h\{Kt_k/2 + H\sqrt{t_k/3}\} + \bar{\sigma}\sqrt{t_k v_{\text{clip}}} + k\xi_h, \quad b_0 = 0, \quad (\text{S.74})$$

$$\begin{aligned} e_0 &= 0, \quad a_k^\Delta = \min\{2\epsilon, L(e_k + v_{\text{obs}}) + 2\delta_\pi\}, \\ e_{k+1} &= (1 + \beta h)e_k + ha_k^\Delta + (b_{k+1} - b_k), \quad 0 \leq k < N. \end{aligned} \quad (\text{S.75})$$

The forcing is nondecreasing. Each term retains the accumulated martingale cancellation; in particular, the clipping term is a finite-horizon second-moment account rather than a sum of cellwise absolute clipping errors.

Define the nonnegative function

$$\begin{aligned} D(t) &= \int_0^t e^{\beta(t-s)} a_{[s/h]}^\Delta ds, \\ Q_T &= s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}. \end{aligned}$$

Here $s_0 \geq \|\Pi Y_0\|_{2,d}$, with Π the projection away from the common-state direction.

For $t \in [t_k, t_{k+1})$, choose $m_{-,k} > 0$ below every action in the common tube and put

$$r_k(t) = \epsilon + |m_0(t) - m_0(t_k)|, \quad C_k(t) = r_k(t) \{[m_0(t)m_{-,k}]^{-1} + \zeta\}.$$

With $W(t) = e^{-\rho t} w(t)$ as before, define

$$B_{h,\nu_{\text{obs}}}^{\text{sens}} = \beta \int_0^T W(t) D(t) dt + \sum_{k=0}^{N-1} a_k^\Delta \int_{t_k}^{t_{k+1}} e^{-\rho t} C_k(t) dt + 2\chi e^{-\rho T} Q_T D(T). \quad (\text{S.76})$$

PROPOSITION S.20—Finite-observation payoff transfer: *Under the stated assumptions,*

$$\|\bar{Y}_{t_k} - Z_k\|_{2,d} \leq e_k,$$

$$\|\bar{a}_k - a_k\|_{2,d} \leq a_k^\Delta,$$

$$\|\bar{Y}_t - Y_t\|_{2,d} \leq D(t),$$

and

$$|J_y(\bar{a}) - J_y(a)| \leq B_{h,\nu_{\text{obs}}}^{\text{sens}}.$$

Consequently, a valid improvement interval $[L_a, U_a]$ for the protected parent at the same time grid transfers to $[L_a - B_{h,\nu_{\text{obs}}}^{\text{sens}}, U_a + B_{h,\nu_{\text{obs}}}^{\text{sens}}]$ for the finite-observation controller. This deterministic operation uses no additional statistical confidence allocation. It requires a certificate for that exact protected parent and grid.

The allowance concerns the original physical diffusion and uses only the stated sensor moment bound. A shared-noise fine-Euler comparison checks the finite-observation implementation and reports its observed payoff difference. Such a diagnostic does not replace the allowance or certify the Euler bias. For fixed weights and horizon, uniformly bounded L, β, ϵ and generator constants, and $m_{-,k} \geq \underline{m} > 0$, the forcing vanishes if $h \rightarrow 0$, $v_{\text{clip}}(h) \rightarrow 0$, $\xi_h/h \rightarrow 0$, $\nu_{\text{obs}} \rightarrow 0$, and $\delta_\pi \rightarrow 0$. Discrete Gronwall then makes the transfer allowance vanish. Fixed clipping and fixed numerical or measurement error leave the corresponding terms in the finite-grid bound.

S.14.2. *Finite-Observation Transfer Proof*S.14.2.1. *Proof of Proposition S.20*

All processes are coupled using the same economic Brownian motion and initial profile; the sensor noise uses the independent randomizer. The real-map Lipschitz bound, uniform implementation error, and nonexpansiveness of the common guard give

$$\|\hat{\pi}_k(\bar{Y}_{t_k} + \eta_k) - \hat{\pi}_k(Z_k)\|_{2,d} \leq L\{\|\bar{Y}_{t_k} - Z_k\|_{2,d} + \nu_{\text{obs}}\} + 2\delta_\pi.$$

The common action tube also bounds this difference by 2ϵ .

Apply Itô's formula to $f(\bar{Y})$ on each cell, where the sensor action has already been chosen. The generator and derivative bounds used in the policy-specific study proof remain valid for this adapted held action. Stochastic Fubini yields

$$\begin{aligned} \int_{t_j}^{t_{j+1}} \{f(\bar{Y}_s) - f(\bar{Y}_{t_j})\} ds &= \int_{t_j}^{t_{j+1}} (t_{j+1} - v) \mathcal{A}^{\bar{a}_j} f(\bar{Y}_v) dv \\ &\quad + \int_{t_j}^{t_{j+1}} (t_{j+1} - v) Df(\bar{Y}_v) \Sigma dW_v. \end{aligned}$$

The L^2 norm of the accumulated first term through t_k is at most $Kht_k/2$. The stochastic integrals are martingale increments over disjoint cells. Itô's isometry bounds their accumulated normalized L^2 norm by $Hh\sqrt{t_k/3}$.

Before representation error, every component of $\Delta W_j - \sqrt{h} \text{clip}(\Delta W_j / \sqrt{h}, -z, z)$ has zero mean and variance at most $h\nu_{\text{clip}}$. Components and cells are independent. Therefore the normalized second moment of the accumulated projected clipping error is at most

$$\frac{1}{d} \text{tr}(\Sigma \Sigma^\top) t_k \nu_{\text{clip}} = \bar{\sigma}^2 t_k \nu_{\text{clip}}.$$

The clipping error and the preceding local-drift martingale need not be independent: the triangle inequality between their already accumulated norms is sufficient. The implementation error contributes at most $k\zeta_h$.

Let $E_k = \|\bar{Y}_{t_k} - Z_k\|_{2,d}$. Subtracting the internal recursion from the integrated sensor state equation thus gives

$$E_k \leq \beta h \sum_{j < k} E_j + h \sum_{j < k} \|\bar{a}_j - a_j\|_{2,d} + b_k.$$

The recursion in (S.75) is equivalent to $e_k = \beta h \sum_{j < k} e_j + h \sum_{j < k} a_j^\Delta + b_k$. Induction, using the action bound already proved, establishes $E_k \leq e_k$ and the asserted action inequalities.

The two physical diffusions have the same Brownian term. Subtracting their integral equations and applying the β -Lipschitz production bound gives

$$\|\bar{Y}_t - Y_t\|_{2,d} \leq \beta \int_0^t \|\bar{Y}_s - Y_s\|_{2,d} ds + \int_0^t a_{[s/h]}^\Delta ds.$$

The integral form of Gronwall's inequality proves the bound by $D(t)$. Projection away from the common-state direction eliminates common drift, the common schedule, and common diffusion. Bounded production, the action tube, and Itô's isometry give, for both physical states, $\sup_{t \leq T} \|\Pi Y_t\|_{2,d} \leq Q_T$.

It remains to compare payoffs. Subtract the two exact identities (S.106); the common schedule terms cancel. Production contributes at most $\beta \int_0^T W(t) D(t) dt$. For the consumption deficit R_t in (S.105), its gradient in the normalized inner product has components

$$-\frac{1}{m_i} + w(t) + \zeta \bar{m} = \frac{1}{m_0(t)} - \frac{1}{m_i} + \zeta \{\bar{m} - m_0(t)\}.$$

Every point on the segment between the two held actions remains in their common coordinate tube at t_k . Its distance from $m_0(t)$ is at most $r_k(t)$, and its coordinates are at least $m_{-,k}$. Thus the normalized gradient norm is bounded by $C_k(t)$. The mean-value inequality and Cauchy-Schwarz bound the absolute expected deficit difference by $C_k(t) a_k^\Delta$.

Finally,

$$\left| \mathbb{E} \{ \|\Pi \bar{Y}_T\|_d^2 - \|\Pi Y_T\|_d^2 \} \right| \leq \|\bar{Y}_T - Y_T\|_{2,d} \{ \|\Pi \bar{Y}_T\|_{2,d} + \|\Pi Y_T\|_{2,d} \} \leq 2Q_T D(T).$$

Adding these three contributions proves (S.76). Adding and subtracting the parent's payoff on the existing confidence event proves the interval transfer.

An outward finite computation. One need not infer a scalar integral error from a refinement plot. The function D is nondecreasing and obeys the exact cell recursion

$$D(t_{k+1}) = e^{\beta h} D(t_k) + a_k^\Delta \int_0^h e^{\beta s} ds.$$

An interval enclosure of m_0 on each cell gives an upper bound C_k^+ for $C_k(t)$. With the already enclosed weights A_k, B_k , a computable upper allowance is

$$\beta \sum_k B_k D(t_{k+1}) + \sum_k A_k C_k^+ a_k^\Delta + 2\chi e^{-\rho T} Q_T D(T),$$

evaluated outward. In the forcing recurrence, use nondecreasing saved upper endpoints $b_k^+ \geq b_k$ and the differences of those exact saved scalars. This preserves the telescoping majorant. The network and recurrence arithmetic assumptions are the same conditional binary64 assumptions used for the protected parent.

S.15. INDEPENDENT NUMERICAL DIAGNOSTICS

S.15.1. A Trained Neural Boundary-Value Problem

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (\text{S.77})$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption interval, positive diffusion variance on the closed domain, and continuous endpoint data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_{\xi}(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_{\xi}\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p = v'(x)$,

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (\text{S.78})$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION S.21—Continuous-domain economic certificate: *For a stopped discounted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissible. Suppose throughout D that*

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_{\infty} \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_{\infty} \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (\text{S.79})$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v , v' , v'' , and the actor on every leaf. Equation (S.78) supplies a global action enclosure. We also use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (\text{S.80})$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

S.15.1.1. *Controlled comparison and failures*

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly.

All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-loss ablation does not: a residual near one half is accompanied by large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uni-

TABLE S.3

MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[0.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[0.02, 0.6]$. All six certificates have uniform evaluation error at most 0.002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

formly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation. Fitting and full-domain verification are separately timed because their computational burdens are different.

S.15.2. Independent Scalar Reference: Equations and Complete Diagnostics

S.15.2.1. The scalar reference and its residual account

The scalar reference isolates policy evaluation and policy improvement in a nonlinear member of the coupled-capital family. It uses the three stored scalar actors selected in the original experiment and the analytical anchor, without retraining. The four state–time–domain specifications are $(401, 256, 6)$, $(801, 512, 6)$, $(1601, 1024, 6)$ and $(1601, 1024, 8)$, where the last coordinate is the half-width L of the log-capital interval. All model coefficients, action limits, terminal values and boundary data remain those of the original reference. The diagnostic program checks the source and input hashes before computation and verifies the input hashes again afterwards.

Write $\eta = (\sigma_{\text{id}}^2 + \sigma_{\text{com}}^2)/2$, $c = \alpha - \eta$, and a for the adjustment-cost coefficient. The realized scalar coupling entry is $B_{11} = 1$. For scalar log capital y and consumption rate $m \in [\underline{m}, \bar{m}] = [0.02, 2]$, the running payoff and drift are

$$u(y, m) = y + \log m - \frac{a}{2}m^2, \quad \mu(y, m) = c + \beta \tanh y - m.$$

The terminal payoff is y ; the cross-sectional dispersion term vanishes in one dimension. Put $t_k = kh$, $h = T/n_t$ and let δ be the uniform state spacing. The original upwind generator is

$$(L_\delta^m v)_i = q_i^-(m)(v_{i-1} - v_i) + q_i^+(m)(v_{i+1} - v_i),$$

$$q_i^- = \frac{\eta}{\delta^2} + \frac{(-\mu_i)^+}{\delta},$$

$$q_i^+ = \frac{\eta}{\delta^2} + \frac{\mu_i^+}{\delta}.$$

Both coefficients are nonnegative. A fixed-policy backward step solves

$$\frac{v_i^k - v_i^{k+1}}{h} + \rho v_i^k - (L_\delta^{m^k} v^k)_i - u(y_i, m_i^k) = 0$$

at every interior node. Boundary values enter neighboring differences exactly once. In particular, residuals are evaluated in this original equation, not by subtracting a boundary-adjusted right-hand side after reintroducing the boundary terms.

For Howard improvement, the Hamiltonian depending on m is

$$\log m - \frac{a}{2}m^2 + (d_i - m)^+ p_i^f + (d_i - m)^- p_i^b,$$

where

$$d_i = c + \beta \tanh y_i, \quad p_i^f = \frac{v_{i+1} - v_i}{\delta}, \quad p_i^b = \frac{v_i - v_{i-1}}{\delta},$$

where $z^- = \min(z, 0)$. On each of the intervals cut by $m = d_i$, the objective is strictly concave. Its unconstrained maximizer has the form $2/(p + \sqrt{p^2 + 4a})$, with the algebraically equivalent stable expression used when $p < 0$. Clipping this root to each nonempty interval and comparing the two attained values gives the global scalar action maximum for the upwind Hamiltonian. Thus the reported HJB residual

uses the full economic action interval, without action-grid enumeration or a neural critic.

The imposed boundary functions have the form $b_{\pm}(t) = \pm Lw(t) + Q_{\pm}(t)$, where

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad \bar{m}(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4a}},$$

and

$$Q_{\pm}(t) = \int_t^T e^{-\rho(s-t)} \left[\log \bar{m}(s) - \frac{a}{2} \bar{m}(s)^2 + w(s) \{c \pm \beta - \bar{m}(s)\} \right] ds.$$

The original cumulative-trapezoid rule and time interpolation evaluate these functions. They represent asymptotic boundary data shared by all policies; they are not independently verified exact boundary values for each learned policy in the unbounded economy. Their satisfaction at grid endpoints is an algebraic check, not a bound on economic boundary bias.

For the optimized grid value, define the original-equation HJB defect

$$r_i^k = \frac{v_i^k - v_i^{k+1}}{h} + \rho v_i^k - \max_m \{u(y_i, m) + (L_{\delta}^m v^k)_i\}.$$

For a fixed policy, omit the maximization and use its stored feedback action. The diagnostic files retain the maximum residual at every time step, the linear residual of every Howard evaluation, every iterate difference, and the iteration count. The original stopping rule is an iterate difference below 10^{-11} , with a cap of 100 iterations. A separate Howard check evaluates $\|\mathcal{T}_k(v^k) - v^k\|_{\infty}/h$, where $\mathcal{T}_k$ solves the linear system for the greedy policy selected from v^k .

The significance of these residuals can be stated directly. Let $\tilde{v}$ be the exact solution of the same finite-grid backward equations, with the same frozen policy (or the same Bellman operator), terminal data and boundary data, and let v be the approximate sequence whose defect is r . In exact arithmetic, the fixed-policy matrix has positive diagonal, nonpositive off-diagonal entries, and row sums at least $h^{-1} + \rho$ after boundary elimination. The discrete maximum principle therefore yields

$$\|\tilde{v}^k - v^k\|_{\infty} \leq \frac{\|\tilde{v}^{k+1} - v^{k+1}\|_{\infty} + hR_k}{1 + \rho h}$$

whenever the boundary data agree and $\|r^k\|_\infty \leq R_k$. The same estimate holds for the Bellman step: at a positive maximum of the difference, select an action maximizing the Hamiltonian of the upper candidate and apply the fixed-action maximum principle; interchange the candidates for the negative part. With exact common terminal data this gives

$$\|\tilde{v}^0 - v^0\|_\infty \leq \sum_{k=0}^{n_t-1} \frac{hR_k}{(1+\rho h)^{k+1}} \leq \frac{\max_k R_k}{\rho} \{1 - (1+\rho h)^{-n_t}\}.$$

The archived propagation diagnostic substitutes the observed floating-point residuals in this formula. These residuals are not outward-rounded enclosures, so that substitution is a numerical diagnostic rather than a certified bound. The estimate concerns the finite-grid equations with the prescribed boundary data; it does not include consistency, state interpolation or boundary bias.

The tables report all four grids, the five initial states $y_0 \in \{-1, -0.5, 0, 0.5, 1\}$, the policies' action-bound frequencies, and changes in values, optimal actions and policy losses. Initial-state values are linearly interpolated from nodal time-zero values. Bound frequencies use an absolute tolerance of 10^{-12} and count equally weighted interior state-time nodes; they are not occupation probabilities. The first two grid comparisons halve both h and δ . Increasing L from 6 to 8 at fixed $(n_y, n_t) = (1601, 1024)$ changes δ from 0.0075 to 0.01. This comparison measures joint domain and spacing sensitivity, not boundary truncation error in isolation. Full common-domain changes and their locations accompany the central-region summaries.

The scalar calculation supplies an independent finite-grid benchmark for non-linear feedback, with a reproducible algebraic error account. In the 10-, 20- and 50-dimensional economies, cross-sectional dispersion, the coupling matrix and the covariance structure remain economically active. Moreover, this scalar benchmark evaluates Markov feedback, whereas the continuous-time deployment theorem specifies a randomized controller with innovation history. Scalar reference losses therefore have their stated one-dimensional meaning; multidimensional conclusions rely on the separate multidimensional policy comparisons and certification arguments.

TABLE S.4

SCALAR FINITE-GRID POLICY LOSSES AT FIVE INITIAL STATES

Grid (n_y, n_t, L)	y_0	NBO	DPO	Affine	Anchor
(401,256,6)	-1	1.02×10^{-5}	6.78×10^{-5}	9.63×10^{-5}	1.16×10^{-4}
	-0.5	7.17×10^{-6}	8.61×10^{-5}	5.67×10^{-5}	5.64×10^{-4}
	0	2.55×10^{-6}	1.09×10^{-4}	1.93×10^{-4}	1.54×10^{-3}
	0.5	3.23×10^{-6}	1.04×10^{-4}	1.00×10^{-4}	1.68×10^{-3}
	1	9.79×10^{-5}	5.30×10^{-5}	1.69×10^{-4}	7.17×10^{-4}
(801,512,6)	-1	1.08×10^{-5}	6.93×10^{-5}	9.51×10^{-5}	1.17×10^{-4}
	-0.5	8.23×10^{-6}	8.96×10^{-5}	5.73×10^{-5}	5.70×10^{-4}
	0	3.12×10^{-6}	1.14×10^{-4}	2.00×10^{-4}	1.56×10^{-3}
	0.5	3.08×10^{-6}	1.04×10^{-4}	9.95×10^{-5}	1.67×10^{-3}
	1	1.02×10^{-4}	5.11×10^{-5}	1.72×10^{-4}	6.99×10^{-4}
(1601,1024,6)	-1	1.11×10^{-5}	7.00×10^{-5}	9.45×10^{-5}	1.17×10^{-4}
	-0.5	8.81×10^{-6}	9.13×10^{-5}	5.77×10^{-5}	5.74×10^{-4}
	0	3.44×10^{-6}	1.16×10^{-4}	2.03×10^{-4}	1.57×10^{-3}
	0.5	2.99×10^{-6}	1.03×10^{-4}	9.89×10^{-5}	1.67×10^{-3}
	1	1.05×10^{-4}	5.02×10^{-5}	1.74×10^{-4}	6.90×10^{-4}
(1601,1024,8)	-1	1.10×10^{-5}	6.98×10^{-5}	9.46×10^{-5}	1.17×10^{-4}
	-0.5	8.62×10^{-6}	9.08×10^{-5}	5.76×10^{-5}	5.72×10^{-4}
	0	3.32×10^{-6}	1.15×10^{-4}	2.02×10^{-4}	1.57×10^{-3}
	0.5	3.02×10^{-6}	1.04×10^{-4}	9.90×10^{-5}	1.67×10^{-3}
	1	1.04×10^{-4}	5.04×10^{-5}	1.73×10^{-4}	6.92×10^{-4}

Entries are the Howard grid value minus the value of the unchanged stored feedback policy, with common asymptotic Dirichlet data. Values at the five stated initial log-capital levels use linear interpolation of time-zero nodal values when necessary. No policy is refitted or chosen using this table. These are scalar finite-grid losses; they do not estimate the optimum in dimensions 10, 20 or 50.

TABLE S.5

SCALAR VALUE AND POLICY CHANGES UNDER REFINEMENT

Comparison	Policy	Core value	Core action	Core loss	Common value
$401 \rightarrow 801, L = 6$	Howard	6.59×10^{-4}	2.87×10^{-4}	0	6.59×10^{-4}
	NBO	6.57×10^{-4}	0	4.28×10^{-6}	6.57×10^{-4}
	DPO	6.61×10^{-4}	0	4.64×10^{-6}	6.61×10^{-4}
	Affine	6.60×10^{-4}	0	7.20×10^{-6}	6.60×10^{-4}
	Anchor	6.78×10^{-4}	0	1.91×10^{-5}	6.78×10^{-4}
$801 \rightarrow 1601, L = 6$	Howard	3.31×10^{-4}	1.45×10^{-4}	0	3.31×10^{-4}
	NBO	3.30×10^{-4}	0	2.22×10^{-6}	3.30×10^{-4}
	DPO	3.32×10^{-4}	0	2.40×10^{-6}	3.32×10^{-4}
	Affine	3.32×10^{-4}	0	3.72×10^{-6}	3.32×10^{-4}
	Anchor	3.40×10^{-4}	0	9.74×10^{-6}	3.40×10^{-4}
$L = 6 \rightarrow 8, n_y = 1601$	Howard	5.21×10^{-5}	4.72×10^{-5}	0	2.72×10^{-4}
	NBO	5.23×10^{-5}	7.59×10^{-7}	6.88×10^{-7}	5.03×10^{-4}
	DPO	5.28×10^{-5}	7.14×10^{-7}	8.13×10^{-7}	2.89×10^{-4}
	Affine	5.25×10^{-5}	1.02×10^{-8}	1.29×10^{-6}	7.37×10^{-3}
	Anchor	5.39×10^{-5}	0	3.63×10^{-6}	2.72×10^{-4}

Entries are maximum absolute changes in values, actions or policy losses. The first three numerical columns use time-zero nodes with $|y| \leq 1$; the final column uses the full common state interval. The second grid is linearly interpolated at nodes of the first. Policy loss is $\ell = V^H - V^m$. The first two comparisons halve both state spacing and time step. The last expands the domain at fixed n_y, n_t , changing state spacing from 0.0075 to 0.01; it is joint domain/spacing sensitivity and does not isolate boundary truncation error. Frozen-policy action changes arise only from interpolation and floating-point evaluation. Full common-interior action changes and their locations are archived.

S.15.3. *Finite-Observation Diagnostic Accounts*

The experiment fixes the same saved NBO checkpoint before drawing the diagnostic bank. The sensor uses finite state measurements and independent private noise; the parent uses a clipped-innovation internal state. The deterministic allowance and the finite-Euler measurement answer different questions. Neither the observed mean nor its simulation standard error supplies the fine-Euler bias.

TABLE S.6: Components of the finite-observation allowance

d	Cells	ν_{obs}	Production	Consumption	Terminal
10	64	0	0.000270	0.000833	0.000169
10	64	0.001	0.000687	0.001876	0.000361
10	64	0.01	0.004433	0.011263	0.002095
10	128	0	0.000137	0.000412	0.000087
10	128	0.001	0.000553	0.001432	0.000282
10	128	0.01	0.004295	0.010610	0.002037
20	64	0	0.000627	0.001756	0.000369
20	64	0.001	0.001535	0.003886	0.000787
20	64	0.01	0.009706	0.023047	0.004553
20	128	0	0.000320	0.000876	0.000191
20	128	0.001	0.001232	0.002973	0.000619
20	128	0.01	0.009442	0.021853	0.004471
50	64	0	0.001754	0.005424	0.001192
50	64	0.001	0.004613	0.013341	0.002877
50	64	0.01	0.022028	0.044889	0.008488
50	128	0	0.000912	0.002769	0.000634
50	128	0.001	0.003840	0.010758	0.002406
50	128	0.01	0.021850	0.043484	0.008503

Each component is displayed upward. The machine-readable records give the full recurrence, network Lipschitz and roundoff bounds, initial-profile bounds, seeds, pathwise diagnostic arrays, and the separate Monte Carlo standard errors.

S.16. THE COMPLETE PREVIOUS COMPUTATIONAL EVIDENCE

The following source-identified sections preserve the previously reviewed computation, including its original protocols, complete method and checkpoint records, scalar and measurement experiments, and paired-transfer refinement. Their numerical statements refer to those completed experiments. The words “current” and “new” inside these retained source sections describe their position in that earlier experiment; the current article’s evidence chronology is specified below.

The first capital comparisons precede the sixteen-stream higher-volatility experiment. That experiment’s original policies, final observations, and stopping decisions were fixed before the paired deterministic transfer was refined. The dimension-ten NBO–HJB conclusion first became positive under that post-freeze analytical reanalysis. Both original and refined intervals appear in the retained tables. The R16 independent bank fixes that paired account in advance and has a separate probability allocation. No retained R15 sentence converts the refinement into a new simulation experiment, a new training design, or a retrospective change in online success.

The source text is retained in full so that its calculations and adverse outcomes can be checked against the new study. The full-class regret bounds retain their anchor gap. The mechanism and critic-risk intervals retain their inconclusive outcomes. The old HJB result refers to its declared stochastic-trace and action-search implementation. The current main article states the direct comparison with the strengthened implementation separately.

S.17. COMPUTATIONAL DESIGN AND THE EXISTING COMPARISON

The numerical question is whether the learned evaluation block improves the economic accuracy or complete work of the NBO map. All comparisons identify the economic model, feasible actions, evaluation target, selected policy, final payoff law, and work boundaries. A policy-specific confidence interval conditions on its fitted objects. A statement about a training method additionally specifies the distribution of training randomness and its sampling uncertainty. An interval containing zero establishes neither superiority nor practical equivalence.

S.17.1. *The executed capital comparison*

The source-bound reference study contains ninety primary fitted policies in dimensions ten, twenty, and fifty. Its primary fitting design uses an announced twenty-second budget, completes the current iteration, and selects the checkpoint using validation data. Each final primary row uses 8,192 independent paths and 1,024 time cells. The comparison concerns a fixed nine-profile population and a separate origin calculation. The simultaneous inference construction covers the declared finite family of selected policies and profiles.

Every fitted policy in the reported primary schedule comparisons has a positive simultaneous lower gain endpoint. For the prespecified population, all thirty NBO policies have this property. These are improvements over the analytical schedule; the direct method comparisons have a different conclusion. Table S.7 retains the findings that motivate the current evaluation experiment.

TABLE S.7
EXECUTED COMPARISON OF THE LEARNED BELLMAN BLOCK

Object	Recorded result
Primary NBO-minus-DPO/affine intervals	All 120 contain zero
Fixed-work direct method intervals	All 42 contain zero
Principal NBO-minus-Raw payoff means	All 12 are negative; intervals contain zero
Raw versus NBO, dimension-checkpoint cells	Higher Raw mean and lower Raw fitting clock in all 6 cells
Critic versus Raw costate MSE	Higher critic MSE in all 6 principal panels; ratio about 1.06–1.99
Critic-greedy versus Raw Hamiltonian gap	Higher critic-greedy gap in all 6 panels

Note: The primary and fixed-work intervals are policy-payoff comparisons conditional on the fitted objects. The twelve means refer to the principal Ubuntu 24 dimension, stream and checkpoint combinations. The six mechanism panels use held-out training-design states and a common independent finite-rollout reference; they do not instantiate a continuous-time candidate-occupation welfare bound. The complete original rows, arrays, work measurements and source identities remain in the reproduction record.

The negative sample means do not prove that Raw dominates NBO, because the simultaneous direct intervals remain inconclusive. The mechanism comparisons nevertheless provide no positive evidence for the fitted critic in that design. They cannot be converted into a favorable mechanism result by observing that the common verifier accepts NBO policies. Likewise, the fixed-work experiment matches simulator visits and checkpoints but has different optimizer update counts; it does not establish equal total computation.

S.17.2. *Accuracy objects and economic scale*

The original finite-population NBO regret upper bounds are approximately 0.02097, 0.02847 and 0.03039 in dimensions ten, twenty and fifty. The corresponding mean schedule improvements are approximately 0.00147, 0.00135 and 0.00161. The regret bounds combine the analytical anchor gap with the measured improvement. They are reported as absolute upper bounds, without interpreting them as tight high-dimensional HJB errors.

The independent scalar computation uses a monotone implicit HJB scheme and full-box action maximization on four original grids. At initial log capital zero, the finest domain-six NBO loss is about 3.44×10^{-6} , compared with 1.16×10^{-4} for direct policy optimization. At initial state one, the corresponding losses are about 1.05×10^{-4} and 5.02×10^{-5} . The ordering is therefore state dependent. These are finite-grid losses of Markov feedbacks in the scalar model; the widening experiment changes both domain and spatial spacing. The residual, refinement and five-state accounts remain in Appendix S.15.

The balance-sheet profiles and the resource fee in (S.118) are transparent model experiments. They are not an empirical calibration. The fee transformation reports what administration cost a verified utility gain can support under the specified log-felicity resource account. It does not provide an additional independent source of evidence for the critic.

S.17.3. *Work and comparison semantics*

The original time-budget and fixed-work records retain fitting clocks, simulator visits, optimizer updates, peak process memory, selected checkpoints and verification components. Their verification clock begins after weight loading and deterministic-constant construction. Their prefix frontier is retrospective. Those quantities remain useful descriptions of the original computation, with those boundaries.

The current method-level comparison specifies its training-randomness estimand, economic target, actual stopping rule, complete cost boundary, and final inferential object before its confirmatory evaluation. It must charge failed checks and retain unattained targets. A direct HJB or policy-iteration comparator isolates the role of a separately learned actor and continuation object under a shared model and verifier. A classical low-dimensional method serves as a policy generator as well as an independent reference. The following study gives the concrete protocol and its executed results.

S.18. EVALUATION AND ECONOMIC PERFORMANCE

The continuation-learning comparison and the finite-observation exercise address separate questions in the capital economy. The former assesses continuation evaluation and feasible improvement under its stated volatility parameters and complete method distribution. The latter evaluates the original saved policies with a protected parent at their original volatility parameters. Their controller identities, information sets, and probability accounts remain distinct.

S.18.1. *The capital experiment*

The experiment uses the capital state equation and objective in (S.99)–(S.100), in dimensions ten and fifty. It changes the volatility parameters to $\sigma_I = 0.6$ and $\sigma_C = 0.3$; the remaining primitives are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\zeta = 0.2$, $\chi = 0.03$, and $[\underline{m}, M] = [0.02, 2]$. The initial population assigns equal mass to nine profiles with means $-0.5, 0, 0.5$ and cross-sectional spreads $0, 0.25, 0.5$. The methods share the declared radius-0.1 feasible tube, coupling matrix, initial population, and

1 payoff evaluator. This is a parameter regime of the same economic model. The 1
 2 earlier comparison, scalar study, and protected-observation experiment retain their 2
 3 original volatility parameters and their separate evidence records. 3

4 Training uses a global grid of 64 cells. Its reference policy is the cell-average 4
 5 analytical schedule and remains fixed through all three work levels. The cumulative 5
 6 occupation samples contain 256, 512, and 1,024 states. Each state comes from an 6
 7 independent reference trajectory at a uniformly selected decision node. Postdecision 7
 8 inputs are the pre-noise mean and the next decision date. NBO and Raw share four- 8
 9 antithetic continuation targets; all member rollouts and all reused labels are counted. 9
 10 These sample sizes define increasing information budgets, not equal floating-point 10
 11 work across methods. 11

12 At each stage NBO fits the spatial continuation derivative by full-replay Sobolev 12
 13 mean-square loss, with 1,200 Adam updates followed by L-BFGS. The latter has 13
 14 nominal iteration and evaluation limits of 300 and 400; the current line search is 14
 15 completed and every actual closure call is counted. A separate time component 15
 16 then receives 400 value-fitting updates with the spatial component fixed. NBO and 16
 17 Raw use the same feasible actor objective and minibatch/update streams, with 200, 17
 18 400, and 800 additional actor updates at the three stages. DPO and HJB use 100, 200, 18
 19 and 400 additional updates. The minibatch size is 128; the actor, spatial-critic, time- 19
 20 component, and HJB learning rates are 0.003, 0.002, 0.003, and 0.002, respectively. 20
 21 Model weights and replay continue across stages. These are declared optimization 21
 22 procedures; a common sample budget does not make their optimization costs 22
 23 identical. 23

24 Online certification and independent final confirmation both use 2,048 decision 24
 25 cells and 8,192 paths per stream. The selected fixed networks are deployed on that 25
 26 finer grid. The learned 64-cell continuation derivative is consequently a predictor in 26
 27 the fine-grid mechanism assessment, whose error account includes that discrepancy; 27
 28 it is not identified with an exact fine-grid or continuous-time costate. The protocol 28
 29 and numerical source were frozen before the confirmation experiment. Exploratory 29
 30 pilots that selected the design remain distinct from the confirmatory observations. 30

31 The measured process clock begins before a fresh worker is launched and ends 31
 32 after its final evidence is durably written. It includes imports, model setup, de- 32

terministic constants, fitting, checks, weight reloading, confirmation, and output. Dependency installation and artifact extraction precede this clock and are not charged as method execution time.

S.18.2. Contribution-isolating comparisons

The comparison separates continuation regression, direct policy optimization, and direct approximation of the Bellman equation. The direct HJB method uses the same two learned scalar components as NBO: a time network of widths (16, 16) and a state–time network of widths (32, 32), each with hyperbolic tangent activations and a scalar output. Its known lift is the full-value lift, so the terminal condition is imposed exactly. NBO’s known lift instead represents its postdecision continuation target. This difference follows from the objects being approximated; the learned residual capacity is the same. Both methods use the prespecified occupation sample and the same available model information. No reference solution enters either training objective.

Write $c = a - (\sigma_I^2 + \sigma_C^2)/2$ and $\Sigma\Sigma' = \sigma_I^2 I + \sigma_C^2 \mathbf{1}\mathbf{1}'$. The direct method minimizes the squared residual of

$$0 = -v_t - \max_{m \in \mathcal{M}_t} \{u(y, m) + (c\mathbf{1} + \kappa \tanh(By) - m)'D_y v\} - \frac{1}{2} \text{tr}(\Sigma\Sigma' D_y^2 v) + \rho v, \quad v(T, y) = g(y), \quad (\text{S.81})$$

where $\mathcal{M}_t$ is the action tube shared by the four primary methods. For $q = dD_y v$, the action-dependent Hamiltonian is

$$H_q(m) = \frac{1}{d} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \frac{1}{d} \sum_i q_i m_i. \quad (\text{S.82})$$

It is strictly concave. Its maximizer is obtained from the scalar consistency equation $z = d^{-1} \sum_i m_i(z)$, with $m_i(z) = \text{proj}_{[\ell_t, u_t]}(q_i + \zeta z)^{-1}$ when $q_i + \zeta z > 0$, and $m_i(z) = u_t$ otherwise. The implementation uses the same 44 monotone bisection steps as the other feasible costate comparisons. The direct HJB policy evaluates a critic gradient and this maximization at every decision. NBO deploys its separately learned actor. The resulting difference in online computation is included in validation, stopping, confirmation, and deployment costs.

Two independent Gaussian probe banks estimate the covariance Hessian trace. If $\hat{R}_1$ and $\hat{R}_2$ are the corresponding residual estimates, then, conditional on a collocation state and the current critic,

$$\mathbb{E}[\hat{R}_1 \hat{R}_2] = R[v]^2. \quad (\text{S.83})$$

The product, rather than the square of one estimate, is the training loss. Its realized value can be negative. Negative batches are retained. This identity specifies an optimization objective; the common payoff verifier supplies the economic comparison.

Classical scalar training. The original volatility calibration also has a scalar comparison. A monotone implicit Howard method solves the one-state problem and saves an action at every time and state node. These saved policies are deployed by interpolation on fresh diffusion paths, together with the analytical schedule and the hash-pinned scalar NBO and DPO policies from the earlier experiment. Full-box and matched-tube classical policies are separate candidates. Three successively refined state–time grids are followed by an enlargement of the state interval at fixed spatial and temporal spacing. The deployment meshes aggregate the same fine Brownian increments. Policy differences are reported both over the declared initial population and at its individual states -1 , 0 , and 1 . Historical neural fitting clocks retain their historical interpretation; the new classical solves and all new deployment work are measured separately.

The scalar reference has a numerical error account at the level of the declared finite-grid equations. Its algebraic error account uses an outward residual calculation. State-domain approximation and diffusion time discretization remain separate from this algebraic account. The scalar comparison supplies an independent low-dimensional accuracy and work benchmark; the high-dimensional method comparison uses its own payoff data.

S.18.3. *The distribution of a computational method*

A computational method includes its initialization, training draws, validation rule, stopping rule, and response to failure. We specify these objects jointly. Let $\mathcal{S}$ be the sixteen complete execution streams listed in the protocol, fixed before

the confirmation experiment, and let $\pi_{m,s}$ denote the policy returned by method m under stream s . The randomized implementation draws s uniformly from $\mathcal{S}$; the experiment executes every member of this distribution. A failed fit returns the feasible analytical schedule and remains in the denominator. Conditional on the stated software environment, the stream specifies the complete training and checking computation. The distribution does not include arbitrary unrecorded initializations or a new draw of the stopping bank on re-execution.

Write $J(\pi)$ for the payoff in the original continuous-time economy, averaged over the specified nine initial-state profiles. The method-level objects are

$$G_m = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \{J(\pi_{m,s}) - J(\bar{\pi})\}, \quad (\text{S.84})$$

$$\Delta_{m,b} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \{J(\pi_{m,s}) - J(\pi_{b,s})\}. \quad (\text{S.85})$$

The first object is improvement on the schedule. The second is the direct comparison of two randomized computational procedures with the same declared stream distribution. Enumeration accounts for the entire training distribution; there is no sampling error from an unobserved subset of these sixteen streams. The remaining uncertainty concerns their independent payoff simulations and the numerical transfer to continuous time. An assertion about this finite distribution is not an assertion about all possible training initializations.

The stopping target is a gain of 0.0005. At each of three declared work checkpoints, fitting pauses for a new, independent certification calculation. Training stops when its lower endpoint attains the target. Failed checks, initialization, action search, checkpoint input and output, and the independent final verification are included in the measured work. The gain 0.001 is a secondary final-attainment threshold; it is not a second measured stopping experiment. A failure to attain either target is retained.

S.18.4. *Simultaneous inference for the finite method distribution*

Each returned policy receives N fresh confirmation paths. The confirmation bank is independent of fitting, validation, and every stopping decision. Methods share

initial profiles and innovations within a stream, whereas the confirmation banks of distinct streams are independent. A direct contrast is formed on each common path. Its common analytical schedule cancels, and its transfer account contains the two policy allowances rather than two copies of the schedule allowance, when both entries are simulated candidate gains. An analytical fallback has the exact gain zero. A comparison with that fallback retains the other candidate's full schedule-relative transfer account; two analytical fallbacks have the exact difference zero.

For one registered endpoint, let X_{sj} be the numerical path statistic, $Y_{sj} = \max\{-C_s, \min(C_s, X_{sj})\}$, and let β_s enclose the difference between the continuous-time endpoint θ_s and $\mathbb{E}Y_{sj}$. Thus β_s includes the appropriate diffusion, implementation, clipping-tail, and arithmetic accounts. Put $C = \max_s C_s$, $n = |\mathcal{S}|N$, $\bar{\beta} = |\mathcal{S}|^{-1} \sum_s \beta_s$, and denote the pooled mean and unbiased sample variance by $\bar{Y}$ and $\hat{v}$.

PROPOSITION S.22—Finite-distribution comparison: *Conditional on the complete executed policy ensemble, suppose the confirmation observations are independent across pairs (s, j) and the stated range and transfer accounts hold. For $0 < \delta < 1$, the interval*

$$\bar{Y} \pm \left\{ \sqrt{\frac{2\hat{v} \log(4/\delta)}{n}} + \frac{14C \log(4/\delta)}{3(n-1)} + \bar{\beta} \right\} \quad (\text{S.86})$$

covers $|\mathcal{S}|^{-1} \sum_s \theta_s$ with probability at least $1 - \delta$. The observations need not be identically distributed across streams.

The confirmation family assigns 0.02 across 238 two-sided events: in each dimension, four schedule comparisons and three direct NBO comparisons, each reported for sixteen streams and their uniform method mean. Online stopping receives a separate 0.01 across its 384 possible checks. The mechanism and protected-observation families each receive 0.01. Their union therefore has error probability at most 0.05 for the new experiment. The historical experiment retains its separately stated confidence event. Unused checks do not release probability for additional comparisons.

The practical-equivalence margin is fixed at 10^{-4} payoff units. For a direct interval $[L, U]$, material superiority requires $L > 10^{-4}$, and practical equivalence requires $-10^{-4} < L \leq U < 10^{-4}$. Statistical superiority, $L > 0$, is reported separately. An

interval containing zero does not establish equivalence. Under log flow utility, the fixed margin corresponds to a proportional consumption fee $1 - \exp\{-10^{-4}/A\}$, where $A = (1 - e^{-\rho T})/\rho$; for the stated primitives this is approximately 1.0201 basis points.

Simultaneous stream intervals also enclose the probability that the randomized implementation attains a target g . If they are $[L_s, U_s]$, the corresponding bounds are

$$\frac{\#\{s : L_s \geq g\}}{|\mathcal{S}|} \leq \frac{\#\{s : \theta_s \geq g\}}{|\mathcal{S}|} \leq \frac{\#\{s : U_s \geq g\}}{|\mathcal{S}|}. \quad (\text{S.87})$$

No binomial sampling interval is attached to a population that has been enumerated in full. Work is likewise reported over all streams, with unattained runs retained in the distribution of work to target. A median over successful runs alone is not the method's median.

The economic decomposition records production, consumption, and terminal dispersion contributions on the same numerical confirmation paths. These components add to the observed numerical payoff gain and help describe the policy's behavior. They have no separate continuous-time confidence statement: the transfer allowance is attached to the total payoff and is not divided among components without additional bounds.

S.18.5. *Method performance, robustness, and complete work*

The confirmation record contains all 128 declared method–dimension–stream executions. Tables S.8 and S.9 report the finite-distribution payoff objects defined in Section S.18.3. Every endpoint is reconstructed from the original final arrays after checking their common-path identities and source manifests.

Tables S.10 and S.11 measure the executed stopping procedures, including unsuccessful checks and final confirmation. Table S.12 describes the numerical channels of the payoff change without assigning the total continuous-time certificate to its separate components.

Economic performance of the declared methods. Continuation evaluation and feasible improvement produce positive continuous-economy gains in both dimensions. NBO's method means are approximately 0.001268 and 0.001342 in dimensions

TABLE S.8

PAYOFF OF THE FINITE-DISTRIBUTION COMPUTATIONAL METHODS

d	Method	Mean gain	Lower	Upper	At least target
10	NBO	0.001268	0.000943	0.001592	16/16
10	Raw	0.001267	0.000942	0.001592	16/16
10	Direct policy	0.001262	0.000938	0.001587	16/16
10	Neural HJB	0.000973	0.000647	0.001299	0/16
50	NBO	0.001342	0.000957	0.001727	16/16
50	Raw	0.001343	0.000958	0.001728	16/16
50	Direct policy	0.001347	0.000962	0.001732	16/16
50	Neural HJB	-0.000156	-0.000558	0.000246	0/16

Notes: Means average all sixteen declared execution streams. Endpoints enclose continuous-time payoff gains on the analytical schedule. The last column counts streams whose unrounded simultaneous final lower endpoint attains 0.0005; it is a lower bound on attainment under the declared finite distribution. No stream is excluded. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE S.9

DIRECT METHOD COMPARISONS AT THE DECLARED ECONOMIC MARGIN

d	Comparison	Mean	Lower	Upper	Decision
10	NBO–Raw	0.000001	-0.000362	0.000364	Unresolved
10	NBO–Direct policy	0.000005	-0.000358	0.000368	Unresolved
10	NBO–Neural HJB	0.000295	-0.000073	0.000663	Unresolved
50	NBO–Raw	-0.000001	-0.000429	0.000426	Unresolved
50	NBO–Direct policy	-0.000005	-0.000432	0.000423	Unresolved
50	NBO–Neural HJB	0.001498	0.001051	0.001945	Superior

Notes: Pathwise differences use common profiles and innovations within each stream. The equivalence margin is 10^{-4} payoff units. Superior and inferior refer to economic differences exceeding that margin. Decisions use the original unrounded endpoints. An unresolved comparison is not evidence of equivalence. The schedule transfer cancels between two simulated candidates; comparison with an exact analytical fallback retains the other candidate's full schedule-relative transfer. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE S.10

COMPLETE MEASURED WORK OF THE STOPPING PROCEDURES

d	Method	Stopped	Mean (s)	Median (s)	Restricted (s)	MiB
10	NBO	0/16	134.82	–	900.00	474.3
10	Raw	0/16	122.67	–	900.00	473.0
10	Direct policy	0/16	285.53	–	900.00	489.4
10	Neural HJB	0/16	373.35	–	900.00	477.3
50	NBO	16/16	357.80	376.03	357.80	530.0
50	Raw	16/16	346.22	365.55	346.22	531.7
50	Direct policy	16/16	596.94	627.10	596.94	545.5
50	Neural HJB	0/16	1690.10	–	900.00	528.4

Stopped counts actual online certificates at the 0.0005 target. Mean is total consumed parent-clock time through independent final confirmation. Median is the lower 0.5 quantile of work to target, with failures assigned infinity; a dash denotes an unattained median. Restricted work is the mean of min(time to target, 900 seconds), retaining failures at 900. Memory is the maximum fresh-worker peak resident size over streams. These clocks describe the recorded execution environment.

TABLE S.11

RECORDED OPERATION COUNTS OF THE COMPLETE STOPPING PROCEDURES

d	Method	Train 10^6	Verify 10^6	Actor	Critic	Grad. 10^6	Search 10^6
10	NBO	0.20	134.22	1400	5700	1.04	0.00
10	Raw	0.20	134.22	1400	0	0.00	0.00
10	Direct policy	23.00	134.22	700	0	0.00	0.00
10	Neural HJB	0.07	134.22	0	700	67.20	2956.73
50	NBO	0.18	127.93	1250	5344	0.93	0.00
50	Raw	0.18	127.93	1250	0	0.00	0.00
50	Direct policy	23.00	134.22	700	0	0.00	0.00
50	Neural HJB	0.07	134.22	0	700	67.20	2956.73

Entries are means over all sixteen streams. Train and verify count simulator transitions; actor and critic count optimizer updates; grad. counts first-derivative rows; search counts action-search iterations, including HJB deployment. These are work proxies, not equivalent FLOPs. Full first- and second-derivative, forward, backward, and other counters are retained in the machine-readable report.

The parent clock additionally includes every failed operation and all I/O.

TABLE S.12

DESCRIPTIVE DECOMPOSITION OF THE NUMERICAL PAYOFF GAIN

d	Method	Production	Consumption	Dispersion	Total
10	NBO	0.002303	-0.001271	0.000236	0.001268
10	Raw	0.002303	-0.001273	0.000237	0.001267
10	Direct policy	0.002279	-0.001248	0.000231	0.001262
10	Neural HJB	0.001830	-0.001009	0.000152	0.000973
50	NBO	0.002297	-0.001343	0.000387	0.001342
50	Raw	0.002297	-0.001342	0.000389	0.001343
50	Direct policy	0.002275	-0.001310	0.000382	0.001347
50	Neural HJB	0.000910	-0.001517	0.000451	-0.000156

Entries are observed finite-step common-path contributions, averaged uniformly over the complete stream set. Consumption is minus the consumption deficit; dispersion is the terminal contribution. Their pathwise sum equals the numerical gain. No component continuous-time confidence interval or causal attribution is asserted; the separate total-payoff account remains authoritative.

ten and fifty; their simultaneous lower endpoints exceed 0.000943 and 0.000957, respectively. These statements average over all sixteen declared complete algorithm streams. All 128 executions returned fitted policies; none used the analytical fallback.

The direct comparison identifies a material advantage over the specified neural HJB procedure in dimension fifty. With the original transfer account, the NBO-minus-HJB interval is contained in $[0.001051, 0.001945]$, whose lower endpoint exceeds the prespecified 10^{-4} economic margin. The original ten-dimensional interval, contained in $[-0.000073, 0.000663]$, is inconclusive. All four direct comparisons with cached Raw and direct policy optimization are also inconclusive under that account. Thus the experiment establishes a domain in which the NBO evaluation-actor construction improves economic performance relative to direct residual fitting with the stated common representation capacity. The finding concerns the declared architectures, training schedules, stopping rule, and finite stream distribution; it is not a ranking of every possible implementation of the three alternatives.

Economic accuracy and complete work. In dimension fifty, all sixteen NBO, Raw, and DPO executions attained the online gain target 0.0005, while no HJB execution

did. Their mean complete work was approximately 357.80, 346.22, 596.94, and 1690.10 seconds, respectively, including unsuccessful checks and final confirmation. NBO therefore attained the same declared economic target with less measured work than DPO on the recorded environment; Raw was cheaper than NBO. HJB's unattained runs remain in the work distribution. The reported 900-second restricted-mean functional for an unattained target is a censoring convention, not the actual work consumed or an assertion that the run stopped at 900 seconds.

No ten-dimensional execution attained the online stopping target. Independent final confirmation nevertheless places every returned NBO, Raw, and DPO policy above 0.0005 in that dimension, as it does in dimension fifty. These final certificates do not alter the earlier stopping decisions. No method has a positively certified stream attaining the secondary 0.001 threshold. The complete stream and checkpoint tables retain this difference between measured stopping and final economic accuracy.

Sources of the observed gain. The numerical decomposition records how the learned decisions change the capital economy. For NBO, production contributes approximately 0.002303 and 0.002297 in dimensions ten and fifty, while the consumption contribution is approximately -0.001271 and -0.001343 . The terminal-dispersion contribution is approximately 0.000236 and 0.000387. Their sums equal the observed numerical gains. These quantities describe the production benefit and the consumption cost of state-dependent decisions on the same paths; the continuous-time confidence statement belongs to the total payoff, not separately to these components.

S.18.6. *Direct comparisons with a shared diffusion-transfer account*

The direct statistic already compares policies on common innovations. The deterministic transfer can use that same coupling. Proposition S.11 in the supplement bounds the error of the payoff difference for two held policies directly, retaining the original decision grid and realized actions. The fitted policies, simulated observations, stopping decisions, clipping, tail account, and simultaneous confidence family remain unchanged.

The following table reports the original and refined endpoints together. The refinement concerns a proved deterministic allowance; it is not an additional simulation experiment or a retrospectively changed stopping rule. Comparisons involving a schedule fallback retain their stated fallback account. The refined comparison

TABLE S.13

ORIGINAL AND REFINED DIRECT METHOD ENDPOINTS ON THE SAME OBSERVATIONS

d	Comparison	Mean	Original interval	Refined interval	Decision
10	NBO–Raw	0.000001	[-0.000362, 0.000364]	[-0.000110, 0.000112]	Unresolved
10	NBO–Direct policy	0.000005	[-0.000358, 0.000368]	[-0.000106, 0.000117]	Unresolved
10	NBO–Neural HJB	0.000295	[-0.000073, 0.000663]	[0.000179, 0.000411]	Superior
50	NBO–Raw	-0.000001	[-0.000429, 0.000426]	[-0.000133, 0.000130]	Unresolved
50	NBO–Direct policy	-0.000005	[-0.000432, 0.000423]	[-0.000137, 0.000127]	Unresolved
50	NBO–Neural HJB	0.001498	[0.001051, 0.001945]	[0.001347, 0.001650]	Superior

The new deterministic common-innovation transfer is proved in Appendix S.11. Both intervals use the identical original clipped statistic, path identities, empirical Bernstein event, and confidence allocation. The refinement was developed after the original source freeze and inspection of the first transfer width.

Only the deterministic transfer changes; mixed analytical fallbacks retain the original allowance. Decisions use the original 10^{-4} economic margin. Original training, stopping, and work records remain unchanged.

establishes economically material NBO gains over the specified neural HJB method in both dimensions. Its direct lower endpoints exceed 0.000179 and 0.001347, respectively, and hence exceed the original 10^{-4} margin. The ten-dimensional comparison changes from unresolved under the original account to materially positive under the proved refinement; the fifty-dimensional conclusion is strengthened. The four NBO–Raw/DPO intervals remain unresolved: they contain zero and extend outside the practical-equivalence band. They also do not certify noninferiority at the stated margin. The conclusion follows from a sharper deterministic account of the same paired observations, with the original statistical and computational records retained.

S.18.7. *Absolute accuracy in the original control problem*

Let ν be the declared uniform distribution over the nine initial profiles, and let J_ν^0 denote the analytical schedule's payoff. The pointwise anchor account in (S.28) implies $V_\nu^* - J_\nu^0 \leq B_{d,\nu}$ after averaging over this initial population. The repeated worker records concern this same bound. We verify their profiles and component weights and retain the maximum of the already averaged upper bounds if their final rounding differs.

For a method m , write $\bar{J}_{d,m} = 16^{-1} \sum_s J_\nu(\pi_{m,s})$. On the simultaneous confirmation event, its existing gain lower endpoint satisfies $\bar{J}_{d,m} - J_\nu^0 \geq L_{d,m}$. Consequently,

$$0 \leq V_\nu^* - \bar{J}_{d,m} \leq B_{d,\nu} - L_{d,m}. \quad (\text{S.88})$$

The optimum in this expression ranges over the original adapted action-box class. The returned methods use the declared common tube, which is a feasible subset. The inequality therefore accounts for absolute loss against the original economic problem. It does not identify the unobserved loss or require another probability allocation.

The reported full-class upper bounds range from 0.025971 to 0.038358 payoff units. Their ratios to the declared margin range from 259.8 to 383.6. Thus the current absolute account gives guarantees at those explicit tolerances. It does not certify proximity to the full optimum within the 10^{-4} margin, even when a relative method comparison is resolved.

Table S.14 puts this absolute account next to the economic margin used for direct method comparisons. A positive gain lower endpoint establishes improvement on the schedule; a direct comparison resolves a difference between two trained methods; and a small right-hand side of (S.88) would establish proximity to the full optimum. These implications concern different quantities. The neural HJB comparator supplies an independent candidate construction, while the scalar calculation supplies a separate finite-grid accuracy account. Neither supplies an exact high-dimensional reference value.

TABLE S.14

FULL-CLASS PAYOFF-GAP UPPER BOUNDS AND THE ECONOMIC MARGIN

d	Method	Gain lower	Regret upper	Margin	Ratio
10	NBO	0.000943	0.025971	0.0001	259.8
10	Raw	0.000942	0.025971	0.0001	259.8
10	Direct policy	0.000938	0.025976	0.0001	259.8
10	Neural HJB	0.000647	0.026266	0.0001	262.7
50	NBO	0.000957	0.036843	0.0001	368.5
50	Raw	0.000958	0.036841	0.0001	368.5
50	Direct policy	0.000962	0.036838	0.0001	368.4
50	Neural HJB	-0.000558	0.038358	0.0001	383.6

The regret upper bound is $B_{d,v} - L_{d,m}$, rounded outward: $B_{d,v}$ bounds the analytical schedule's loss against the original full adapted control class, and $L_{d,m}$ is the existing lower endpoint for the uniform mean over all sixteen returned policies. The ratio divides the regret upper bound by the prespecified 10^{-4} payoff margin. All 128 source-bound confirmation records are audited. This deterministic translation uses the existing confirmation event and no additional alpha.

S.18.8. Continuation accuracy where the policy changes the allocation

The mechanism assessment follows the policy that each registered NBO stream actually returns. It draws fresh occupation paths and uniform action bridges after all policy selection. The reference remains the cell-average analytical schedule on one global assessment grid. The saved scalar critic may have been fitted on a coarser grid; its assessment error includes this discrepancy.

Table S.15 compares the fitted costate with a direct four-path antithetic bank on the identical continuation queries. The calculation uses independent cross-products to remove reference-bank variance from the critic's risk estimand. It also reports the paired difference in risk, retaining negative sample contributions and all registered streams. In dimension 10, the costate-risk contrast has endpoints -0.0344447 and 0.0343934; this interval does not determine which predictor has smaller risk.

In dimension 50, the costate-risk contrast has endpoints -0.0225684 and 0.0225961; this interval does not determine which predictor has smaller risk.

Table S.16 carries the same continuation account through the feasible actor's actual action change and the quantitative continuous-payoff transfer of Theorem S.34. The approximate continuation's scalar advantage is evaluated at both action endpoints. The actor's gap is bounded from the saved global curvature certificate, including the endpoint calculation when strong concavity is unavailable. The combined Bellman-bridge lower endpoint in dimension 10 is -0.006778. The retained costate, sampling, and implementation allowances do not establish improvement through this sufficient bound.

The combined Bellman-bridge lower endpoint in dimension 50 is -0.004800. The retained costate, sampling, and implementation allowances do not establish improvement through this sufficient bound.

The complete six-statistic intervals and numerical cushions appear in Table S.22. Table S.23 reports the additional fresh-process assessment work. These calculations distinguish accuracy of the evaluation stage, quality of the feasible action update, and the independent payoff comparison between complete algorithms.

TABLE S.15

COSTATE PREDICTION ON THE CANDIDATE'S OCCUPATION LAW

d	Critic risk mean	Raw risk mean	Contrast lower	Contrast upper
10	0.045	0.302	-344.447	343.934
50	0.398	0.259	-225.684	225.961

Notes: All displayed risk entries are multiplied by 10^4 . Each dimension pools 256 independent action bridges from every one of the 16 registered NBO streams, or 4,096 bridges. Each bridge has two independent four-path antithetic continuation banks on the 2,048-cell global grid. The signed critic statistic removes reference Monte Carlo variance in expectation. The raw statistic is the risk of one four-path bank on the identical represented queries. The paired contrast estimates critic risk minus raw risk. Its endpoints use the registered family of 12 two-sided statements with total error probability 0.01 and include interval sample arithmetic and proved clipping tails. These are evaluation-subproblem comparisons; the returned Raw policy's welfare is assessed separately.

TABLE S.16

FROM THE CONTINUATION ACCOUNT TO ECONOMIC GAIN

d	L_M	$\sqrt{U_A U_E}$	U_D	Holding/transfer	Gain lower
10	-0.000849	0.005630	0.004113	0.000300	-0.006778
50	-0.000153	0.004285	0.002962	0.000363	-0.004800

Notes: The costate upper bound includes the exact-state representation allowance. The holding and transfer column sums the outward holding deficit and the quantitative paired payoff transfer to the original continuous economy. The last column is the larger of the product bound and the optional quadratic actor-gap bound when its positive modulus is verified. A negative endpoint records the limit of this sufficient mechanism account. Actual method welfare remains assessed by the independent direct payoff experiment; the mechanism assessment cannot change selection or stopping.

S.18.9. Independent scalar training and numerical accuracy

The separate scalar study uses the original volatility calibration, $(\sigma_I, \sigma_C) = (0.15, 0.10)$, and the same one-state capital economy. Eight independent Howard solves generate full-box and matched-tube policies on all four prespecified grids. The policies are then evaluated by actual interpolation-based deployment on 8,192 fresh paths at both 1,024 and 2,048 time steps. The analytical schedule and both frozen scalar neural policies use the same initial states and Brownian paths.

All 8 classical solves reached the stated stopping criterion, using at most 3 Howard updates at each date. Their finite-grid algebraic bounds range from 1.072×10^{-12} to 8.938×10^{-12} . The last joint state–time refinement changes the three initial-state values by amounts between 1.724×10^{-4} and 3.250×10^{-4} . The subsequent enlargement of the domain changes these values by at most 4.441×10^{-16} . Thus the algebraic solve, the training-grid approximation, and the finite-domain approximation have separate, observable numerical accounts. The small domain change does not bound the remaining time or spatial discretization error.

At 2,048 deployment steps, the largest-domain classical policy has a mean schedule gain of 7.819×10^{-4} , compared with 7.412×10^{-4} for NBO and 7.065×10^{-4} for DPO. The paired classical advantages are 4.069×10^{-5} (Monte Carlo standard error 6.841×10^{-7}) against NBO and 7.541×10^{-5} (4.063×10^{-7}) against DPO. Both advantages are positive at each of the three reported initial states. The numerical

TABLE S.17

INDEPENDENT SCALAR TRAINING AND ALGEBRAIC ACCURACY

Policy	$N_y/N_t/L$	$v(-1)$	$v(0)$	$v(1)$	E_0	Seconds
F1	401/256/6	-3.393994	-1.314658	0.816406	1.08E-12	3.76
F2	801/512/6	-3.393648	-1.314316	0.817061	2.28E-12	9.30
F3	1601/1024/6	-3.393474	-1.314143	0.817386	5.93E-12	24.54
F4	2401/1024/9	-3.393474	-1.314143	0.817386	8.94E-12	30.26
T1	401/256/6	-3.393994	-1.314658	0.816406	1.08E-12	3.77
T2	801/512/6	-3.393648	-1.314316	0.817061	2.28E-12	9.33
T3	1601/1024/6	-3.393474	-1.314143	0.817386	5.93E-12	24.64
T4	2401/1024/9	-3.393474	-1.314143	0.817386	8.94E-12	30.50

F1–F4 solve the full primitive action box; T1–T4 solve the common schedule tube of radius 0.1. E_0 is the outward finite-grid algebraic error bound in Proposition S.23. The displayed error endpoints are rounded upward; the full outward endpoints are retained in the artifact. Seconds include each new solve, its interval residual audit, and policy serialization. The last grid enlarges the domain from $[-6, 6]$ to $[-9, 9]$ at the same declared spatial and temporal spacing. Boundary data are the stated far-state approximation.

comparison therefore identifies an independently computed feasible classical competitor with higher observed payoffs than both frozen neural policies in this scalar economy.

The neural ordering varies across initial states. NBO minus DPO has population mean 3.472×10^{-5} , but its conditional mean at $y = 1$ is -6.206×10^{-5} . The same reversal occurs on the 1,024-step deployment mesh. This state-specific outcome is retained alongside the population comparison. Corresponding full-box and tube policies have identical payoffs path by path on both reported deployment meshes. The maximum recorded nodal deviation from the schedule is 0.046600, below the tube radius. This observation concerns the computed scalar policies and does not establish that the tube restriction is harmless in every dimension.

The complete full-box and matched-tube value arrays coincide bit for bit on each of the four grids. Because the action sets are nested and the terminal and boundary data are identical, Corollary S.24 converts the two algebraic enclosures into a bound on the loss from the tube restriction in the exact finite-grid problem.

TABLE S.18

DEPLOYED SCALAR POLICY GAINS ON COMMON DIFFUSION PATHS

Policy	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	0.0000	0.0000	0.0000	0.0000	0.0000
F1	0.1137	1.5586	0.6768	0.7818	0.0069
F2	0.1137	1.5587	0.6771	0.7819	0.0069
F3	0.1136	1.5586	0.6772	0.7819	0.0069
F4	0.1136	1.5586	0.6772	0.7819	0.0069
T1	0.1137	1.5586	0.6768	0.7818	0.0069
T2	0.1137	1.5587	0.6771	0.7819	0.0069
T3	0.1136	1.5586	0.6772	0.7819	0.0069
T4	0.1136	1.5586	0.6772	0.7819	0.0069
NBO (R12)	0.1036	1.5557	0.5683	0.7412	0.0071
DPO (R12)	0.0468	1.4460	0.6303	0.7065	0.0066

All payoff gains and standard errors are multiplied by 10^3 . The comparator is the analytical schedule evaluated on the same numerical mesh and Brownian path. Population denotes the mean over independent uniform initial-state draws; the last column is its paired Monte Carlo standard error. Initial-state columns retain their separate rankings. NBO and DPO are the prespecified frozen R12 scalar policies. These are conditional Euler-deployment diagnostics; Monte Carlo standard errors do not include continuous-time or boundary approximation error.

Across the four grids, this loss at every initial-date node is at most 1.788×10^{-11} . This deterministic conclusion concerns the stated finite-grid problem; the domain and diffusion approximation accounts retain their separate scope.

The complete scalar process consumed 195.90 seconds, including fresh-process setup, all eight solves, interval audits, serialization, both common-path deployments, and output. The individual new classical solves consumed 3.76–30.50 seconds. Within the fine-mesh deployment, the measured action-evaluation totals were 1.26–1.36 seconds for the classical tables and 9.38–9.60 seconds for the frozen neural maps. These are operation-specific measurements on the recorded worker. The historical neural fitting clocks are not included in a new end-to-end training comparison.

The algebraic error bound concerns the exact equations with their declared boundary data. The independent domain enlargement and paired deployment-mesh comparisons quantify distinct numerical sensitivities. The complete tables retain all trained policies, both meshes, all initial-state strata, and all 110 pairwise numerical comparisons. They supply a scalar accuracy and classical-work benchmark without identifying an exact value for the high-dimensional diffusion.

S.18.10. *Implementation from finite state measurements*

The implementation question has a direct economic interpretation. A manager observes a vector of log capital stocks at specified dates, computes consumption rates with the saved network, and holds those rates until the next observation. The manager's input consists of the measured physical state. The finite-observation controller does not require a Brownian coordinate or the intervening drift integral. Proposition S.20 applies to this controller through a protected parent with the same saved weights and inward action guard.

The experiment verifies that parent afresh at 1,024 and 4,096 decision dates in dimensions 10 and 50. It retains the original volatility calibration and the previously fixed iteration-80 NBO weights; no outcome in the new observation experiment selects a policy. Table S.19 subtracts each deterministic sensing allowance from the corresponding parent guarantee. All four parents have positive lower endpoints. Six of the twelve finite-observation lower endpoints remain positive. With 4,096 dates and exact measurements, the continuous-economy gain is at least 0.001050 in dimension 10 and 0.001054 in dimension 50. In dimension 10, a measurement error of 0.0001 retains a lower gain of 0.000889 at the finer grid.

The dependence on measurement precision is economically material. At dimension 50, an error of 0.0001 already exceeds the available parent margin under the current global network bound, and an error of 0.001 removes a positive guarantee in both dimensions. These endpoints give an implementation requirement for the specific measured-state rule. They do not infer the sign of its actual gain when the allowance is too large. The full intervals, allowance-to-gain ratios, and fresh-process costs appear in Tables S.28 and S.29.

TABLE S.19
CONTINUOUS-ECONOMY GAIN BOUNDS WITH FINITE OBSERVATIONS

d	N	Parent lower	$\nu = 0$	$\nu = 0.0001$	$\nu = 0.001$
10	1024	0.000766	0.000687	0.000526	-0.000925
10	4096	0.001070	0.001050	0.000889	-0.000559
50	1024	0.000807	0.000252	-0.001038	-0.012651
50	4096	0.001193	0.001054	-0.000239	-0.011873

Notes: All lower endpoints are printed downward to six decimal places. The last three columns are lower bounds for the finite-observation controller's payoff gain over the analytical schedule. Each parent is freshly verified at the same decision grid as its sensor transfer, using 8,192 independent paths. The four parent intervals share error probability 0.01; the transfers are deterministic. The fixed weights and original volatility calibration are retained from the reviewed study. A negative endpoint means that this error account does not certify improvement at that observation precision.

S.19. COMPLETE COMPUTATIONAL OUTCOMES

The following tables retain every declared stream and saved checkpoint.

TABLE S.20: Every Declared Stream Payoff and Direct Contrast

d	Stream	Endpoint	Mean	Lower	Upper
10	2095568532	NBO	0.001273	0.000537	0.002008
10	2095568532	Raw	0.001273	0.000537	0.002009
10	2095568532	Direct policy	0.001267	0.000531	0.002002
10	2095568532	Neural HJB	0.001120	0.000386	0.001855
10	2095568532	NBO–Raw	0.000000	-0.001152	0.001152
10	2095568532	NBO–Direct policy	0.000006	-0.001146	0.001159
10	2095568532	NBO–Neural HJB	0.000152	-0.001007	0.001311
10	2133755523	NBO	0.001257	0.000521	0.001992
10	2133755523	Raw	0.001257	0.000521	0.001993
10	2133755523	Direct policy	0.001249	0.000513	0.001984
10	2133755523	Neural HJB	0.001071	0.000337	0.001805
10	2133755523	NBO–Raw	0.000000	-0.001153	0.001152
10	2133755523	NBO–Direct policy	0.000008	-0.001145	0.001161

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2							2
3	10	2133755523	NBO-Neural HJB	0.000186	-0.000974	0.001346	3
4	10	153722668	NBO	0.001262	0.000527	0.001998	4
5	10	153722668	Raw	0.001261	0.000526	0.001997	5
6	10	153722668	Direct policy	0.001256	0.000520	0.001991	6
7	10	153722668	Neural HJB	0.001135	0.000401	0.001869	7
8	10	153722668	NBO-Raw	0.000001	-0.001151	0.001153	8
9	10	153722668	NBO-Direct policy	0.000007	-0.001146	0.001159	9
10	10	153722668	NBO-Neural HJB	0.000127	-0.001030	0.001285	10
11	10	603655094	NBO	0.001272	0.000536	0.002007	11
12	10	603655094	Raw	0.001269	0.000533	0.002004	12
13	10	603655094	Direct policy	0.001267	0.000531	0.002002	13
14	10	603655094	Neural HJB	0.001079	0.000343	0.001815	14
15	10	603655094	NBO-Raw	0.000003	-0.001149	0.001155	15
16	10	603655094	NBO-Direct policy	0.000005	-0.001147	0.001158	16
17	10	603655094	NBO-Neural HJB	0.000193	-0.000968	0.001354	17
18	10	216592699	NBO	0.001266	0.000531	0.002002	18
19	10	216592699	Raw	0.001264	0.000529	0.002000	19
20	10	216592699	Direct policy	0.001260	0.000525	0.001995	20
21	10	216592699	Neural HJB	0.001156	0.000420	0.001892	21
22	10	216592699	NBO-Raw	0.000002	-0.001150	0.001154	22
23	10	216592699	NBO-Direct policy	0.000006	-0.001146	0.001159	23
24	10	216592699	NBO-Neural HJB	0.000110	-0.001046	0.001266	24
25	10	1950831757	NBO	0.001266	0.000530	0.002001	25
26	10	1950831757	Raw	0.001264	0.000528	0.002000	26
27	10	1950831757	Direct policy	0.001260	0.000524	0.001995	27
28	10	1950831757	Neural HJB	0.001055	0.000322	0.001788	28
29	10	1950831757	NBO-Raw	0.000001	-0.001151	0.001153	29
30	10	1950831757	NBO-Direct policy	0.000006	-0.001147	0.001158	30
31	10	1950831757	NBO-Neural HJB	0.000211	-0.000952	0.001373	31
32	10	107486742	NBO	0.001271	0.000536	0.002007	32
	10	107486742	Raw	0.001272	0.000536	0.002007	
	10	107486742	Direct policy	0.001266	0.000530	0.002001	
	10	107486742	Neural HJB	0.000876	0.000143	0.001609	

	d	Stream	Endpoint	Mean	Lower	Upper	
1							1
2	10	107486742	NBO–Raw	0.000000	-0.001153	0.001152	2
3	10	107486742	NBO–Direct policy	0.000006	-0.001147	0.001158	3
4	10	107486742	NBO–Neural HJB	0.000395	-0.000770	0.001561	4
5	10	1142235589	NBO	0.001269	0.000534	0.002005	5
6	10	1142235589	Raw	0.001268	0.000532	0.002004	6
7	10	1142235589	Direct policy	0.001263	0.000528	0.001999	7
8	10	1142235589	Neural HJB	0.000034	-0.000700	0.000768	8
9	10	1142235589	NBO–Raw	0.000001	-0.001151	0.001153	9
10	10	1142235589	NBO–Direct policy	0.000006	-0.001147	0.001159	10
11	10	1142235589	NBO–Neural HJB	0.001235	0.000054	0.002417	11
12	10	1824517164	NBO	0.001269	0.000533	0.002005	12
13	10	1824517164	Raw	0.001269	0.000534	0.002005	13
14	10	1824517164	Direct policy	0.001267	0.000531	0.002002	14
15	10	1824517164	Neural HJB	0.001130	0.000395	0.001866	15
16	10	1824517164	NBO–Raw	0.000000	-0.001152	0.001152	16
17	10	1824517164	NBO–Direct policy	0.000002	-0.001151	0.001155	17
18	10	1824517164	NBO–Neural HJB	0.000139	-0.001018	0.001295	18
19	10	1965147083	NBO	0.001267	0.000532	0.002003	19
20	10	1965147083	Raw	0.001267	0.000532	0.002003	20
21	10	1965147083	Direct policy	0.001263	0.000528	0.001999	21
22	10	1965147083	Neural HJB	0.001063	0.000326	0.001799	22
23	10	1965147083	NBO–Raw	0.000000	-0.001152	0.001152	23
24	10	1965147083	NBO–Direct policy	0.000004	-0.001149	0.001156	24
25	10	1965147083	NBO–Neural HJB	0.000204	-0.000957	0.001365	25
26	10	1860620983	NBO	0.001270	0.000535	0.002006	26
27	10	1860620983	Raw	0.001270	0.000534	0.002005	27
28	10	1860620983	Direct policy	0.001268	0.000532	0.002003	28
29	10	1860620983	Neural HJB	0.001149	0.000414	0.001884	29
30	10	1860620983	NBO–Raw	0.000001	-0.001151	0.001153	30
31	10	1860620983	NBO–Direct policy	0.000003	-0.001150	0.001155	31
32	10	1860620983	NBO–Neural HJB	0.000121	-0.001036	0.001279	32
	10	1618144748	NBO	0.001275	0.000539	0.002010	
	10	1618144748	Raw	0.001275	0.000540	0.002010	

	d	Stream	Endpoint	Mean	Lower	Upper	
1							1
2							2
3	10	1618144748	Direct policy	0.001268	0.000532	0.002003	3
4	10	1618144748	Neural HJB	0.001119	0.000385	0.001854	4
5	10	1618144748	NBO–Raw	0.000000	-0.001152	0.001152	5
6	10	1618144748	NBO–Direct policy	0.000007	-0.001145	0.001160	6
7	10	1618144748	NBO–Neural HJB	0.000156	-0.001004	0.001315	7
8	10	592131000	NBO	0.001265	0.000530	0.002001	8
9	10	592131000	Raw	0.001264	0.000528	0.001999	9
10	10	592131000	Direct policy	0.001259	0.000523	0.001994	10
11	10	592131000	Neural HJB	0.001076	0.000342	0.001811	11
12	10	592131000	NBO–Raw	0.000001	-0.001151	0.001153	12
13	10	592131000	NBO–Direct policy	0.000007	-0.001146	0.001159	13
14	10	592131000	NBO–Neural HJB	0.000189	-0.000972	0.001349	14
15	10	964082955	NBO	0.001275	0.000540	0.002011	15
16	10	964082955	Raw	0.001273	0.000538	0.002009	16
17	10	964082955	Direct policy	0.001271	0.000535	0.002006	17
18	10	964082955	Neural HJB	0.001106	0.000372	0.001840	18
19	10	964082955	NBO–Raw	0.000002	-0.001150	0.001154	19
20	10	964082955	NBO–Direct policy	0.000005	-0.001148	0.001157	20
21	10	964082955	NBO–Neural HJB	0.000169	-0.000990	0.001328	21
22	10	887230079	NBO	0.001265	0.000530	0.002000	22
23	10	887230079	Raw	0.001265	0.000530	0.002001	23
24	10	887230079	Direct policy	0.001260	0.000525	0.001995	24
25	10	887230079	Neural HJB	0.001081	0.000343	0.001818	25
26	10	887230079	NBO–Raw	0.000000	-0.001152	0.001152	26
27	10	887230079	NBO–Direct policy	0.000005	-0.001148	0.001157	27
28	10	887230079	NBO–Neural HJB	0.000184	-0.000974	0.001343	28
29	10	1818435806	NBO	0.001264	0.000528	0.001999	29
30	10	1818435806	Raw	0.001262	0.000526	0.001997	30
31	10	1818435806	Direct policy	0.001258	0.000523	0.001994	31
32	10	1818435806	Neural HJB	0.000320	-0.000410	0.001050	32
	10	1818435806	NBO–Raw	0.000002	-0.001150	0.001154	
	10	1818435806	NBO–Direct policy	0.000006	-0.001147	0.001158	
	10	1818435806	NBO–Neural HJB	0.000944	-0.000230	0.002118	

	d	Stream	Endpoint	Mean	Lower	Upper	
1							1
2	50	2095568532	NBO	0.001341	0.000558	0.002124	2
3	50	2095568532	Raw	0.001343	0.000560	0.002125	3
4	50	2095568532	Direct policy	0.001343	0.000561	0.002126	4
5	50	2095568532	Neural HJB	0.001127	0.000343	0.001911	5
6	50	2095568532	NBO–Raw	-0.000002	-0.001204	0.001201	6
7	50	2095568532	NBO–Direct policy	-0.000002	-0.001206	0.001201	7
8	50	2095568532	NBO–Neural HJB	0.000214	-0.000995	0.001423	8
9	50	2133755523	NBO	0.001345	0.000562	0.002128	9
10	50	2133755523	Raw	0.001347	0.000564	0.002130	10
11	50	2133755523	Direct policy	0.001346	0.000563	0.002129	11
12	50	2133755523	Neural HJB	-0.001572	-0.002492	-0.000652	12
13	50	2133755523	NBO–Raw	-0.000002	-0.001204	0.001201	13
14	50	2133755523	NBO–Direct policy	-0.000001	-0.001204	0.001202	14
15	50	2133755523	NBO–Neural HJB	0.002917	0.001566	0.004269	15
16	50	153722668	NBO	0.001353	0.000570	0.002135	16
17	50	153722668	Raw	0.001354	0.000571	0.002137	17
18	50	153722668	Direct policy	0.001351	0.000568	0.002134	18
19	50	153722668	Neural HJB	-0.001096	-0.001986	-0.000205	19
20	50	153722668	NBO–Raw	-0.000001	-0.001204	0.001201	20
21	50	153722668	NBO–Direct policy	0.000002	-0.001201	0.001205	21
22	50	153722668	NBO–Neural HJB	0.002449	0.001128	0.003769	22
23	50	603655094	NBO	0.001346	0.000563	0.002129	23
24	50	603655094	Raw	0.001349	0.000566	0.002132	24
25	50	603655094	Direct policy	0.001349	0.000566	0.002132	25
26	50	603655094	Neural HJB	0.000703	-0.000091	0.001497	26
27	50	603655094	NBO–Raw	-0.000002	-0.001205	0.001200	27
28	50	603655094	NBO–Direct policy	-0.000002	-0.001206	0.001201	28
29	50	603655094	NBO–Neural HJB	0.000643	-0.000580	0.001866	29
30	50	216592699	NBO	0.001345	0.000562	0.002129	30
31	50	216592699	Raw	0.001346	0.000563	0.002130	31
32	50	216592699	Direct policy	0.001344	0.000561	0.002127	32
	50	216592699	Neural HJB	-0.000866	-0.001738	0.000006	
	50	216592699	NBO–Raw	-0.000001	-0.001204	0.001201	

	<i>d</i>	Stream	Endpoint	Mean	Lower	Upper	
1							1
2							2
3	50	216592699	NBO-Direct policy	0.000002	-0.001202	0.001205	3
4	50	216592699	NBO-Neural HJB	0.002211	0.000909	0.003513	4
5	50	1950831757	NBO	0.001344	0.000561	0.002127	5
6	50	1950831757	Raw	0.001347	0.000564	0.002129	6
7	50	1950831757	Direct policy	0.001348	0.000565	0.002131	7
8	50	1950831757	Neural HJB	0.000737	-0.000051	0.001526	8
9	50	1950831757	NBO-Raw	-0.000003	-0.001205	0.001200	9
10	50	1950831757	NBO-Direct policy	-0.000004	-0.001207	0.001199	10
11	50	1950831757	NBO-Neural HJB	0.000607	-0.000612	0.001825	11
12	50	107486742	NBO	0.001328	0.000545	0.002111	12
13	50	107486742	Raw	0.001330	0.000547	0.002113	13
14	50	107486742	Direct policy	0.001350	0.000567	0.002133	14
15	50	107486742	Neural HJB	-0.000616	-0.001469	0.000237	15
16	50	107486742	NBO-Raw	-0.000002	-0.001204	0.001201	16
17	50	107486742	NBO-Direct policy	-0.000022	-0.001225	0.001182	17
18	50	107486742	NBO-Neural HJB	0.001944	0.000659	0.003230	18
19	50	1142235589	NBO	0.001344	0.000561	0.002127	19
20	50	1142235589	Raw	0.001345	0.000562	0.002128	20
21	50	1142235589	Direct policy	0.001348	0.000566	0.002131	21
22	50	1142235589	Neural HJB	-0.000003	-0.000825	0.000819	22
23	50	1142235589	NBO-Raw	-0.000001	-0.001203	0.001202	23
24	50	1142235589	NBO-Direct policy	-0.000004	-0.001208	0.001199	24
25	50	1142235589	NBO-Neural HJB	0.001347	0.000096	0.002598	25
26	50	1824517164	NBO	0.001348	0.000565	0.002131	26
27	50	1824517164	Raw	0.001349	0.000566	0.002132	27
28	50	1824517164	Direct policy	0.001346	0.000564	0.002129	28
29	50	1824517164	Neural HJB	-0.000298	-0.001130	0.000534	29
30	50	1824517164	NBO-Raw	-0.000001	-0.001204	0.001201	30
31	50	1824517164	NBO-Direct policy	0.000001	-0.001202	0.001205	31
32	50	1824517164	NBO-Neural HJB	0.001646	0.000384	0.002907	32
33	50	1965147083	NBO	0.001323	0.000540	0.002107	33
34	50	1965147083	Raw	0.001324	0.000540	0.002107	34
35	50	1965147083	Direct policy	0.001345	0.000562	0.002128	35

	d	Stream	Endpoint	Mean	Lower	Upper	
1							1
2	50	1965147083	Neural HJB	-0.000361	-0.001197	0.000475	2
3	50	1965147083	NBO–Raw	-0.000001	-0.001203	0.001202	3
4	50	1965147083	NBO–Direct policy	-0.000022	-0.001225	0.001181	4
5	50	1965147083	NBO–Neural HJB	0.001684	0.000418	0.002950	5
6	50	1860620983	NBO	0.001346	0.000563	0.002129	6
7	50	1860620983	Raw	0.001347	0.000564	0.002130	7
8	50	1860620983	Direct policy	0.001349	0.000566	0.002131	8
9	50	1860620983	Neural HJB	0.000779	-0.000005	0.001564	9
10	50	1860620983	NBO–Raw	-0.000001	-0.001204	0.001202	10
11	50	1860620983	NBO–Direct policy	-0.000002	-0.001205	0.001201	11
12	50	1860620983	NBO–Neural HJB	0.000567	-0.000648	0.001782	12
13	50	1618144748	NBO	0.001330	0.000547	0.002113	13
14	50	1618144748	Raw	0.001331	0.000548	0.002114	14
15	50	1618144748	Direct policy	0.001350	0.000567	0.002132	15
16	50	1618144748	Neural HJB	-0.000491	-0.001334	0.000353	16
17	50	1618144748	NBO–Raw	-0.000001	-0.001203	0.001201	17
18	50	1618144748	NBO–Direct policy	-0.000019	-0.001223	0.001184	18
19	50	1618144748	NBO–Neural HJB	0.001821	0.000545	0.003096	19
20	50	592131000	NBO	0.001346	0.000563	0.002129	20
21	50	592131000	Raw	0.001346	0.000563	0.002129	21
22	50	592131000	Direct policy	0.001346	0.000563	0.002129	22
23	50	592131000	Neural HJB	-0.000211	-0.001040	0.000617	23
24	50	592131000	NBO–Raw	0.000000	-0.001203	0.001202	24
25	50	592131000	NBO–Direct policy	0.000000	-0.001203	0.001203	25
26	50	592131000	NBO–Neural HJB	0.001557	0.000298	0.002816	26
27	50	964082955	NBO	0.001346	0.000563	0.002129	27
28	50	964082955	Raw	0.001348	0.000565	0.002131	28
29	50	964082955	Direct policy	0.001347	0.000564	0.002129	29
30	50	964082955	Neural HJB	-0.000332	-0.001165	0.000502	30
31	50	964082955	NBO–Raw	-0.000002	-0.001204	0.001201	31
32	50	964082955	NBO–Direct policy	-0.000001	-0.001204	0.001203	32
	50	964082955	NBO–Neural HJB	0.001678	0.000414	0.002942	
	50	887230079	NBO	0.001343	0.000560	0.002126	

<i>d</i>	Stream	Endpoint	Mean	Lower	Upper
50	887230079	Raw	0.001345	0.000562	0.002128
50	887230079	Direct policy	0.001345	0.000562	0.002128
50	887230079	Neural HJB	-0.000282	-0.001110	0.000547
50	887230079	NBO–Raw	-0.000002	-0.001204	0.001201
50	887230079	NBO–Direct policy	-0.000002	-0.001205	0.001202
50	887230079	NBO–Neural HJB	0.001625	0.000366	0.002884
50	1818435806	NBO	0.001344	0.000562	0.002127
50	1818435806	Raw	0.001345	0.000562	0.002127
50	1818435806	Direct policy	0.001346	0.000563	0.002128
50	1818435806	Neural HJB	0.000280	-0.000525	0.001085
50	1818435806	NBO–Raw	-0.000001	-0.001203	0.001202
50	1818435806	NBO–Direct policy	-0.000001	-0.001205	0.001202
50	1818435806	NBO–Neural HJB	0.001064	-0.000172	0.002301

Notes: Every declared stream and endpoint is retained. These conditional-policy intervals and the method averages belong to the single registered confirmation family; their simulation banks are independent of stopping. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE S.21: Actual online certification by work checkpoint

<i>d</i>	Method	Checkpoint	Attempted	Completed	Certified by then
10	NBO	1	16	16	0/16
10	NBO	2	16	16	0/16
10	NBO	3	16	16	0/16
10	Raw	1	16	16	0/16
10	Raw	2	16	16	0/16
10	Raw	3	16	16	0/16
10	Direct policy	1	16	16	0/16
10	Direct policy	2	16	16	0/16
10	Direct policy	3	16	16	0/16

d	Method	Checkpoint	Attempted	Completed	Certified by then
10	Neural HJB	1	16	16	0/16
10	Neural HJB	2	16	16	0/16
10	Neural HJB	3	16	16	0/16
50	NBO	1	16	16	0/16
50	NBO	2	16	16	3/16
50	NBO	3	13	13	16/16
50	Raw	1	16	16	0/16
50	Raw	2	16	16	3/16
50	Raw	3	13	13	16/16
50	Direct policy	1	16	16	0/16
50	Direct policy	2	16	16	0/16
50	Direct policy	3	16	16	16/16
50	Neural HJB	1	16	16	0/16
50	Neural HJB	2	16	16	0/16
50	Neural HJB	3	16	16	0/16

Every listed check was executed before the continuation or stopping decision. The last column counts all streams that had attained the primary 0.0005 lower-gain target by the checkpoint, retaining earlier stopped streams. Missing later checks represent actual stopping or recorded failures. This table does not infer an unexecuted fixed-budget payoff frontier.

S.20. COMPLETE MECHANISM ASSESSMENT

The following records retain every candidate and the work used in its independent mechanism assessment.

S.21. AN OUTWARD ALGEBRAIC ERROR ACCOUNT FOR THE SCALAR COMPARATOR

Fix the declared spatial nodes, common spacing Δy , time step h , action endpoints, and terminal and boundary data. Each stored binary64 input denotes an exact real in this finite-grid problem. Let $\eta = (\sigma_I^2 + \sigma_C^2)/2$, $d_0(y) = a - \eta + \kappa \tanh y$, and define the forward and backward slopes $p_f = (v_{i+1} - v_i)/\Delta y$ and $p_b = (v_i - v_{i-1})/\Delta y$.

TABLE S.22

COMPLETE PROTECTED BELLMAN-BRIDGE STATISTICS

d	Statistic	Raw mean	Lower	Upper	Clip bound	Arithmetic
10	M	0.0012636	-0.0008484	0.0033757	0.21174	0.00000019
10	A	0.0008803	0.0007515	0.0010091	0.01003	0.00000001
10	E	0.0000045	-0.0313992	0.0314082	3.25082	0.00000001
10	D	0.0000272	-0.0040585	0.0041129	0.42285	0.00000001
10	E^{raw}	0.0000302	-0.0029862	0.0030465	0.31182	0.00000001
10	$E - E^{\text{raw}}$	-0.0000257	-0.0344447	0.0343934	3.56264	0.00000001
50	M	0.0013307	-0.0001528	0.0028143	0.14696	0.00000012
50	A	0.0009390	0.0008149	0.0010631	0.01003	0.00000001
50	E	0.0000398	-0.0171787	0.0172583	1.78228	0.00000001
50	D	0.0001221	-0.0027173	0.0029614	0.29347	0.00000001
50	E^{raw}	0.0000259	-0.0053392	0.0053911	0.55513	0.00000001
50	$E - E^{\text{raw}}$	0.0000139	-0.0225684	0.0225961	2.33741	0.00000001

Notes: Every registered statistic and dimension appears. These intervals concern the exact represented-query sample expectations before the additional state-representation conversion used in the welfare bound. The arithmetic column is the outward sample-mean and sample-standard-deviation per-perturbation allowance, not an empirical population range. The source records each stream's scalar weights, verified network norms, curvature, clipping tail and state/adjoint error constants. There is no per-stream inferential claim.

TABLE S.23

COMPLETE WORK FOR THE INDEPENDENT MECHANISM ASSESSMENT

d	Process seconds	CPU seconds	Peak MiB	Future updates	Padded rows
10	339.05	334.24	461.1	33,646,832	66,803,712
50	561.96	558.12	810.1	33,517,664	66,828,288

Notes: Process and CPU seconds sum all 16 fresh stream processes in each dimension; peak memory is their maximum. The assessment includes loading the already selected policy, rebuilding constants, candidate occupation paths, both continuation banks, reverse derivatives, interval arithmetic and output. Future updates count mathematically active continuation transitions; padded rows additionally disclose vectorized inactive arithmetic. This work is reported in addition to the primary algorithms' complete stopping and independent payoff-assessment costs.

The action-dependent upwind Hamiltonian is

$$\mathcal{H}_i(m; v) = \log m - \frac{\zeta}{2} m^2 + \max\{d_0(y_i) - m, 0\} p_f + \min\{d_0(y_i) - m, 0\} p_b. \quad (\text{S.89})$$

The positive-drift and negative-drift branches are each concave in m . Their feasible intervals are, respectively, $[\ell, \min\{u, d_0(y_i)\}]$ and $[\max\{\ell, d_0(y_i)\}, u]$, when nonempty. On either branch, the unconstrained positive critical point solves $1/m - \zeta m = p_j$. The Howard update clips this point to the branch interval and compares the two resulting objective values. Thus the sign of the drift selects the appropriate difference operator throughout maximization.

The floating-point Howard solve proposes values and actions. A separate outward calculation encloses the Bellman residual, including the error in the proposed maximizer. For a concave branch H on $[a, b]$ and any $m_o \in [a, b]$,

$$\max_{m \in [a, b]} H(m) \leq H(m_o) + \max\{0, H'(m_o)(a - m_o), H'(m_o)(b - m_o)\}. \quad (\text{S.90})$$

Interval evaluations of log, tanh, the slopes, and the arithmetic operations enclose the right-hand side. Interval uncertainty in $d_0(y_i)$ is handled by enlarging each branch interval before constructing its upper bound. An arbitrary globally feasible action supplies a lower bound for the maximized Hamiltonian. Combining the two bounds encloses

$$r_{k,i} = \frac{v_{k,i} - v_{k+1,i}}{h} + \rho v_{k,i} - \eta \frac{v_{k,i+1} - 2v_{k,i} + v_{k,i-1}}{(\Delta y)^2} - y_i - \max_m \mathcal{H}_i(m; v_k). \quad (\text{S.91})$$

The artifact stores every proposed value and action, the boundary arrays, the action endpoints, and the outward residual bound at every time step.

PROPOSITION S.23—Algebraic error in the declared scalar boundary problem: Fix finitely many ordered spatial coordinate labels and a common positive difference spacing Δy , $h > 0$, $\rho \geq 0$, $\eta \geq 0$, compact action intervals $[\ell_k, u_k]$ with $0 < \ell_k \leq u_k < \infty$, and finite terminal and boundary data. Let $\tilde{v}$ denote the exact solution of the backward Bellman equations (S.91) with zero residual. Let v have identical terminal and boundary data and residuals $r_{k,i}$. For any $R_k \geq \max_i |r_{k,i}|$, define $E_N = 0$ and

$$E_k = \frac{E_{k+1} + hR_k}{1 + \rho h}. \quad (\text{S.92})$$

Then the exact finite-grid solution exists uniquely, and $\|v_k - \tilde{v}_k\|_\infty \leq E_k$ at every time step.

PROOF: For a fixed action m , write the discrete generator at an interior node as

$$(L_m w)_i = \lambda_i^-(m)(w_{i-1} - w_i) + \lambda_i^+(m)(w_{i+1} - w_i),$$

$$\lambda_i^-(m) = \frac{\eta}{(\Delta y)^2} + \frac{\max\{m - d_0(y_i), 0\}}{\Delta y},$$

$$\lambda_i^+(m) = \frac{\eta}{(\Delta y)^2} + \frac{\max\{d_0(y_i) - m, 0\}}{\Delta y}.$$

Both rates are nonnegative. Each frozen-policy matrix therefore has nonpositive off-diagonal entries and diagonal dominance at least $h^{-1} + \rho$ after the boundary values are moved to the right-hand side. This is the M -matrix comparison property used below.

For completeness, existence can be established at each backward step by uniformizing these finitely many bounded rates. Choose $\Lambda \geq \max_{i,m} \{\lambda_i^-(m) + \lambda_i^+(m)\}$. When $\Lambda > 0$, $L_m = \Lambda(P_m - I)$, where P_m has nonnegative entries and row sums one before the fixed boundary values are substituted. The Bellman equation is a fixed-point equation whose dependence on the unknown interior values has supremum-norm Lipschitz constant at most $\Lambda/(h^{-1} + \rho + \Lambda) < 1$. The fixed-point theorem gives a unique solution. If $\Lambda = 0$, the equation is pointwise and its coefficient $h^{-1} + \rho$ is strictly positive. Backward recursion starts from the specified terminal values.

To establish the error bound, suppose $\|v_{k+1} - \tilde{v}_{k+1}\|_\infty \leq E_{k+1}$ and set $e_i = v_{k,i} - \tilde{v}_{k,i}$. If $M = \max_i e_i > 0$, the maximum occurs at an interior node i , because the boundary error is zero. Nonnegative rates give $(L_m e)_i \leq 0$ for every feasible m . Writing $f_i(m) = y_i + \log m - \zeta m^2/2$, it follows that

$$\sup_m \{(L_m v_k)_i + f_i(m)\} - \sup_m \{(L_m \tilde{v}_k)_i + f_i(m)\} \leq 0.$$

Subtract the exact and approximate Bellman equations at this node. The preceding inequality implies

$$(h^{-1} + \rho)M \leq h^{-1}(v_{k+1,i} - \tilde{v}_{k+1,i}) + r_{k,i} \leq h^{-1}E_{k+1} + R_k.$$

Interchanging v_k and $\tilde{v}_k$ gives the same bound for the largest negative error. Equation (S.92) follows, and induction from $E_N = 0$ completes the proof. *Q.E.D.*

The implementation evaluates this recurrence outward under the inherited arithmetic contract. It controls algebraic solution error for a specified finite-grid boundary problem. The separate grid, domain, and deployed-policy mesh comparisons measure the remaining numerical sensitivities.

COROLLARY S.24—Loss from a nested scalar action restriction: *Consider two problems in Proposition S.23 with identical spatial labels, difference and time steps, primitives, and terminal and boundary data. Suppose the action intervals of the second problem are contained in those of the first at every date. Denote their exact values by $\tilde{v}^F$ and $\tilde{v}^T$, their computed values by v^F and v^T , and their algebraic error bounds by E^F and E^T . Then, at every initial-date node,*

$$0 \leq \tilde{v}_{0,i}^F - \tilde{v}_{0,i}^T \leq \|v_0^F - v_0^T\|_\infty + E_0^F + E_0^T.$$

In particular, identical stored initial-date values reduce the upper bound to $E_0^F + E_0^T$.

PROOF: At each backward step, use the common uniformization constant from the preceding proof for both action sets. The resulting Bellman maps are monotone in continuation and current interior values. Enlarging the maximization set increases the map pointwise. The contraction fixed points therefore inherit this order; backward induction from the common terminal data gives $\tilde{v}^F \geq \tilde{v}^T$. The upper inequality follows by adding and subtracting $v_{0,i}^F$ and $v_{0,i}^T$ and applying the two algebraic error bounds. *Q.E.D.*

S.22. COMPLETE SCALAR COMPARISON TABLES

TABLE S.24: Deployed scalar policy gains on common diffusion paths

Policy	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	1024	0.0000	0.0000	0.0000	0.0000	0.0000
F1	1024	0.1103	1.5441	0.6666	0.7725	0.0068
F2	1024	0.1103	1.5441	0.6670	0.7726	0.0068
F3	1024	0.1103	1.5440	0.6672	0.7726	0.0068

Policy	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
F4	1024	0.1103	1.5440	0.6672	0.7726	0.0068
T1	1024	0.1103	1.5441	0.6666	0.7725	0.0068
T2	1024	0.1103	1.5441	0.6670	0.7726	0.0068
T3	1024	0.1103	1.5440	0.6672	0.7726	0.0068
T4	1024	0.1103	1.5440	0.6672	0.7726	0.0068
NBO (R12)	1024	0.1017	1.5418	0.5541	0.7312	0.0070
DPO (R12)	1024	0.0471	1.4365	0.6231	0.7010	0.0066
Schedule	2048	0.0000	0.0000	0.0000	0.0000	0.0000
F1	2048	0.1137	1.5586	0.6768	0.7818	0.0069
F2	2048	0.1137	1.5587	0.6771	0.7819	0.0069
F3	2048	0.1136	1.5586	0.6772	0.7819	0.0069
F4	2048	0.1136	1.5586	0.6772	0.7819	0.0069
T1	2048	0.1137	1.5586	0.6768	0.7818	0.0069
T2	2048	0.1137	1.5587	0.6771	0.7819	0.0069
T3	2048	0.1136	1.5586	0.6772	0.7819	0.0069
T4	2048	0.1136	1.5586	0.6772	0.7819	0.0069
NBO (R12)	2048	0.1036	1.5557	0.5683	0.7412	0.0071
DPO (R12)	2048	0.0468	1.4460	0.6303	0.7065	0.0066

All payoff gains and standard errors are multiplied by 10^3 . The comparator is the analytical schedule evaluated on the same numerical mesh and Brownian path. Population denotes the mean over independent uniform initial-state draws; the last column is its paired Monte Carlo standard error. Initial-state columns retain their separate rankings. NBO and DPO are the prespecified frozen R12 scalar policies. These are conditional Euler-deployment diagnostics; Monte Carlo standard errors do not include continuous-time or boundary approximation error.

TABLE S.25

SCALAR TRAINING-GRID AND DOMAIN SENSITIVITY

Comparison	Change	$y = -1$	$y = 0$	$y = 1$
F1-F2	Mesh	3.454e-04	3.424e-04	6.555e-04
F2-F3	Mesh	1.742e-04	1.724e-04	3.250e-04
F3-F4	Domain	4.441e-16	-4.441e-16	-2.220e-16
T1-T2	Mesh	3.454e-04	3.424e-04	6.555e-04
T2-T3	Mesh	1.742e-04	1.724e-04	3.250e-04
T3-T4	Domain	4.441e-16	-4.441e-16	-2.220e-16

Entries are later-grid minus earlier-grid values, retaining their signs. Domain changes keep the declared difference spacing and time step fixed. The differences are numerical sensitivities, separate from the algebraic residual bound.

TABLE S.26

SCALAR DEPLOYMENT-MESH SENSITIVITY

Policy	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	-1.564e-04	-1.702e-04	-1.323e-04	-1.529e-04	7.858e-07
F1	-1.531e-04	-1.557e-04	-1.221e-04	-1.436e-04	7.856e-07
F2	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.854e-07
F3	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
F4	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
T1	-1.531e-04	-1.557e-04	-1.221e-04	-1.436e-04	7.856e-07
T2	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.854e-07
T3	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
T4	-1.530e-04	-1.556e-04	-1.222e-04	-1.436e-04	7.853e-07
NBO (R12)	-1.544e-04	-1.563e-04	-1.181e-04	-1.429e-04	7.934e-07
DPO (R12)	-1.567e-04	-1.608e-04	-1.250e-04	-1.475e-04	7.884e-07

Entries are payoff at 2,048 steps minus payoff at 1,024 steps using the same fine Brownian increments and initial draws. The frozen policy table or neural weights are unchanged between deployments. These paired numerical changes are not a time-discretization error bound.

Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
TABLE S.27: Every paired scalar policy comparison						
Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule-F1	1024	-0.1103	-1.5441	-0.6666	-0.7725	0.0068
Schedule-F2	1024	-0.1103	-1.5441	-0.6670	-0.7726	0.0068
Schedule-F3	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-F4	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-T1	1024	-0.1103	-1.5441	-0.6666	-0.7725	0.0068
Schedule-T2	1024	-0.1103	-1.5441	-0.6670	-0.7726	0.0068
Schedule-T3	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-T4	1024	-0.1103	-1.5440	-0.6672	-0.7726	0.0068
Schedule-NBO (R12)	1024	-0.1017	-1.5418	-0.5541	-0.7312	0.0070
Schedule-DPO (R12)	1024	-0.0471	-1.4365	-0.6231	-0.7010	0.0066
F1-F2	1024	0.0000	0.0000	-0.0004	-0.0001	0.0000
F1-F3	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000
F1-F4	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000
F1-T1	1024	0.0000	0.0000	0.0000	0.0000	0.0000
F1-T2	1024	0.0000	0.0000	-0.0004	-0.0001	0.0000
F1-T3	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000
F1-T4	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000
F1-NBO (R12)	1024	0.0087	0.0023	0.1125	0.0412	0.0007
F1-DPO (R12)	1024	0.0633	0.1076	0.0436	0.0714	0.0004
F2-F3	1024	0.0000	0.0001	-0.0001	0.0000	0.0000
F2-F4	1024	0.0000	0.0001	-0.0001	0.0000	0.0000
F2-T1	1024	0.0000	0.0000	0.0004	0.0001	0.0000
F2-T2	1024	0.0000	0.0000	0.0000	0.0000	0.0000
F2-T3	1024	0.0000	0.0001	-0.0001	0.0000	0.0000
F2-T4	1024	0.0000	0.0001	-0.0001	0.0000	0.0000
F2-NBO (R12)	1024	0.0086	0.0023	0.1129	0.0414	0.0007
F2-DPO (R12)	1024	0.0632	0.1076	0.0440	0.0715	0.0004
F3-F4	1024	0.0000	0.0000	0.0000	0.0000	0.0000
F3-T1	1024	-0.0001	-0.0001	0.0005	0.0001	0.0000

	Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.	
1								1
2								2
3	F3-T2	1024	0.0000	-0.0001	0.0001	0.0000	0.0000	3
4	F3-T3	1024	0.0000	0.0000	0.0000	0.0000	0.0000	4
5	F3-T4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	5
6	F3-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	6
7	F3-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	7
8	F4-T1	1024	-0.0001	-0.0001	0.0005	0.0001	0.0000	8
9	F4-T2	1024	0.0000	-0.0001	0.0001	0.0000	0.0000	9
10	F4-T3	1024	0.0000	0.0000	0.0000	0.0000	0.0000	10
11	F4-T4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	11
12	F4-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	12
13	F4-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	13
14	T1-T2	1024	0.0000	0.0000	-0.0004	-0.0001	0.0000	14
15	T1-T3	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	15
16	T1-T4	1024	0.0001	0.0001	-0.0005	-0.0001	0.0000	16
17	T1-NBO (R12)	1024	0.0087	0.0023	0.1125	0.0412	0.0007	17
18	T1-DPO (R12)	1024	0.0633	0.1076	0.0436	0.0714	0.0004	18
19	T2-T3	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	19
20	T2-T4	1024	0.0000	0.0001	-0.0001	0.0000	0.0000	20
21	T2-NBO (R12)	1024	0.0086	0.0023	0.1129	0.0414	0.0007	21
22	T2-DPO (R12)	1024	0.0632	0.1076	0.0440	0.0715	0.0004	22
23	T3-T4	1024	0.0000	0.0000	0.0000	0.0000	0.0000	23
24	T3-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	24
25	T3-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	25
26	T4-NBO (R12)	1024	0.0086	0.0022	0.1131	0.0414	0.0007	26
27	T4-DPO (R12)	1024	0.0632	0.1075	0.0441	0.0716	0.0004	27
28	NBO (R12)-DPO (R12)	1024	0.0546	0.1053	-0.0690	0.0302	0.0010	28
29	Schedule-F1	2048	-0.1137	-1.5586	-0.6768	-0.7818	0.0069	29
30	Schedule-F2	2048	-0.1137	-1.5587	-0.6771	-0.7819	0.0069	30
31	Schedule-F3	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	31
32	Schedule-F4	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	32
	Schedule-T1	2048	-0.1137	-1.5586	-0.6768	-0.7818	0.0069	
	Schedule-T2	2048	-0.1137	-1.5587	-0.6771	-0.7819	0.0069	
	Schedule-T3	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	

	Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.	
1								1
2								2
3	Schedule-T4	2048	-0.1136	-1.5586	-0.6772	-0.7819	0.0069	3
4	Schedule-NBO (R12)	2048	-0.1036	-1.5557	-0.5683	-0.7412	0.0071	4
5	Schedule-DPO (R12)	2048	-0.0468	-1.4460	-0.6303	-0.7065	0.0066	5
6	F1-F2	2048	0.0000	-0.0001	-0.0003	-0.0001	0.0000	6
7	F1-F3	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	7
8	F1-F4	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	8
9	F1-T1	2048	0.0000	0.0000	0.0000	0.0000	0.0000	9
10	F1-T2	2048	0.0000	-0.0001	-0.0003	-0.0001	0.0000	10
11	F1-T3	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	11
12	F1-T4	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000	12
13	F1-NBO (R12)	2048	0.0100	0.0029	0.1085	0.0406	0.0007	13
14	F1-DPO (R12)	2048	0.0669	0.1126	0.0465	0.0753	0.0004	14
15	F2-F3	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	15
16	F2-F4	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	16
17	F2-T1	2048	0.0000	0.0001	0.0003	0.0001	0.0000	17
18	F2-T2	2048	0.0000	0.0000	0.0000	0.0000	0.0000	18
19	F2-T3	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	19
20	F2-T4	2048	0.0000	0.0000	-0.0001	0.0000	0.0000	20
21	F2-NBO (R12)	2048	0.0100	0.0030	0.1088	0.0407	0.0007	21
22	F2-DPO (R12)	2048	0.0669	0.1127	0.0468	0.0754	0.0004	22
23	F3-F4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	23
24	F3-T1	2048	0.0000	0.0001	0.0004	0.0001	0.0000	24
25	F3-T2	2048	0.0000	0.0000	0.0001	0.0000	0.0000	25
26	F3-T3	2048	0.0000	0.0000	0.0000	0.0000	0.0000	26
27	F3-T4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	27
28	F3-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007	28
29	F3-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004	29
30	F4-T1	2048	0.0000	0.0001	0.0004	0.0001	0.0000	30
31	F4-T2	2048	0.0000	0.0000	0.0001	0.0000	0.0000	31
32	F4-T3	2048	0.0000	0.0000	0.0000	0.0000	0.0000	32
	F4-T4	2048	0.0000	0.0000	0.0000	0.0000	0.0000	
	F4-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007	
	F4-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004	

Left minus right	Steps	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
T1-T2	2048	0.0000	-0.0001	-0.0003	-0.0001	0.0000
T1-T3	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000
T1-T4	2048	0.0000	-0.0001	-0.0004	-0.0001	0.0000
T1-NBO (R12)	2048	0.0100	0.0029	0.1085	0.0406	0.0007
T1-DPO (R12)	2048	0.0669	0.1126	0.0465	0.0753	0.0004
T2-T3	2048	0.0000	0.0000	-0.0001	0.0000	0.0000
T2-T4	2048	0.0000	0.0000	-0.0001	0.0000	0.0000
T2-NBO (R12)	2048	0.0100	0.0030	0.1088	0.0407	0.0007
T2-DPO (R12)	2048	0.0669	0.1127	0.0468	0.0754	0.0004
T3-T4	2048	0.0000	0.0000	0.0000	0.0000	0.0000
T3-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007
T3-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004
T4-NBO (R12)	2048	0.0100	0.0030	0.1089	0.0407	0.0007
T4-DPO (R12)	2048	0.0669	0.1127	0.0469	0.0754	0.0004
NBO (R12)-DPO (R12)	2048	0.0569	0.1097	-0.0621	0.0347	0.0010

All 55 pairs on both meshes are retained; means and standard errors are multiplied by 10^3 . The full machine-readable account additionally retains every initial-state standard error and raw payoff array. These conditional numerical comparisons spend no confidence allocation and do not claim continuous-time coverage.

S.22.1. Complete protected-parent and finite-observation results

The source package fixes the two saved policies, the four parent grids, the population, all observation precisions, and the confidence allocation before simulation. Each cell uses an independent process with pinned numerical dependencies. The archive retains all raw paired payoffs, final states, per-cell transfer recurrences, and the source and output digests.

TABLE S.28

COMPLETE FINITE-OBSERVATION PAYOFF TRANSFERS

d	N	ν	Allowance	Lower	Upper	Allowance/parent lower
10	1024	0.0000	0.000080	0.000687	0.002205	0.103
10	1024	0.0001	0.000241	0.000526	0.002366	0.314
10	1024	0.0010	0.001692	-0.000925	0.003817	2.206
10	4096	0.0000	0.000020	0.001050	0.001822	0.019
10	4096	0.0001	0.000181	0.000889	0.001983	0.169
10	4096	0.0010	0.001630	-0.000559	0.003432	1.522
50	1024	0.0000	0.000555	0.000252	0.002920	0.687
50	1024	0.0001	0.001845	-0.001038	0.004211	2.285
50	1024	0.0010	0.013458	-0.012651	0.015824	16.673
50	4096	0.0000	0.000139	0.001054	0.002122	0.116
50	4096	0.0001	0.001432	-0.000239	0.003415	1.200
50	4096	0.0010	0.013066	-0.011873	0.015049	10.949

Notes: Every declared dimension, grid, and observation precision appears. Lower endpoints are printed downward; upper endpoints and allowances are printed upward to six decimal places. Ratios are descriptive rounded values. The allowance is subtracted from the parent lower endpoint and added to its upper endpoint with outward arithmetic. The final column compares implementation error with the positive parent guarantee; a ratio above one removes that guarantee. The candidate's physical state remains the original unbounded diffusion. Its measurements arrive only at the N decision dates and have normalized root mean square error at most ν .

TABLE S.29

MEASURED WORK FOR PROTECTED-PARENT VERIFICATION AND SENSOR TRANSFERS

d	N	Complete seconds	Peak MiB	State transitions
10	1024	37.334	383.4	16,777,216
10	4096	126.355	434.6	67,108,864
50	1024	55.194	405.0	16,777,216
50	4096	263.063	451.6	67,108,864

Notes: Each row is a fresh CPU process. Its elapsed time includes imports, weight loading, interval constants, protected simulation, all three sensor allowances, raw-file writes, and the final result record. Environment provisioning is reported separately. No policy is refitted in this experiment.

d	Stream	NBO minus	Mean	Original interval	Refined interval	Case
TABLE S.30: Every registered direct stream comparison before and after deterministic refinement						
d	Stream	NBO minus	Mean	Original interval	Refined interval	Case
10	2095568532	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	2133755523	Raw	-0.000000	[-0.001153, 0.001152]	[-0.000901, 0.000900]	Normal
10	153722668	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	603655094	Raw	0.000003	[-0.001149, 0.001155]	[-0.000898, 0.000903]	Normal
10	216592699	Raw	0.000002	[-0.001150, 0.001154]	[-0.000899, 0.000902]	Normal
10	1950831757	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	107486742	Raw	-0.000000	[-0.001153, 0.001152]	[-0.000901, 0.000900]	Normal
10	1142235589	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	1824517164	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	1965147083	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000900, 0.000900]	Normal
10	1860620983	Raw	0.000001	[-0.001151, 0.001153]	[-0.000900, 0.000901]	Normal
10	1618144748	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	592131000	Raw	0.000001	[-0.001151, 0.001153]	[-0.000899, 0.000902]	Normal
10	964082955	Raw	0.000002	[-0.001150, 0.001154]	[-0.000898, 0.000903]	Normal
10	887230079	Raw	-0.000000	[-0.001152, 0.001152]	[-0.000901, 0.000900]	Normal
10	1818435806	Raw	0.000002	[-0.001150, 0.001154]	[-0.000898, 0.000903]	Normal
10	2095568532	Direct policy	0.000006	[-0.001146, 0.001159]	[-0.000895, 0.000907]	Normal
10	2133755523	Direct policy	0.000008	[-0.001145, 0.001161]	[-0.000893, 0.000909]	Normal
10	153722668	Direct policy	0.000007	[-0.001146, 0.001159]	[-0.000894, 0.000907]	Normal
10	603655094	Direct policy	0.000005	[-0.001147, 0.001158]	[-0.000896, 0.000906]	Normal
10	216592699	Direct policy	0.000006	[-0.001146, 0.001159]	[-0.000895, 0.000907]	Normal
10	1950831757	Direct policy	0.000006	[-0.001147, 0.001158]	[-0.000895, 0.000907]	Normal
10	107486742	Direct policy	0.000006	[-0.001147, 0.001158]	[-0.000895, 0.000906]	Normal
10	1142235589	Direct policy	0.000006	[-0.001147, 0.001159]	[-0.000895, 0.000907]	Normal
10	1824517164	Direct policy	0.000002	[-0.001151, 0.001155]	[-0.000899, 0.000903]	Normal
10	1965147083	Direct policy	0.000004	[-0.001149, 0.001156]	[-0.000897, 0.000905]	Normal
10	1860620983	Direct policy	0.000003	[-0.001150, 0.001155]	[-0.000898, 0.000904]	Normal
10	1618144748	Direct policy	0.000007	[-0.001145, 0.001160]	[-0.000894, 0.000908]	Normal

	<i>d</i>	Stream	NBO minus	Mean	Original interval	Refined interval	Case	
1								1
2	10	592131000	Direct policy	0.000007	[-0.001146, 0.001159]	[-0.000894, 0.000907]	Normal	2
3	10	964082955	Direct policy	0.000005	[-0.001148, 0.001157]	[-0.000896, 0.000905]	Normal	3
4	10	887230079	Direct policy	0.000005	[-0.001148, 0.001157]	[-0.000896, 0.000906]	Normal	4
5	10	1818435806	Direct policy	0.000006	[-0.001147, 0.001158]	[-0.000895, 0.000906]	Normal	5
6	10	2095568532	Neural HJB	0.000152	[-0.001007, 0.001311]	[-0.000755, 0.001060]	Normal	6
7	10	2133755523	Neural HJB	0.000186	[-0.000974, 0.001346]	[-0.000723, 0.001094]	Normal	7
8	10	153722668	Neural HJB	0.000127	[-0.001030, 0.001285]	[-0.000779, 0.001034]	Normal	8
9	10	603655094	Neural HJB	0.000193	[-0.000968, 0.001354]	[-0.000717, 0.001102]	Normal	9
10	10	216592699	Neural HJB	0.000110	[-0.001046, 0.001266]	[-0.000795, 0.001015]	Normal	10
11	10	1950831757	Neural HJB	0.000211	[-0.000952, 0.001373]	[-0.000700, 0.001122]	Normal	11
12	10	107486742	Neural HJB	0.000395	[-0.000770, 0.001561]	[-0.000519, 0.001309]	Normal	12
13	10	1142235589	Neural HJB	0.001235	[0.000054, 0.002417]	[0.000306, 0.002165]	Normal	13
14	10	1824517164	Neural HJB	0.000139	[-0.001018, 0.001295]	[-0.000767, 0.001044]	Normal	14
15	10	1965147083	Neural HJB	0.000204	[-0.000957, 0.001365]	[-0.000705, 0.001114]	Normal	15
16	10	1860620983	Neural HJB	0.000121	[-0.001036, 0.001279]	[-0.000785, 0.001027]	Normal	16
17	10	1618144748	Neural HJB	0.000156	[-0.001004, 0.001315]	[-0.000752, 0.001063]	Normal	17
18	10	592131000	Neural HJB	0.000189	[-0.000972, 0.001349]	[-0.000720, 0.001098]	Normal	18
19	10	964082955	Neural HJB	0.000169	[-0.000990, 0.001328]	[-0.000738, 0.001077]	Normal	19
20	10	887230079	Neural HJB	0.000184	[-0.000974, 0.001343]	[-0.000722, 0.001091]	Normal	20
21	10	1818435806	Neural HJB	0.000944	[-0.000230, 0.002118]	[0.000022, 0.001867]	Normal	21
22	50	2095568532	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal	22
23	50	2133755523	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000909, 0.000905]	Normal	23
24	50	153722668	Raw	-0.000001	[-0.001204, 0.001201]	[-0.000908, 0.000906]	Normal	24
25	50	603655094	Raw	-0.000002	[-0.001205, 0.001200]	[-0.000909, 0.000905]	Normal	25
26	50	216592699	Raw	-0.000001	[-0.001204, 0.001201]	[-0.000908, 0.000906]	Normal	26
27	50	1950831757	Raw	-0.000003	[-0.001205, 0.001200]	[-0.000909, 0.000904]	Normal	27
28	50	107486742	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal	28
29	50	1142235589	Raw	-0.000001	[-0.001203, 0.001202]	[-0.000908, 0.000906]	Normal	29
30	50	1824517164	Raw	-0.000001	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal	30
31	50	1965147083	Raw	-0.000001	[-0.001203, 0.001202]	[-0.000907, 0.000906]	Normal	31
32	50	1860620983	Raw	-0.000001	[-0.001204, 0.001202]	[-0.000908, 0.000906]	Normal	32
	50	1618144748	Raw	-0.000001	[-0.001203, 0.001201]	[-0.000908, 0.000906]	Normal	
	50	592131000	Raw	-0.000000	[-0.001203, 0.001202]	[-0.000907, 0.000907]	Normal	

<i>d</i>	Stream	NBO minus	Mean	Original interval	Refined interval	Case
50	964082955	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000909, 0.000905]	Normal
50	887230079	Raw	-0.000002	[-0.001204, 0.001201]	[-0.000908, 0.000905]	Normal
50	1818435806	Raw	-0.000001	[-0.001203, 0.001202]	[-0.000907, 0.000906]	Normal
50	2095568532	Direct policy	-0.000002	[-0.001206, 0.001201]	[-0.000910, 0.000905]	Normal
50	2133755523	Direct policy	-0.000001	[-0.001204, 0.001202]	[-0.000909, 0.000906]	Normal
50	153722668	Direct policy	0.000002	[-0.001201, 0.001205]	[-0.000906, 0.000909]	Normal
50	603655094	Direct policy	-0.000002	[-0.001206, 0.001201]	[-0.000910, 0.000905]	Normal
50	216592699	Direct policy	0.000002	[-0.001202, 0.001205]	[-0.000906, 0.000909]	Normal
50	1950831757	Direct policy	-0.000004	[-0.001207, 0.001199]	[-0.000911, 0.000904]	Normal
50	107486742	Direct policy	-0.000022	[-0.001225, 0.001182]	[-0.000929, 0.000886]	Normal
50	1142235589	Direct policy	-0.000004	[-0.001208, 0.001199]	[-0.000912, 0.000903]	Normal
50	1824517164	Direct policy	0.000001	[-0.001202, 0.001205]	[-0.000906, 0.000909]	Normal
50	1965147083	Direct policy	-0.000022	[-0.001225, 0.001181]	[-0.000930, 0.000886]	Normal
50	1860620983	Direct policy	-0.000002	[-0.001205, 0.001201]	[-0.000910, 0.000905]	Normal
50	1618144748	Direct policy	-0.000019	[-0.001223, 0.001184]	[-0.000927, 0.000888]	Normal
50	592131000	Direct policy	-0.000000	[-0.001203, 0.001203]	[-0.000908, 0.000907]	Normal
50	964082955	Direct policy	-0.000001	[-0.001204, 0.001203]	[-0.000908, 0.000907]	Normal
50	887230079	Direct policy	-0.000002	[-0.001205, 0.001202]	[-0.000909, 0.000906]	Normal
50	1818435806	Direct policy	-0.000001	[-0.001205, 0.001202]	[-0.000909, 0.000906]	Normal
50	2095568532	Neural HJB	0.000214	[-0.000995, 0.001423]	[-0.000699, 0.001127]	Normal
50	2133755523	Neural HJB	0.002917	[0.001566, 0.004269]	[0.001862, 0.003973]	Normal
50	153722668	Neural HJB	0.002449	[0.001128, 0.003769]	[0.001424, 0.003474]	Normal
50	603655094	Neural HJB	0.000643	[-0.000580, 0.001866]	[-0.000284, 0.001570]	Normal
50	216592699	Neural HJB	0.002211	[0.000909, 0.003513]	[0.001205, 0.003217]	Normal
50	1950831757	Neural HJB	0.000607	[-0.000612, 0.001825]	[-0.000316, 0.001529]	Normal
50	107486742	Neural HJB	0.001944	[0.000659, 0.003230]	[0.000955, 0.002934]	Normal
50	1142235589	Neural HJB	0.001347	[0.000096, 0.002598]	[0.000392, 0.002303]	Normal
50	1824517164	Neural HJB	0.001646	[0.000384, 0.002907]	[0.000680, 0.002611]	Normal
50	1965147083	Neural HJB	0.001684	[0.000418, 0.002950]	[0.000714, 0.002654]	Normal
50	1860620983	Neural HJB	0.000567	[-0.000648, 0.001782]	[-0.000352, 0.001486]	Normal
50	1618144748	Neural HJB	0.001821	[0.000545, 0.003096]	[0.000841, 0.002800]	Normal
50	592131000	Neural HJB	0.001557	[0.000298, 0.002816]	[0.000594, 0.002521]	Normal
50	964082955	Neural HJB	0.001678	[0.000414, 0.002942]	[0.000710, 0.002647]	Normal

d	Stream	NBO minus	Mean	Original interval	Refined interval	Case
50	887230079	Neural HJB	0.001625	[0.000366, 0.002884]	[0.000662, 0.002588]	Normal
50	1818435806	Neural HJB	0.001064	[-0.000172, 0.002301]	[0.000123, 0.002005]	Normal

All declared streams are retained. Normal means two simulated total-held policies; Mixed means one analytical fallback; Both means two exact analytical references. Only Normal rows are eligible for the new transfer. No payoff vector is regenerated and no skipped stopping event is recycled.

S.23. DESIGN OF THE REUSABLE CAPITAL DECISION MENU

This experiment evaluates the finite Bellman object in (S.1). A current task changes the consumption utility weight w , the coefficient η on squared mean consumption, or the current upper consumption bound u . All future primitives and the feasible future schedule remain fixed within a calibration. The continuation identity in Proposition S.14 therefore holds across tasks. A change to the future horizon, technology, preferences, innovation law, terminal dispersion cost, or schedule defines a different continuation and requires a different cache or fitted object.

S.23.1. Calibrations and the query population

Table S.31 records all four calibrations. The drift uses the original fixed dense coupling matrix in dimensions ten and fifty, with $f(y) = \kappa \tanh(By)$. Productivity is 0.1, discounting is 0.04, the future consumption penalty coefficient is 0.2, and the original consumption interval is $[0.02, 2]$. The terminal penalty is proportional to cross-sector dispersion. The period weights, feasible reference schedule, drift constant, innovation amplitudes, and terminal coefficients are stored before fitting. Their binary64 values, together with the stored states and coupling matrices, are the exact real coefficients of the finite economic problem under study. The continuous-diffusion primitives motivate these coefficients; they do not make the coarse finite value an unqualified continuous-time approximation.

Nine tasks are the Cartesian product $w \in \{0.8, 1, 1.2\}$ and $\eta \in \{0.1, 0.2, 0.4\}$ with $u = 2$. The other three are $(w, \eta, u) = (0.9, 0.3, 0.6)$, $(1.1, 0.15, 0.8)$, and $(1, 0.25, 1)$.

TABLE S.31

COMPLETE CAPITAL-MENU CALIBRATION DESIGN

Calibration	T	K	h	σ_I	σ_C	κ	χ
Original	1	4	0.25	0.15	0.10	0.15	0.03
Quarterly	2	8	0.25	0.60	0.30	0.30	0.10
Long	4	8	0.50	0.60	0.30	0.30	0.10
Intermediate	3	12	0.25	0.375	0.20	0.225	0.06

K is the number of finite decision periods and $h = T/K$. Each row is run at $d = 10, 50$ with sixteen fixed streams. Original retains the original low-volatility primitive anchor. Quarterly was used in one disclosed exploratory five-method pilot; Long and Intermediate had no method-development or outcome pilot. Long increases calendar time and the length of a period relative to Quarterly while retaining eight periods.

Every task retains the lower bound 0.02. The first nine appeared in the disclosed pilot; the three tighter-cap tasks did not. Each dimension has thirty-two stored states crossed with all twelve tasks. The inferential population is the uniform finite collection of these 384 queries and the sixteen declared streams, separately in each calibration and dimension. No query, stream, task, or work stage is removed because of its realized performance. Training inputs, method failures, completed prefixes, and feasible reference fallbacks remain recorded.

S.23.2. *Five reusable procedures*

The NBO scalar critic and the vector costate predictor both have width thirty-two. At each work stage they share the same replay states, uniform feasible training actions, postdecision arguments, and four-path antithetic continuation labels. The scalar critic fits gradients of one differentiable continuation; an intercept fits its level using the value labels. The vector predictor fits costate components without the restriction that they be the gradient of a scalar. These procedures consequently share observations while using different fitted function classes and optimization objectives. Their economic contrast does not by itself isolate a single statistical mechanism.

The materialized Raw procedure stores costate labels at the reference current consumption. Its task-conditioned actor maximizes the exact current reward and the corresponding linearized continuation in the action. Direct policy optimization also uses a task-conditioned actor, but retains innovations and recomputes the nonlinear future payoff at each proposed postdecision argument. Both actors have width thirty-two. Query-level Raw instead optimizes each of the 384 task-state decisions directly against a sample-average continuation. Its innovation cache grows from four to sixteen to sixty-four antithetic paths, and its previous action initializes the next stage. Retaining an innovation does not eliminate the state-recursion work required by a new argument. All five methods follow the same cache-validity rule.

For the fitted procedures, the cumulative replay budgets are 128, 512, and 2,048 states. Each field stage has 1,200 Adam updates with learning rate 0.003, followed by L-BFGS with at most 300 iterations and 400 function evaluations. Each actor stage has 1,200 Adam updates with learning rate 0.003 and batch size 128. Query optimization uses 180 Adam updates with learning rate 0.06. NBO and sample-average Raw then apply L-BFGS to their scalar objective, with at most sixty iterations and ninety evaluations. The vector costate procedure retains its 180 gradient-driven action updates; it has no additional scalar-objective L-BFGS stage. All three completed stages are assessed. The final payoff bank is inaccessible to fitting and cannot select a method, checkpoint, work stage, query, or replacement stream.

S.23.3. *Independent payoff and scalar-accuracy statements*

There are $4 \times 2 \times 16 = 128$ trials, 640 method processes, and 1,920 stage candidates. Conditional on these fitted objects, each task-state query has 512 independent antithetic confirmation pairs. Common innovations pair the method contrasts. Proposition S.16 supplies the query-conditional range after removing known analytic centers. The numerical account also retains the fixed Gaussian truncation and arithmetic allowances. The primary family allocates error 0.018 across all 216 events: 120 method-versus-reference gains and 96 direct NBO-versus-comparator contrasts. Thus every comparator has twenty-four direct statements across all calibrations,

dimensions, and stages. There are no additional per-stream confidence events or continuation-risk estimates.

The economic superiority margin is 10^{-4} . A protected lower contrast above zero establishes a positive difference; material superiority requires the lower contrast to reach 10^{-4} . Equivalence requires the complete interval to lie in $[-10^{-4}, 10^{-4}]$. Positive differences and equivalence can therefore both be established for the same comparison. These categories are reported separately. Training loss and an observed economic payoff gain do not establish a reduction in independent critic mean-squared error.

The separate scalar family allocates error 0.002 across 128 statements. For the central task $(w, \eta, u) = (1, 0.2, 2)$ and the first stored state (index zero), a successful fit selects its candidate using the final NBO critic on 257 fixed grid points in the interval consisting of the reference current consumption plus or minus 0.1, intersected with the action box. A failed fit retains the predeclared feasible reference fallback. All sectors receive this same scalar action. The independent verifier uses 65,536 antithetic pairs and the proved derivative and curvature account to cover the entire continuous interval, including points between the candidate grid. These bounds concern this restricted finite scalar intervention. They do not bound regret over all vector actions, future adapted policies, or the continuous diffusion.

S.23.4. *Complete work and development history*

The final-stage process clock covers startup, imports, input checking, all three fitting stages, caches, derivative and optimization work, query deployment, candidate output, and finalization. Earlier-stage clocks combine the executed prefix with all measured outer overhead and finalization; they are conservative allocations of this run. They are not observations from separate online stopping experiments. The independent confirmation process is a shared workload covering all five methods, all three stages, and the scalar assessment. Its full bill is reported separately. Charging that bill to each method is an explicit upper allocation, not a measurement of five independently executed verification jobs. Timing summaries describe these

runs; the confidence family makes no claim about an unseen runtime or optimizer distribution.

The development archive retains the single exploratory five-method menu pilot, its interrupted preliminary attempt, and all twelve fixed-random-feature explorations, including their adverse results. Their payoff observations are not confirmation observations. The final source fixes new streams, the complete task population, every budget, both margins, failure behavior, power calculations, and all inference code before the registered fitting and confirmation. The independent fixed-dictionary identity in Proposition S.15 describes a precise conditional statistical benefit and its approximation cost. It neither identifies the nonlinear fitted network with that projection nor makes the observed five-procedure economic comparison a causal decomposition of critic risk.

S.23.5. Complete continuation reuse evidence

TABLE S.32: Final-Stage Menu Gains over the Common Reference

Economy	d	Method	Mean	Lower	Upper
Original	10	NBO	0.0033940	0.0033920	0.0033960
Original	10	Vector	0.0033939	0.0033919	0.0033959
Original	10	Raw actor	0.0033872	0.0033852	0.0033892
Original	10	DPO	0.0033873	0.0033854	0.0033893
Original	10	Cached SAA	0.0033940	0.0033920	0.0033960
Original	50	NBO	0.0033263	0.0033245	0.0033281
Original	50	Vector	0.0033223	0.0033205	0.0033241
Original	50	Raw actor	0.0033010	0.0032992	0.0033029
Original	50	DPO	0.0033019	0.0033000	0.0033037
Original	50	Cached SAA	0.0033287	0.0033269	0.0033305
Quarterly	10	NBO	0.0109756	0.0109582	0.0109929
Quarterly	10	Vector	0.0109755	0.0109581	0.0109929
Quarterly	10	Raw actor	0.0109526	0.0109350	0.0109702
Quarterly	10	DPO	0.0109610	0.0109438	0.0109782
Quarterly	10	Cached SAA	0.0109790	0.0109616	0.0109964

Economy	d	Method	Mean	Lower	Upper
Quarterly	50	NBO	0.0102941	0.0102772	0.0103110
Quarterly	50	Vector	0.0102697	0.0102528	0.0102865
Quarterly	50	Raw actor	0.0102080	0.0101907	0.0102253
Quarterly	50	DPO	0.0102148	0.0101982	0.0102313
Quarterly	50	Cached SAA	0.0103326	0.0103156	0.0103497
Long	10	NBO	0.0528323	0.0526929	0.0529718
Long	10	Vector	0.0527792	0.0526452	0.0529132
Long	10	Raw actor	0.0522140	0.0520610	0.0523670
Long	10	DPO	0.0528933	0.0527598	0.0530267
Long	10	Cached SAA	0.0529927	0.0528601	0.0531253
Long	50	NBO	0.0520777	0.0519142	0.0522411
Long	50	Vector	0.0503524	0.0501869	0.0505179
Long	50	Raw actor	0.0501904	0.0499908	0.0503900
Long	50	DPO	0.0517015	0.0515360	0.0518670
Long	50	Cached SAA	0.0523063	0.0521340	0.0524785
Intermediate	10	NBO	0.0155805	0.0155529	0.0156081
Intermediate	10	Vector	0.0155797	0.0155521	0.0156072
Intermediate	10	Raw actor	0.0155516	0.0155237	0.0155794
Intermediate	10	DPO	0.0155626	0.0155351	0.0155900
Intermediate	10	Cached SAA	0.0155812	0.0155534	0.0156089
Intermediate	50	NBO	0.0145140	0.0144820	0.0145459
Intermediate	50	Vector	0.0144413	0.0144098	0.0144729
Intermediate	50	Raw actor	0.0143586	0.0143258	0.0143914
Intermediate	50	DPO	0.0143768	0.0143454	0.0144083
Intermediate	50	Cached SAA	0.0145578	0.0145245	0.0145911

Notes: Every cell averages all sixteen declared streams and all 384 fixed current-period queries. Each query has 512 independent antithetic future pairs; opposite signs form one observation. Five methods share each pair. All three stages and all eight economic-dimension cells contribute to one 216-event family with error probability 0.018. Current reward and a known terminal quadratic are removed before the random residual is bounded. The mean column is the clipped residual estimator plus this known offset. Numerical arithmetic, clipping and the unclipped Gaussian law are included. The materiality margin is 10^{-4} .

TABLE S.33: All Registered Menu Payoff Intervals

Economy	d	Stage	Gain or contrast	Mean	Lower	Upper
Original	10	1	NBO	0.0033939	0.0033919	0.0033959
Original	10	1	Vector	0.0033931	0.0033911	0.0033951
Original	10	1	Raw actor	0.0033393	0.0033373	0.0033413
Original	10	1	DPO	0.0033392	0.0033372	0.0033412
Original	10	1	Cached SAA	0.0033939	0.0033919	0.0033959
Original	10	1	NBO – Vector	0.0000008	0.0000007	0.0000009
Original	10	1	NBO – Raw actor	0.0000546	0.0000541	0.0000552
Original	10	1	NBO – DPO	0.0000547	0.0000542	0.0000552
Original	10	1	NBO – Cached SAA	-0.0000000	-0.0000001	0.0000001
Original	10	2	NBO	0.0033939	0.0033920	0.0033959
Original	10	2	Vector	0.0033939	0.0033919	0.0033959
Original	10	2	Raw actor	0.0033845	0.0033825	0.0033865
Original	10	2	DPO	0.0033845	0.0033825	0.0033865
Original	10	2	Cached SAA	0.0033939	0.0033919	0.0033959
Original	10	2	NBO – Vector	0.0000001	0.0000000	0.0000001
Original	10	2	NBO – Raw actor	0.0000095	0.0000092	0.0000097
Original	10	2	NBO – DPO	0.0000094	0.0000092	0.0000097
Original	10	2	NBO – Cached SAA	0.0000000	-0.0000001	0.0000001
Original	10	3	NBO	0.0033940	0.0033920	0.0033960
Original	10	3	Vector	0.0033939	0.0033919	0.0033959
Original	10	3	Raw actor	0.0033872	0.0033852	0.0033892
Original	10	3	DPO	0.0033873	0.0033854	0.0033893
Original	10	3	Cached SAA	0.0033940	0.0033920	0.0033960
Original	10	3	NBO – Vector	0.0000000	0.0000000	0.0000001
Original	10	3	NBO – Raw actor	0.0000068	0.0000066	0.0000069
Original	10	3	NBO – DPO	0.0000066	0.0000064	0.0000068
Original	10	3	NBO – Cached SAA	-0.0000000	-0.0000001	0.0000001
Original	50	1	NBO	0.0033193	0.0033175	0.0033211
Original	50	1	Vector	0.0033105	0.0033087	0.0033123
Original	50	1	Raw actor	0.0031705	0.0031687	0.0031724
Original	50	1	DPO	0.0031712	0.0031693	0.0031730

	Economy	d	Stage	Gain or contrast	Mean	Lower	Upper	
1								1
2								2
3	Original	50	1	Cached SAA	0.0033287	0.0033269	0.0033305	3
4	Original	50	1	NBO – Vector	0.0000088	0.0000085	0.0000090	4
5	Original	50	1	NBO – Raw actor	0.0001488	0.0001482	0.0001495	5
6	Original	50	1	NBO – DPO	0.0001481	0.0001475	0.0001488	6
7	Original	50	1	NBO – Cached SAA	-0.0000094	-0.0000095	-0.0000092	7
8	Original	50	2	NBO	0.0033249	0.0033231	0.0033267	8
9	Original	50	2	Vector	0.0033185	0.0033167	0.0033203	9
10	Original	50	2	Raw actor	0.0032975	0.0032957	0.0032993	10
11	Original	50	2	DPO	0.0032978	0.0032960	0.0032996	11
12	Original	50	2	Cached SAA	0.0033287	0.0033269	0.0033305	12
13	Original	50	2	NBO – Vector	0.0000064	0.0000062	0.0000066	13
14	Original	50	2	NBO – Raw actor	0.0000274	0.0000270	0.0000277	14
15	Original	50	2	NBO – DPO	0.0000271	0.0000268	0.0000274	15
16	Original	50	2	NBO – Cached SAA	-0.0000038	-0.0000039	-0.0000037	16
17	Original	50	3	NBO	0.0033263	0.0033245	0.0033281	17
18	Original	50	3	Vector	0.0033223	0.0033205	0.0033241	18
19	Original	50	3	Raw actor	0.0033010	0.0032992	0.0033029	19
20	Original	50	3	DPO	0.0033019	0.0033000	0.0033037	20
21	Original	50	3	Cached SAA	0.0033287	0.0033269	0.0033305	21
22	Original	50	3	NBO – Vector	0.0000040	0.0000038	0.0000041	22
23	Original	50	3	NBO – Raw actor	0.0000253	0.0000250	0.0000255	23
24	Original	50	3	NBO – DPO	0.0000244	0.0000242	0.0000247	24
25	Original	50	3	NBO – Cached SAA	-0.0000024	-0.0000025	-0.0000023	25
26	Quarterly	10	1	NBO	0.0108321	0.0108142	0.0108501	26
27	Quarterly	10	1	Vector	0.0106818	0.0106625	0.0107011	27
28	Quarterly	10	1	Raw actor	0.0108685	0.0108501	0.0108868	28
29	Quarterly	10	1	DPO	0.0108832	0.0108652	0.0109012	29
30	Quarterly	10	1	Cached SAA	0.0108782	0.0108602	0.0108963	30
31	Quarterly	10	1	NBO – Vector	0.0001504	0.0001428	0.0001579	31
32	Quarterly	10	1	NBO – Raw actor	-0.0000363	-0.0000434	-0.0000292	32
	Quarterly	10	1	NBO – DPO	-0.0000511	-0.0000581	-0.0000441	
	Quarterly	10	1	NBO – Cached SAA	-0.0000461	-0.0000524	-0.0000398	
	Quarterly	10	2	NBO	0.0109392	0.0109213	0.0109572	

	Economy	d	Stage	Gain or contrast	Mean	Lower	Upper	
1								1
2	Quarterly	10	2	Vector	0.0109177	0.0108998	0.0109355	2
3	Quarterly	10	2	Raw actor	0.0109386	0.0109209	0.0109563	3
4	Quarterly	10	2	DPO	0.0109460	0.0109288	0.0109632	4
5	Quarterly	10	2	Cached SAA	0.0109592	0.0109416	0.0109767	5
6	Quarterly	10	2	NBO – Vector	0.0000216	0.0000178	0.0000254	6
7	Quarterly	10	2	NBO – Raw actor	0.0000007	-0.0000034	0.0000047	7
8	Quarterly	10	2	NBO – DPO	-0.0000068	-0.0000107	-0.0000029	8
9	Quarterly	10	2	NBO – Cached SAA	-0.0000199	-0.0000238	-0.0000161	9
10	Quarterly	10	3	NBO	0.0109756	0.0109582	0.0109929	10
11	Quarterly	10	3	Vector	0.0109755	0.0109581	0.0109929	11
12	Quarterly	10	3	Raw actor	0.0109526	0.0109350	0.0109702	12
13	Quarterly	10	3	DPO	0.0109610	0.0109438	0.0109782	13
14	Quarterly	10	3	Cached SAA	0.0109790	0.0109616	0.0109964	14
15	Quarterly	10	3	NBO – Vector	0.0000000	-0.0000015	0.0000015	15
16	Quarterly	10	3	NBO – Raw actor	0.0000230	0.0000205	0.0000254	16
17	Quarterly	10	3	NBO – DPO	0.0000145	0.0000123	0.0000168	17
18	Quarterly	10	3	NBO – Cached SAA	-0.0000034	-0.0000052	-0.0000017	18
19	Quarterly	50	1	NBO	0.0100177	0.0099990	0.0100365	19
20	Quarterly	50	1	Vector	0.0097747	0.0097560	0.0097933	20
21	Quarterly	50	1	Raw actor	0.0099934	0.0099748	0.0100120	21
22	Quarterly	50	1	DPO	0.0100152	0.0099973	0.0100330	22
23	Quarterly	50	1	Cached SAA	0.0102226	0.0102049	0.0102402	23
24	Quarterly	50	1	NBO – Vector	0.0002430	0.0002345	0.0002516	24
25	Quarterly	50	1	NBO – Raw actor	0.0000243	0.0000157	0.0000329	25
26	Quarterly	50	1	NBO – DPO	0.0000026	-0.0000056	0.0000108	26
27	Quarterly	50	1	NBO – Cached SAA	-0.0002048	-0.0002117	-0.0001980	27
28	Quarterly	50	2	NBO	0.0102491	0.0102318	0.0102665	28
29	Quarterly	50	2	Vector	0.0101577	0.0101404	0.0101750	29
30	Quarterly	50	2	Raw actor	0.0101564	0.0101384	0.0101743	30
31	Quarterly	50	2	DPO	0.0101680	0.0101508	0.0101852	31
32	Quarterly	50	2	Cached SAA	0.0103124	0.0102952	0.0103297	32
	Quarterly	50	2	NBO – Vector	0.0000914	0.0000874	0.0000954	
	Quarterly	50	2	NBO – Raw actor	0.0000927	0.0000880	0.0000975	

	Economy	d	Stage	Gain or contrast	Mean	Lower	Upper	
1								1
2								2
3	Quarterly	50	2	NBO – DPO	0.0000811	0.0000767	0.0000856	3
4	Quarterly	50	2	NBO – Cached SAA	-0.0000633	-0.0000662	-0.0000603	4
5	Quarterly	50	3	NBO	0.0102941	0.0102772	0.0103110	5
6	Quarterly	50	3	Vector	0.0102697	0.0102528	0.0102865	6
7	Quarterly	50	3	Raw actor	0.0102080	0.0101907	0.0102253	7
8	Quarterly	50	3	DPO	0.0102148	0.0101982	0.0102313	8
9	Quarterly	50	3	Cached SAA	0.0103326	0.0103156	0.0103497	9
10	Quarterly	50	3	NBO – Vector	0.0000245	0.0000222	0.0000268	10
11	Quarterly	50	3	NBO – Raw actor	0.0000861	0.0000828	0.0000894	11
12	Quarterly	50	3	NBO – DPO	0.0000794	0.0000763	0.0000824	12
13	Quarterly	50	3	NBO – Cached SAA	-0.0000385	-0.0000403	-0.0000367	13
14	Long	10	1	NBO	0.0495449	0.0493567	0.0497330	14
15	Long	10	1	Vector	0.0488354	0.0486557	0.0490152	15
16	Long	10	1	Raw actor	0.0512017	0.0510309	0.0513725	16
17	Long	10	1	DPO	0.0522829	0.0521347	0.0524311	17
18	Long	10	1	Cached SAA	0.0516542	0.0515012	0.0518071	18
19	Long	10	1	NBO – Vector	0.0007095	0.0005682	0.0008507	19
20	Long	10	1	NBO – Raw actor	-0.0016568	-0.0017850	-0.0015287	20
21	Long	10	1	NBO – DPO	-0.0027380	-0.0028558	-0.0026202	21
22	Long	10	1	NBO – Cached SAA	-0.0021093	-0.0022408	-0.0019777	22
23	Long	10	2	NBO	0.0518962	0.0517355	0.0520568	23
24	Long	10	2	Vector	0.0513189	0.0511589	0.0514790	24
25	Long	10	2	Raw actor	0.0519224	0.0517625	0.0520823	25
26	Long	10	2	DPO	0.0527086	0.0525711	0.0528460	26
27	Long	10	2	Cached SAA	0.0527162	0.0525788	0.0528536	27
28	Long	10	2	NBO – Vector	0.0005772	0.0004797	0.0006748	28
29	Long	10	2	NBO – Raw actor	-0.0000262	-0.0001089	0.0000565	29
30	Long	10	2	NBO – DPO	-0.0008124	-0.0008863	-0.0007385	30
31	Long	10	2	NBO – Cached SAA	-0.0008200	-0.0009051	-0.0007349	31
32	Long	10	3	NBO	0.0528323	0.0526929	0.0529718	32
	Long	10	3	Vector	0.0527792	0.0526452	0.0529132	
	Long	10	3	Raw actor	0.0522140	0.0520610	0.0523670	
	Long	10	3	DPO	0.0528933	0.0527598	0.0530267	

	Economy	<i>d</i>	Stage	Gain or contrast	Mean	Lower	Upper	
1								1
2	Long	10	3	Cached SAA	0.0529927	0.0528601	0.0531253	2
3	Long	10	3	NBO – Vector	0.0000531	0.0000149	0.0000914	3
4	Long	10	3	NBO – Raw actor	0.0006184	0.0005676	0.0006691	4
5	Long	10	3	NBO – DPO	-0.0000609	-0.0001025	-0.0000194	5
6	Long	10	3	NBO – Cached SAA	-0.0001604	-0.0001984	-0.0001224	6
7	Long	50	1	NBO	0.0465819	0.0463756	0.0467881	7
8	Long	50	1	Vector	0.0419222	0.0417457	0.0420987	8
9	Long	50	1	Raw actor	0.0477739	0.0475594	0.0479884	9
10	Long	50	1	DPO	0.0497426	0.0495600	0.0499253	10
11	Long	50	1	Cached SAA	0.0508312	0.0506506	0.0510118	11
12	Long	50	1	NBO – Vector	0.0046597	0.0044614	0.0048580	12
13	Long	50	1	NBO – Raw actor	-0.0011920	-0.0013358	-0.0010482	13
14	Long	50	1	NBO – DPO	-0.0031608	-0.0032923	-0.0030292	14
15	Long	50	1	NBO – Cached SAA	-0.0042493	-0.0043762	-0.0041224	15
16	Long	50	2	NBO	0.0503603	0.0501658	0.0505548	16
17	Long	50	2	Vector	0.0479168	0.0477480	0.0480856	17
18	Long	50	2	Raw actor	0.0493996	0.0491956	0.0496037	18
19	Long	50	2	DPO	0.0511171	0.0509450	0.0512891	19
20	Long	50	2	Cached SAA	0.0520075	0.0518319	0.0521831	20
21	Long	50	2	NBO – Vector	0.0024435	0.0023206	0.0025664	21
22	Long	50	2	NBO – Raw actor	0.0009607	0.0008434	0.0010780	22
23	Long	50	2	NBO – DPO	-0.0007567	-0.0008605	-0.0006529	23
24	Long	50	2	NBO – Cached SAA	-0.0016472	-0.0017417	-0.0015526	24
25	Long	50	3	NBO	0.0520777	0.0519142	0.0522411	25
26	Long	50	3	Vector	0.0503524	0.0501869	0.0505179	26
27	Long	50	3	Raw actor	0.0501904	0.0499908	0.0503900	27
28	Long	50	3	DPO	0.0517015	0.0515360	0.0518670	28
29	Long	50	3	Cached SAA	0.0523063	0.0521340	0.0524785	29
30	Long	50	3	NBO – Vector	0.0017253	0.0016381	0.0018124	30
31	Long	50	3	NBO – Raw actor	0.0018872	0.0018026	0.0019719	31
32	Long	50	3	NBO – DPO	0.0003761	0.0003273	0.0004250	32
	Long	50	3	NBO – Cached SAA	-0.0002286	-0.0002678	-0.0001894	
	Intermediate	10	1	NBO	0.0155003	0.0154718	0.0155288	

	Economy	d	Stage	Gain or contrast	Mean	Lower	Upper	
1								1
2								2
3	Intermediate	10	1	Vector	0.0154165	0.0153872	0.0154457	3
4	Intermediate	10	1	Raw actor	0.0154536	0.0154252	0.0154820	4
5	Intermediate	10	1	DPO	0.0154718	0.0154437	0.0154999	5
6	Intermediate	10	1	Cached SAA	0.0155071	0.0154786	0.0155356	6
7	Intermediate	10	1	NBO – Vector	0.0000838	0.0000760	0.0000916	7
8	Intermediate	10	1	NBO – Raw actor	0.0000467	0.0000378	0.0000557	8
9	Intermediate	10	1	NBO – DPO	0.0000285	0.0000201	0.0000368	9
10	Intermediate	10	1	NBO – Cached SAA	-0.0000068	-0.0000147	0.0000011	10
11	Intermediate	10	2	NBO	0.0155599	0.0155321	0.0155878	11
12	Intermediate	10	2	Vector	0.0155528	0.0155250	0.0155806	12
13	Intermediate	10	2	Raw actor	0.0155307	0.0155026	0.0155588	13
14	Intermediate	10	2	DPO	0.0155429	0.0155151	0.0155707	14
15	Intermediate	10	2	Cached SAA	0.0155678	0.0155398	0.0155958	15
16	Intermediate	10	2	NBO – Vector	0.0000071	0.0000030	0.0000112	16
17	Intermediate	10	2	NBO – Raw actor	0.0000292	0.0000240	0.0000344	17
18	Intermediate	10	2	NBO – DPO	0.0000170	0.0000123	0.0000217	18
19	Intermediate	10	2	NBO – Cached SAA	-0.0000079	-0.0000124	-0.0000034	19
20	Intermediate	10	3	NBO	0.0155805	0.0155529	0.0156081	20
21	Intermediate	10	3	Vector	0.0155797	0.0155521	0.0156072	21
22	Intermediate	10	3	Raw actor	0.0155516	0.0155237	0.0155794	22
23	Intermediate	10	3	DPO	0.0155626	0.0155351	0.0155900	23
24	Intermediate	10	3	Cached SAA	0.0155812	0.0155534	0.0156089	24
25	Intermediate	10	3	NBO – Vector	0.0000008	-0.0000009	0.0000025	25
26	Intermediate	10	3	NBO – Raw actor	0.0000289	0.0000256	0.0000322	26
27	Intermediate	10	3	NBO – DPO	0.0000179	0.0000153	0.0000206	27
28	Intermediate	10	3	NBO – Cached SAA	-0.0000007	-0.0000029	0.0000016	28
29	Intermediate	50	1	NBO	0.0141970	0.0141627	0.0142313	29
30	Intermediate	50	1	Vector	0.0139889	0.0139552	0.0140226	30
31	Intermediate	50	1	Raw actor	0.0140369	0.0140014	0.0140724	31
32	Intermediate	50	1	DPO	0.0140905	0.0140567	0.0141242	32
	Intermediate	50	1	Cached SAA	0.0144745	0.0144403	0.0145086	
	Intermediate	50	1	NBO – Vector	0.0002081	0.0001941	0.0002222	
	Intermediate	50	1	NBO – Raw actor	0.0001601	0.0001424	0.0001778	

1	Economy	d	Stage	Gain or contrast	Mean	Lower	Upper	1
2	Intermediate	50	1	NBO – DPO	0.0001066	0.0000901	0.0001230	2
3	Intermediate	50	1	NBO – Cached SAA	-0.0002774	-0.0002892	-0.0002656	3
4	Intermediate	50	2	NBO	0.0144695	0.0144367	0.0145022	4
5	Intermediate	50	2	Vector	0.0143166	0.0142844	0.0143488	5
6	Intermediate	50	2	Raw actor	0.0142777	0.0142445	0.0143110	6
7	Intermediate	50	2	DPO	0.0143040	0.0142720	0.0143359	7
8	Intermediate	50	2	Cached SAA	0.0145400	0.0145065	0.0145735	8
9	Intermediate	50	2	NBO – Vector	0.0001529	0.0001440	0.0001617	9
10	Intermediate	50	2	NBO – Raw actor	0.0001917	0.0001816	0.0002018	10
11	Intermediate	50	2	NBO – DPO	0.0001655	0.0001561	0.0001749	11
12	Intermediate	50	2	NBO – Cached SAA	-0.0000706	-0.0000764	-0.0000647	12
13	Intermediate	50	3	NBO	0.0145140	0.0144820	0.0145459	13
14	Intermediate	50	3	Vector	0.0144413	0.0144098	0.0144729	14
15	Intermediate	50	3	Raw actor	0.0143586	0.0143258	0.0143914	15
16	Intermediate	50	3	DPO	0.0143768	0.0143454	0.0144083	16
17	Intermediate	50	3	Cached SAA	0.0145578	0.0145245	0.0145911	17
18	Intermediate	50	3	NBO – Vector	0.0000726	0.0000669	0.0000784	18
19	Intermediate	50	3	NBO – Raw actor	0.0001553	0.0001479	0.0001628	19
20	Intermediate	50	3	NBO – DPO	0.0001371	0.0001309	0.0001433	20
21	Intermediate	50	3	NBO – Cached SAA	-0.0000438	-0.0000482	-0.0000395	21

22 *Notes:* Every cell averages all sixteen declared streams and all 384 fixed current-period queries. Each
23 query has 512 independent antithetic future pairs; opposite signs form one observation. Five methods
24 share each pair. All three stages and all eight economic-dimension cells contribute to one 216-event
25 family with error probability 0.018. Current reward and a known terminal quadratic are removed
26 before the random residual is bounded. The mean column is the clipped residual estimator plus this
27 known offset. Numerical arithmetic, clipping and the unclipped Gaussian law are included. The
28 materiality margin is 10^{-4} .

TABLE S.34: Every Menu Work Stage and Retained Fallback

Economy	d	Stage	Method	Construction/query s	Fallbacks
Original	10	1	NBO	5.28	0
Original	10	1	Vector	4.45	0
Original	10	1	Raw actor	4.14	0
Original	10	1	DPO	7.21	0
Original	10	1	Cached SAA	3.47	0
Original	10	2	NBO	8.01	0
Original	10	2	Vector	6.31	0
Original	10	2	Raw actor	5.48	0
Original	10	2	DPO	11.67	0
Original	10	2	Cached SAA	5.88	0
Original	10	3	NBO	12.52	0
Original	10	3	Vector	8.73	0
Original	10	3	Raw actor	6.86	0
Original	10	3	DPO	16.21	0
Original	10	3	Cached SAA	15.60	0
Original	50	1	NBO	5.25	0
Original	50	1	Vector	4.49	0
Original	50	1	Raw actor	3.97	0
Original	50	1	DPO	8.42	0
Original	50	1	Cached SAA	3.97	0
Original	50	2	NBO	8.21	0
Original	50	2	Vector	6.61	0
Original	50	2	Raw actor	5.38	0
Original	50	2	DPO	14.21	0
Original	50	2	Cached SAA	8.76	0
Original	50	3	NBO	12.67	0
Original	50	3	Vector	9.85	0
Original	50	3	Raw actor	7.07	0
Original	50	3	DPO	20.24	0
Original	50	3	Cached SAA	27.68	0
Quarterly	10	1	NBO	4.71	0

	Economy	d	Stage	Method	Construction/query s	Fallbacks	
1							1
2							2
3	Quarterly	10	1	Vector	4.00	0	3
4	Quarterly	10	1	Raw actor	3.76	0	4
5	Quarterly	10	1	DPO	8.85	0	5
6	Quarterly	10	1	Cached SAA	3.82	0	6
7	Quarterly	10	2	NBO	7.05	0	7
8	Quarterly	10	2	Vector	5.59	0	8
9	Quarterly	10	2	Raw actor	4.85	0	9
10	Quarterly	10	2	DPO	15.22	0	10
11	Quarterly	10	2	Cached SAA	8.09	0	11
12	Quarterly	10	3	NBO	10.36	0	12
13	Quarterly	10	3	Vector	7.69	0	13
14	Quarterly	10	3	Raw actor	6.09	0	14
15	Quarterly	10	3	DPO	21.85	0	15
16	Quarterly	10	3	Cached SAA	25.23	0	16
17	Quarterly	50	1	NBO	5.83	0	17
18	Quarterly	50	1	Vector	4.98	0	18
19	Quarterly	50	1	Raw actor	4.38	0	19
20	Quarterly	50	1	DPO	14.58	0	20
21	Quarterly	50	1	Cached SAA	5.96	0	21
22	Quarterly	50	2	NBO	9.15	0	22
23	Quarterly	50	2	Vector	7.39	0	23
24	Quarterly	50	2	Raw actor	6.05	0	24
25	Quarterly	50	2	DPO	26.40	0	25
26	Quarterly	50	2	Cached SAA	17.39	0	26
27	Quarterly	50	3	NBO	14.34	0	27
28	Quarterly	50	3	Vector	11.15	0	28
29	Quarterly	50	3	Raw actor	8.28	0	29
30	Quarterly	50	3	DPO	38.93	0	30
31	Quarterly	50	3	Cached SAA	62.90	0	31
32	Long	10	1	NBO	5.38	0	32
	Long	10	1	Vector	4.57	0	
	Long	10	1	Raw actor	4.13	0	
	Long	10	1	DPO	10.61	0	

	Economy	d	Stage	Method	Construction/query s	Fallbacks	
1							1
2							2
3	Long	10	1	Cached SAA	4.68	0	3
4	Long	10	2	NBO	8.23	0	4
5	Long	10	2	Vector	6.41	0	5
6	Long	10	2	Raw actor	5.47	0	6
7	Long	10	2	DPO	18.45	0	7
8	Long	10	2	Cached SAA	9.76	0	8
9	Long	10	3	NBO	12.05	0	9
10	Long	10	3	Vector	8.87	0	10
11	Long	10	3	Raw actor	6.96	0	11
12	Long	10	3	DPO	26.53	0	12
13	Long	10	3	Cached SAA	30.42	0	13
14	Long	50	1	NBO	5.81	0	14
15	Long	50	1	Vector	4.93	0	15
16	Long	50	1	Raw actor	4.30	0	16
17	Long	50	1	DPO	14.18	0	17
18	Long	50	1	Cached SAA	7.01	0	18
19	Long	50	2	NBO	9.26	0	19
20	Long	50	2	Vector	7.29	0	20
21	Long	50	2	Raw actor	5.95	0	21
22	Long	50	2	DPO	25.78	0	22
23	Long	50	2	Cached SAA	23.28	0	23
24	Long	50	3	NBO	14.59	0	24
25	Long	50	3	Vector	11.06	0	25
26	Long	50	3	Raw actor	8.16	0	26
27	Long	50	3	DPO	38.03	0	27
28	Long	50	3	Cached SAA	67.04	0	28
29	Intermediate	10	1	NBO	5.40	0	29
30	Intermediate	10	1	Vector	4.60	0	30
31	Intermediate	10	1	Raw actor	4.15	0	31
32	Intermediate	10	1	DPO	13.85	0	32
	Intermediate	10	1	Cached SAA	4.97	0	
	Intermediate	10	2	NBO	8.19	0	
	Intermediate	10	2	Vector	6.52	0	

1	Economy	d	Stage	Method	Construction/query s	Fallbacks	1
2	Intermediate	10	2	Raw actor	5.52	0	2
3	Intermediate	10	2	DPO	25.08	0	3
4	Intermediate	10	2	Cached SAA	12.69	0	4
5	Intermediate	10	3	NBO	11.96	0	5
6	Intermediate	10	3	Vector	9.16	0	6
7	Intermediate	10	3	Raw actor	7.01	0	7
8	Intermediate	10	3	DPO	36.55	0	8
9	Intermediate	10	3	Cached SAA	44.71	0	9
10	Intermediate	50	1	NBO	5.08	0	10
11	Intermediate	50	1	Vector	4.24	0	11
12	Intermediate	50	1	Raw actor	3.80	0	12
13	Intermediate	50	1	DPO	16.43	0	13
14	Intermediate	50	1	Cached SAA	7.82	0	14
15	Intermediate	50	2	NBO	8.01	0	15
16	Intermediate	50	2	Vector	6.25	0	16
17	Intermediate	50	2	Raw actor	5.19	0	17
18	Intermediate	50	2	DPO	30.64	0	18
19	Intermediate	50	2	Cached SAA	22.00	0	19
20	Intermediate	50	3	NBO	12.70	0	20
21	Intermediate	50	3	Vector	9.62	0	21
22	Intermediate	50	3	Raw actor	7.18	0	22
23	Intermediate	50	3	DPO	45.58	0	23
24	Intermediate	50	3	Cached SAA	79.46	0	24

Notes: Stage three is an actual full process measurement. Earlier stages use their observed saved-candidate prefix plus all outer overhead and finalization, an explicitly conservative allocation; no early-stopping run is invented. The complete JSON additionally retains every stream clock, CPU and memory record, counters, and the full common confirmation bill. No failed stream is replaced.

TABLE S.35: Every Conditional Scalar Accuracy Endpoint

Economy	d	Stream	Candidate s	Gradient lower	Gradient upper	Gap upper
Original	10	3443860912	0.52344	-0.0000028	0.0000771	0.0000002
Original	10	2063512792	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	3176718348	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	782199465	0.52344	-0.0000027	0.0000771	0.0000002
Original	10	3487691819	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	2023679796	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	1635636351	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	2427697457	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	3308926306	0.52344	-0.0000028	0.0000771	0.0000002
Original	10	4219929003	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	1264292074	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	3089473238	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	527701303	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	3641281291	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	194396786	0.52344	-0.0000027	0.0000772	0.0000002
Original	10	320343828	0.52344	-0.0000027	0.0000772	0.0000002
Original	50	3443860912	0.52344	-0.0001025	-0.0000147	0.0000002
Original	50	2063512792	0.52344	-0.0001024	-0.0000147	0.0000002
Original	50	3176718348	0.51953	0.0000474	0.0001351	0.0000004
Original	50	782199465	0.52344	-0.0001024	-0.0000147	0.0000002
Original	50	3487691819	0.52344	-0.0001025	-0.0000147	0.0000002
Original	50	2023679796	0.52344	-0.0001024	-0.0000147	0.0000002
Original	50	1635636351	0.51953	0.0000474	0.0001352	0.0000004
Original	50	2427697457	0.52344	-0.0001024	-0.0000147	0.0000002
Original	50	3308926306	0.52344	-0.0001025	-0.0000147	0.0000002
Original	50	4219929003	0.52344	-0.0001025	-0.0000147	0.0000002
Original	50	1264292074	0.52344	-0.0001025	-0.0000147	0.0000002
Original	50	3089473238	0.51953	0.0000474	0.0001351	0.0000004
Original	50	527701303	0.52344	-0.0001024	-0.0000147	0.0000002
Original	50	3641281291	0.52344	-0.0001024	-0.0000147	0.0000002
Original	50	194396786	0.52344	-0.0001025	-0.0000147	0.0000002

	Economy	d	Stream	Candidate s	Gradient lower	Gradient upper	Gap upper	
1								1
2								2
3	Original	50	320343828	0.52344	-0.0001025	-0.0000147	0.0000002	3
4	Quarterly	10	3443860912	0.57422	-0.0001014	0.0003308	0.0000017	4
5	Quarterly	10	2063512792	0.57422	-0.0001039	0.0003284	0.0000017	5
6	Quarterly	10	3176718348	0.57422	-0.0001023	0.0003300	0.0000017	6
7	Quarterly	10	782199465	0.57422	-0.0001030	0.0003291	0.0000017	7
8	Quarterly	10	3487691819	0.57422	-0.0001061	0.0003260	0.0000017	8
9	Quarterly	10	2023679796	0.57812	-0.0003918	0.0000405	0.0000024	9
10	Quarterly	10	1635636351	0.57422	-0.0001092	0.0003228	0.0000016	10
11	Quarterly	10	2427697457	0.57422	-0.0001024	0.0003298	0.0000017	11
12	Quarterly	10	3308926306	0.57812	-0.0003953	0.0000370	0.0000024	12
13	Quarterly	10	4219929003	0.57422	-0.0001040	0.0003281	0.0000017	13
14	Quarterly	10	1264292074	0.57812	-0.0003937	0.0000386	0.0000024	14
15	Quarterly	10	3089473238	0.57812	-0.0003931	0.0000391	0.0000024	15
16	Quarterly	10	527701303	0.57422	-0.0001055	0.0003269	0.0000017	16
17	Quarterly	10	3641281291	0.57812	-0.0003934	0.0000388	0.0000024	17
18	Quarterly	10	194396786	0.57812	-0.0003969	0.0000354	0.0000024	18
19	Quarterly	10	320343828	0.57422	-0.0001032	0.0003291	0.0000017	19
20	Quarterly	50	3443860912	0.55078	-0.0001083	0.0004012	0.0000140	20
21	Quarterly	50	2063512792	0.55078	-0.0001088	0.0004007	0.0000139	21
22	Quarterly	50	3176718348	0.55078	-0.0001099	0.0003995	0.0000138	22
23	Quarterly	50	782199465	0.55078	-0.0001065	0.0004030	0.0000141	23
24	Quarterly	50	3487691819	0.55078	-0.0001081	0.0004013	0.0000140	24
25	Quarterly	50	2023679796	0.55078	-0.0001097	0.0003998	0.0000139	25
26	Quarterly	50	1635636351	0.55078	-0.0001098	0.0003997	0.0000138	26
27	Quarterly	50	2427697457	0.55078	-0.0001069	0.0004026	0.0000140	27
28	Quarterly	50	3308926306	0.55078	-0.0001096	0.0003999	0.0000139	28
29	Quarterly	50	4219929003	0.55078	-0.0001091	0.0004003	0.0000139	29
30	Quarterly	50	1264292074	0.55078	-0.0001076	0.0004019	0.0000140	30
31	Quarterly	50	3089473238	0.55078	-0.0001071	0.0004023	0.0000140	31
32	Quarterly	50	527701303	0.55078	-0.0001085	0.0004011	0.0000139	32
	Quarterly	50	3641281291	0.55078	-0.0001086	0.0004008	0.0000139	
	Quarterly	50	194396786	0.55078	-0.0001082	0.0004012	0.0000139	
	Quarterly	50	320343828	0.55078	-0.0001083	0.0004012	0.0000139	

	Economy	d	Stream	Candidate s	Gradient lower	Gradient upper	Gap upper	
1								1
2	Long	10	3443860912	0.58594	-0.0009560	0.0026157	0.0440599	2
3	Long	10	2063512792	0.58594	-0.0010004	0.0025721	0.0440859	3
4	Long	10	3176718348	0.58594	-0.0009790	0.0025936	0.0440733	4
5	Long	10	782199465	0.58984	-0.0023279	0.0012434	0.0454547	5
6	Long	10	3487691819	0.58984	-0.0023348	0.0012368	0.0454588	6
7	Long	10	2023679796	0.58594	-0.0009692	0.0026018	0.0440676	7
8	Long	10	1635636351	0.58594	-0.0009680	0.0026036	0.0440669	8
9	Long	10	2427697457	0.58594	-0.0010081	0.0025636	0.0440904	9
10	Long	10	3308926306	0.58594	-0.0010031	0.0025686	0.0440875	10
11	Long	10	4219929003	0.58594	-0.0010039	0.0025680	0.0440879	11
12	Long	10	1264292074	0.58984	-0.0023151	0.0012581	0.0454472	12
13	Long	10	3089473238	0.58984	-0.0023328	0.0012395	0.0454576	13
14	Long	10	527701303	0.58984	-0.0022921	0.0012787	0.0454337	14
15	Long	10	3641281291	0.58594	-0.0009615	0.0026118	0.0440631	15
16	Long	10	194396786	0.58984	-0.0023099	0.0012607	0.0454441	16
17	Long	10	320343828	0.58984	-0.0023248	0.0012475	0.0454529	16
17	Long	50	3443860912	0.55078	-0.0025015	0.0025874	0.1963134	17
18	Long	50	2063512792	0.55078	-0.0024957	0.0025930	0.1963101	18
19	Long	50	3176718348	0.55078	-0.0024728	0.0026164	0.1962975	19
20	Long	50	782199465	0.55078	-0.0024783	0.0026111	0.1963006	20
21	Long	50	3487691819	0.55078	-0.0024820	0.0026070	0.1963026	21
22	Long	50	2023679796	0.55078	-0.0024855	0.0026038	0.1963045	22
23	Long	50	1635636351	0.55078	-0.0024729	0.0026165	0.1962976	23
24	Long	50	2427697457	0.55078	-0.0024793	0.0026102	0.1963011	24
25	Long	50	3308926306	0.55078	-0.0025078	0.0025811	0.1963168	24
26	Long	50	4219929003	0.55078	-0.0024891	0.0026001	0.1963065	25
27	Long	50	1264292074	0.55078	-0.0024941	0.0025950	0.1963093	26
28	Long	50	3089473238	0.55078	-0.0025058	0.0025837	0.1963157	27
29	Long	50	527701303	0.55078	-0.0024863	0.0026030	0.1963050	28
30	Long	50	3641281291	0.55078	-0.0024890	0.0026003	0.1963065	29
31	Long	50	194396786	0.55078	-0.0024825	0.0026065	0.1963029	30
32	Long	50	320343828	0.55078	-0.0024798	0.0026090	0.1963014	31
32	Intermediate	10	3443860912	0.58203	-0.0004604	0.0002138	0.0000023	32

	Economy	d	Stream	Candidate s	Gradient lower	Gradient upper	Gap upper	
1								1
2								2
3	Intermediate	10	2063512792	0.58203	-0.0004660	0.0002081	0.0000024	3
4	Intermediate	10	3176718348	0.58203	-0.0004671	0.0002073	0.0000024	4
5	Intermediate	10	782199465	0.58203	-0.0004684	0.0002059	0.0000024	5
6	Intermediate	10	3487691819	0.58203	-0.0004639	0.0002106	0.0000023	6
7	Intermediate	10	2023679796	0.58203	-0.0004652	0.0002091	0.0000024	7
8	Intermediate	10	1635636351	0.58203	-0.0004595	0.0002146	0.0000023	8
9	Intermediate	10	2427697457	0.58203	-0.0004642	0.0002102	0.0000023	9
10	Intermediate	10	3308926306	0.58203	-0.0004647	0.0002097	0.0000023	10
11	Intermediate	10	4219929003	0.58203	-0.0004598	0.0002145	0.0000023	11
12	Intermediate	10	1264292074	0.58203	-0.0004638	0.0002105	0.0000023	12
13	Intermediate	10	3089473238	0.58203	-0.0004689	0.0002053	0.0000024	13
14	Intermediate	10	527701303	0.58203	-0.0004629	0.0002112	0.0000023	14
15	Intermediate	10	3641281291	0.58203	-0.0004653	0.0002090	0.0000024	15
16	Intermediate	10	194396786	0.58203	-0.0004611	0.0002132	0.0000023	16
17	Intermediate	10	320343828	0.58203	-0.0004635	0.0002109	0.0000023	17
18	Intermediate	50	3443860912	0.55078	-0.0004447	0.0003769	0.0002831	18
19	Intermediate	50	2063512792	0.55078	-0.0004433	0.0003783	0.0002823	19
20	Intermediate	50	3176718348	0.55078	-0.0004428	0.0003789	0.0002821	20
21	Intermediate	50	782199465	0.55078	-0.0004454	0.0003762	0.0002835	21
22	Intermediate	50	3487691819	0.55078	-0.0004425	0.0003790	0.0002819	22
23	Intermediate	50	2023679796	0.55078	-0.0004469	0.0003748	0.0002843	23
24	Intermediate	50	1635636351	0.55078	-0.0004431	0.0003785	0.0002822	24
25	Intermediate	50	2427697457	0.55078	-0.0004451	0.0003765	0.0002833	25
26	Intermediate	50	3308926306	0.55078	-0.0004454	0.0003762	0.0002835	26
27	Intermediate	50	4219929003	0.55078	-0.0004441	0.0003775	0.0002828	27
28	Intermediate	50	1264292074	0.55078	-0.0004461	0.0003756	0.0002838	28
29	Intermediate	50	3089473238	0.55078	-0.0004474	0.0003742	0.0002846	29
30	Intermediate	50	527701303	0.55078	-0.0004442	0.0003775	0.0002828	30
31	Intermediate	50	3641281291	0.55078	-0.0004428	0.0003788	0.0002820	31
32	Intermediate	50	194396786	0.55078	-0.0004445	0.0003772	0.0002830	32
	Intermediate	50	320343828	0.55078	-0.0004425	0.0003792	0.0002819	

Notes: All prescribed scalar events appear. Their error budget is 0.002/128 each. The secant-to-gradient error, global curvature, arithmetic and Gaussian clipping allowances enter before the global scalar regret upper bound. A large bound is retained without choosing another action class.

TABLE S.36

COMPLETE SHARED CONFIRMATION WORK

Economy	d	Mean seconds	Minimum seconds	Maximum seconds
Original	10	10.73	7.80	11.41
Original	50	29.50	20.81	33.83
Quarterly	10	15.90	11.21	19.22
Quarterly	50	63.46	41.23	71.25
Long	10	17.76	11.81	21.62
Long	50	64.37	54.64	86.26
Intermediate	10	24.82	16.61	30.22
Intermediate	50	82.20	60.44	102.48

Each observation is the actual complete confirmation-process clock for one of the sixteen declared streams. The shared bundle evaluates five methods, all three work stages, the common reference, and the scalar accuracy exercise. Its mean is an additional bill to the construction and query clocks in Table III. Charging that complete bill to an individual method gives a conservative allocation; it does not measure a separate method-specific verification job. All eight economic-dimension cells and every stream are retained. These are descriptive timings for the executed runs, without a runtime-distribution confidence claim.

S.24. INDEPENDENT CONFIRMATION OF THE ORIGINAL POLICIES

The new family conditions on all 128 original policy weights. It adds 32 independent confirmation banks, each with 8,192 paths on 2,048 cells, and retains all 238 declared events. The main article reports all eight pooled method gains and six direct NBO contrasts. The complete per-stream endpoint array, original-transfer counterparts, allowance reconstruction, and finite target-attainment account are retained in the source-bound numerical report.

The additional verification clocks include process startup, imports, weight reload, constants, simulation, and durable output. The original training, checkpoint assessment, and stopping work remain in their earlier accounts. This experiment was

TABLE S.37

NEW FINAL CERTIFICATES OVER THE COMPLETE SIXTEEN-STREAM POLICY POPULATION

d	Method	Gain 0.0005: definite–possible	Gain 0.001: definite–possible
10	NBO	14–16	0–16
10	Raw	13–16	0–16
10	Direct policy	10–16	0–16
10	Neural HJB	0–16	0–15
50	NBO	16–16	0–16
50	Raw	16–16	0–16
50	Direct policy	16–16	0–16
50	Neural HJB	0–11	0–5

The definite count uses protected lower endpoints; the possible count uses protected upper endpoints. Both refer to the complete declared population of sixteen already fitted policies per method and dimension. They neither estimate an unseen training-seed population nor change any original online stopping decision. All counts derive from the same fixed family as Tables V and VI.

TABLE S.38

ADDITIONAL WORK FOR INDEPENDENT POLICY REPLICATION

d	Method	Mean seconds	Total seconds
10	NBO	29.15	466.37
10	Raw	28.95	463.15
10	Direct policy	29.10	465.64
10	Neural HJB	88.36	1413.78
50	NBO	88.41	1414.55
50	Raw	87.98	1407.61
50	Direct policy	88.29	1412.57
50	Neural HJB	396.86	6349.80

Complete fresh-process confirmation clocks include imports, frozen weight reload, constants, simulation and durable output. These additional assessment costs do not replace the original R15 training or stopping work.

specified to confirm fixed-policy payoffs; its new final certificates do not constitute a new online work-to-target experiment.

S.24.1. Complete fresh signed mechanism evidence

TABLE S.39

EVERY SIGNED BELLMAN INTERVAL AND NUMERICAL ACCOUNT

d	Statistic	Mean	Lower	Upper	Clip radius	Arithmetic
10	M	0.0012435	0.0007427	0.0017443	0.211739	0.00000018
10	C	-0.0000022	-0.0006220	0.0006175	0.266829	0.00002456
10	G	0.0012457	0.0010673	0.0014242	0.055090	0.00002439
50	M	0.0013538	0.0009967	0.0017109	0.146951	0.00000012
50	C	0.0000006	-0.0005105	0.0005117	0.214706	0.00003080
50	G	0.0013532	0.0011389	0.0015675	0.067755	0.00003068

Notes: All six registered intervals appear, including inconclusive correction intervals. Endpoints include the additional shared-prefix representation allowance. The centered clipping intervals are fixed before new confirmation and can be asymmetric about zero. The arithmetic column is the interval-sample mean and standard-deviation perturbation allowance, not an empirical population range. Every endpoint is rounded outward.

TABLE S.40

ADDITIONAL WORK FOR THE FRESH SIGNED ASSESSMENT

d	Process seconds	CPU seconds	Peak MiB	Endpoint updates
10	527.70	523.02	455.4	132,789,688
50	1046.35	1039.37	589.9	134,903,120

Notes: Process and CPU times sum all sixteen fresh processes per dimension; memory is their maximum. The clock includes imports, original weight reload, constants, candidate occupation paths, paired future paths, interval arithmetic and durable evidence. These are additional scientific assessment costs and do not alter the original method work-to-target records.

S.25. ECONOMIC ROBUSTNESS AND THE STRENGTHENED HJB PROCEDURE

This experiment changes the continuous-diffusion capital economy and trains all procedures anew. It is separate from both the confirmation of 128 previously selected policies and the finite-period continuation menu. Its four economic calibrations, dimensions ten and fifty, and eight fixed training streams give 64 trials. The report retains all 256 construction processes and 320 deployed policy outputs. All

procedures completed without reference-policy fallback. A fresh bank of 8,192 paths on 2,048 cells assesses each output, with common innovations within a trial. The 648-event family has allocation $\alpha = 0.01$ and includes method averages, individual-policy endpoints, and the declared direct comparisons. No calibration or stream is selected on its outcome.

S.25.1. *Economic variation*

Table S.41 gives the complete economic grid. The high calibration retains the original high-volatility diffusion; low changes its diffusion risk. The longer-horizon economy has horizon two years. The joint-stress economy changes coupling, terminal dispersion costs, initial heterogeneity, and the actor's feasible radius together. Its comparison therefore describes that economic design rather than a single-primitive causal effect. These names are local to the continuous economy: in particular, the finite menu's Long calibration has a four-year horizon and half-year periods.

TABLE S.41

CONTINUOUS-ECONOMY ROBUSTNESS DESIGNS

Design	T	σ_I	σ_C	κ	χ	Actor radius
high	1	.60	.30	.15	.03	.10
low	1	.15	.10	.15	.03	.10
long	2	.30	.15	.15	.03	.10
stress	1	.30	.15	.30	.10	.20

All designs use dimensions 10 and 50, eight fixed streams per dimension, discount rate .04, productivity .10, adjustment parameter .20, and action bounds $[-.02, 2]$. The first three designs use nine initial profiles from means $\{-.5, 0, .5\}$ and dispersions $\{0, .25, .5\}$. Stress uses means $\{-.75, 0, .75\}$ and dispersions $\{0, .375, .75\}$. The feasible actor radius is measured around the reference schedule.

S.25.2. *HJB differentiation, selection, and deployment*

The two-layer tanh value network propagates its value, gradient, and covariance-contracted Hessian by the chain rule. The diffusion trace is exact for this network in

both dimensions, so training uses no random trace estimate. Independent automatic-differentiation checks compare both the network derivatives and their parameter gradients. The source validation records discrepancies at floating-point precision. Stochastic traces with 2, 8, and 32 probes are evaluated only on an untouched finite diagnostic bank after selection, for both covariance-Gaussian probes and idiosyncratic Rademacher probes with the common-noise term treated exactly.

Four candidates combine learning rates .0015 and .0005 with either reference occupation collocation or an equal mixture of reference occupation and broad collocation. Each uses up to 1,200 Adam updates and 150 L-BFGS iterations. Checkpoints are evaluated on independent occupation and broad residual banks; the combined residual criterion chooses the candidate and checkpoint. Its prescribed residual threshold is .001. Final payoff observations play no role in selection. Every candidate, including an unsuccessful or unselected candidate, is charged to the procedure.

The selected value network yields two retained actions. The greedy action uses the explicit first-derivative calculation and a bracketed scalar root solver with Newton updates; the prescribed bisection fallback remains part of the implementation. A width-32 actor is separately fitted to these actions. Independent diagnostic states record its action difference and Hamiltonian difference from the greedy action. These are finite-state diagnostics. They do not establish the economic loss of distillation, since the protected family does not contain a direct greedy-minus-distilled payoff contrast.

The diagnostic tables below retain all four configurations in all eight economic cells. Their means and ranges use all eight streams in each cell. The complete checkpoint histories, time-stratified residuals, selected network identities, root counters, and action diagnostics remain in the unchanged machine-readable report. A zero terminal discrepancy follows from the specified terminal lift. Residuals on finite independent banks and changes between iterates are reported with their corresponding scope; they are not uniform PDE residual bounds or errors against the unknown optimal value and policy.

TABLE S.42: All prespecified HJB candidate configurations

Design	d	Config.	Mean RMS	RMS range	Chosen	Stop	Fit sec.
high	10	O15	0.006102	[0.005151, 0.006967]	0/8	0/8	14.10
high	10	O05	0.006280	[0.005199, 0.007189]	0/8	0/8	14.08
high	10	M15	0.005500	[0.004842, 0.006109]	6/8	0/8	14.10
high	10	M05	0.005936	[0.005037, 0.006863]	2/8	0/8	14.19
high	50	O15	0.004162	[0.003410, 0.005158]	1/8	0/8	35.96
high	50	O05	0.006985	[0.004926, 0.010107]	0/8	0/8	35.75
high	50	M15	0.003714	[0.003368, 0.004140]	7/8	0/8	34.98
high	50	M05	0.004732	[0.003850, 0.007696]	0/8	0/8	34.61
low	10	O15	0.002043	[0.001504, 0.002651]	2/8	0/8	15.88
low	10	O05	0.002567	[0.001879, 0.003504]	0/8	0/8	15.90
low	10	M15	0.001684	[0.001263, 0.002372]	4/8	0/8	15.61
low	10	M05	0.002085	[0.001273, 0.002628]	2/8	0/8	15.75
low	50	O15	0.002916	[0.002202, 0.003680]	3/8	0/8	37.97
low	50	O05	0.004182	[0.003633, 0.004568]	0/8	0/8	38.28
low	50	M15	0.002770	[0.001780, 0.004655]	5/8	0/8	36.42
low	50	M05	0.004554	[0.003358, 0.006472]	0/8	0/8	37.06
long	10	O15	0.006106	[0.004142, 0.007295]	0/8	0/8	15.73
long	10	O05	0.006880	[0.005568, 0.009664]	0/8	0/8	15.97
long	10	M15	0.004367	[0.003966, 0.004945]	8/8	0/8	15.50
long	10	M05	0.005465	[0.004689, 0.006388]	0/8	0/8	15.82
long	50	O15	0.004367	[0.003918, 0.005369]	0/8	0/8	35.13
long	50	O05	0.009539	[0.006777, 0.011023]	0/8	0/8	35.27
long	50	M15	0.003349	[0.002680, 0.003739]	8/8	0/8	32.72
long	50	M05	0.010590	[0.005321, 0.013372]	0/8	0/8	33.10
stress	10	O15	0.005248	[0.004654, 0.006116]	0/8	0/8	16.57
stress	10	O05	0.006521	[0.005367, 0.008260]	0/8	0/8	16.55
stress	10	M15	0.004643	[0.003880, 0.005171]	7/8	0/8	16.35
stress	10	M05	0.005211	[0.004528, 0.005862]	1/8	0/8	16.22
stress	50	O15	0.004790	[0.004261, 0.005632]	0/8	0/8	34.63
stress	50	O05	0.006495	[0.005779, 0.007554]	0/8	0/8	34.49
stress	50	M15	0.003732	[0.003409, 0.004341]	8/8	0/8	33.85

Design	d	Config.	Mean RMS	RMS range	Chosen	Stop	Fit sec.
stress	50	M05	0.005291	[0.004535, 0.006117]	0/8	0/8	33.73

Each row retains all eight declared streams. O15 and O05 use occupation training collocation with learning rates 0.0015 and 0.0005; M15 and M05 use equal occupation and broad training collocation at the same two rates. Selection always gives equal weight to the separate occupation and broad selection banks. Mean RMS and its observed range use the best retained checkpoint of each candidate. Chosen counts residual-only family winners; Stop counts residual-only early stopping. Fit seconds are per-candidate components, already included in the complete shared HJB work in Table S.46. All 256 candidate fits and 2048 saved checkpoints are retained; no candidate failed and no final payoff selected a checkpoint or configuration.

TABLE S.43: Untouched HJB residual audits and selected-checkpoint iterate changes

Design	d	Occupation RMS	Broad RMS	ΔV RMS	Δu RMS
high	10	[0.003482, 0.004507]	[0.005814, 0.007266]	[0.002822, 0.008041]	[0.006681, 0.014383]
high	50	[0.002561, 0.003223]	[0.003911, 0.005019]	[0.001268, 0.003759]	[0.007258, 0.019404]
low	10	[0.000969, 0.001512]	[0.001476, 0.002424]	[0.002016, 0.013412]	[0.012218, 0.032167]
low	50	[0.001409, 0.002877]	[0.002093, 0.004012]	[0.002262, 0.007162]	[0.029766, 0.046553]
long	10	[0.002812, 0.003629]	[0.005056, 0.006162]	[0.005533, 0.011884]	[0.009387, 0.019914]
long	50	[0.002297, 0.003004]	[0.003169, 0.004497]	[0.005593, 0.015276]	[0.020734, 0.040882]
stress	10	[0.002724, 0.003714]	[0.004856, 0.006165]	[0.002291, 0.007751]	[0.010873, 0.027689]
stress	50	[0.002594, 0.003421]	[0.003962, 0.005011]	[0.001553, 0.002530]	[0.014736, 0.022117]

Each entry is the observed minimum and maximum over all eight streams in that economic cell. Residuals use the selected critic on untouched audit banks of 1024 occupation and 1024 broad states; these audit outcomes did not select a candidate. The ΔV and Δu columns use the saved selection-bank RMS changes from Adam step 1200 to the selected final L-BFGS checkpoint, which was selected in all 64 trials. They measure differences between successive saved iterates. They do not measure error against the unknown optimal value or policy. Ranges are descriptive, not confidence intervals or uniform residual bounds.

TABLE S.44: Finite-bank trace-probe variance for both retained estimators

Design	d	Probe law	Probes	Mean variance	Variance range
high	10	Gaussian	2	7.274e-05	[6.700e-05, 7.862e-05]
high	10	Gaussian	8	1.849e-05	[1.671e-05, 1.972e-05]
high	10	Gaussian	32	4.639e-06	[4.152e-06, 5.020e-06]
high	10	Rademacher	2	1.222e-05	[1.049e-05, 1.492e-05]
high	10	Rademacher	8	3.018e-06	[2.602e-06, 3.691e-06]
high	10	Rademacher	32	7.678e-07	[6.502e-07, 9.196e-07]
high	50	Gaussian	2	1.276e-05	[1.099e-05, 1.494e-05]
high	50	Gaussian	8	3.176e-06	[2.696e-06, 3.804e-06]
high	50	Gaussian	32	8.017e-07	[6.780e-07, 9.383e-07]
high	50	Rademacher	2	2.607e-06	[1.509e-06, 3.877e-06]
high	50	Rademacher	8	6.708e-07	[3.858e-07, 9.657e-07]
high	50	Rademacher	32	1.614e-07	[9.272e-08, 2.294e-07]
low	10	Gaussian	2	4.242e-07	[3.127e-07, 6.243e-07]
low	10	Gaussian	8	1.044e-07	[7.834e-08, 1.548e-07]
low	10	Gaussian	32	2.632e-08	[1.934e-08, 3.989e-08]
low	10	Rademacher	2	9.637e-08	[4.980e-08, 1.793e-07]
low	10	Rademacher	8	2.416e-08	[1.230e-08, 4.672e-08]
low	10	Rademacher	32	5.992e-09	[3.043e-09, 1.190e-08]
low	50	Gaussian	2	1.157e-07	[6.848e-08, 1.806e-07]
low	50	Gaussian	8	2.856e-08	[1.705e-08, 4.046e-08]
low	50	Gaussian	32	7.193e-09	[4.243e-09, 1.027e-08]
low	50	Rademacher	2	2.588e-08	[8.015e-09, 4.475e-08]
low	50	Rademacher	8	6.200e-09	[2.034e-09, 1.148e-08]
low	50	Rademacher	32	1.556e-09	[4.686e-10, 2.983e-09]
long	10	Gaussian	2	1.407e-05	[1.231e-05, 1.676e-05]
long	10	Gaussian	8	3.522e-06	[2.959e-06, 4.388e-06]
long	10	Gaussian	32	8.806e-07	[7.080e-07, 1.069e-06]
long	10	Rademacher	2	3.836e-06	[3.107e-06, 4.673e-06]
long	10	Rademacher	8	9.719e-07	[8.110e-07, 1.213e-06]
long	10	Rademacher	32	2.331e-07	[1.977e-07, 2.797e-07]
long	50	Gaussian	2	2.434e-06	[1.748e-06, 3.182e-06]
long	50	Gaussian	8	6.223e-07	[4.397e-07, 7.653e-07]
long	50	Gaussian	32	1.547e-07	[1.070e-07, 1.950e-07]

Design	d	Probe law	Probes	Mean variance	Variance range
long	50	Rademacher	2	1.075e-06	[6.689e-07, 1.605e-06]
long	50	Rademacher	8	2.716e-07	[1.780e-07, 3.970e-07]
long	50	Rademacher	32	6.835e-08	[4.474e-08, 1.046e-07]
stress	10	Gaussian	2	3.350e-05	[3.051e-05, 3.649e-05]
stress	10	Gaussian	8	8.262e-06	[7.296e-06, 9.575e-06]
stress	10	Gaussian	32	2.050e-06	[1.827e-06, 2.352e-06]
stress	10	Rademacher	2	4.510e-06	[4.023e-06, 5.437e-06]
stress	10	Rademacher	8	1.141e-06	[9.669e-07, 1.351e-06]
stress	10	Rademacher	32	2.806e-07	[2.440e-07, 3.421e-07]
stress	50	Gaussian	2	7.471e-06	[6.926e-06, 7.885e-06]
stress	50	Gaussian	8	1.887e-06	[1.790e-06, 2.059e-06]
stress	50	Gaussian	32	4.733e-07	[4.469e-07, 5.197e-07]
stress	50	Rademacher	2	7.426e-07	[5.281e-07, 9.103e-07]
stress	50	Rademacher	8	1.868e-07	[1.362e-07, 2.293e-07]
stress	50	Rademacher	32	4.708e-08	[3.412e-08, 5.920e-08]

Each row retains all eight streams. Within each saved trial, the unbiased variance across 64 independent probe banks is averaged over 64 fixed broad-audit states; the table then reports the mean and observed range across the eight trial statistics. The traced quantity is $\text{tr}(\Sigma \Sigma^\top D_x^2 V)$, before the generator factor $1/2$. Gaussian uses covariance-distributed directions. Rademacher uses idiosyncratic directions and evaluates the common-shock contraction exactly. Probe counts 2, 8 and 32 use nested prefixes within each bank. The fitted HJB models use exact diffusion contractions; neither these probe outcomes nor the untouched audit selected a checkpoint. The numbers describe the frozen finite banks and introduce no additional probability event.

S.25.3. Economic gains and complete process work

Table S.45 gives all forty mean method gains and their full-adapted-class regret upper bounds. Every mean lower endpoint exceeds .0005. The complete direct comparison is Table VII in the main article. Only the two low-volatility, fifty-dimensional NBO–HJB comparisons have lower endpoints above 10^{-4} ; the other thirty direct intervals cross zero. The intervals are conditional on the complete declared eight-

stream policy populations. They do not assert a law for unseen optimizer randomness or pool across economic designs.

TABLE S.45: Prospective robustness over economic calibrations

Design	d	Method	Mean gain	Simultaneous interval	Regret bound
high	10	NBO	0.001270	[0.000908, 0.001632]	0.026006
high	10	Raw	0.001270	[0.000908, 0.001632]	0.026006
high	10	Direct policy	0.001265	[0.000903, 0.001627]	0.026011
high	10	HJB greedy	0.001262	[0.000900, 0.001624]	0.026014
high	10	HJB actor	0.001259	[0.000898, 0.001621]	0.026016
high	50	NBO	0.001347	[0.000927, 0.001768]	0.036873
high	50	Raw	0.001349	[0.000929, 0.001770]	0.036871
high	50	Direct policy	0.001349	[0.000928, 0.001769]	0.036872
high	50	HJB greedy	0.001368	[0.000947, 0.001789]	0.036853
high	50	HJB actor	0.001308	[0.000887, 0.001729]	0.036913
low	10	NBO	0.001463	[0.001196, 0.001731]	0.020406
low	10	Raw	0.001463	[0.001196, 0.001731]	0.020406
low	10	Direct policy	0.001462	[0.001194, 0.001729]	0.020408
low	10	HJB greedy	0.001440	[0.001172, 0.001707]	0.020430
low	10	HJB actor	0.001440	[0.001172, 0.001708]	0.020430
low	50	NBO	0.001604	[0.001289, 0.001918]	0.029761
low	50	Raw	0.001604	[0.001289, 0.001918]	0.029761
low	50	Direct policy	0.001601	[0.001287, 0.001916]	0.029763
low	50	HJB greedy	0.001088	[0.000771, 0.001406]	0.030280
low	50	HJB actor	0.001082	[0.000764, 0.001400]	0.030286
long	10	NBO	0.008497	[0.006871, 0.010123]	0.191272
long	10	Raw	0.008495	[0.006869, 0.010121]	0.191274
long	10	Direct policy	0.008477	[0.006851, 0.010103]	0.191292
long	10	HJB greedy	0.008437	[0.006811, 0.010064]	0.191332
long	10	HJB actor	0.008425	[0.006799, 0.010052]	0.191344
long	50	NBO	0.008272	[0.006298, 0.010245]	0.293097
long	50	Raw	0.008275	[0.006301, 0.010248]	0.293094

Design	d	Method	Mean gain	Simultaneous interval	Regret bound
long	50	Direct policy	0.008266	[0.006292, 0.010239]	0.293103
long	50	HJB greedy	0.008008	[0.006032, 0.009985]	0.293363
long	50	HJB actor	0.007948	[0.005972, 0.009924]	0.293423
stress	10	NBO	0.005733	[0.004616, 0.006850]	0.176182
stress	10	Raw	0.005732	[0.004615, 0.006849]	0.176183
stress	10	Direct policy	0.005726	[0.004609, 0.006842]	0.176188
stress	10	HJB greedy	0.005712	[0.004595, 0.006829]	0.176202
stress	10	HJB actor	0.005695	[0.004578, 0.006811]	0.176219
stress	50	NBO	0.007444	[0.006119, 0.008770]	0.247977
stress	50	Raw	0.007448	[0.006122, 0.008773]	0.247974
stress	50	Direct policy	0.007425	[0.006100, 0.008750]	0.247996
stress	50	HJB greedy	0.007468	[0.006143, 0.008794]	0.247953
stress	50	HJB actor	0.007242	[0.005917, 0.008568]	0.248179

All four economic designs and all eight pre-drawn streams are retained. Each method receives 8192 independent confirmation paths on 2048 cells. Intervals share a prospectively fixed familywise allocation $\alpha = 0.01$ over 648 two-sided events. The full-adapted-class regret bound is displayed beside each gain; positive improvement does not imply near optimality. Endpoints are rounded outward.

Table S.46 reports the complete process work for all outputs and the number of individual-policy final lower endpoints exceeding .0005. A pooled positive mean is distinct from an individual policy's target certificate. In the high-volatility, ten-dimensional cell, all five pooled lower endpoints exceed .0005 while no individual policy's protected final lower endpoint does. In dimension fifty, the corresponding individual counts are 2 of 8 for NBO, 4 for Raw, 4 for direct policy, 8 for HJB's greedy deployment, and zero for its distilled actor. All of these outcomes remain in the comparison; a policy is not dropped because its final certificate does not attain the target.

The recorded costs include every fixed-envelope fit, independent selection, action construction, confirmation, and outer process overhead. Both HJB outputs retain their shared prerequisites and their own deployment cost. Their common construc-

tion is not divided to manufacture a cheaper actor. These are environment-specific observed times, with complete operation counters available in the report. They are not equal-FLOP comparisons, runtime confidence intervals, or newly observed online stopping times.

TABLE S.46: Complete fixed-envelope work and final target attainment

Design	d	Method	Mean seconds	Final target	Fallbacks
high	10	NBO	38.98	0/8	0
high	10	Raw	29.94	0/8	0
high	10	Direct policy	163.36	0/8	0
high	10	HJB greedy	108.91	0/8	0
high	10	HJB actor	85.70	0/8	0
high	50	NBO	96.38	2/8	0
high	50	Raw	86.11	4/8	0
high	50	Direct policy	279.87	4/8	0
high	50	HJB greedy	287.72	8/8	0
high	50	HJB actor	227.84	0/8	0
low	10	NBO	40.19	8/8	0
low	10	Raw	30.57	8/8	0
low	10	Direct policy	170.31	8/8	0
low	10	HJB greedy	117.10	8/8	0
low	10	HJB actor	92.77	8/8	0
low	50	NBO	102.27	8/8	0
low	50	Raw	91.29	8/8	0
low	50	Direct policy	310.13	8/8	0
low	50	HJB greedy	297.44	5/8	0
low	50	HJB actor	240.86	5/8	0
long	10	NBO	41.85	8/8	0
long	10	Raw	32.38	8/8	0
long	10	Direct policy	179.86	8/8	0
long	10	HJB greedy	119.14	8/8	0
long	10	HJB actor	93.89	8/8	0

Design	d	Method	Mean seconds	Final target	Fallbacks
long	50	NBO	99.13	8/8	0
long	50	Raw	88.43	8/8	0
long	50	Direct policy	304.95	8/8	0
long	50	HJB greedy	277.63	8/8	0
long	50	HJB actor	225.88	8/8	0
stress	10	NBO	42.27	8/8	0
stress	10	Raw	32.58	8/8	0
stress	10	Direct policy	185.96	8/8	0
stress	10	HJB greedy	124.05	8/8	0
stress	10	HJB actor	97.34	8/8	0
stress	50	NBO	100.17	8/8	0
stress	50	Raw	89.80	8/8	0
stress	50	Direct policy	306.69	8/8	0
stress	50	HJB greedy	282.68	8/8	0
stress	50	HJB actor	226.49	8/8	0

All streams enter the mean, including failed training and nonattainment. Target is 0.0005 at the single final decision. HJB greedy and its distilled actor share the complete four-candidate construction; each is charged that prerequisite and its own deployment, with all launch and finalization overhead. These are environment-specific measured costs, not equal FLOPs.

S.26. ECONOMIC ACTION CONTRASTS AND INDEPENDENT RISK ASSESSMENT

This section concerns the finite continuation economy defined in Section S.1.2. All expectations below use its unclipped Gaussian law and its stored economic coefficients. Conditioning on a fitted continuation is different from assuming a fixed feature dictionary before training. The results in this section permit nonlinear fitting, data-dependent features, and actions chosen by the fitted critic, provided the assessment innovations are independent of these choices.

Let $\mathcal{F}$ contain the complete training record, a finite collection of queries indexed by j , two feasible actions at each query, and every deployed prediction to be assessed.

Let x_j^+ and x_j^0 denote their postdecision states. Write

$$c_j = S(x_j^+) - S(x_j^0), \quad p_{Nj} = \widehat{S}_N(x_j^+) - \widehat{S}_N(x_j^0), \quad p_{Rj} \in \mathbb{R}.$$

Here S is the true continuation of the reference policy in the finite economy; p_{Rj} may be a realized cached simulation predictor. Conditional on $\mathcal{F}$, all these quantities are fixed. The risk uses the same action pair and the same query weights for both predictors:

$$\mathcal{R}_M = \frac{1}{J} \sum_{j=1}^J (p_{Mj} - c_j)^2, \quad M \in \{N, R\}. \quad (\text{S.93})$$

This definition does not compare the payoffs of two different actors. It does not average over unrealized Raw caches or unexecuted training seeds.

PROPOSITION S.25—Action contrasts and economic decisions: *Let r_j be the known current-reward difference between the two actions. The selector that chooses the first action when $r_j + p_{Mj} \geq 0$ has loss, relative to the better of these two actions, bounded by $|p_{Mj} - c_j|$. Consequently its average loss is at most $\sqrt{\mathcal{R}_M}$. An additive error of at most η_j in maximizing the predicted two-action objective adds η_j to the pointwise bound. For a finite action catalog with reference contrast $c_{j0} = p_{Mj0} = 0$, an η_j -approximate predicted maximizer instead has loss at most*

$$2 \max_a |p_{Mja} - c_{ja}| + \eta_j.$$

If the catalog risk at query j is $\sum_a w_{ja} (p_{Mja} - c_{ja})^2$ and $w_{ja} \geq \underline{w} > 0$ for every action, the last bound is at most twice the square root of this risk divided by $\underline{w}$, plus η_j .

PROOF: Put $v = r_j + c_j$ and $\widehat{v} = r_j + p_{Mj}$. When the selectors based on v and $\widehat{v}$ agree, the loss is zero. When they disagree, zero lies between v and $\widehat{v}$, so the loss $|v|$ is at most $|v - \widehat{v}|$. For an approximate choice, the same argument bounds a wrong choice by $|v - \widehat{v}| + \eta_j$. Averaging and applying Cauchy–Schwarz proves the risk bound. For a catalog let a^* maximize the true objective and let $\widehat{a}$ be the approximate predicted maximizer. Insert the predicted objective at a^* and $\widehat{a}$ into their true difference; the middle term is at most η_j and each of the other terms

is at most the maximum contrast error. Finally, each squared error is at most the weighted risk divided by its weight. This proves every assertion. *Q.E.D.*

The first assertion provides an economic unit for the risk without asserting that risk ordering determines payoff ordering. Such an assertion is false: when the true net contrast is .01, predictions $-.01$ and $.2$ have squared errors .0004 and .0361, respectively, but only the less accurate predictor chooses the better action. A risk comparison and an actual payoff comparison therefore remain separate empirical results. The two-action bound is not a certificate over all vector actions or over the full adapted diffusion class.

PROPOSITION S.26—Cancellation of common continuation noise: *Conditional on $\mathcal{F}$, let Z_{jk} be independent across queries and replicates, with finite second moments and $\mathbb{E}[Z_{jk} | \mathcal{F}] = c_j$. Put $d_j = p_{Nj} - p_{Rj}$ and*

$$T_{jk} = d_j(p_{Nj} + p_{Rj} - 2Z_{jk}).$$

Then the uniform average of T_{jk} is unbiased for $\mathcal{R}_N - \mathcal{R}_R$, and

$$\text{Var}(T_{jk} | \mathcal{F}) = 4d_j^2 \text{Var}(Z_{jk} | \mathcal{F}). \quad (\text{S.94})$$

An identical pair of predictions has identically zero comparison statistic. Neither an independent estimate of c_j^2 nor two independent assessment banks are required for this risk difference. For absolute risk, two independent replicates Z_{j1}, Z_{j2} give the unbiased statistic $(p_{Mj} - Z_{j1})(p_{Mj} - Z_{j2})$.

PROOF: The algebraic identity

$$(p_{Nj} - z)^2 - (p_{Rj} - z)^2 = (p_{Nj} - p_{Rj})(p_{Nj} + p_{Rj} - 2z)$$

removes the common z^2 term. Take conditional expectations and average. The only random component of the right side is $-2d_j Z_{jk}$, proving (S.94). For the last assertion, conditional independence factors the expectation of the product into $(p_{Mj} - c_j)^2$. Reusing one replicate instead would add its conditional variance. No assumption about how the predictions were fitted enters this argument, because fitting is included in $\mathcal{F}$. *Q.E.D.*

COROLLARY S.27—An economic contrast training objective: *For fixed training-query weights, the conditional expectation of $\sum_j w_j \{f(x_j^+) - f(x_j^0) - Z_j\}^2$ equals the corresponding true contrast risk plus $\sum_j w_j \text{Var}(Z_j \mid \mathcal{F})$. The latter does not depend on f . Thus minimizing the population loss over scalar functions minimizes contrast risk, while preserving the common scalar continuation and its path-consistent differences. The loss is invariant to an additive constant in f . A fresh assessment bank is required after empirical fitting or model selection; a training loss is not an independent risk certificate.*

PROOF: Expand the square around the conditional mean c_j . The cross term has zero conditional mean. Summing gives the identity. Additive constants cancel in each difference, and differences of a single scalar function telescope around every finite cycle. These are algebraic properties, not statistical claims. *Q.E.D.*

PROPOSITION S.28—Centered finite-sample comparison: *Fix q_j, b_j, t_j, e_j before assessment, with $b_j, t_j, e_j \geq 0$. Suppose the exact residual $U_{jk} = Z_{jk} - q_j$ obeys $\mathbb{E}(|U_{jk}| - b_j)_+ \leq t_j$. Suppose the executed residual $\tilde{U}_{jk}$ can be coupled with U_{jk} so that $\mathbb{E}|\tilde{U}_{jk} - U_{jk}| \leq e_j$. Define the known offset and the bounded random statistic by*

$$K_j = d_j(p_{Nj} + p_{Rj} - 2q_j), \quad W_{jk} = -2d_j \text{clip}_{b_j}(\tilde{U}_{jk}).$$

Then $|W_{jk}| \leq B := \max_j 2|d_j|b_j$ and the difference between the risk contrast and $J^{-1} \sum_j K_j + \mathbb{E}\bar{W}$ is at most $\varepsilon = J^{-1} \sum_j 2|d_j|(e_j + t_j)$. For $n = JK \geq 2$ independent observations with K replicates per query, a simultaneous two-sided interval for one comparison is

$$\frac{1}{J} \sum_j K_j + \bar{W} \pm \left\{ \sqrt{\frac{2s_W^2 \log(4/\alpha_*)}{n}} + \frac{14B \log(4/\alpha_*)}{3(n-1)} + \varepsilon \right\}, \quad (\text{S.95})$$

with coverage at least $1 - \alpha_$. For H prespecified comparisons, $\alpha_* \leq \alpha/H$ yields simultaneous coverage at least $1 - \alpha$. The observations need not be identically distributed.*

PROOF: Clipping is one-Lipschitz. The triangle inequality gives

$$\mathbb{E}|U_{jk} - \text{clip}_{b_j}(\tilde{U}_{jk})| \leq t_j + e_j.$$

Multiply by $2|d_j|$ and average to obtain the expectation allowance. The range is immediate. For $B > 0$, transform W from $[-B, B]$ to $[0, 1]$, apply [Maurer and Pon-](#)

1 [til](#) (2009, Theorem 11) to each tail with probability $\alpha_*/2$, and transform back. Its 1
 2 sample variance is the usual unbiased sample variance; the theorem allows indepen- 2
 3 dent nonidentical observations. The union bound proves the family assertion. For 3
 4 $B = 0$ the statistic is identically zero and only the known offset and deterministic 4
 5 allowance remain. Q.E.D. 5

6 The implementation takes q_j to be a representative of the known terminal 6
 7 quadratic contrast. The inherited finite-economy geometric proof supplies b_j, t_j 7
 8 before observation; its protected-arithmetic and Gaussian-clipping accounts supply 8
 9 e_j . Antithetic signs are averaged before they enter the sample: they are not counted 9
 10 as independent draws. Outward interval arithmetic also encloses the known offset, 10
 11 coefficient subtraction, coefficient multiplication, and final endpoint addition. The 11
 12 sample moments are calculated as exact rational moments of the stored binary64 12
 13 observations before directed Decimal evaluation of the radius. Identical predictors 13
 14 are assigned their exact zero identity, not an artificially positive comparison width. 14
 15

16 S.27. SHARP ACCURACY BOUNDS FROM SIMULTANEOUS PAYOFF CONTRASTS 16

17 Fix a finite nonempty set I of computed procedures, all evaluated on the same 17
 18 declared economic population. For each supplied pair (a, b) , let $[L_{ab}, U_{ab}]$ be a 18
 19 finite interval for $V_a - V_b$. All intervals are assumed to hold on one event $\mathcal{E}$ with 19
 20 probability at least $1 - \alpha$. No independence across intervals is required. Define 20
 21

$$22 \quad \mathcal{U} = \{v \in \mathbb{R}^I : L_{ab} \leq v_a - v_b \leq U_{ab} \text{ for every supplied pair}\}. 22$$

23 The graph has an edge $b \rightarrow a$ of length U_{ab} and an edge $a \rightarrow b$ of length $-L_{ab}$ for 23
 24 every supplied pair. It is connected in both directions in the application, because 24
 25 every candidate has a comparison with the reference. The convention for the empty 25
 26 path is length zero. 26
 27

28 PROPOSITION S.29—Exact catalogue closure: *Suppose the graph is strongly con-* 28
 29 *connected and contains no negative cycle. Let D_{ij} be its shortest-path lengths. Then $\mathcal{U}$ is* 29
 30 *nonempty, $D_{ii} = 0$, and* 30

$$31 \quad \sup_{v \in \mathcal{U}} (v_j - v_i) = D_{ij}, \quad (S.96) 31$$

$$\inf_{v \in \mathcal{U}} \{ \max_j v_j - v_i \} = \max_j (-D_{ji}), \quad (\text{S.97})$$

$$\sup_{v \in \mathcal{U}} \{ \max_j v_j - v_i \} = \max_j D_{ij}. \quad (\text{S.98})$$

All three extrema are attained. Normalizing the reference value to zero does not change them. On $\mathcal{E}$, these bounds hold simultaneously for every candidate and every pair, including pairs not directly compared in the original report.

PROOF: Every edge $a \rightarrow b$ of length w imposes $v_b - v_a \leq w$. Summing along a path and then minimizing yields $v_j - v_i \leq D_{ij}$ for every $v \in \mathcal{U}$. There are finite shortest paths because the graph is finite, strongly connected, and has no negative cycle. They satisfy the triangle inequality $D_{ib} \leq D_{ia} + D_{ab}$ and $D_{ii} = 0$. For fixed i , the vector $v_k^+ = D_{ik}$ satisfies each original edge constraint: if $a \rightarrow b$ has length w , then $D_{ib} \leq D_{ia} + w$. It therefore belongs to $\mathcal{U}$ and attains $v_j^+ - v_i^+ = D_{ij}$ simultaneously for all j . This proves (S.96) and (S.98).

For the lower bound, reversing the pairwise inequality gives $v_j - v_i \geq -D_{ji}$. The vector $v_k^- = -D_{ki}$ is also feasible, since the same edge implies $D_{ai} \leq w + D_{bi}$ and hence $v_b^- - v_a^- = D_{ai} - D_{bi} \leq w$. It simultaneously attains $v_j^- - v_i^- = -D_{ji}$, proving (S.97). Both extrema include $j = i$, so both regret bounds are nonnegative. Subtracting the reference coordinate leaves every difference unchanged. Finally, on $\mathcal{E}$ the true payoff vector belongs to $\mathcal{U}$; all deterministic conclusions therefore hold on that same event. Q.E.D.

Sharpness here is relative to the interval-defined uncertainty set. It does not assert that every compatible vector can be generated by the underlying capital model, or that the original statistical intervals are optimal. A negative cycle makes $\mathcal{U}$ empty. The implementation stops with an inconsistency diagnostic; it does not delete an unfavorable interval, truncate a negative cycle, or report an accuracy certificate in that case.

COROLLARY S.30—Selection with an economic tolerance: Let $c_i \geq 0$ be observed accounting costs and $\tau \geq 0$. Define $\mathcal{C}_\tau = \{i : \max_j D_{ij} \leq \tau\}$. When this set is nonempty, choose a deterministic tie-broken minimizer $\hat{i}$ of c_i over $\mathcal{C}_\tau$. Then on $\mathcal{E}$, $\max_j V_j - V_{\hat{i}} \leq \tau$

and $c_{\hat{i}} \leq c_k$ for every $k \in \mathcal{C}_\tau$. The accuracy assertion remains valid for a tolerance chosen after inspecting the intervals, interpreted as a simultaneous assertion for all tolerances. A common nonnegative assessment charge preserves the cost ordering.

PROOF: On $\mathcal{E}$ all candidate bounds hold at once. Membership of the chosen candidate in $\mathcal{C}_\tau$ gives its regret bound irrespective of how the index or tolerance was selected. The cost inequality is the definition of the minimizer; addition of the same charge to both sides preserves it. If $\mathcal{C}_\tau$ is empty, the rule returns no certified candidate. *Q.E.D.*

The corollary gives no assertion about c_k for an uncertified candidate with unknown true regret below τ . Observed cost is not an expectation over future machines or training runs. Selecting after the full computation does not save the work already spent computing the competing candidates. In particular, a retrospective least certified prefix allocation is not a measured work-to-target stopping time. These distinctions are necessary when the same continuation is amortized across an economic decision menu.

S.27.1. Exact arithmetic and the complete input family

The computation reads all 216 original protected intervals: eight economic cells, three stages, five gains relative to the common reference, and four direct NBO contrasts at each stage. The stored binary64 endpoints are interpreted as exact rational numbers. Floyd–Warshall updates use only rational addition, ordering, and minima. Final lower endpoints are rounded downward and upper endpoints upward. The reference plus fifteen method-stage candidates are retained in every cell, including costly and dominated candidates. All means use the same sixteen streams; no stream is selected on the basis of the analysis. The calculation adds no observations and inherits the original .018 payoff-family failure probability.

S.28. INDEPENDENT RISK AND CATALOGUE EVIDENCE

The continuation-risk assessment was generated from frozen source and retained in evidence commit . The present revision independently reconstructs its complete report from the stored predictions, audit statistics, and

pre-audit receipts. The replay is byte-for-byte identical to the 24-event report. It does not retrain the policies or replace the original innovations. Each of the eight cells has $16 \times 384 \times 256 = 1,572,864$ independent antithetic-pair observations. All three Raw budgets use the same assessment observations within a cell. The registered geometric bounds, known-offset enclosures, empirical Bernstein allowances, and exact moment calculations remain in the machine-readable report, together with all original cell work records.

The catalogue calculation uses the complete original menu report, with SHA256 beginning `2a729aff851dc706`. Every original gain and direct contrast remains current. It produces 128 candidate records and 8×16^2 closed pairwise records. The main table rounds regret bounds upward; the following complete table rounds lower endpoints downward and upper endpoints upward. Costs are displayed descriptively and are not statistical confidence bounds. The reference's zero construction allocation does not claim that evaluating it is computationally free.

TABLE S.47: Complete Candidate Accuracy and Accounted Work

Design	Candidate	Lower	Upper	Seconds	Clock
Low 10	Reference	33.920	33.960	0.000	Base
Low 10	NBO (1)	0	0.039412	5.284	Prefix
Low 10	Vector (1)	0.0074173	0.047303	4.450	Prefix
Low 10	Raw actor (1)	0.54142	0.58579	4.139	Prefix
Low 10	DPO (1)	0.54214	0.58636	7.207	Prefix
Low 10	SAA (1)	0	0.039348	3.474	Prefix
Low 10	NBO (2)	0	0.039098	8.006	Prefix
Low 10	Vector (2)	0.00053842	0.039808	6.314	Prefix
Low 10	Raw actor (2)	0.092878	0.13402	5.481	Prefix
Low 10	DPO (2)	0.092228	0.13320	11.670	Prefix
Low 10	SAA (2)	0	0.039122	5.883	Prefix
Low 10	NBO (3)	0	0.038830	12.524	Full
Low 10	Vector (3)	0.00011639	0.039052	8.734	Full
Low 10	Raw actor (3)	0.066305	0.10653	6.856	Full

Design	Candidate	Lower	Upper	Seconds	Clock
Low 10	DPO (3)	0.064921	0.10496	16.213	Full
Low 10	SAA (3)	0	0.038822	15.602	Full
Low 50	Reference	33.269	33.305	0.000	Base
Low 50	NBO (1)	0.092481	0.12931	5.252	Prefix
Low 50	Vector (1)	0.17847	0.21704	4.491	Prefix
Low 50	Raw actor (1)	1.5744	1.6178	3.971	Prefix
Low 50	DPO (1)	1.5677	1.6109	8.424	Prefix
Low 50	SAA (1)	0	0.035621	3.966	Prefix
Low 50	NBO (2)	0.037405	0.073230	8.211	Prefix
Low 50	Vector (2)	0.099862	0.13694	6.609	Prefix
Low 50	Raw actor (2)	0.30830	0.34709	5.379	Prefix
Low 50	DPO (2)	0.30561	0.34422	14.213	Prefix
Low 50	SAA (2)	0	0.035241	8.764	Prefix
Low 50	NBO (3)	0.023969	0.059449	12.673	Full
Low 50	Vector (3)	0.062895	0.099169	9.846	Full
Low 50	Raw actor (3)	0.27437	0.31227	7.069	Full
Low 50	DPO (3)	0.26619	0.30390	20.242	Full
Low 50	SAA (3)	0	0.035037	27.685	Full
Quarterly 10	Reference	109.61	109.97	0.000	Base
Quarterly 10	NBO (1)	1.1163	1.8209	4.714	Prefix
Quarterly 10	Vector (1)	2.6062	3.3381	4.001	Prefix
Quarterly 10	Raw actor (1)	0.74872	1.4617	3.762	Prefix
Quarterly 10	DPO (1)	0.60477	1.3107	8.847	Prefix
Quarterly 10	SAA (1)	0.65412	1.3613	3.822	Prefix
Quarterly 10	NBO (2)	0.16161	0.75034	7.050	Prefix
Quarterly 10	Vector (2)	0.33971	0.96446	5.588	Prefix
Quarterly 10	Raw actor (2)	0.12810	0.75423	4.854	Prefix
Quarterly 10	DPO (2)	0.055258	0.67495	15.217	Prefix
Quarterly 10	SAA (2)	0	0.54670	8.088	Prefix
Quarterly 10	NBO (3)	0.017217	0.18475	10.358	Full
Quarterly 10	Vector (3)	0.0029464	0.18530	7.694	Full
Quarterly 10	Raw actor (3)	0.22295	0.41669	6.090	Full

Design	Candidate	Lower	Upper	Seconds	Clock
Quarterly 10	DPO (3)	0.14048	0.32821	21.853	Full
Quarterly 10	SAA (3)	0	0.15040	25.232	Full
Quarterly 50	Reference	103.15	103.50	0.000	Base
Quarterly 50	NBO (1)	2.7918	3.5059	5.830	Prefix
Quarterly 50	Vector (1)	5.2230	5.9356	4.982	Prefix
Quarterly 50	Raw actor (1)	3.0362	3.7480	4.384	Prefix
Quarterly 50	DPO (1)	2.8267	3.5226	14.580	Prefix
Quarterly 50	SAA (1)	0.75411	1.4471	5.959	Prefix
Quarterly 50	NBO (2)	0.60384	1.1779	9.149	Prefix
Quarterly 50	Vector (2)	1.4782	2.0914	7.390	Prefix
Quarterly 50	Raw actor (2)	1.4840	2.1116	6.050	Prefix
Quarterly 50	DPO (2)	1.3713	1.9877	26.400	Prefix
Quarterly 50	SAA (2)	0	0.54412	17.394	Prefix
Quarterly 50	NBO (3)	0.36728	0.52341	14.340	Full
Quarterly 50	Vector (3)	0.58949	0.76773	11.147	Full
Quarterly 50	Raw actor (3)	1.1959	1.3885	8.282	Full
Quarterly 50	DPO (3)	1.1310	1.3136	38.928	Full
Quarterly 50	SAA (3)	0	0.14011	62.898	Full
Long 10	Reference	528.60	531.26	0.000	Base
Long 10	NBO (1)	31.272	37.685	5.379	Prefix
Long 10	Vector (1)	38.450	44.696	4.573	Prefix
Long 10	Raw actor (1)	14.877	20.944	4.135	Prefix
Long 10	DPO (1)	4.2917	9.9053	10.606	Prefix
Long 10	SAA (1)	10.530	16.241	4.680	Prefix
Long 10	NBO (2)	8.0339	13.897	8.230	Prefix
Long 10	Vector (2)	13.812	19.664	6.413	Prefix
Long 10	Raw actor (2)	7.7795	13.627	5.473	Prefix
Long 10	DPO (2)	0.14222	5.5406	18.455	Prefix
Long 10	SAA (2)	0.066603	5.4645	9.757	Prefix
Long 10	NBO (3)	1.2243	1.9834	12.047	Full
Long 10	Vector (3)	1.3742	2.8964	8.870	Full
Long 10	Raw actor (3)	6.9010	8.6738	6.956	Full

Design	Candidate	Lower	Upper	Seconds	Clock
Long 10	DPO (3)	0.20001	1.7890	26.533	Full
Long 10	SAA (3)	0	0	30.423	Full
Long 50	Reference	521.34	524.79	0.000	Base
Long 50	NBO (1)	53.460	61.028	5.807	Prefix
Long 50	Vector (1)	100.35	107.33	4.927	Prefix
Long 50	Raw actor (1)	41.457	49.190	4.300	Prefix
Long 50	DPO (1)	22.088	29.185	14.180	Prefix
Long 50	SAA (1)	11.223	18.279	7.010	Prefix
Long 50	NBO (2)	15.793	23.126	9.257	Prefix
Long 50	Vector (2)	40.485	47.305	7.292	Prefix
Long 50	Raw actor (2)	25.304	32.829	5.953	Prefix
Long 50	DPO (2)	8.4504	15.334	25.776	Prefix
Long 50	SAA (2)	0	6.4653	23.279	Prefix
Long 50	NBO (3)	1.8944	2.6878	14.589	Full
Long 50	Vector (3)	18.276	20.802	11.060	Full
Long 50	Raw actor (3)	19.920	22.397	8.158	Full
Long 50	DPO (3)	5.1676	6.9273	38.029	Full
Long 50	SAA (3)	0	0.48925	67.039	Full
Intermediate 10	Reference	155.53	156.09	0.000	Base
Intermediate 10	NBO (1)	0.24743	1.3706	5.399	Prefix
Intermediate 10	Vector (1)	1.0785	2.2161	4.599	Prefix
Intermediate 10	Raw actor (1)	0.71554	1.8367	4.154	Prefix
Intermediate 10	DPO (1)	0.53629	1.6510	13.847	Prefix
Intermediate 10	SAA (1)	0.17938	1.3026	4.966	Prefix
Intermediate 10	NBO (2)	0.034494	0.76770	8.187	Prefix
Intermediate 10	Vector (2)	0.065200	0.83813	6.519	Prefix
Intermediate 10	Raw actor (2)	0.27506	1.0623	5.520	Prefix
Intermediate 10	DPO (2)	0.15846	0.93711	25.075	Prefix
Intermediate 10	SAA (2)	0	0.69081	12.686	Prefix
Intermediate 10	NBO (3)	0	0.42816	11.960	Full
Intermediate 10	Vector (3)	0	0.43595	9.160	Full
Intermediate 10	Raw actor (3)	0.25689	0.72000	7.007	Full

Design	Candidate	Lower	Upper	Seconds	Clock
Intermediate 10	DPO (3)	0.15336	0.60586	36.546	Full
Intermediate 10	SAA (3)	0	0.42297	44.710	Full
Intermediate 50	Reference	145.24	145.92	0.000	Base
Intermediate 50	NBO (1)	2.9324	4.2832	5.085	Prefix
Intermediate 50	Vector (1)	5.0197	6.3588	4.240	Prefix
Intermediate 50	Raw actor (1)	4.5218	5.8964	3.801	Prefix
Intermediate 50	DPO (1)	4.0032	5.3436	16.429	Prefix
Intermediate 50	SAA (1)	0.15973	1.5070	7.825	Prefix
Intermediate 50	NBO (2)	0.64766	1.5434	8.010	Prefix
Intermediate 50	Vector (2)	2.0881	3.0663	6.247	Prefix
Intermediate 50	Raw actor (2)	2.4644	3.4656	5.188	Prefix
Intermediate 50	DPO (2)	2.2092	3.1901	30.636	Prefix
Intermediate 50	SAA (2)	0	0.84542	22.003	Prefix
Intermediate 50	NBO (3)	0.39510	0.91391	12.699	Full
Intermediate 50	Vector (3)	1.0642	1.6368	9.623	Full
Intermediate 50	Raw actor (3)	1.8743	2.4760	7.182	Full
Intermediate 50	DPO (3)	1.7050	2.2800	45.584	Full
Intermediate 50	SAA (3)	0	0.48982	79.459	Full

Notes: Regret bounds are in units of 10^{-4} . Full denotes the actual complete construction/query process; Prefix denotes its allocated early-stage component; Base denotes the zero reference construction allocation. No cost confidence interval is claimed.

S.28.1. Confidence families and chronology

The original-policy confirmation, stronger-HJB robustness, and signed Bellman assessments each have allocation .01; the finite menu has .02, split into .018 for payoff and .002 for scalar accuracy. The independent contrast-risk family adds .01. Thus the entire five-family collection has a union-bound failure probability at most .06, not .05. The central menu payoff and risk statements together have failure

TABLE S.48

COMPLETE COMPARATIVE WORK RETAINED AFTER SELECTION

Design	d	Shared assessment	All construction plus assessment
Low	10	10.733	70.662
Low	50	29.497	107.012
Quarterly	10	15.899	87.127
Quarterly	50	63.458	199.054
Long	10	17.763	102.593
Long	50	64.373	203.247
Intermediate	10	24.818	134.201
Intermediate	50	82.196	236.741

Notes: Mean seconds over the complete sixteen-stream population. The shared assessment is the original complete five-method, three-stage confirmation process. The last column also includes all five final construction/query processes. It is not replaced by the selected procedure's construction cost.

probability at most .028. The catalogue closure spends no additional probability because it holds deterministically whenever all original menu payoff intervals hold.

The original dimension-ten HJB refinement remains a post-freeze analysis of old observations. The later original-policy confirmation fixed the refined analysis before new observations. The contrast-risk assessment also fixed its source and within-cell prediction record before its own fresh audit innovations. The catalogue closure was developed after the menu and risk outcomes were available. Its validity follows from the already simultaneous family, not from describing this later selection as prospectively registered. No reported training run, failure, contrary comparison, or earlier stopping decision is replaced by the new analysis.

S.29. SOURCE PRESERVATION AND EVIDENCE INDEX

The reviewed R15 publication is repository commit

1cb3cc9efe135966ec228dfaf51c6841a6dfc97c.

The subsequent referee report is pinned to commit

cb4595bbcc7147e47e44ba40cb2f5034510ae19f.

The report was commissioned by the repository owner as an AI-assisted advisory review. It is not represented as a journal decision.

The exact R15 main, supplement, and repository guide are preserved in this revision's archive. The main and supplement have Git blob identities

32953ff2500f3af8a491cf1dd710276d2d77bdbf,

50eac8d04b25b6521780d4e07b9617b377168e88.

The archive manifest checks their SHA256 and Git blob identities. The source inventory records the 70 files in the reviewed article's complete input closure. The editorial map locates every one of its 285 equation, result, section, algorithm, and table labels in the current main article, technical supplement, or applications companion. No reviewed label is assigned only to the archive.

The mathematical preservation check compares every explicit theorem, proposition, assumption, corollary, and proof environment and every proof section identified by its heading. It also checks whole retained components. Its normalization permits comments, whitespace, relocation of input paths, and an enumerated set of location and grammar corrections; it does not permit changes in mathematical content. Introductory synthesis, literature organization, conclusion, and the source index are edited anew, while the mathematical statements and complete evidence to which they refer remain current. The single current NBO algorithm is retained.

The machine-readable response map assigns all eight blocking and ten major referee comments to specific work, evidence, and scientific closure conditions. Completion of a reply is not itself evidence that a scientific comparison is resolved. The compilation record identifies the four current PDFs, their source roots, page counts, and digests. Independent numerical reports preserve their source freezes and bank identities at their original repository paths.

S.30. DETAILED CAPITAL AND IMPLEMENTATION ACCOUNTS

The following four components are retained without textual edits from the previous main article. They supply the full capital specification, operational Bellman bridge, implementation account, and fee normalization. Their relocation places the common economic question and complete primary comparisons first; it removes no theorem, proof block, economic model, or numerical observation.

S.31. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule. Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z}1$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - \nu^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (\text{S.99})$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{\overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2\} dt + e^{-\rho T} \{\bar{Y}_T - \chi \|\Pi Y_T\|_d^2\} \right], \quad (\text{S.100})$$

where $\nu^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\sigma_I = 0.15$, $\sigma_C = 0.10$, $\zeta = 0.2$, $\chi = 0.03$, and $[\underline{m}, M] = [0.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (\text{S.101})$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. For the analytical comparison, keep m_0 interior and assume $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, and let $\beta \geq \kappa \|B\|_2$ and $s_0 \geq \|\Pi y\|_d$. Choose $\epsilon \geq 0$ so that $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$ at every date. With $V^*(y) = \sup_m J_y(m)$ over the full adapted admissible class, the schedule gap obeys $0 \leq V^*(y) - J_y(m_0) \leq G_d$. The exact scalar integral defining G_d is (S.28); Appendix S.9 gives its curvature and dispersion constants, Theorem S.9, and the complete proof. This analytical bound supplies a common economic reference. The policy-specific endpoint below measures what the fitted actor adds to that reference.

S.31.1. The deployed neural map and its numerical guard

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (\text{S.102})$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (\text{S.103})$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its

interval midpoint and is not evidence of successful training. This scalar feasibility guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

S.31.2. Policy-Specific Verification after Bellman Improvement

The bound in Theorem S.9 is a useful starting point, but not a measure of what training accomplishes. Every policy in the same tube receives the same allowance, including a poorly trained policy. We now replace that allowance by an independently verified, policy-specific payoff difference. The economic primitives, initial state, and full adapted comparison class in (S.99)–(S.100) are unchanged. The implementation to which the new bound applies is specified explicitly below. It is not identified with continuously observed-state feedback.

S.31.2.1. The deployed controller

Fix a deterministic grid $t_k = kh$, $Nh = T$. The controller has an internal state Z_k , initially y , and observes the Brownian innovations at the grid dates. At date t_k it selects

$$a_k = \mathcal{P}_k\{m_\phi(t_k, Z_k)\}, \quad Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\Delta\tilde{W}_k, \quad (\text{S.104})$$

where $c = a - v^2/2$, $f(z) = \kappa \tanh(Bz)$, and $\mathcal{P}_k$ projects inward onto the declared action interval around $m_0(t_k)$. The vector $\Delta\tilde{W}_k$ clips each standardized innovation at ten before rescaling by $\sqrt{h}$. The numerical implementation and its rounding convention form part of the controller. Between dates it holds a_k fixed. Its economic state, denoted Y^ϕ , nevertheless satisfies

$$dY_t^\phi = \{c\mathbf{1} + f(Y_t^\phi) - a_k\} dt + \Sigma dW_t, \quad t_k \leq t < t_{k+1}.$$

In particular, the economic Brownian motion and state are not clipped, stopped, or replaced by Z . The controller is adapted to the original innovation filtration and remains feasible. Observing innovations is an implementation assumption, not a conclusion inferred from a state-only observation model. The continuously evaluated absolute-bound study policies remain separate objects in the supplement.

This specification has an important analytical consequence. When the numerical state and the economic state are compared, they use exactly the same chosen actions. Their state discrepancy therefore depends on the smoothness of the economic drift, not on a global Lipschitz bound for the learned network. The following construction uses this fact rather than inserting sampled residuals into a uniform verification theorem.

S.31.2.2. An exact improvement identity

Write $W(t) = e^{-\rho t}w(t)$ and define the nonnegative consumption deficit

$$R_t(m) = \log m_0(t) - \overline{\log m} + w(t)\{\bar{m} - m_0(t)\} + \frac{\zeta}{2}\{\bar{m}^2 - m_0(t)^2\}. \quad (\text{S.105})$$

The first-order identity $m_0^{-1} = w + \zeta m_0$ and concavity of the logarithm imply $R_t(m) \geq 0$. The exact mean-state identity in the proof of Theorem S.9 gives

$$\begin{aligned} \Delta_\phi := J_y(a^\phi) - J_y(m_0) &= \mathbb{E} \int_0^T W(t) \overline{f(Y_t^\phi) - f(Y_t^0)} dt - \mathbb{E} \int_0^T e^{-\rho t} R_t(a_t^\phi) dt \\ &\quad - \chi e^{-\rho T} \mathbb{E} \{ \|\Pi Y_T^\phi\|_d^2 - \|\Pi Y_T^0\|_d^2 \}. \end{aligned} \quad (\text{S.106})$$

Thus state-dependent consumption has to earn its utility-curvature cost through production or terminal dispersion. The identity is also an effective variance reduction: the common mean-state martingale cancels before simulation statistics are formed. Positive neural output is not itself evidence of positive economic improvement.

For each cell, calculate the scalar integrals

$$\begin{aligned} A_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, & B_k &= \int_{t_k}^{t_{k+1}} W(t) dt, \\ C_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} \{ \log m_0(t) - w(t)m_0(t) - \zeta m_0(t)^2/2 \} dt. \end{aligned}$$

The consumption-deficit integral for a held action is exactly $C_k - A_k \overline{\log a_k} + B_k \bar{a}_k + \zeta A_k \bar{a}_k^2/2$. Consequently, consumption need not be approximated by a left-point utility sum. Let $X_{\phi,h}$ be the paired statistic obtained from (S.106) by using the internal Euler states in the production and terminal terms and these integrated

consumption terms. The reference internal state uses the integrated common action $\int_{t_k}^{t_{k+1}} m_0(t)dt$. All scalar integral enclosures and their midpoint errors are saved.

S.31.2.3. *Transferring the computed payoff to the economy*

The statistic $X_{\phi,h}$ is evaluated for the held controller in (S.104), using the same actions when comparing its internal state with its economic diffusion. Proposition S.10 in Appendix S.9 bounds $|\Delta_\phi - \mathbb{E}X_{\phi,h}|$ by the saved allowance $E_{\phi,h}^{\text{all}}$. The exact-arithmetic diffusion allowance is $E_\epsilon(h) + E_0(h)$, with E_ϵ defined in (S.37). The appendix gives the additional clipped-innovation, forward-arithmetic, and enclosed-integration terms. The $O(h)$ diffusion bound uses bounded actions, the global drift and derivative bounds of the specified hyperbolic-tangent production function, and additive diffusion; it requires no derivative bound for the trained network.

S.31.2.4. *A post-training certificate*

Use the deterministic clipping threshold b_ϕ and expectation-tail allowance τ_ϕ constructed in Appendix S.9.4. They account for the unbounded Gaussian terminal-dispersion term and are not estimated from the final sample range. Let $X_{\phi,h}^c$ be $X_{\phi,h}$ clipped to $[-b_\phi, b_\phi]$, so that $|\mathbb{E}X_{\phi,h} - \mathbb{E}X_{\phi,h}^c| \leq \tau_\phi$.

For n fresh independent paths, let $\bar{X}_\phi^c$ and s_ϕ^2 be the clipped empirical mean and unbiased variance. Fix a family of at most F one-sided statements before inspecting this noise bank, and set $\ell = \log(2F/\alpha)$. Define

$$L_\phi = \bar{X}_\phi^c - \sqrt{2s_\phi^2 \ell / n} - \frac{14b_\phi \ell}{3(n-1)} - E_{\phi,h}^{\text{all}} - \tau_\phi. \quad (\text{S.107})$$

In the implementation the empirical mean is enclosed downward and the variance and every subtracted quantity upward. The upper payoff endpoint uses the same terms with reversed signs. The factor F counts both tails when both are reported.

THEOREM S.31 —Policy-specific continuous-time certificate: *Condition on the training and validation data, and suppose the specified final simulation paths are independent draws from the declared innovation law. Under the arithmetic contract and transfer assumptions above, with probability at least $1 - \alpha$, simultaneously for all declared policies and initial states,*

$$J_y(a^\phi) - J_y(m_0) \geq L_\phi, \quad 0 \leq V^*(y) - J_y(a^\phi) \leq G_d - L_\phi. \quad (\text{S.108})$$

The optimum V^ still ranges over all adapted feasible controls in the original continuous economy, not merely the networks, schedules, or sampled controllers used in the experiment. Common shocks may be reused across policies; independence across those policies is not required. The guarantee is pointwise in each declared initial state and conditional on training, rather than uniform on an uncountable state domain.*

PROOF: Apply the empirical Bernstein inequality of [Maurer and Pontil \(2009, Theorem 4\)](#) to $(X_{\phi,h}^c + b_\phi)/(2b_\phi)$ and use a union bound over the declared one-sided statements. Rescaling gives the first two error terms in (S.107). The clipping expectation bound and Proposition S.10 then give the first inequality in (S.108). Subtract it from the deterministic anchor bound (S.28). Feasibility yields nonnegative regret. An observed negative upper regret endpoint would indicate failure of the confidence event or of an implementation assumption, not a negative optimality gap. Q.E.D.

Unlike $G_d + D_{d,\epsilon}$, the new endpoint depends on the deployed policy through its paired outcomes and sample variance. It can distinguish seeds and unsuccessful policies in the same architectural tube. When $L_\phi > 0$, the post-training guarantee is tighter than the analytical schedule’s guarantee. This is a statement about the verified policy, not a deterministic promise that every training run improves welfare. The concentration inequality, the identity that transfers a lower improvement bound into a regret bound, and conservative policy improvement are established ideas ([Kakade and Langford, 2002](#), [Pirotta et al., 2013](#), [Scherrer et al., 2015](#)). The contribution here is their operational connection to the continuous capital economy through (S.106) and the transfer bound (S.37), without a state-covering certificate or a global bound on the trained network’s derivatives.

S.31.3. *An operational evaluation–improvement account*

The costate in a rollout is the derivative of a specified continuation problem. Its approximation error has to be measured where the new policy uses that continuation. We now make both objects explicit. The construction uses the capital economy, including its dense production interaction, common and idiosyncratic shocks, consumption curvature, and terminal dispersion preference. It also accommodates a critic fitted on a coarser grid than the one used for assessment.

Fix the assessment grid $t_k = kh$, $Nh = T$. Write $\pi_k = \bar{M}_k/h$, where $\bar{M}_k$ is the disclosed numerical representative of $\int_{t_k}^{t_{k+1}} m_0(t)dt$. In the following finite problem, these representatives are fixed real numbers. They specify a feasible, state-independent reference. The internal transition and transformed reward are

$$Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\sqrt{h} \text{clip}_z(G_k), \quad G_k \sim N(0, I_{d+1}), \quad (\text{S.109})$$

$$\ell_k(z, a) = B_k \overline{f(z)} + A_k \overline{\log a} - B_k \bar{a} - \frac{\zeta A_k}{2} \bar{a}^2, \quad g_N(z) = -\chi e^{-\rho T} \|\Pi z\|_d^2. \quad (\text{S.110})$$

Here A_k, B_k are the integrated weights defined above; all discounting is included in them. Let $\mathcal{R}_h(a)$ be the expected sum of these rewards and g_N . The candidate may depend on its entire observed innovation history. Its actions and the reference belong to the same declared convex action box $\mathcal{A}_k$ at each date.

Let V_k^π be the value of this finite problem under π , and define its continuation after choosing the deterministic drift by

$$S_k^\pi(x) = \mathbb{E}[V_{k+1}^\pi\{x + \Sigma\sqrt{h} \text{clip}_z(G_k)\}], \quad q_k^\pi = dD_x S_k^\pi. \quad (\text{S.111})$$

The argument x is the conditional mean before the current innovation. The current reward ℓ_k is excluded from S_k^π ; subsequent rewards start at $k+1$ on the same global grid. In particular, $q_{N-1}^\pi(x) = -2\chi e^{-\rho T} \Pi x$. Smoothness of f , bounded actions, finite moments, and the quadratic terminal reward justify the derivatives and expectations in this finite problem.

After fitting, freeze an arbitrary continuously differentiable scalar predictor $\hat{S}_k$ and its costate $\hat{q}_k = dD\hat{S}_k$. Set

$$x_b(z) = z + h\{c\mathbf{1} + f(z) - b\}, \quad \hat{Q}_k(z, b) = \ell_k(z, b) + \hat{S}_k\{x_b(z)\}.$$

For U uniform on $[0, 1]$, independent of the candidate's past, the *action bridge* is $X_{k,U} = x_{\pi_k}(Z_k) + U\{x_{a_k}(Z_k) - x_{\pi_k}(Z_k)\}$. Define the following quantities under the candidate's internal-state law:

$$\begin{aligned}\mathcal{M}_h &= \sum_{k=0}^{N-1} \mathbb{E} [\hat{Q}_k(Z_k, a_k) - \hat{Q}_k(Z_k, \pi_k)], \\ \mathcal{A}_h &= \sum_{k=0}^{N-1} h \mathbb{E} \|a_k - \pi_k \mathbf{1}\|_d^2, \quad \mathcal{E}_h = \sum_{k=0}^{N-1} h \mathbb{E} \|\hat{q}_k(X_{k,U}) - q_k^\pi(X_{k,U})\|_d^2.\end{aligned}\tag{S.112}$$

$\mathcal{M}_h$ records the improvement predicted by the fitted scalar continuation, $\mathcal{A}_h$ records the size of the implemented change, and $\mathcal{E}_h$ measures the error that can affect this particular change. Training-state accuracy alone does not identify $\mathcal{E}_h$.

PROPOSITION S.32—Finite Bellman bridge: *If the displayed quantities are integrable, then*

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) \geq \mathcal{M}_h - \sqrt{\mathcal{A}_h \mathcal{E}_h}.\tag{S.113}$$

The assertion holds for every feasible candidate. It requires neither exact maximization of $\hat{Q}$ nor a continuous-time costate approximation. A predictor fitted on another mesh enters $\mathcal{E}_h$ with its full discrepancy from the assessment-grid continuation.

The proposition follows from the finite-horizon performance-difference identity and a line integral of the costate error along each action bridge. It is useful even when action optimization is incomplete. The next result additionally quantifies that incompleteness with a finite, checkable calculation.

PROPOSITION S.33—A feasible-actor account: *Suppose $\hat{S}_k$ is twice continuously differentiable and $dD^2\hat{S}_k(x) \preceq \Lambda_k I_d$ everywhere. Let $m_{+,k}$ be the upper endpoint of $\mathcal{A}_k$, and put*

$$\Gamma_k = h^2 \Lambda_k - A_k / m_{+,k}^2, \quad g_{k,i} = A_k \{a_{k,i}^{-1} - \zeta \bar{a}_k\} - B_k - h \hat{q}_{k,i} \{x_{a_k}(Z_k)\}.$$

An upper bound on the actor's $\widehat{Q}_k$ maximization gap is

$$\delta_k = \max_{b \in \mathcal{A}_k} \left\{ \langle g_k, b - a_k \rangle_d + \frac{\Gamma_k}{2} \|b - a_k\|_d^2 \right\}. \quad (\text{S.114})$$

This is a separable scalar calculation. If $\Gamma_k < 0$, its maximizer is the coordinate projection of $a_k - g_k/\Gamma_k$. If $\Gamma_k \geq 0$, one compares the two endpoints in each coordinate. Thus (S.114) remains valid when global strong concavity of the learned continuation problem cannot be verified.

If $\mu_0 = \min_k (-\Gamma_k/h) > 0$ is verified and $\mathcal{D}_h = \sum_k \mathbb{E} \delta_k$, then

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) \geq \frac{1}{2} \mathcal{M}_h - \mathcal{D}_h - 2\mathcal{E}_h/\mu_0. \quad (\text{S.115})$$

For exact $\widehat{Q}$ maximizers, $\mathcal{D}_h = 0$ and the coefficient of $\mathcal{E}_h/\mu_0$ improves from two to one. These conclusions apply to the actual feasible action; a small training loss is not substituted for δ_k .

Independent occupation and continuation samples. Draw an independent candidate trajectory, K uniformly from $\{0, \dots, N-1\}$, and U uniformly from $[0, 1]$. At its bridge state, draw two independent continuation banks Z_1, Z_2 with conditional mean $q_K^\pi(X_{K,U})$. A bank may contain antithetic pairs; independence is required between the two banks, not between the members of an antithetic pair. The statistics

$$\begin{aligned} M &= N \{ \widehat{Q}_K(Z_K, a_K) - \widehat{Q}_K(Z_K, \pi_K) \}, & A &= T \|a_K - \pi_K \mathbf{1}\|_d^2, \\ E &= T \langle \widehat{q}_K(X_{K,U}) - Z_1, \widehat{q}_K(X_{K,U}) - Z_2 \rangle_d, & D &= N \delta_K \end{aligned} \quad (\text{S.116})$$

have expectations $\mathcal{M}_h, \mathcal{A}_h, \mathcal{E}_h, \mathcal{D}_h$. Individual values of E may be negative. Replacing them by their positive parts would change the estimand. Moreover, $E^{\text{raw}} = T \|Z_1 - Z_2\|_d^2/2$ estimates the integrated mean-square error of one raw bank on this same occupation law. A confidence bound for $E - E^{\text{raw}}$ therefore assesses the critic's reusable costate prediction against a direct bank with the declared number of rollouts. It concerns this evaluation subproblem; the direct method comparison continues to use the returned policies' independently assessed payoffs.

The supplement gives explicit clipping ranges and expectation allowances for these statistics. They exploit the common quadratic terminal component and anti-

thetic cancellation. They do not use an empirical maximum as a population bound. Independent, equally sized samples for every member of a declared finite stream distribution may be pooled using the maximum proved range and the mean numerical and tail allowances. The resulting target is the uniform mean over all declared streams.

THEOREM S.34—From fitted continuation to continuous economic gain: *Condition on the fitted objects and their selection. Suppose a fresh, simultaneous assessment gives $L_M \leq \mathcal{M}_h$, $\mathcal{A}_h \leq U_A$, $\mathcal{E}_h \leq U_E$, and, when used, $\mathcal{D}_h \leq U_D$. Include the disclosed arithmetic and state-representation errors in these endpoints. Define the deterministic holding deficit*

$$D_{\text{hold}} = \sum_k \left\{ C_k - A_k \log \pi_k + B_k \pi_k + \zeta A_k \pi_k^2 / 2 \right\} \geq 0.$$

For the sampled continuous-economy implementation of Proposition S.10, let $E_{a,h}^{\text{all}}$ be its saved paired payoff-transfer allowance. Then, on that same event,

$$J(a) - J(m_0) \geq L_M - \sqrt{\max(0, U_A) \max(0, U_E)} - D_{\text{hold}} - E_{a,h}^{\text{all}}. \quad (\text{S.117})$$

When the positive curvature modulus in Proposition S.33 is verified, one may also use $L_M/2 - U_D - 2 \max(0, U_E) / \mu_0$ before subtracting the same two allowances. Taking the larger of these two bounds spends no additional probability. Subtracting a valid lower gain bound from the analytical reference regret bound gives a bound against all adapted feasible controls in the original economy.

This account separates three observable economic requirements: a continuation that predicts a useful feasible change, an error small enough under the candidate's occupation law, and a sufficiently accurate implementation. The finite Bellman identity is exact; the transfer of payoffs to the diffusion is quantitative. Neither equality of discrete and continuous costates nor global concavity of the full adapted-control problem is assumed.

S.31.4. *Information and the implemented decision rule*

The economic institution is a controller that receives capital observations at specified dates, stores an internal state, and holds the selected consumption share until the next decision date. Its economic primitive is the original capital diffusion. Its internal numerical state need not equal the physical state. When the controller uses private randomization, that randomization is part of its specified information and is included in the law of the evaluated policy.

The exact-history representation in Proposition S.18 assumes continuous noiseless observation of every capital coordinate, known coefficients and actions, exact drift integration, and an independent private Brownian motion. Under those assumptions the observation history supplies a Brownian representation with the law required by the innovation-driven controller. It does not recover the original latent null-space shock path by path. Proposition S.19 states why the added independent randomization leaves the economic optimum unchanged. These are representation results for a randomized history-dependent policy.

Finite observations require a different implementation account. Proposition S.20 bounds the difference between the measured-state controller and its specified protected parent, retaining sensor error, the internal recurrence, action-map error, clipping, and arithmetic. Its allowance belongs to that parent and that time grid. The empirical comparison reports it alongside the gain measured for the same parent and population; a certificate for a different mesh is not substituted. Complete assumptions and proofs are in Appendix S.14.

S.31.5. *When is state-dependent management worth its fee?*

To give the policy differences a decision interpretation, consider a proportional resource cost $\tau \in [0, 1)$ for administering a state-dependent consumption rule. Gross withdrawals from capital remain $m_i e^{Y_i}$, and the service absorbs $\tau m_i e^{Y_i}$. Household consumption is $(1 - \tau)m_i e^{Y_i}$. Production, gross depletion, the gross-withdrawal adjustment cost, and terminal capital are unchanged. This is a self-financed resource experiment rather than an externally financed utility supplement. It changes the specified fee parameter, not the model used to train or benchmark the gross policy.

Let $A_T = \int_0^T e^{-\rho t} dt$. Log felicity gives the exact identity

$$J_{\mathcal{P}}^{\tau}(a) - J_{\mathcal{P}}(m_0) = J_{\mathcal{P}}(a) - J_{\mathcal{P}}(m_0) + A_T \log(1 - \tau). \quad (\text{S.118})$$

Hence a positive certified improvement L_a supports every fee τ satisfying

$$\tau < 1 - \exp(-L_a / A_T).$$

A negative upper endpoint after the fee rules out an improvement for that candidate under the confidence event; an interval spanning zero is inconclusive. The fee grid is fixed before the final simulation. We do not equate wall-clock cost to a monetary fee or call these deliberately stylized balance sheets a calibrated household population. The experiment instead states precisely which adoption decisions are supported by the verified utility differences.

S.32. DECISION ACCURACY, TRANSPORT, AND STOPPING: PROOFS

This section gives the complete arguments for the new economic error chain. The inherited verification, Bellman-bridge, and diffusion proofs are retained in the preceding sections. The constants below pertain to the new stored finite-law experiment unless a more general hypothesis is stated.

S.32.1. *Proof of Proposition 1*

First suppose an optimal action a^* exists. The approximate maximizing property implies

$$r_{\theta}(s, a^*) + \widehat{S}_0(T_{\theta}(s, a^*)) - r_{\theta}(s, \widehat{a}) - \widehat{S}_0(T_{\theta}(s, \widehat{a})) \leq \delta_{\theta}(s).$$

Substituting $S_{\theta} = \widehat{S}_0 - e + \Delta_{\theta}$ gives

$$\begin{aligned} Q_{\theta}(s, a^*) - Q_{\theta}(s, \widehat{a}) &\leq \delta_{\theta}(s) + e(T_{\theta}(s, \widehat{a})) - e(T_{\theta}(s, a^*)) \\ &\quad + \Delta_{\theta}(T_{\theta}(s, a^*)) - \Delta_{\theta}(T_{\theta}(s, \widehat{a})). \end{aligned}$$

Each difference is at most its action-set oscillation. For an unattained finite supremum, take an ϵ -maximizer and let ϵ decrease to zero. This proves (3) without a differentiability assumption.

For finite actions let $v_j = e(T_\theta(s, a_j)) - \bar{e}_\pi(s)$. In the weighted inner product $\langle x, y \rangle_\pi = \sum_j \pi_j x_j y_j$, the representer of $v_i - v_j$ has entries $1/\pi_i$ at i , $-1/\pi_j$ at j , and zero elsewhere. Its squared norm is $1/\pi_i + 1/\pi_j$. Cauchy–Schwarz gives

$$(v_i - v_j)^2 \leq (1/\pi_i + 1/\pi_j) \sum_k \pi_k v_k^2.$$

Maximizing over the finite pair set proves $\text{osc } e \leq \sqrt{C_\pi R_\pi}$. The transport difference is at most $2b_\theta$, proving (4). For random fitted objects the inequality is pathwise. With deterministic C_π , for example, $\mathbb{E} \sqrt{C_\pi R_\pi} \leq \sqrt{C_\pi \mathbb{E} R_\pi}$ whenever the expectation is finite. Repetition of the same random seed does not create a new draw for this expectation. Finally, if every nonoptimal action loses at least g and the displayed bound is below g , a nonoptimal implemented action would contradict the bound. $\square$

S.32.2. Proof of Theorem 2

Couple both economies by the same innovation sequence and the same initial postdecision state. If $\|X_k^\theta - X_k^0\| \leq d_k$, then the triangle inequality and (5) give

$$\|X_{k+1}^\theta - X_{k+1}^0\| \leq \epsilon_{\theta,k} + L_k d_k = d_{k+1}.$$

Induction starts at $d_0 = 0$. At each reward date, $|g_{\theta,k}(X_k^\theta) - g_{0,k}(X_k^0)| \leq \eta_{\theta,k} + H_k d_k$. Multiply by the nonnegative weights, sum including the terminal reward, and take expectations. The bound is uniform because each assumed bound is uniform on the invariant domain. No interchange of a supremum over future controls with a fixed-policy expectation is used.

For recursive evaluations write $D_k = \|V_{\theta,k} - V_{0,k}\|_\infty$. Add and subtract $\mathcal{G}_{\theta,k}(V_{0,k+1})$ to obtain $D_k \leq \nu_{\theta,k} + \beta_k D_{k+1}$. Backward substitution from $D_K \leq b_K$ gives (7). Both arguments require that every coupled state and every recursively evaluated utility belong to the domains on which the bounds were verified. The theorem makes no extension past these domains. If a domain is localized, an exit contribution must be bounded separately before using the result as a global guarantee. $\square$

S.32.3. Proof of Proposition 4

Strong concavity gives, for any feasible b ,

$$Q(b) - Q(a) \leq Q'(a)(b - a) - \kappa(b - a)^2/2.$$

In the interior, maximize this upper bound over all real $b - a$ to obtain $|Q'(a)|^2/(2\kappa)$.

At the lower endpoint restrict the maximization to $b - a \geq 0$. Its value is

$$\frac{(\max\{Q'(a), 0\})^2}{2\kappa}.$$

At the upper endpoint the analogous residual is $\max\{-Q'(a), 0\}$. Enlarging a bounded feasible half-interval to the half-line is conservative. This proves (10).

An outward derivative interval bounds the applicable residual, and a positive lower curvature bound enlarges the resulting nonnegative fraction. Thus each task certificate holds deterministically at every attempted stage. A first successful average certificate implies the average economic target; previous failures do not affect validity. If checks are stochastic, let B_j denote the event that attempted check j falsely certifies its candidate. The conditional hypothesis gives $\Pr(B_j) \leq \mathbb{E}\alpha_j$; an unattempted check contributes zero. The union bound yields $\Pr(\bigcup_j B_j) \leq \mathbb{E}\sum_j \alpha_j \leq \alpha$. This proof requires conditional validity for the data actually used after past decisions. A fixed-time interval repeatedly queried without such a property does not meet the hypothesis.

The work identity follows by summing the disjoint measured operations through the stop, including unsuccessful fits and tests. Under the stated exact linear cost model, subtraction gives $W_N(q) - W_R(q) = F_N - F_R - q(c_R - c_N)$, establishing the strict inequality in (11). Integer query counts use that strict inequality; a tie at an integer threshold is not a strict saving. Measured costs in the experiment are used directly and are not assumed to satisfy a linear model. $\square$

S.32.4. Proof of Corollary 3

Condition on the feature map and design, which are independent of the fitting noises. Least squares gives $\hat{\beta} - \beta = G_n^{-1}\Phi^\top \zeta$. For a fixed queried argument x , the

prediction error is a linear combination of the independent noises. Its sub-Gaussian variance parameter is at most

$$\sigma^2 \|\Phi G_n^{-1} \phi(x)\|^2 = \sigma^2 \phi(x)^\top G_n^{-1} \phi(x) \leq \sigma^2 \ell / n.$$

The two-sided sub-Gaussian bound and a union bound over the K distinct arguments therefore give

$$\Pr \left(\max_x |\hat{S}(x) - S(x)| > \sigma \sqrt{\frac{2\ell}{n} \log \frac{4K}{\alpha}} \right) \leq \alpha/2.$$

Uniformly on that event, the oscillation in every feasible finite action menu is at most twice the displayed bound. Proposition 1 with zero transport and (8) gives loss at most ε at every query. Full column rank and the leverage bound are separate hypotheses and must actually hold at the selected integer n . They are not consequences of the scalar sample-size inequality.

For the direct method, each prediction average has sub-Gaussian variance parameter at most σ^2/t . Dependence between predictions caused by common innovations across different arguments does not affect the union bound; independence is required only for the observations within an average, or an equivalent sub-Gaussian bound for that average. The same calculation with ℓ/n replaced by $1/t$ proves the second guarantee. A union bound over the two methods gives joint coverage $1 - \alpha$, without assuming independence between them. Reusing a cached prediction at exactly the same continuation identity and argument incurs no additional simulation count. This is why K , rather than the number of repeated requests, enters the work comparison.

Subtract the complete specified costs to obtain (9). The fixed term $F(n)$ includes dictionary construction, all labels, the least-squares solve and its validation; it is not just a neural update clock. Common selection and verification work cancels only under the explicit equal-cost hypothesis. Otherwise its difference remains. These are sufficient budgets and actual costs for the declared procedures, not statistical lower bounds on arbitrary algorithms. If $|\mathbb{E}\hat{S}(x) - S(x)| \leq B$ uniformly under a misspecified dictionary, the prediction bound acquires B ; if future continuation changes by at most b , the decision bound acquires a further $2b$. Thus the same

argument uses margin $\varepsilon - \delta - 2B - 2b > 0$, provided the stated bias bound is verified. A pointwise approximation error of an arbitrary basis expansion is not automatically a bound on the bias of its fitted least-squares projection. $\square$

S.32.5. Uniform curvature in the stored capital economy

Let $M = (1 - \mu)I + \mu \mathbf{1}\mathbf{1}^\top / d$. It is row stochastic and has infinity operator norm one. For $f(y) = \lambda \tanh(My)$, the inequalities $|\tanh'| \leq 1$ and $|\tanh''| \leq 2$ imply, along a scalar path with $\|y'\|_\infty \leq P$ and $\|y''\|_\infty \leq D$,

$$\|(f(y))''\|_\infty \leq \lambda(D + 2P^2). \quad (\text{S.119})$$

The postdecision map has first derivative $-h\mathbf{1}$ and second derivative zero, so initialize $P = h$, $D = 0$. Adding the first innovation does not change these derivatives. For each of the three future transitions set

$$B_k = \lambda(D + 2P^2), \quad D \leftarrow D + hB_k, \quad P \leftarrow (1 + h\lambda)P.$$

The order matters: B_k uses the preceding P, D . The second derivative of the future production reward is at most $W_k B_k$, since every stored W_k is positive. All future terms involving the fixed withdrawal are constant with respect to the current action.

For a vector v write $\text{range}(v) = \max_i v_i - \min_i v_i$. At the first shocked postdecision state a valid range bound for each stored path is

$$R_0 = \text{range}(y) + 2h\lambda + \text{range}(z_0).$$

Every subsequent transition satisfies $R_k = R_{k-1} + 2h\lambda + \text{range}(z_k)$ as an upper bound. The scalar drift and common withdrawal cancel from a coordinate range. For the terminal payoff $g(y) = -\gamma \text{Var}_d(y)/2$,

$$(g(y))'' = -\gamma \text{Var}_d(y') - \gamma \overline{(y - \bar{y}\mathbf{1})y''} \leq \gamma R_3 D_3.$$

The first term is nonpositive; dropping it is an upper bound, not a sign reversal. Combining these inequalities gives a uniform upper bound on the continuation

curvature,

$$S_0(x(y, a))'' \leq M(y) := \sum_{k=1}^3 W_k B_k + \gamma \mathbb{E}_{16}[R_3 D_3].$$

Since the current reward has second derivative $-A_0(w/a^2 + \eta)$, a valid lower bound for true strong concavity on the whole interval is

$$\kappa(y, \theta) = A_0\{w/\bar{a}^2 + \eta\} - M(y). \quad (\text{S.120})$$

The verifier evaluates this expression with outward arithmetic and requires a strictly positive lower endpoint for every task. A nonpositive result would be a verification failure, not a reason to delete that task. The published records retain every returned lower endpoint.

The true derivative is enclosed separately rather than inferred from the surrogate. For each stored path initialize $p_0 = -h1$. At a future state,

$$f'(y) = \lambda\{1 - \tanh^2(My)\}Mp, \quad p \leftarrow p + hf'(y).$$

Accumulate $W_k \overline{f'(y)}$ and the terminal derivative $-\gamma \overline{(y - \bar{y}1)p}$. Average over all sixteen atoms and add the current derivative $A_0(w/a - \eta a - \tau) - W_0$. Every operation is evaluated on enclosing intervals. Dependency between occurrences of the same variable may widen an interval, but cannot invalidate the enclosure.

S.32.6. Arithmetic enclosure and its implementation

The stored binary64 coefficients and innovation increments define the exact finite economy. Floating-point evaluation of a Gaussian generator is not being asserted to be exact sampling from a continuous Gaussian economy. The generator is used to construct and freeze a finite array; the subsequent scientific statement concerns that array with equal masses.

The arithmetic module encloses each binary operation by the adjacent representable numbers toward minus and plus infinity. It checks finite values, IEEE round-to-nearest behavior and gradual underflow. Products and quotients take extrema over endpoint combinations, with divisors bounded away from zero. Squaring includes zero when the argument interval contains zero. Balanced reductions

avoid interpreting a non-enclosing ordinary mean as an interval sum. Induction on an expression's operation graph proves inclusion for arithmetic expressions from enclosed inputs.

The nonlinear enclosure does not assume that a library's hyperbolic tangent is correctly rounded. For $0 \leq t \leq 1/2$, the exponential is enclosed by its degree-18 Taylor polynomial and the nonnegative tail bound

$$0 \leq e^t - \sum_{j=0}^{18} t^j / j! \leq \frac{t^{19} / 19!}{1 - t/20}.$$

All terms are enclosed using the same arithmetic. Exact binary scaling (checked on every replayed argument) reduces the argument to this interval; repeated squaring and reciprocation recover positive and negative arguments. The identity $\tanh x = (e^{2x} - 1) / (e^{2x} + 1)$ and monotonicity provide endpoint enclosures. For the auxiliary logarithm check, scaling to $[1, 2)$ gives $z = (x - 1) / (x + 1) \in [0, 1/3)$ and

$$\log x = 2 \sum_{j=0}^{31} \frac{z^{2j+1}}{2j+1} + R, \quad 0 \leq R \leq \frac{2z^{65}}{65(1-z^2)}.$$

The stored enclosing endpoints for $\log 2$ is independently checked with exact rational series bounds. Synthetic high-precision derivative tests check the implemented recursion on separate small-dimensional cases. These tests complement the analytical enclosure argument; they are not substitutes for it or additional economic observations.

S.32.7. Charge monotonicity and the economic tolerance

For an interior maximizing action, differentiate $Q'_\theta(y, a^*(\tau)) = 0$. Because the charge enters the derivative as $-A_0\tau$, the implicit-function identity is $Q'' \partial a^* / \partial \tau - A_0 = 0$, proving (29). For the constrained case take $\tau_2 > \tau_1$ and respective optimal actions a_2, a_1 . Adding their optimality inequalities leaves $A_0(\tau_2 - \tau_1)(a_2 - a_1) \leq 0$, hence $a_2 \leq a_1$. Uniqueness follows from strict concavity.

An external grant in the utility account satisfies $A_0 w \log(1 + g) = U$ exactly when $g = \exp(U / (A_0 w)) - 1$. This is a monotone normalization of a certified payoff loss. The published grant table uses ordinary binary64 evaluation and is explicitly

descriptive; the primary rigorous statement is the outward payoff-loss bound. No claim is made that increasing actual endogenous consumption by this amount leaves capital, future rewards, or the reference policy unchanged.

S.33. PROSPECTIVE EXPERIMENT: COMPLETE RECORD

The execution modules and the two finite economic inputs match the original implementation freeze. This integrated execution replays that fixed design; it is not an additional independent experiment. The original descriptive design file is not reconstructed byte for byte. Its replacement documents the unchanged executable choices and is explicitly identified as such. Three fitting/cache streams, two dimensions, six query volumes and seven procedures give 252 actual service runs. Their task prefixes are deliberately shared, so the 252 runs are not claimed to be statistically independent. Synthetic dimension-three algebra checks are not economic observations. No registered seed, budget, tolerance or architecture was changed using integrated outcomes.

The stored JSON records contain every selected action, per-task lower curvature bound, outward derivative endpoints, individual regret upper bounds, cache indices, attempted stage, time component, failure status, and post-stop diagnostic. The data report generator checks unique identities, the first-success stopping rule, positive curvature, arithmetic summaries, and that reported service time covers all timed components. These record-consistency checks supplement, rather than replace, the mathematical enclosure proof.

TABLE S.51: All complete procedure cells

d	Q	Procedure	Passes	Mean seconds	Max. regret upper
10	1	Current only	0/3	0.0268	4.43×10^{-4}
10	1	Quadratic	3/3	0.0363	4.98×10^{-6}
10	1	Radial basis	3/3	0.0992	1.14×10^{-7}
10	1	NBO	3/3	0.3361	4.51×10^{-7}
10	1	Shared actor	1/3	0.4602	2.60×10^{-4}

<i>d</i>	<i>Q</i>	Procedure	Passes	Mean seconds	Max. regret upper
10	1	Cached SAA	3/3	0.0429	2.30×10^{-7}
10	1	Enumerated	3/3	0.0894	1.86×10^{-18}
10	4	Current only	0/3	0.0281	2.76×10^{-4}
10	4	Quadratic	3/3	0.0588	2.62×10^{-6}
10	4	Radial basis	3/3	0.0558	5.47×10^{-7}
10	4	NBO	3/3	0.1759	3.78×10^{-7}
10	4	Shared actor	1/3	0.5178	1.89×10^{-4}
10	4	Cached SAA	3/3	0.0555	3.18×10^{-7}
10	4	Enumerated	3/3	0.0751	6.43×10^{-19}
10	16	Current only	0/3	0.0635	3.61×10^{-4}
10	16	Quadratic	3/3	0.0779	3.14×10^{-6}
10	16	Radial basis	3/3	0.0523	4.40×10^{-7}
10	16	NBO	3/3	0.1278	5.77×10^{-7}
10	16	Shared actor	0/3	0.6222	2.09×10^{-4}
10	16	Cached SAA	3/3	0.0614	1.74×10^{-7}
10	16	Enumerated	3/3	0.0727	1.26×10^{-18}
10	64	Current only	0/3	0.0838	3.76×10^{-4}
10	64	Quadratic	3/3	0.0982	2.58×10^{-6}
10	64	Radial basis	3/3	0.1552	3.24×10^{-7}
10	64	NBO	3/3	0.2071	5.42×10^{-7}
10	64	Shared actor	0/3	0.7488	1.71×10^{-4}
10	64	Cached SAA	3/3	0.1177	1.68×10^{-7}
10	64	Enumerated	3/3	0.1236	9.32×10^{-19}
10	256	Current only	0/3	0.2770	3.69×10^{-4}
10	256	Quadratic	3/3	0.2963	2.07×10^{-6}
10	256	Radial basis	3/3	0.3052	3.55×10^{-7}
10	256	NBO	3/3	0.3556	4.67×10^{-7}
10	256	Shared actor	0/3	1.1744	1.81×10^{-4}
10	256	Cached SAA	3/3	0.3300	1.91×10^{-7}
10	256	Enumerated	3/3	0.3766	9.34×10^{-19}

d	Q	Procedure	Passes	Mean seconds	Max. regret upper
10	1024	Current only	0/3	1.0979	3.72×10^{-4}
10	1024	Quadratic	3/3	1.1419	1.91×10^{-6}
10	1024	Radial basis	3/3	1.3165	3.34×10^{-7}
10	1024	NBO	3/3	1.1964	4.58×10^{-7}
10	1024	Shared actor	0/3	2.7992	2.02×10^{-4}
10	1024	Cached SAA	3/3	1.1947	1.79×10^{-7}
10	1024	Enumerated	3/3	1.4544	9.43×10^{-19}
50	1	Current only	0/3	0.0302	4.49×10^{-4}
50	1	Quadratic	3/3	0.0447	2.55×10^{-6}
50	1	Radial basis	3/3	0.0590	2.48×10^{-8}
50	1	NBO	3/3	0.0998	5.67×10^{-7}
50	1	Shared actor	1/3	1.4862	2.85×10^{-4}
50	1	Cached SAA	3/3	0.0496	5.80×10^{-8}
50	1	Enumerated	3/3	0.0490	1.41×10^{-18}
50	4	Current only	0/3	0.0435	4.22×10^{-4}
50	4	Quadratic	3/3	0.0602	2.24×10^{-6}
50	4	Radial basis	3/3	0.0858	2.56×10^{-8}
50	4	NBO	3/3	0.1145	8.26×10^{-7}
50	4	Shared actor	0/3	1.5160	9.92×10^{-4}
50	4	Cached SAA	3/3	0.0893	2.32×10^{-8}
50	4	Enumerated	3/3	0.0823	1.01×10^{-18}
50	16	Current only	0/3	0.0981	4.26×10^{-4}
50	16	Quadratic	3/3	0.1173	1.61×10^{-6}
50	16	Radial basis	3/3	0.1150	2.19×10^{-7}
50	16	NBO	3/3	0.1684	1.47×10^{-6}
50	16	Shared actor	0/3	1.6177	1.12×10^{-3}
50	16	Cached SAA	3/3	0.1206	4.52×10^{-8}
50	16	Enumerated	3/3	0.1614	1.10×10^{-18}
50	64	Current only	0/3	0.3343	4.04×10^{-4}
50	64	Quadratic	3/3	0.3358	1.76×10^{-6}

d	Q	Procedure	Passes	Mean seconds	Max. regret upper
50	64	Radial basis	3/3	0.3737	1.82×10^{-7}
50	64	NBO	3/3	0.4150	1.12×10^{-6}
50	64	Shared actor	0/3	2.0360	1.27×10^{-3}
50	64	Cached SAA	3/3	0.3505	4.83×10^{-8}
50	64	Enumerated	3/3	0.3898	8.26×10^{-19}
50	256	Current only	0/3	1.3214	3.70×10^{-4}
50	256	Quadratic	3/3	1.3370	1.47×10^{-6}
50	256	Radial basis	3/3	1.4488	1.94×10^{-7}
50	256	NBO	3/3	1.4148	1.09×10^{-6}
50	256	Shared actor	0/3	4.0480	1.24×10^{-3}
50	256	Cached SAA	3/3	1.4007	4.99×10^{-8}
50	256	Enumerated	3/3	1.6894	8.21×10^{-19}
50	1024	Current only	0/3	5.6282	3.73×10^{-4}
50	1024	Quadratic	3/3	5.8102	1.68×10^{-6}
50	1024	Radial basis	3/3	5.9134	2.27×10^{-7}
50	1024	NBO	3/3	5.7318	1.32×10^{-6}
50	1024	Shared actor	0/3	12.7202	1.12×10^{-3}
50	1024	Cached SAA	3/3	5.9296	5.19×10^{-8}
50	1024	Enumerated	3/3	6.7602	9.53×10^{-19}

TABLE S.49

COMPLETE STAGE-COST DECOMPOSITION AT 1,024 QUERIES

d	Procedure	Fixed/cache/fit	Queries	All checks	Total
10	Current only	0.002	0.000	1.085	1.098
10	Quadratic	0.003	0.016	1.098	1.142
10	Radial basis	0.003	0.113	1.095	1.316
10	NBO	0.057	0.035	1.094	1.196
10	Shared actor	0.542	0.001	2.183	2.799
10	Cached SAA	0.002	0.072	1.095	1.195
10	Enumerated	0.002	0.338	1.098	1.454
50	Current only	0.003	0.000	5.613	5.628
50	Quadratic	0.007	0.034	5.684	5.810
50	Radial basis	0.007	0.283	5.581	5.913
50	NBO	0.062	0.052	5.604	5.732
50	Shared actor	1.359	0.001	11.227	12.720
50	Cached SAA	0.003	0.249	5.635	5.930
50	Enumerated	0.003	1.168	5.577	6.760

Seconds, averaged over all three streams. Fixed cost includes model and task initialization. Total additionally includes durable stage records and measured timing metadata. Multiple declared stages are all charged; the table is not a selected-fit clock.

TABLE S.50

TEMPORARY-CHARGE DECISIONS AND EXTERNAL COMPENSATION

d	Procedure	Withdrawal	Change vs. zero charge	Grant bound
10	Current only	0.51494	-0.01567	1.54×10^{-3}
10	Quadratic	0.49789	-0.01469	7.65×10^{-6}
10	Radial basis	0.49800	-0.01469	7.85×10^{-7}
10	NBO	0.49805	-0.01469	1.78×10^{-6}
10	Shared actor	0.49785	-0.01476	8.22×10^{-4}
10	Cached SAA	0.49798	-0.01469	3.35×10^{-7}
10	Enumerated	0.49785	-0.01468	3.99×10^{-18}
50	Current only	0.51503	-0.01597	1.55×10^{-3}
50	Quadratic	0.49804	-0.01498	6.97×10^{-6}
50	Radial basis	0.49815	-0.01498	7.34×10^{-7}
50	NBO	0.49829	-0.01499	4.88×10^{-6}
50	Shared actor	0.49621	-0.01275	3.75×10^{-3}
50	Cached SAA	0.49812	-0.01498	1.28×10^{-7}
50	Enumerated	0.49803	-0.01497	4.03×10^{-18}

All 1,024 tasks and three streams. The zero-charge query uses the same fitted object and changes only the current charge. The external grant uses $A_0 w \log(1 + g)$ and does not change production or the future policy. Grant entries are descriptive binary64 conversions of the certified regret upper bounds; the primary machine certificate remains the regret bound itself.

S.34. PROOFS FOR POLICY COMPOSITION AND ECONOMIC CONTRASTS

PROOF OF THEOREM 5: Backward induction and order preservation give $V_t^* \geq V_t^\pi$. Suppose $0 \leq V_{t+1}^* - V_{t+1}^\pi \leq b$ on $\mathcal{D}_{t+1}$. The elementary supremum inequality and the uniform Lipschitz condition give

$$0 \leq T_t V_{t+1}^* - T_t V_{t+1}^\pi \leq \beta_t b.$$

Adding $T_t V_{t+1}^\pi - V_t^\pi = g_t^\pi$ shows that the sup-norm error is at most $\varepsilon_t + \beta_t b$. Iteration from the common terminal value proves (14), including the case of a zero Lipschitz constant. No assumption that π itself attains the optimal action is used. For the state envelope, suppose by induction that $V_{t+1}^* - V_{t+1}^\pi \leq B_{t+1}$. Then

$$\begin{aligned} V_t^*(s) - V_t^\pi(s) &\leq \sup_a \{r_t(s, a) + \beta_t P_t^a V_{t+1}^\pi(s) + \beta_t P_t^a B_{t+1}(s)\} - V_t^\pi(s) \\ &\leq g_t^\pi(s) + \beta_t \sup_a P_t^a B_{t+1}(s) \leq B_t(s). \end{aligned}$$

Invariance ensures that every expectation is taken on a domain on which the induction applies. These are pathwise deductions on the common certificate event, so no additional probability statement is introduced by the proof. *Q.E.D.*

S.34.0.0.1. *Why catalogue averages cannot be substituted.* Consider a two-date economy with zero first-date rewards. A deviation at the initial state can reach a state with a unit second-date gain that the implemented policy never visits. All local advantages at the implemented policy's visited states can be zero while the initial optimality gap equals one. Thus a bound only on that occupancy distribution cannot be substituted for (13). The same example shows why changing the supremum in (15) to the transition of π is generally invalid. Weighted sampling-norm results require their own distribution-shift or coverage assumptions. We make no such inference from the finite query banks.

PROOF OF THE BRIDGE (16): For the returned approximating policy $\pi_h \in \Pi_h$, write

$$\sup_{\pi} J(\pi) - J(\hat{\pi}) \leq \eta_{\text{class}} + \sup_{\pi \in \Pi_h} J(\pi) - J(\pi_h) + \eta_{\text{impl}}$$

$$\leq \eta_{\text{class}} + 2\eta_{\text{value}} + \sup_{\pi \in \Pi_h} J_h(\pi) - J_h(\pi_h) + \eta_{\text{impl}}.$$

Apply Theorem 5 to the last difference with the specified policy class and action information. If an enlarged approximating action class is used, its optimum is an upper bound and the inequality continues to hold. The conclusion for fixed opponents follows by taking this same supremum over each player's admissible deviations. A simultaneous profile of these inequalities is precisely the stated approximate best-response condition; no existence or fixed-point claim is inferred. *Q.E.D.*

PROOF OF PROPOSITION 6: For any positive feasible allowances, Hölder's inequality implies

$$H = \sum_t \left(\frac{a_t}{e_t^2} \right)^{1/3} (w_t e_t)^{2/3} \leq \left(\sum_t \frac{a_t}{e_t^2} \right)^{1/3} \left(\sum_t w_t e_t \right)^{2/3}.$$

Hence the objective is at least H^3/E^2 . The displayed allowances satisfy $\sum_t w_t e_t = E$ and attain equality. Since $0 \leq \lceil A_t/e_t^2 \rceil - A_t/e_t^2 < 1$, their integer observation cost exceeds the continuous bill by less than $\sum_t c_t$. At any date whose statistical design requires a minimum sample size, that requirement must also hold at the rounded budget; otherwise the design must be charged and the constrained allocation recomputed. Dates with zero propagation weight are omitted from this positive-weight allocation problem. *Q.E.D.*

S.34.1. Optimal-action intervals and comparative statics

The economic bands in Table XV use more information than a mean value-loss certificate. Let Q be continuously differentiable and κ -strongly concave on $[\ell, u]$, with unique maximizer a^* . For feasible a , let $r(a)$ be the directional residual in Proposition 4. Then

$$|a - a^*| \leq r(a)/\kappa. \quad (\text{S.121})$$

To see this for $a < a^*$, the optimality condition gives $Q'(a^*) \geq 0$ when $a^* = u$, and gives zero at an interior optimum. Strong concavity then implies $Q'(a) \geq \kappa(a^* - a)$. For $a > a^*$, the reverse argument gives $-Q'(a) \geq \kappa(a - a^*)$. At an endpoint the

positive-part residual gives exactly the appropriate one-sided inequality. The case $a = a^*$ is immediate.

Use the interval upper bound $\bar{r}_i$ and the strictly positive curvature lower bound $\underline{\kappa}_i$ for each task. Outward arithmetic and intersection with feasibility give

$$I_i = \left[a_i - \frac{\bar{r}_i}{\underline{\kappa}_i}, a_i + \frac{\bar{r}_i}{\underline{\kappa}_i} \right] \cap [\ell_i, u_i], \quad a_i^* \in I_i.$$

For the same tasks under a changed future θ and the anchor, the mean optimal response belongs to $q^{-1} \sum_i (I_{\theta,i} - I_{0,i})$. No independent-noise assumption is needed: these are deterministic inclusions for fixed finite laws. We use the final NBO actions, not the enumeration actions, to obtain the NBO response bands. The envelope over three fitting streams contains every corresponding inclusion; it does not turn three fixed streams into a population confidence interval. Printed endpoints are rounded outwards. A negative upper endpoint establishes a negative mean optimal response in that declared economy.

S.34.2. *Source and arithmetic audit*

The R21 audit reads every original R20 service record, verifies its stored SHA256 digest, reestablishes the inherited source identities, regenerates only the fixed task arguments, and recomputes the certificates at the saved actions. It does not retrain any predictor or replace any clock. All 984 attempted certificates and 120 additional zero-charge certificates are compared field for field with their original values. The attempted checks cover 319,062 task actions. The audit also verifies feasibility, stopping order, recorded success flags, every cost decomposition, and the exact binary scaling used in the interval exponential implementation. This is an arithmetic and source audit of the specified finite model, not an empirical assessment of external economic calibration.

S.35. COMPLETE FUTURE-CHANGE SERVICE RECORD

TABLE S.52

CERTIFIED CURRENT WITHDRAWALS AND CHARGE RESPONSES

d	Future	Mean optimal withdrawal	Reduction due to charge
10	Anchor	[0.4962333, 0.4962334]	[0.0149696, 0.0149697]
10	Lower future withdrawal	[0.4958117, 0.4958118]	[0.0149393, 0.0149394]
10	Higher future withdrawal	[0.4967919, 0.4967920]	[0.0150070, 0.0150071]
10	Higher future production	[0.4644415, 0.4644416]	[0.0131771, 0.0131772]
10	Higher future payoff weight	[0.4803394, 0.4803395]	[0.0140592, 0.0140593]
50	Anchor	[0.4992611, 0.4992612]	[0.0152073, 0.0152074]
50	Lower future withdrawal	[0.4988337, 0.4988338]	[0.0151774, 0.0151775]
50	Higher future withdrawal	[0.4998262, 0.4998263]	[0.0152447, 0.0152448]
50	Higher future production	[0.4673148, 0.4673149]	[0.0133774, 0.0133775]
50	Higher future payoff weight	[0.4832926, 0.4832927]	[0.0142782, 0.0142783]

The 1,024-task catalogue is fixed in each dimension. Intervals enclose the finite-law optimizers, not a population mean. Endpoints are rounded outwards. The zero-charge comparison holds the future and every other task primitive fixed. Enumeration gives the same deterministic target in every stream; one identical certificate is printed rather than treating duplicates as observations.

TABLE S.53: All frozen future-change services

d	q	Stream	Procedure	Seconds	Futures	Attempts	Bound
10	1	200501	Reuse	1.374	3/5	5	2.71×10^{-3}
10	1	200501	Adapt	1.037	5/5	7	9.99×10^{-7}
10	1	200501	Refit	0.678	5/5	5	4.90×10^{-6}
10	1	200501	Quad	0.381	5/5	7	5.44×10^{-5}
10	1	200501	RBF	0.432	5/5	7	1.90×10^{-7}
10	1	200501	SAA	0.334	5/5	5	1.21×10^{-7}
10	1	200501	Enum	0.336	5/5	5	1.20×10^{-18}
10	1	200502	Reuse	0.350	3/5	5	2.69×10^{-3}
10	1	200502	Adapt	1.041	5/5	7	2.37×10^{-6}
10	1	200502	Refit	0.681	5/5	5	2.47×10^{-6}
10	1	200502	Quad	0.382	5/5	7	4.97×10^{-5}
10	1	200502	RBF	0.395	5/5	7	4.08×10^{-7}

	d	q	Stream	Procedure	Seconds	Futures	Attempts	Bound	
1									1
2	10	1	200502	SAA	0.334	5/5	5	3.74×10^{-7}	2
3	10	1	200502	Enum	0.336	5/5	5	1.20×10^{-18}	3
4	10	1	200503	Reuse	0.351	3/5	5	2.68×10^{-3}	4
5	10	1	200503	Adapt	1.035	5/5	7	1.64×10^{-6}	5
6	10	1	200503	Refit	0.679	5/5	5	4.06×10^{-7}	6
7	10	1	200503	Quad	0.375	5/5	7	5.10×10^{-5}	7
8	10	1	200503	RBF	0.389	5/5	7	4.99×10^{-7}	8
9	10	1	200503	SAA	0.334	5/5	5	1.98×10^{-7}	9
10	10	1	200503	Enum	0.335	5/5	5	1.20×10^{-18}	10
11	10	16	200501	Reuse	0.462	3/5	5	1.86×10^{-3}	11
12	10	16	200501	Adapt	1.189	5/5	7	1.65×10^{-6}	12
13	10	16	200501	Refit	0.790	5/5	5	4.63×10^{-6}	13
14	10	16	200501	Quad	0.535	5/5	7	2.12×10^{-5}	14
15	10	16	200501	RBF	0.547	5/5	7	2.09×10^{-6}	15
16	10	16	200501	SAA	0.449	5/5	5	7.23×10^{-8}	16
17	10	16	200501	Enum	0.463	5/5	5	9.85×10^{-19}	17
18	10	16	200502	Reuse	0.468	3/5	5	1.84×10^{-3}	18
19	10	16	200502	Adapt	1.190	5/5	7	2.33×10^{-6}	19
20	10	16	200502	Refit	0.788	5/5	5	3.97×10^{-6}	20
21	10	16	200502	Quad	0.530	5/5	7	2.05×10^{-5}	21
22	10	16	200502	RBF	0.548	5/5	7	2.96×10^{-6}	22
23	10	16	200502	SAA	0.447	5/5	5	3.59×10^{-7}	23
24	10	16	200502	Enum	0.463	5/5	5	9.85×10^{-19}	24
25	10	16	200503	Reuse	0.462	3/5	5	1.82×10^{-3}	25
26	10	16	200503	Adapt	1.193	5/5	7	1.15×10^{-6}	26
27	10	16	200503	Refit	0.786	5/5	5	2.46×10^{-6}	27
28	10	16	200503	Quad	0.533	5/5	7	2.09×10^{-5}	28
29	10	16	200503	RBF	0.546	5/5	7	1.05×10^{-6}	29
30	10	16	200503	SAA	0.452	5/5	5	7.30×10^{-8}	30
31	10	16	200503	Enum	0.462	5/5	5	9.85×10^{-19}	31
32	10	256	200501	Reuse	2.393	3/5	5	1.71×10^{-3}	32
	10	256	200501	Adapt	3.637	5/5	7	1.79×10^{-6}	

	d	q	Stream	Procedure	Seconds	Futures	Attempts	Bound	
1									1
2									2
3	10	256	200501	Refit	2.544	5/5	5	5.07×10^{-6}	3
4	10	256	200501	Quad	2.945	5/5	7	1.83×10^{-5}	4
5	10	256	200501	RBF	2.973	5/5	7	2.04×10^{-6}	5
6	10	256	200501	SAA	2.215	5/5	5	1.54×10^{-7}	6
7	10	256	200501	Enum	2.758	5/5	5	1.29×10^{-18}	7
8	10	256	200502	Reuse	2.192	3/5	5	1.70×10^{-3}	8
9	10	256	200502	Adapt	3.606	5/5	7	2.40×10^{-6}	9
10	10	256	200502	Refit	2.534	5/5	5	4.63×10^{-6}	10
11	10	256	200502	Quad	2.912	5/5	7	1.80×10^{-5}	11
12	10	256	200502	RBF	2.959	5/5	7	2.58×10^{-6}	12
13	10	256	200502	SAA	2.226	5/5	5	8.40×10^{-7}	13
14	10	256	200502	Enum	2.725	5/5	5	1.29×10^{-18}	14
15	10	256	200503	Reuse	2.174	3/5	5	1.69×10^{-3}	15
16	10	256	200503	Adapt	3.623	5/5	7	1.67×10^{-6}	16
17	10	256	200503	Refit	2.515	5/5	5	3.28×10^{-6}	17
18	10	256	200503	Quad	2.908	5/5	7	1.80×10^{-5}	18
19	10	256	200503	RBF	2.876	5/5	7	9.70×10^{-7}	19
20	10	256	200503	SAA	2.152	5/5	5	9.17×10^{-8}	20
21	10	256	200503	Enum	2.602	5/5	5	1.29×10^{-18}	21
22	10	1024	200501	Reuse	8.235	3/5	5	1.73×10^{-3}	22
23	10	1024	200501	Adapt	11.987	5/5	7	1.61×10^{-6}	23
24	10	1024	200501	Refit	8.423	5/5	5	4.91×10^{-6}	24
25	10	1024	200501	Quad	10.997	5/5	7	1.90×10^{-5}	25
26	10	1024	200501	RBF	12.495	5/5	7	1.90×10^{-6}	26
27	10	1024	200501	SAA	8.244	5/5	5	1.49×10^{-7}	27
28	10	1024	200501	Enum	10.423	5/5	5	1.38×10^{-18}	28
29	10	1024	200502	Reuse	8.302	3/5	5	1.72×10^{-3}	29
30	10	1024	200502	Adapt	11.835	5/5	7	2.14×10^{-6}	30
31	10	1024	200502	Refit	8.372	5/5	5	4.06×10^{-6}	31
32	10	1024	200502	Quad	10.917	5/5	7	1.89×10^{-5}	32
	10	1024	200502	RBF	12.223	5/5	7	2.34×10^{-6}	
	10	1024	200502	SAA	8.104	5/5	5	8.20×10^{-7}	

	d	q	Stream	Procedure	Seconds	Futures	Attempts	Bound	
1									1
2									2
3	10	1024	200502	Enum	10.224	5/5	5	1.38×10^{-18}	3
4	10	1024	200503	Reuse	8.141	3/5	5	1.70×10^{-3}	4
5	10	1024	200503	Adapt	11.814	5/5	7	1.86×10^{-6}	5
6	10	1024	200503	Refit	8.434	5/5	5	3.11×10^{-6}	6
7	10	1024	200503	Quad	11.064	5/5	7	1.89×10^{-5}	7
8	10	1024	200503	RBF	12.295	5/5	7	8.78×10^{-7}	8
9	10	1024	200503	SAA	8.205	5/5	5	9.73×10^{-8}	9
10	10	1024	200503	Enum	10.067	5/5	5	1.38×10^{-18}	10
11	50	1	200501	Reuse	0.406	3/5	5	2.32×10^{-3}	11
12	50	1	200501	Adapt	1.207	5/5	7	1.49×10^{-6}	12
13	50	1	200501	Refit	0.756	5/5	5	1.45×10^{-6}	13
14	50	1	200501	Quad	0.487	5/5	7	2.58×10^{-5}	14
15	50	1	200501	RBF	0.492	5/5	7	1.44×10^{-6}	15
16	50	1	200501	SAA	0.390	5/5	5	2.82×10^{-8}	16
17	50	1	200501	Enum	0.389	5/5	5	3.35×10^{-18}	17
18	50	1	200502	Reuse	0.405	3/5	5	2.21×10^{-3}	18
19	50	1	200502	Adapt	1.198	5/5	7	5.29×10^{-6}	19
20	50	1	200502	Refit	0.755	5/5	5	8.34×10^{-6}	20
21	50	1	200502	Quad	0.480	5/5	7	2.52×10^{-5}	21
22	50	1	200502	RBF	0.488	5/5	7	3.57×10^{-6}	22
23	50	1	200502	SAA	0.385	5/5	5	3.66×10^{-6}	23
24	50	1	200502	Enum	0.389	5/5	5	3.35×10^{-18}	24
25	50	1	200503	Reuse	0.404	3/5	5	2.31×10^{-3}	25
26	50	1	200503	Adapt	1.204	5/5	7	3.36×10^{-6}	26
27	50	1	200503	Refit	0.760	5/5	5	3.76×10^{-7}	27
28	50	1	200503	Quad	0.488	5/5	7	3.33×10^{-5}	28
29	50	1	200503	RBF	0.498	5/5	7	1.69×10^{-6}	29
30	50	1	200503	SAA	0.390	5/5	5	5.10×10^{-8}	30
31	50	1	200503	Enum	0.392	5/5	5	3.35×10^{-18}	31
32	50	16	200501	Reuse	0.906	3/5	5	1.76×10^{-3}	32
	50	16	200501	Adapt	1.916	5/5	7	1.89×10^{-6}	
	50	16	200501	Refit	1.252	5/5	5	5.47×10^{-6}	

	d	q	Stream	Procedure	Seconds	Futures	Attempts	Bound	
1									1
2									2
3	50	16	200501	Quad	1.165	5/5	7	2.26×10^{-5}	3
4	50	16	200501	RBF	1.184	5/5	7	1.40×10^{-6}	4
5	50	16	200501	SAA	0.890	5/5	5	3.35×10^{-8}	5
6	50	16	200501	Enum	0.959	5/5	5	1.34×10^{-18}	6
7	50	16	200502	Reuse	0.914	3/5	5	1.73×10^{-3}	7
8	50	16	200502	Adapt	1.902	5/5	7	2.90×10^{-6}	8
9	50	16	200502	Refit	1.257	5/5	5	8.42×10^{-6}	9
10	50	16	200502	Quad	1.156	5/5	7	1.99×10^{-5}	10
11	50	16	200502	RBF	1.182	5/5	7	2.92×10^{-6}	11
12	50	16	200502	SAA	0.891	5/5	5	2.60×10^{-6}	12
13	50	16	200502	Enum	0.958	5/5	5	1.34×10^{-18}	13
14	50	16	200503	Reuse	0.897	3/5	5	1.76×10^{-3}	14
15	50	16	200503	Adapt	1.891	5/5	7	3.09×10^{-6}	15
16	50	16	200503	Refit	1.247	5/5	5	1.02×10^{-5}	16
17	50	16	200503	Quad	1.159	5/5	7	2.06×10^{-5}	17
18	50	16	200503	RBF	1.182	5/5	7	1.89×10^{-6}	18
19	50	16	200503	SAA	0.889	5/5	5	3.80×10^{-8}	19
20	50	16	200503	Enum	0.955	5/5	5	1.34×10^{-18}	20
21	50	256	200501	Reuse	9.053	3/5	5	1.77×10^{-3}	21
22	50	256	200501	Adapt	13.465	5/5	7	2.60×10^{-6}	22
23	50	256	200501	Refit	9.510	5/5	5	7.35×10^{-6}	23
24	50	256	200501	Quad	12.394	5/5	7	1.99×10^{-5}	24
25	50	256	200501	RBF	12.963	5/5	7	1.97×10^{-6}	25
26	50	256	200501	SAA	9.429	5/5	5	4.41×10^{-8}	26
27	50	256	200501	Enum	11.484	5/5	5	1.37×10^{-18}	27
28	50	256	200502	Reuse	9.212	3/5	5	1.77×10^{-3}	28
29	50	256	200502	Adapt	13.364	5/5	7	2.98×10^{-6}	29
30	50	256	200502	Refit	9.340	5/5	5	1.22×10^{-5}	30
31	50	256	200502	Quad	12.384	5/5	7	1.99×10^{-5}	31
32	50	256	200502	RBF	12.746	5/5	7	3.30×10^{-6}	32
	50	256	200502	SAA	9.475	5/5	5	2.07×10^{-6}	
	50	256	200502	Enum	11.758	5/5	5	1.37×10^{-18}	

d	q	Stream	Procedure	Seconds	Futures	Attempts	Bound
50	256	200503	Reuse	9.161	3/5	5	1.78×10^{-3}
50	256	200503	Adapt	13.339	5/5	7	2.53×10^{-6}
50	256	200503	Refit	9.520	5/5	5	9.90×10^{-6}
50	256	200503	Quad	12.733	5/5	7	2.05×10^{-5}
50	256	200503	RBF	12.817	5/5	7	1.49×10^{-6}
50	256	200503	SAA	9.425	5/5	5	3.98×10^{-8}
50	256	200503	Enum	11.663	5/5	5	1.37×10^{-18}
50	1024	200501	Reuse	41.590	3/5	5	1.74×10^{-3}
50	1024	200501	Adapt	60.744	5/5	7	2.30×10^{-6}
50	1024	200501	Refit	42.633	5/5	5	6.50×10^{-6}
50	1024	200501	Quad	56.906	5/5	7	2.03×10^{-5}
50	1024	200501	RBF	60.718	5/5	7	1.69×10^{-6}
50	1024	200501	SAA	44.279	5/5	5	4.23×10^{-8}
50	1024	200501	Enum	51.981	5/5	5	1.37×10^{-18}
50	1024	200502	Reuse	42.006	3/5	5	1.73×10^{-3}
50	1024	200502	Adapt	58.318	5/5	7	2.71×10^{-6}
50	1024	200502	Refit	41.533	5/5	5	1.05×10^{-5}
50	1024	200502	Quad	58.127	5/5	7	2.00×10^{-5}
50	1024	200502	RBF	63.043	5/5	7	2.88×10^{-6}
50	1024	200502	SAA	41.933	5/5	5	2.10×10^{-6}
50	1024	200502	Enum	48.720	5/5	5	1.37×10^{-18}
50	1024	200503	Reuse	41.230	3/5	5	1.74×10^{-3}
50	1024	200503	Adapt	58.550	5/5	7	2.58×10^{-6}
50	1024	200503	Refit	43.815	5/5	5	8.89×10^{-6}
50	1024	200503	Quad	58.705	5/5	7	2.06×10^{-5}
50	1024	200503	RBF	61.957	5/5	7	1.52×10^{-6}
50	1024	200503	SAA	41.510	5/5	5	4.12×10^{-8}
50	1024	200503	Enum	49.673	5/5	5	1.37×10^{-18}

S.35.1. Mechanism and complete work components

The following finite-law diagnostics average the same fixed tasks and streams within each method and future. Absolute prediction risk, action-centered risk,

and own-action risk are distinct from decision loss. Every stored per-service field remains available in the original execution records.

TABLE S.54: Common-action and own-action prediction diagnostics

d	Procedure	Future	Absolute	Centered	Own action	Menu loss
10	Reuse	Anchor	1.48×10^{-6}	2.50×10^{-8}	1.46×10^{-6}	4.29×10^{-8}
10	Reuse	Low	7.08×10^{-2}	3.53×10^{-8}	7.08×10^{-2}	5.63×10^{-7}
10	Reuse	High	1.98×10^{-6}	9.15×10^{-8}	2.00×10^{-6}	2.82×10^{-7}
10	Reuse	Production	2.89×10^{-3}	1.17×10^{-4}	2.94×10^{-3}	5.32×10^{-4}
10	Reuse	Utility	1.04×10^0	2.73×10^{-5}	1.04×10^0	1.45×10^{-4}
10	Adapt	Anchor	1.48×10^{-6}	2.50×10^{-8}	1.46×10^{-6}	4.29×10^{-8}
10	Adapt	Low	7.08×10^{-2}	3.53×10^{-8}	7.08×10^{-2}	5.63×10^{-7}
10	Adapt	High	1.98×10^{-6}	9.15×10^{-8}	2.00×10^{-6}	2.82×10^{-7}
10	Adapt	Production	4.43×10^{-6}	1.27×10^{-7}	4.36×10^{-6}	7.38×10^{-7}
10	Adapt	Utility	2.21×10^{-6}	4.54×10^{-8}	2.17×10^{-6}	6.31×10^{-8}
10	Refit	Anchor	1.48×10^{-6}	2.50×10^{-8}	1.46×10^{-6}	4.29×10^{-8}
10	Refit	Low	1.68×10^{-6}	2.52×10^{-8}	1.67×10^{-6}	4.85×10^{-9}
10	Refit	High	1.26×10^{-6}	2.38×10^{-8}	1.24×10^{-6}	5.49×10^{-8}
10	Refit	Production	1.58×10^{-5}	2.77×10^{-7}	1.57×10^{-5}	1.82×10^{-6}
10	Refit	Utility	6.90×10^{-6}	9.53×10^{-8}	6.84×10^{-6}	1.47×10^{-7}
10	Quad	Anchor	2.21×10^{-6}	1.36×10^{-7}	2.11×10^{-6}	6.03×10^{-7}
10	Quad	Low	7.08×10^{-2}	1.62×10^{-7}	7.08×10^{-2}	1.13×10^{-6}
10	Quad	High	2.51×10^{-6}	1.87×10^{-7}	2.40×10^{-6}	9.11×10^{-7}
10	Quad	Production	1.32×10^{-5}	1.31×10^{-6}	1.29×10^{-5}	1.26×10^{-6}
10	Quad	Utility	5.57×10^{-6}	4.92×10^{-7}	5.47×10^{-6}	1.31×10^{-6}
10	RBF	Anchor	1.30×10^{-6}	1.16×10^{-8}	1.29×10^{-6}	1.39×10^{-8}
10	RBF	Low	7.07×10^{-2}	4.05×10^{-8}	7.07×10^{-2}	3.04×10^{-7}
10	RBF	High	2.10×10^{-6}	6.11×10^{-8}	2.14×10^{-6}	5.99×10^{-7}
10	RBF	Production	3.58×10^{-6}	1.16×10^{-7}	3.55×10^{-6}	6.32×10^{-7}
10	RBF	Utility	1.91×10^{-6}	4.46×10^{-8}	1.89×10^{-6}	1.42×10^{-7}
10	SAA	Anchor	1.09×10^{-6}	2.34×10^{-9}	1.08×10^{-6}	7.60×10^{-9}
10	SAA	Low	1.18×10^{-6}	1.76×10^{-9}	1.18×10^{-6}	0.00×10^0

d	Procedure	Future	Absolute	Centered	Own action	Menu loss
10	SAA	High	9.70×10^{-7}	3.03×10^{-9}	9.52×10^{-7}	0.00×10^0
10	SAA	Production	7.16×10^{-6}	2.37×10^{-8}	7.12×10^{-6}	7.09×10^{-8}
10	SAA	Utility	4.35×10^{-6}	9.38×10^{-9}	4.32×10^{-6}	8.03×10^{-8}
10	Enum	Anchor	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
10	Enum	Low	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
10	Enum	High	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
10	Enum	Production	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
10	Enum	Utility	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
50	Reuse	Anchor	3.32×10^{-6}	6.84×10^{-8}	3.31×10^{-6}	3.83×10^{-7}
50	Reuse	Low	7.11×10^{-2}	8.01×10^{-8}	7.11×10^{-2}	2.49×10^{-7}
50	Reuse	High	3.01×10^{-6}	1.32×10^{-7}	3.05×10^{-6}	5.73×10^{-7}
50	Reuse	Production	2.90×10^{-3}	1.19×10^{-4}	2.97×10^{-3}	5.04×10^{-4}
50	Reuse	Utility	1.04×10^0	2.78×10^{-5}	1.04×10^0	1.25×10^{-4}
50	Adapt	Anchor	3.32×10^{-6}	6.84×10^{-8}	3.31×10^{-6}	3.83×10^{-7}
50	Adapt	Low	7.11×10^{-2}	8.01×10^{-8}	7.11×10^{-2}	2.49×10^{-7}
50	Adapt	High	3.01×10^{-6}	1.32×10^{-7}	3.05×10^{-6}	5.73×10^{-7}
50	Adapt	Production	1.35×10^{-5}	1.71×10^{-7}	1.33×10^{-5}	6.41×10^{-7}
50	Adapt	Utility	5.94×10^{-6}	6.40×10^{-8}	5.85×10^{-6}	1.08×10^{-7}
50	Refit	Anchor	3.32×10^{-6}	6.84×10^{-8}	3.31×10^{-6}	3.83×10^{-7}
50	Refit	Low	3.43×10^{-6}	6.23×10^{-8}	3.43×10^{-6}	3.64×10^{-7}
50	Refit	High	3.12×10^{-6}	7.32×10^{-8}	3.11×10^{-6}	4.24×10^{-7}
50	Refit	Production	2.51×10^{-5}	5.76×10^{-7}	2.50×10^{-5}	9.14×10^{-7}
50	Refit	Utility	1.24×10^{-5}	1.74×10^{-7}	1.24×10^{-5}	6.82×10^{-7}
50	Quad	Anchor	4.03×10^{-6}	1.38×10^{-7}	3.91×10^{-6}	5.97×10^{-7}
50	Quad	Low	7.11×10^{-2}	1.64×10^{-7}	7.11×10^{-2}	9.25×10^{-7}
50	Quad	High	3.40×10^{-6}	1.87×10^{-7}	3.26×10^{-6}	2.72×10^{-7}
50	Quad	Production	2.09×10^{-5}	1.38×10^{-6}	2.06×10^{-5}	2.47×10^{-6}
50	Quad	Utility	8.81×10^{-6}	5.21×10^{-7}	8.68×10^{-6}	4.59×10^{-7}
50	RBF	Anchor	2.96×10^{-6}	1.73×10^{-8}	2.96×10^{-6}	0.00×10^0
50	RBF	Low	7.11×10^{-2}	2.88×10^{-8}	7.11×10^{-2}	0.00×10^0
50	RBF	High	2.92×10^{-6}	8.45×10^{-8}	2.95×10^{-6}	1.26×10^{-7}
50	RBF	Production	1.22×10^{-5}	1.37×10^{-7}	1.22×10^{-5}	1.99×10^{-7}

d	Procedure	Future	Absolute	Centered	Own action	Menu loss
50	RBF	Utility	5.49×10^{-6}	5.35×10^{-8}	5.44×10^{-6}	0.00×10^0
50	SAA	Anchor	2.87×10^{-6}	4.55×10^{-9}	2.86×10^{-6}	0.00×10^0
50	SAA	Low	2.99×10^{-6}	4.17×10^{-9}	2.99×10^{-6}	4.62×10^{-8}
50	SAA	High	2.70×10^{-6}	4.87×10^{-9}	2.68×10^{-6}	0.00×10^0
50	SAA	Production	2.55×10^{-5}	4.82×10^{-8}	2.55×10^{-5}	2.58×10^{-7}
50	SAA	Utility	1.15×10^{-5}	1.82×10^{-8}	1.15×10^{-5}	4.71×10^{-8}
50	Enum	Anchor	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
50	Enum	Low	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
50	Enum	High	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
50	Enum	Production	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0
50	Enum	Utility	0.00×10^0	0.00×10^0	0.00×10^0	0.00×10^0

REFERENCES

- BARLES, GUY AND PANAGIOTIS E. SOUGANIDIS (1991): "Convergence of Approximation Schemes for Fully Nonlinear Second Order Equations," *Asymptotic Analysis*, 4, 271–283. [28]
- BORKAR, VIVEK S. (2024): "Stochastic Approximation with Two Time Scales: The General Case," ArXiv:2412.19872v1. [19]
- CRANDALL, MICHAEL G., HITOSHI ISHII, AND PIERRE-LOUIS LIONS (1992): "User's Guide to Viscosity Solutions of Second Order Partial Differential Equations," *Bulletin of the American Mathematical Society*, 27 (1), 1–67. [28]
- KAKADE, SHAM AND JOHN LANGFORD (2002): "Approximately Optimal Approximate Reinforcement Learning," in *Proceedings of the Nineteenth International Conference on Machine Learning*, Morgan Kaufmann, 267–274. [202]
- MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): "Empirical Bernstein Bounds and Sample Variance Penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [51, 57, 71, 186, 202]
- PIROTTA, MATTEO, MARCELLO RESTELLI, ALESSIO PECORINO, AND DANIELE CALANDRIELLO (2013): "Safe Policy Iteration," in *Proceedings of the 30th International Conference on Machine Learning*, vol. 28 of *Proceedings of Machine Learning Research*, 307–315. [202]
- SCHERRER, BRUNO, MOHAMMAD GHAVAMZADEH, VICTOR GABILLON, BORIS LESNER, AND MATTHIEU GEIST (2015): "Approximate Modified Policy Iteration and its Application to the Game of Tetris," *Journal of Machine Learning Research*, 16 (49), 1629–1676. [202]